

GATES LEARJET CORPORATION
maintenance manual

CHAPTER

32

LANDING GEAR

Gates Learjet Corporation

RECORD OF TEMPORARY REVISIONS

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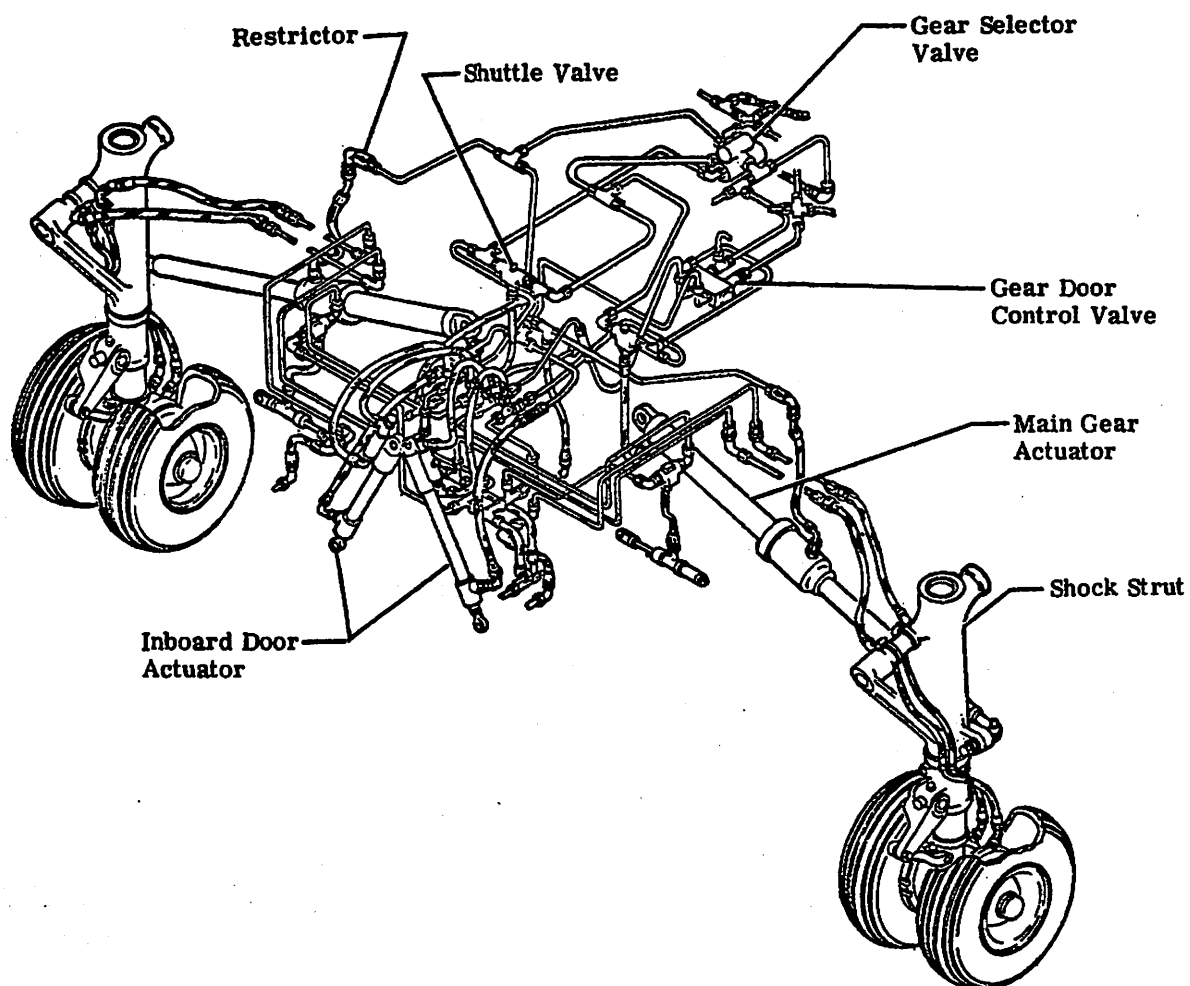
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LANDING GEAR - DESCRIPTION AND OPERATION

1. Description (See figures 1 and 2.)
 - A. The aircraft is equipped with a retractable tricycle-type landing gear with dual main gear wheels.
 - B. The nose gear consists of the nose wheel and chined tire mounted on a conventional air-hydraulic shock strut which is housed in and attached to the fuselage structure by bearing plates. The nose gear is steerable through an electrical steering actuator. The nose gear wheel well is covered by two hinge mounted doors which are opened and closed mechanically through linkage connected to the nose gear strut.
 - C. The main landing gear consists of a conventional air-hydraulic shock strut, dual wheels, and an anti-skid brake system. Each main gear strut is attached to the wing at two points. Additional support is obtained from the main gear actuator. Each main gear is enclosed during flight by an inboard and an outboard door. The outboard door is mechanically connected to the landing gear strut by an adjustable link. The inboard door is opened and closed by a hydraulic actuator.
 - D. Landing gear extension and retraction is accomplished through hydraulic system pressure. The landing gear is held in the retracted position by up-latch hooks. An emergency system permits pneumatic blowdown of the landing gear.
 - E. The primary brake system includes four power brake valves, six shuttle valves, two parking brake valves, an anti-skid system and four hydraulic fuses. An integral braking system is incorporated to stop the wheel spinning before the gear is fully retracted. An emergency system permits pneumatic pressure for emergency braking.
2. Operation
 - A. Normal Landing Gear Operation (See figure 3.)
 - (1) The landing gear is retracted or extended by operating the landing gear selector switch. Setting the switch to UP initiates a sequence of all three gear systems. The inboard main gear doors are opened, the gear is retracted, and the doors are then closed. The nose gear doors are connected by linkage to the nose gear and close when the gear retracts. Setting the switch to DN reverses the sequence.

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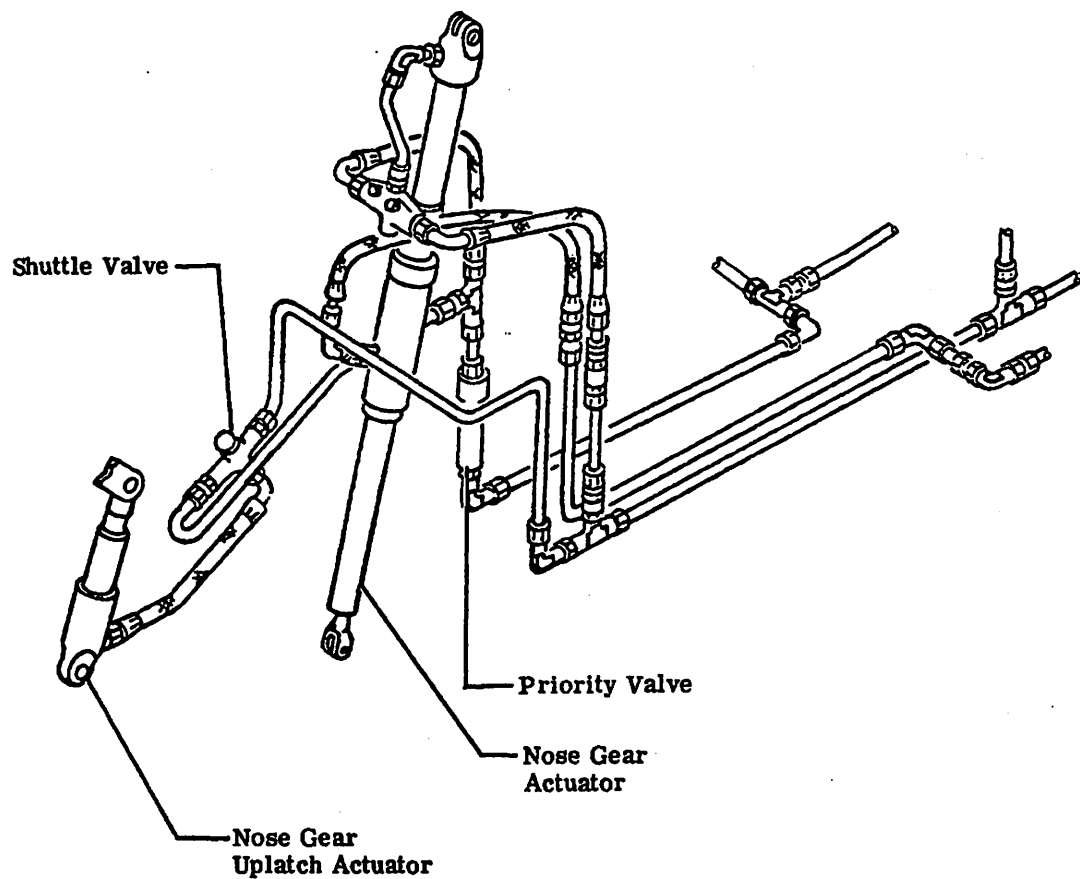
Main Landing Gear Hydraulic Components Installation
Figure 1

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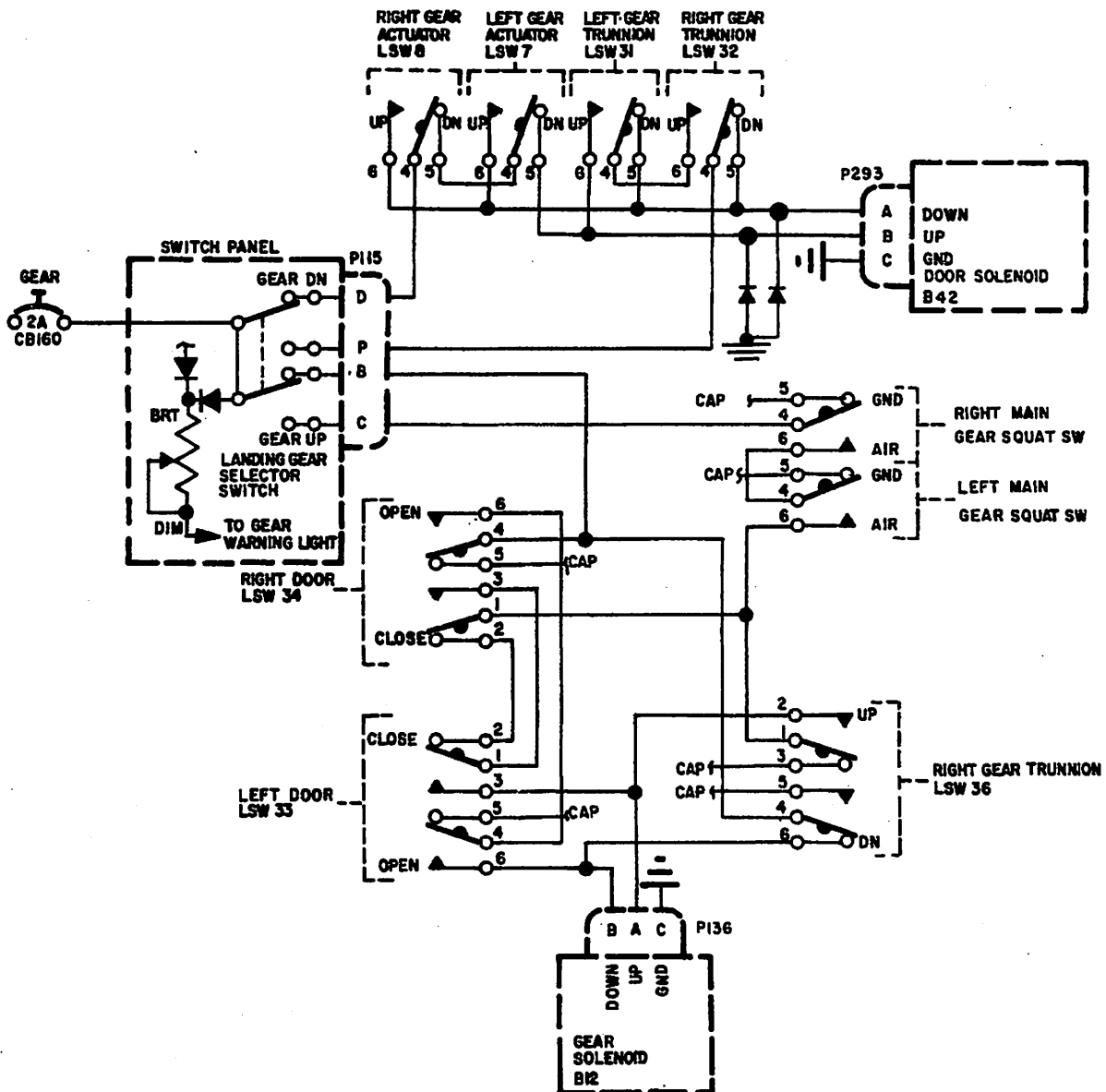
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**Nose Landing Gear Hydraulic Components Installation
Figure 2**

EFFECTIVITY: ALL

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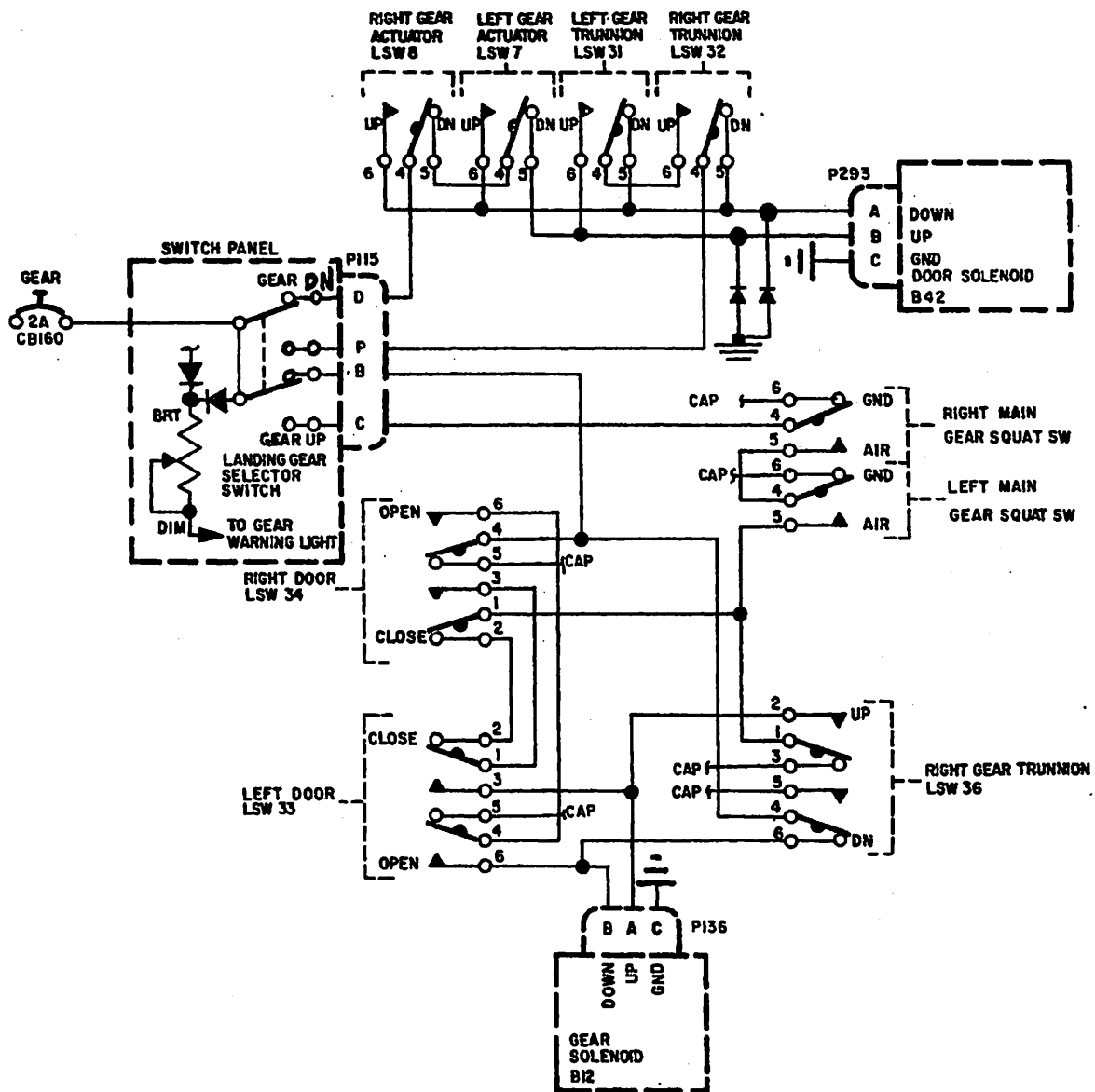
Landing Gear Electrical Control Schematic
 Figure 3 (Sheet 1 of 2)

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Landing Gear Electrical Control Schematic
Figure 3 (Sheet 2 of 2)

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B. Emergency Landing Gear Operation

CAUTION: THE LANDING GEAR SELECTOR SWITCH IS FIRST SET TO THE GEAR DOWN POSITION TO AVOID SUBSEQUENT ACCIDENTAL RETRACTION OF THE GEAR AND TO OBTAIN PROPER SAFE INDICATIONS FROM THE GEAR POSITION LIGHTS; ALSO THE RED LEFT AND RIGHT UNSAFE LIGHTS WILL BE ILLUMINATED DUE TO THE MAIN GEAR INBOARD DOORS REMAINING OPEN IN THE EMERGENCY EXTENSION MODE.

- (1) In the event of main hydraulic system failure, landing gear selector system malfunction, or electrical system failure, the gear can be extended pneumatically. Emergency extension is initiated by depressing the emergency gear extension handle on the left side of the pedestal. Air under pressure is then directed from the emergency air bottle through lines and shuttle valves to:
 - (a) The down side of the gear selector and door control valves.
 - (b) The nose gear uplatch actuator.
 - (c) The down side of the nose gear actuator.
 - (d) The main gear inboard door uplatch actuators.
 - (e) The down side of the main gear inboard door actuators.
 - (f) The down side of the main gear actuators. After landing, the EMERG BRAKE handle, located on the left side of the pedestal, is pushed down until adequate braking is achieved.
- (2) The emergency air system to the landing gear is operated by the emergency gear extension handle, located on the left side of the pedestal. The handle remains in the down position when depressed. Air pressure is bled from the gear system through an overboard vent by pulling the ratchet spring release on the emergency gear extension handle. This allows the handle to return to the off position.

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LANDING GEAR - TROUBLE SHOOTING

1. Trouble Shooting

CAUTION: IF EMERGENCY GEAR BLOWDOWN WAS REQUIRED, ASSURE THAT THE LANDING GEAR SELECTOR SWITCH IS IN THE GEAR DOWN POSITION TO AVOID SUBSEQUENT RETRACTION OF THE GEAR.

A. Perform trouble shooting procedure.

NOTE: The procedures indicated in Figure 101 must be accomplished in conjunction with these trouble shooting procedures.

- (1) Perform following landing gear switch adjustments:
 - (a) Adjustment of gear up switches. Refer to 32-30-01.
 - (b) Adjustment of right gear down safety switch. Refer to 32-30-02.
 - (c) Adjustment of main gear down and locked switches. Refer to 32-60-01.
 - (d) Adjustment of door down switches. Refer to 32-30-03.
 - (e) Adjustment of main gear squat switches. Refer to 32-30-04.
 - (f) Adjustment of main gear door uplatch switches. Refer to 32-60-02.
 - (g) Adjustment of nose gear up and locked switch. Refer to 32-60-04.
 - (h) Adjustment of nose gear down and locked switch. Refer to 32-60-03.
 - (i) Adjustment of throttle aural warning switches. Refer to Chapter 76.
- (2) Place aircraft on jacks.
- (3) Pull GEAR circuit breaker and depressurize hydraulic system. Refer to Chapter 29.

NOTE: If emergency gear blowdown was required, the main gear inboard doors will be open. Pull GEAR circuit breaker and omit steps (4) and (5).

- (4) Using a screwdriver to override the uplock actuator, release uplock hook from roller.

CAUTION: THE BRAKES MUST BE PUMPED DURING MANUAL LOWERING OF THE INBOARD GEAR DOOR TO PREVENT DOOR ACTUATOR SEAL DAMAGE AND POSSIBLE NOSE GEAR VACUUMING TO UP CYCLE.

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- (5) To lower main gear inboard door, station a man at the door and a man in the cockpit to operate brakes. With one man pumping brakes, the man stationed at the gear door must pull it to its down position.
- (6) Disconnect main gear inboard doors by removing bolts at lower end of each actuator.
- (7) Disconnect main gear outboard doors and nose gear door linkage.
- (8) When checking a particular electrical system circuit, check voltage and ground points at all valve and switch locations.

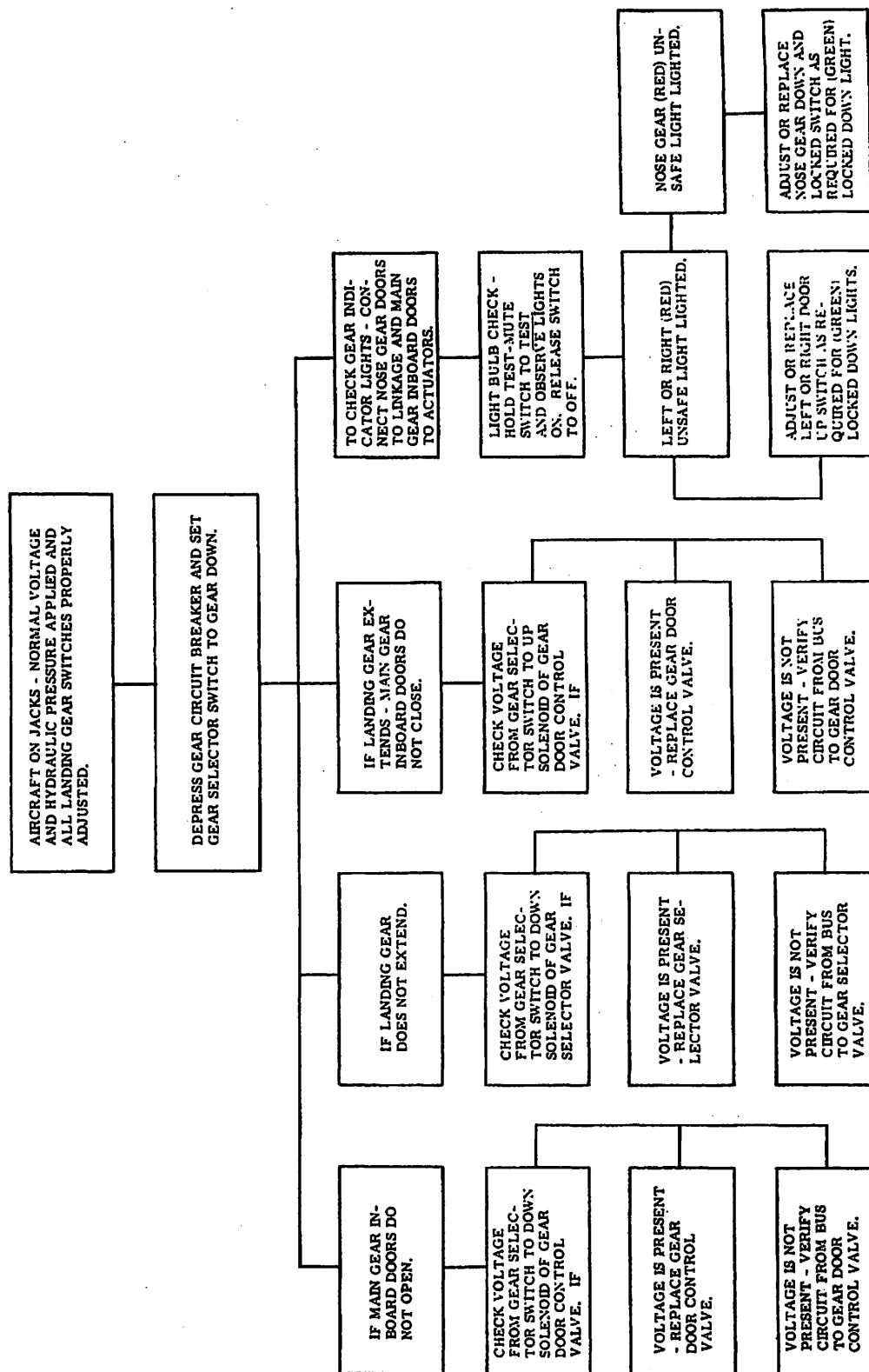
CAUTION: IF EMERGENCY GEAR BLOWDOWN HAS OCCURRED, ASSURE THAT ALL AIR IS RELEASED FROM THE LANDING GEAR SYSTEM LINES. TO PREVENT POSSIBLE RESERVOIR DAMAGE, THIS MUST BE ACCOMPLISHED BEFORE PERFORMING NORMAL HYDRAULIC SYSTEM BLEEDING PROCEDURES.

- (9) If emergency air gear extension has occurred, bleed air from hydraulic system (refer to Chapter 29), and perform extension system Adjustment/Test (return to normal operation), (refer to 32-33-00).
- (10) If emergency air braking has occurred, bleed air from hydraulic system (refer to Chapter 29), and perform brake system Adjustment/Test (return to normal operation), (refer to 32-44-00).
- (11) Connect main gear outboard doors after completion of trouble shooting.

CAUTION: DO NOT REMOVE AIRCRAFT FROM JACKS UNTIL THE LANDING GEAR IS FULLY EXTENDED TO THE DOWN AND LOCKED POSITION WITH ALL GREEN LOCKED DN LIGHTS ILLUMINATED.

- (12) Cycle gear for proper operation. During cycling, observe main gear inboard door actuators for proper operation.
- (13) Remove aircraft from jacks.

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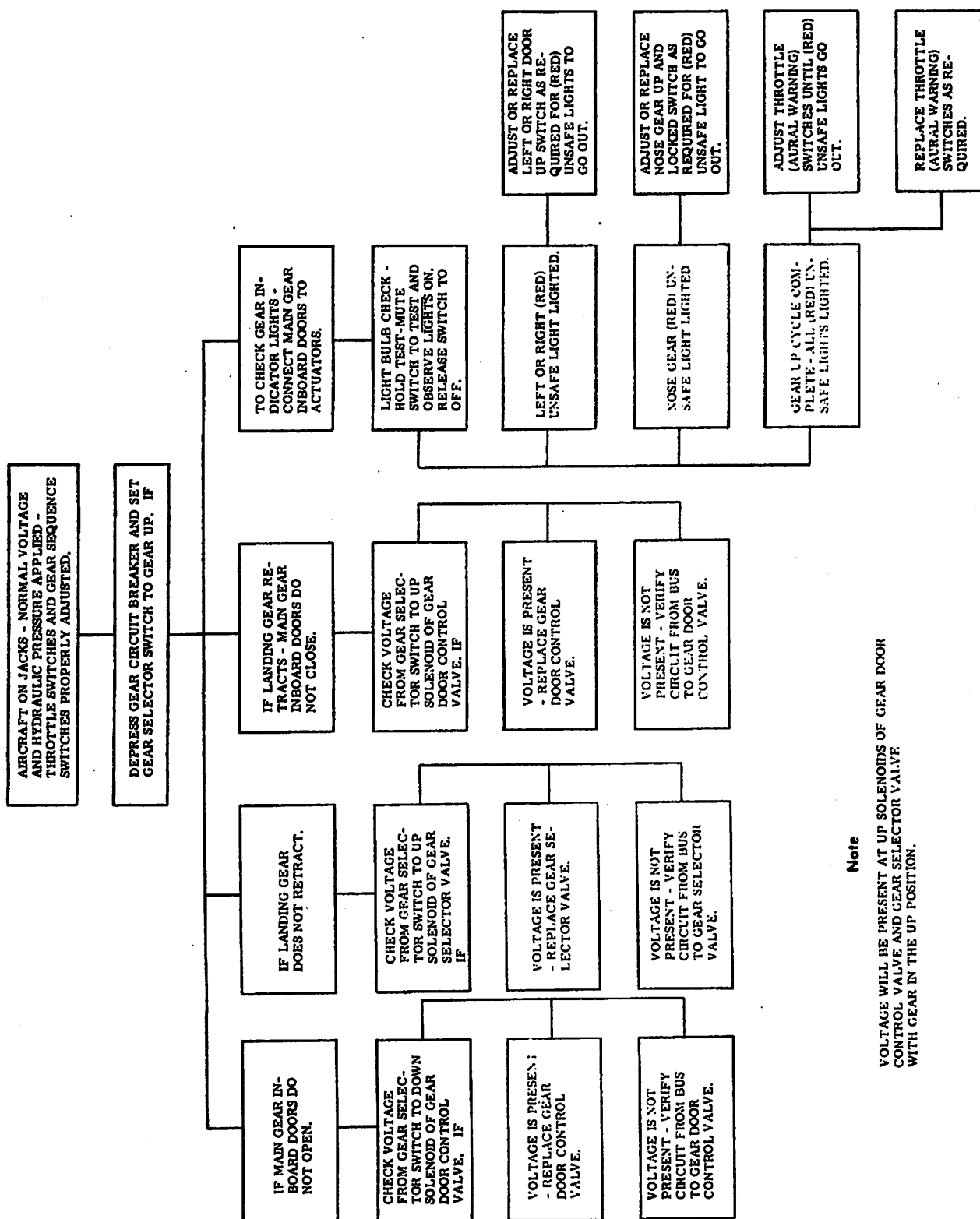


Note
WITH GEAR IN THE DOWN POSITION, VOLTAGE WILL BE PRESENT AT THE UP SOLENOID OF THE GEAR DOOR CONTROL VALVE AND AT THE DOWN SOLENOID OF THE GEAR SELECTOR VALVE.

Landing Gear System Trouble Shooting
Figure 101 (Sheet 1 of 2)

EFFECTIVITY: ALL

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Note

VOLTAGE WILL BE PRESENT AT UP SOLENOIDS OF GEAR DOOR CONTROL VALVE AND GEAR SELECTOR VALVE WITH GEAR IN THE UP POSITION.

Landing Gear System Trouble Shooting
Figure 101 (Sheet 2 of 2)

EFFECTIVITY: ALL

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MAIN GEAR AND DOORS - DESCRIPTION AND OPERATION

1. Description

- A. Each main landing gear consists of dual wheels and brakes attached to a shock strut.
- B. Each main gear is attached to the wing at two points. Additional support is obtained through the main gear actuator attach points.
- C. Normally the gear is extended and retracted hydraulically. A pneumatic system is installed for emergency gear blowdown and emergency braking.
- D. Multiple disc brakes are installed on each main landing gear and are operated by valves connected to the rudder pedals.
- E. The inboard main gear doors are operated by hydraulic actuators while the outboard main gear doors are operated mechanically by linkage connected to the gear.
- F. Component Description.
 - (1) Each shock strut is a conventional air-hydraulic unit designed to absorb the shock of landing, takeoff, and taxiing. It is essentially a steel piston riding in a cylinder and utilizing O-rings to prevent leakage between the piston and the outer cylinder. An air-oil filler valve is installed on the strut for servicing purposes.
 - (2) Each main gear is enclosed by an inboard and an outboard door. The inboard door is hydraulically operated. The outboard door is mechanically connected to the landing gear strut by an adjustable link. The doors are hinged from the structure by continuous-type hinges.

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MAIN GEAR SHOCK STRUT - MAINTENANCE PRACTICES

1. Removal/Installation

- NOTE:
- Removal and installation of right and left main gear shock struts is identical.
 - During disassembly, tag and keep all parts together for assembly into the same unit. Discard all O-rings, backup rings, seal rings, felt wiper and rod scraper.
 - It is not necessary to remove the anti-skid generator during disassembly of main landing gear shock strut.

A. Remove Main Gear Shock Strut. (See figures 201 and 202.)

- (1) Place aircraft on jacks.
- (2) Extend flaps and spoilers.
- (3) Remove electrical power from aircraft.
- (4) Depressurize hydraulic system.
- (5) Remove main gear actuator. Refer to 32-31-01.
- (6) Remove clamps securing electrical wiring and hydraulic hoses to gear strut.
- (7) Remove squat switch and landing light from strut and tie them to aircraft structure out of the immediate work area. This will eliminate cutting and splicing of electrical wiring.
- (8) Remove nut and bolt attaching outboard door link to trunnion.
- (9) Disconnect hydraulic brake hoses from brake assemblies. Cap lines and plug ports in brakes.
- (10) Remove hydraulic lines and clamp assembly from main gear strut. Tie hydraulic lines and clamp assembly to structure out of the immediate work area.

NOTE: If desired, wheels may be removed at this point (refer to 32-42-00), and brakes may be removed at this point, (refer to 32-43-01). If wheel and brake removal is performed, provide protection for axle.

- (11) Disconnect anti-skid electrical wiring.
- (12) Remove access door, inboard of gear hinge point, on spar 7 to gain access to aft trunnion pin.
- (13) Remove bolt from aft side of trunnion, releasing switch striker.

WARNING: REMOVAL OF AFT TRUNNION PIN RELEASES THE MAIN GEAR FROM THE AIRCRAFT. ADEQUATE SUPPORT MUST BE GIVEN THE GEAR TO PREVENT IT FROM FALLING AND CAUSING BODILY INJURY TO PERSONNEL OR STRUCTURAL DAMAGE TO THE AIRCRAFT.

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- (14) Provide protection for plumbing in vicinity of aft trunnion pin. Remove round access cover located on aft side of spar 8. This access cover is located on wing spar 8 just below inboard end of spoiler and is secured by six screws.
- (15) Fabricate puller block and slide hammer fork similar to that shown in figure 201.
- (16) Screw an AN8-20 bolt into the puller block. Screw bolt into hole in trunnion pin a few threads. Place spacers between puller block and trunnion bearing and screw block down against spacers. Tighten bolt to draw trunnion pin out. As bolt bottoms into trunnion pin hole, back bolt out, screw puller block down against spacers, and retighten bolt. Continue this process until lip on trunnion pin is exposed enough to put a pry bar under.

NOTE:

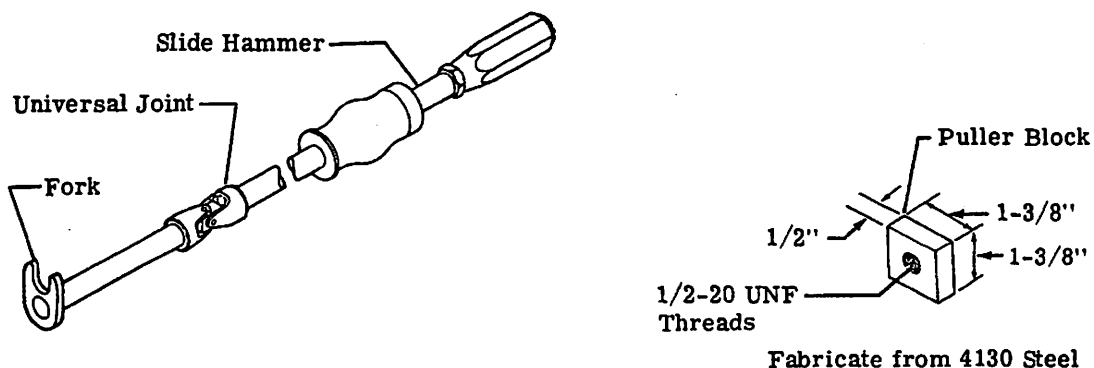
- When tightening bolt it will be helpful if a socket, extension, universal joint, and ratchet are used through the access hole in spar 8.
- Proper alignment (strut support) of aft trunnion pin to strut fitting will allow easier pin removal.

- (17) Position slidehammer fork through access hole in wing spar 8 and hook fork over head of bolt which is screwed into trunnion pin.
- (18) Extract trunnion pin by alternately using slidehammer and prying on trunnion pin lip.

NOTE: As trunnion pin is removed, shims will be released. Be sure to retain shims for reinstallation.

B. Install Main Gear Shock Strut. (See figure 202.)

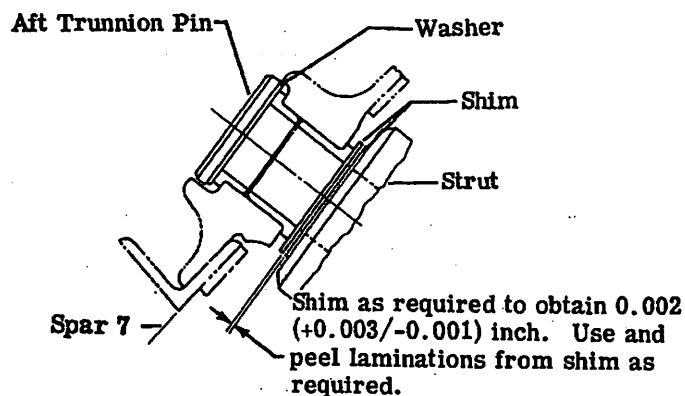
- (1) If forward trunnion pin was removed from shock strut, install and secure with nut and bolt.



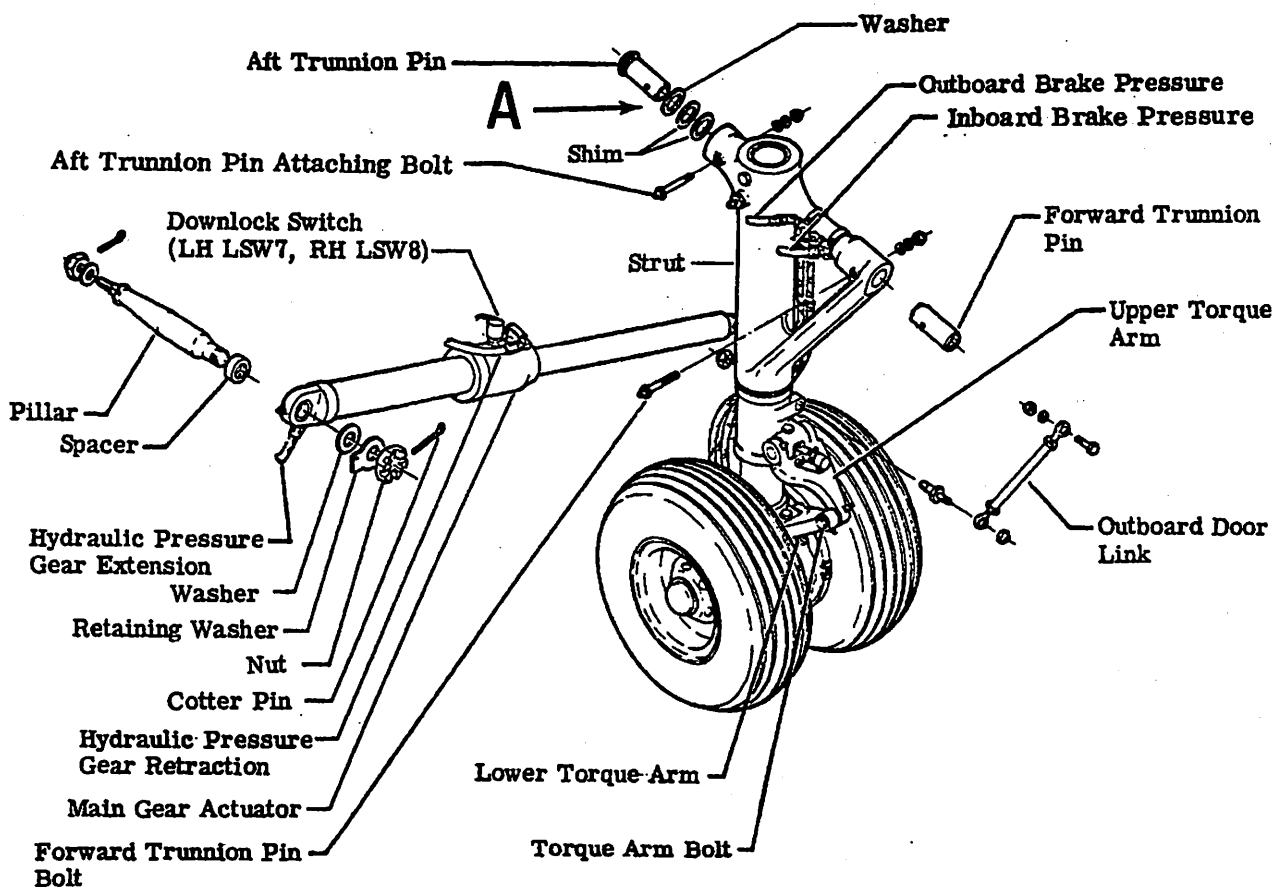
Aft Trunnion Pin Removal Tools
Figure 201

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Detail A



Main Gear Shock Strut Installation Figure 202

EFFECTIVITY: ALL

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- (2) Screw an AN8-20 bolt temporarily into aft trunnion pin to assist in trunnion pin installation.
- (3) Raise shock strut and insert forward trunnion pin into forward trunnion bearing.
- (4) Align shock strut with aft trunnion fitting, place shims on aft trunnion pin, and insert aft trunnion pin from aft side of spar 7 through trunnion bearing. Place same number of shims as previously removed on aft trunnion pin (between bearing plate and strut).
- (5) Insert aft trunnion pin through aft trunnion fitting into shock strut, assuring that shims are still in place and not damaged.
- (6) Align bolt hole in aft trunnion pin with bolt hole in shock strut and temporarily install trunnion pin attaching bolt. Check tolerances as shown in figure 202, detail A.
- (7) Remove trunnion pin attaching bolt and align holes in switch striker with holes aligned in step (6). Reinsert trunnion pin attaching bolt and secure with washers and nut.
- (8) Remove AN8-20 bolt from aft trunnion pin.
- (9) Install access door on spars 7 and 8.
- (10) Install brake hydraulic hoses and clamps on main gear strut and connect hoses to brake ports.
- (11) Connect anti-skid electrical wiring.
- (12) Attach outboard door link to main gear strut and secure with bolt and nut.
- (13) Install and secure main gear actuator. (Refer to 32-31-01.)

NOTE: A gap of 0.001 to 0.036 inch must exist between bearing surface of actuator pin bushing and actuator pin to assure pins' freedom of movement.

- (14) Install main gear squat switch and landing light.
- (15) Adjust squat switch. (Refer to 32-30-05.)
- (16) Install clamps securing electrical wiring to gear strut.
- (17) Restore electrical power to aircraft.
- (18) Remove aircraft from jacks.
- (19) Service main gear shock strut. (Refer to Chapter 12.)

2. Approved Repairs

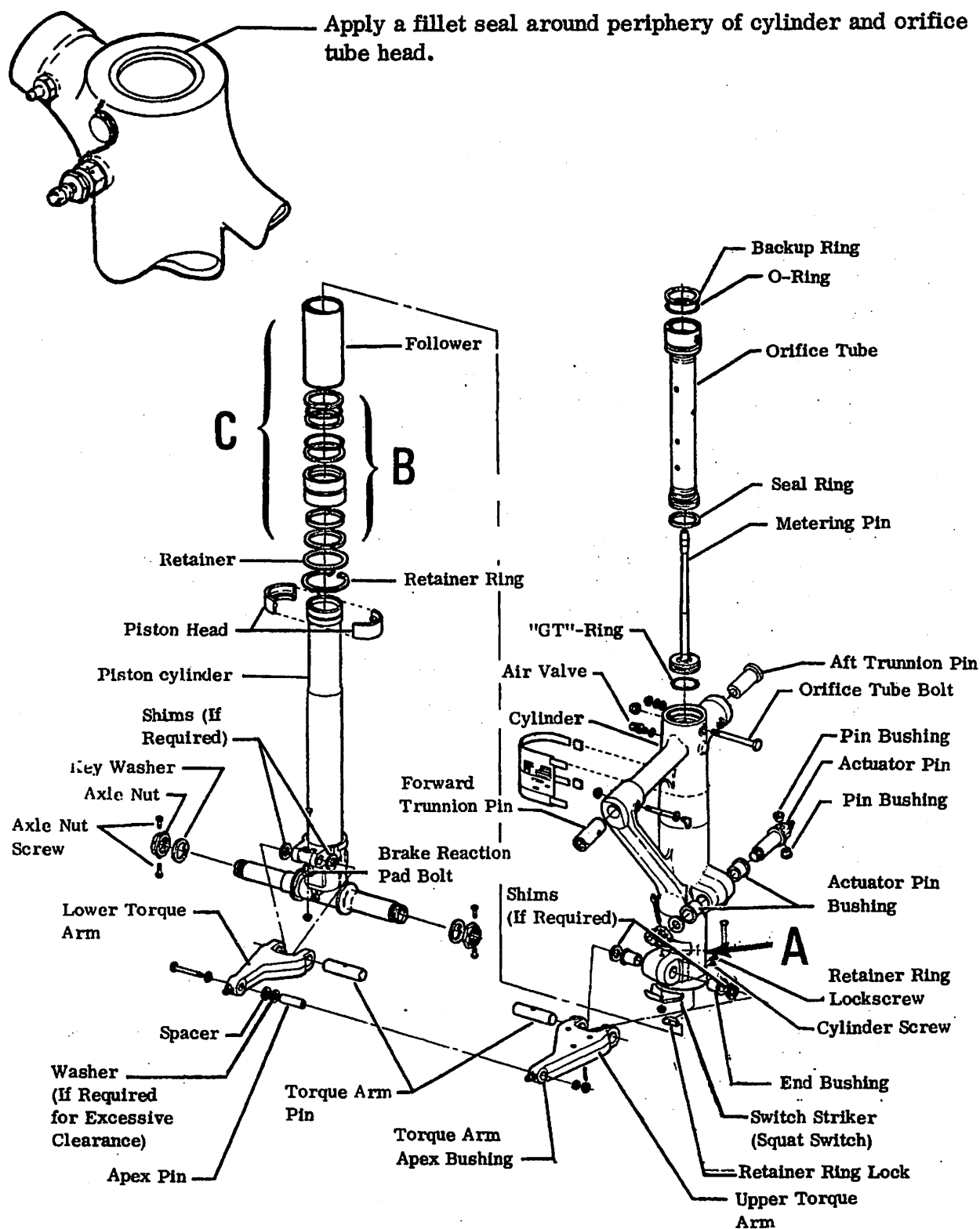
A. Disassemble Main Gear Shock Strut. (See figure 203.)

WARNING: DO NOT RELEASE AIR BY LOOSENING OR REMOVING AIR VALVE. FAILURE TO COMPLY CAN CAUSE BODILY INJURY.

EFFECTIVITY: ALL

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Main Gear Shock Strut Assembly
Figure 203 (Sheet 1 of 4)

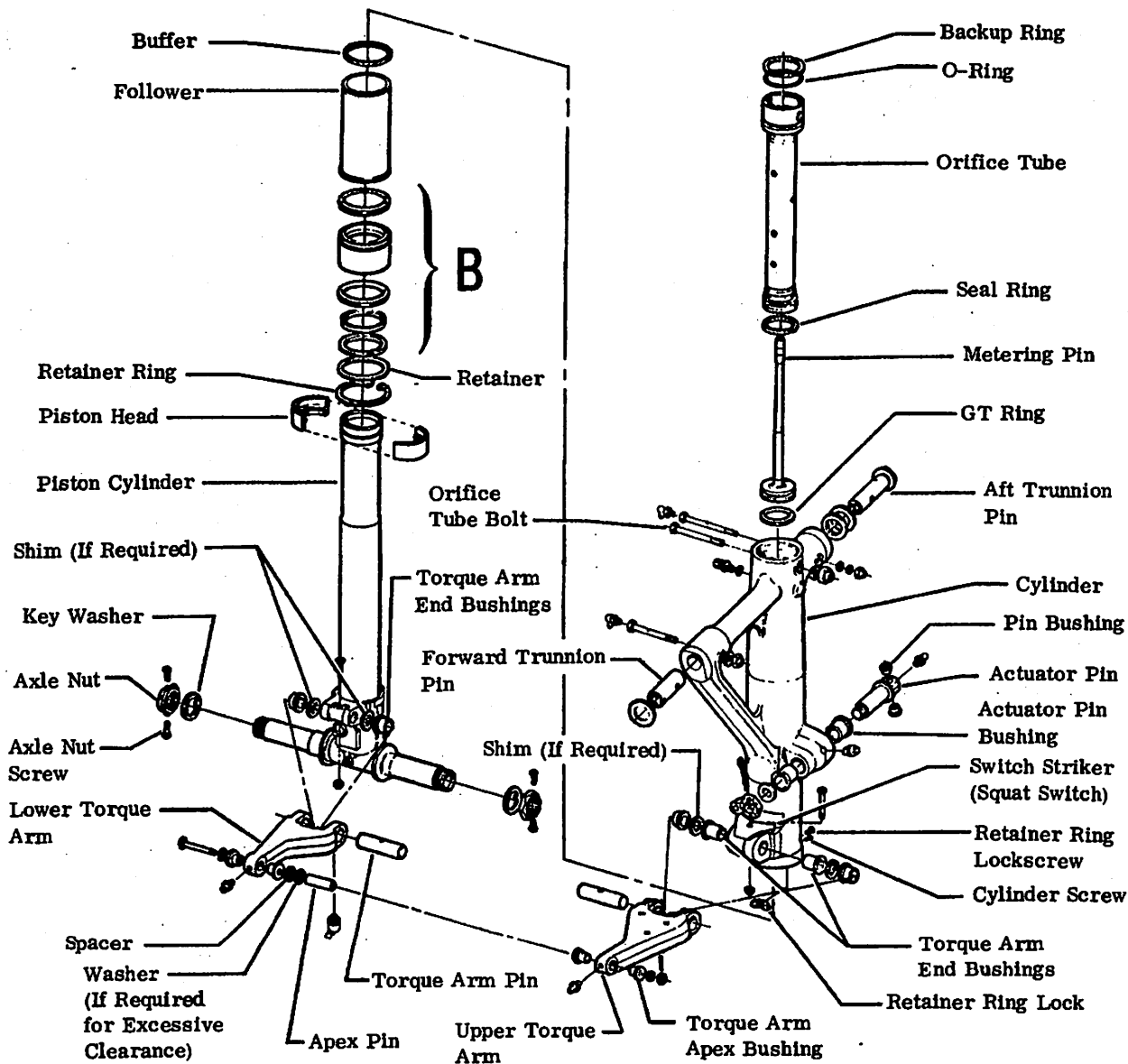
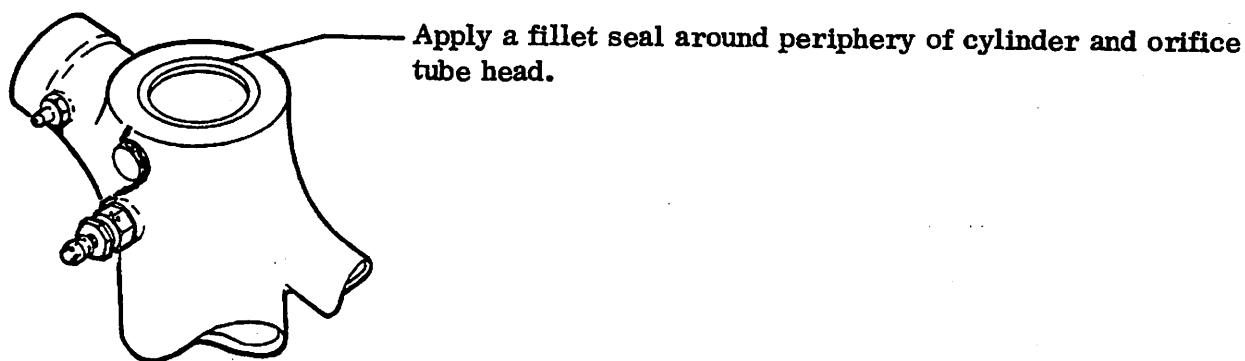
EFFECTIVITY: 25-061, 25-070 thru 25-277

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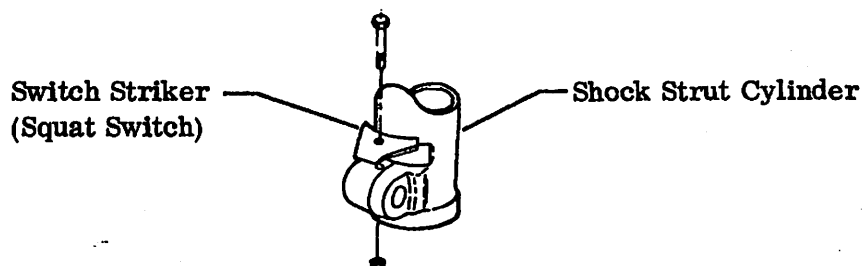


Main Gear Shock Strut Assembly

Figure 203 (Sheet 2 of 4)

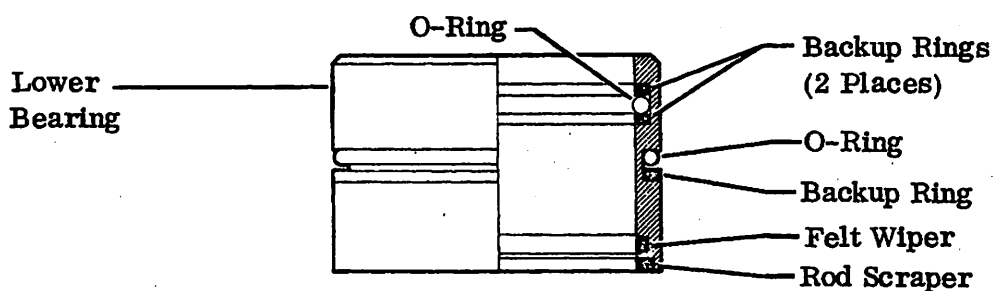
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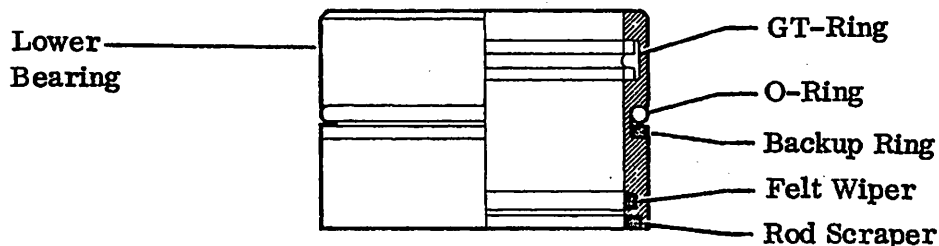


Aircraft 25-206 and Subsequent and earlier Aircraft modified per AMK 75-12

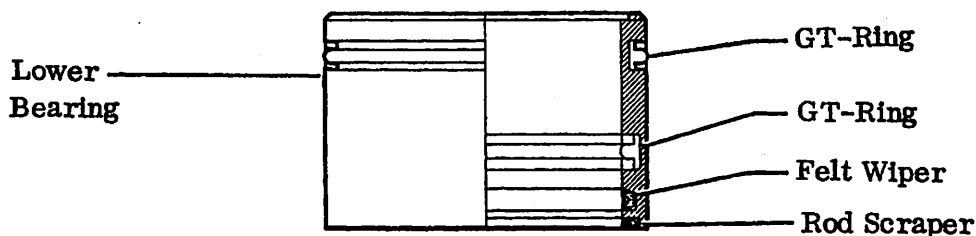
Detail A



Aircraft 25-061, 25-070 thru 25-154 Equipped with 2394101-2
(Strut Lower Bearing Repair Kit)



Aircraft 25-155 thru 25-278 and Aircraft Equipped with 2394101-4
(Strut Lower Bearing Repair Kit)



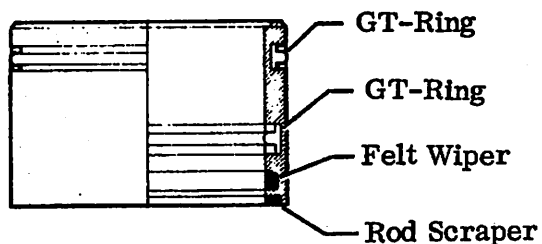
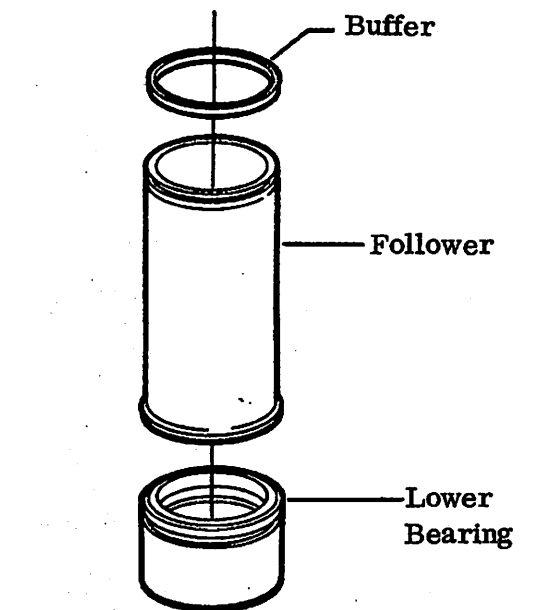
Aircraft 25-278 and Subsequent and Aircraft
modified per AAK 78-3

Detail B

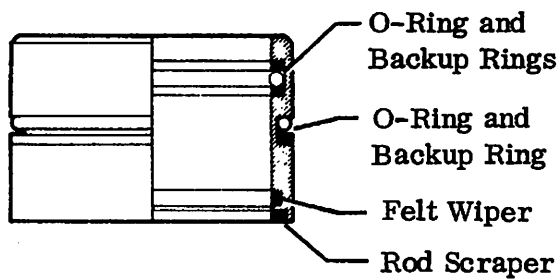
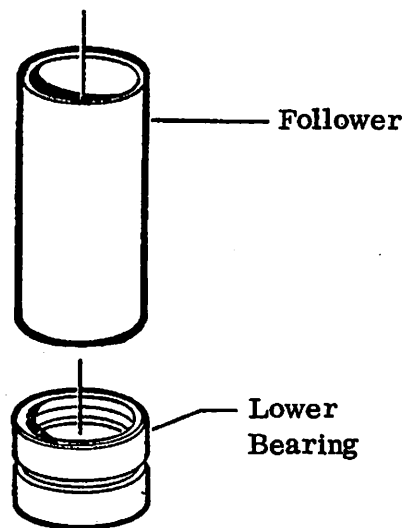
Main Gear Shock Strut Assembly
Figure 203 (Sheet 3 of 4)

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Configuration when modified
by AAK 78-3 (Main Landing
Gear Snubber Kit)



Original Configuration

Detail C

Main Gear Shock Strut Assembly
Figure 203 (Sheet 4 of 4)

6-13C
6-42A

EFFECTIVITY: 25-061, 25-070 thru 25-277

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- (1) Remove cap from air valve and loosen end nut 3/4 turn to release all air from strut.
- (2) Remove wheels if not already accomplished. (Refer to 32-42-00.)
- (3) Remove brakes if not already accomplished. (Refer to 32-43-01.)
- (4) Remove air valve and drain hydraulic fluid from strut through air valve port. Move piston through stroke to aid draining. Retain fluid for contamination inspection.
- (5) Remove safety wire from retainer ring lock screw and cylinder screw. Remove retainer ring lock screw and retainer ring lock from strut.
- (6) Remove bolt, releasing orifice tube, and remove orifice tube from cylinder.
- (7) Remove O-ring, backup ring, and seal ring from orifice tube.
- (8) Fabricate a metering pin removal tool per figure 204 and remove metering pin as follows:
 - (a) Remove bolt through brake reaction pads.
 - (b) Install metering pin removal tool in bolt hole.
 - (c) Attach a hydraulic pressure source to removal tool end fitting.

WARNING: • ALL AIR PRESSURE MUST BE BLED FROM CAVITY BEFORE HYDRAULIC PRESSURE IS APPLIED TO REMOVE METERING PIN. HYDRAULIC PRESSURE APPLIED TO THE METERING PIN REMOVAL TOOL MUST NOT EXCEED 3750 PSI.

• USE NECESSARY PRECAUTIONS TO CONTAIN THE METERING PIN WHEN IT RELEASES FROM STRUT OR BODILY INJURY CAN RESULT. THE METERING PIN COULD EJECT AT A HIGH VELOCITY.

- (d) Slowly apply hydraulic pressure until metering pin rises sufficiently in cylinder for removal.
- (e) Disconnect hydraulic pressure source and replace removal tool with bolt.
- (f) Remove GT-ring from metering pin.
- (9) Remove bolt, washer, and nut securing torque arms together.
- (10) Remove apex pin and retain spacer and washer (if installed).
- (11) Inspect torque arms for fit on torque arm pins. Maximum allowable side movement is 0.010 inch. Maximum allowable bushing/pin clearance is 0.009 inch. If torque arm removal is necessary, refer to 32-11-01.

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- (12) Disengage retainer ring from its groove in cylinder and remove piston head from piston.
- (13) Remove follower, lower bearing, retainer, and retainer ring from piston.
- (14) Remove O-rings and/or GT-rings, backup rings, felt wiper and rod scraper from lower bearing.
- (15) Clean all metal parts with Stoddard solvent or equivalent.
- (16) Remove all fingerprints from machined surfaces not plated or painted using a suitable fingerprint neutralizer.
- (17) Dry metal parts after cleaning with clean, dry, compressed air.
- (18) Coat internal metal parts with MIL-H-5606 hydraulic fluid.
- (19) Wrap cleaned metal parts in lint-free cloth or clean waxed paper while awaiting repair or reassembly.

NOTE: When storing unplated machine surface parts more than a few hours, coat with MIL-C-11796 preservative. Remove preservative prior to assembly of parts.

B. Inspect Main Gear Shock Strut for Wear (See figure 205.)

- (1) Screen drained hydraulic fluid for metal particles which may be used as an aid in determining trouble source.
- (2) Inspect all metal parts including welds, machined and forged parts for evidence of structural failure, nicks, burrs, cracks, corrosion, worn threads, and excessive wear.
- (3) Perform a magnetic particle inspection on all ferrous parts. Perform a fluorescent penetrant check on all aluminum parts. After ferrous parts inspection, demagnetize to remove all metal particles.
- (4) Check all friction surfaces of bushings and strut holes for out of round and taper using an internal micrometer, plug gages, or similar equipment.
- (5) Inspect OD and ID surfaces of piston and ID surfaces of cylinder for galls, scratches, scoring, corrosion, and worn plating.
- (6) Buff out minor scratches in piston, cylinder, piston head, lower bearing, and trunnion pins. Do not create low spots by excessive buffing or polishing. Use oil stone to remove the scratches and polish with crocus cloth. Check tolerances after polishing.
- (7) Replace any part which does not meet inspection requirements or tolerances.
- (8) On Aircraft 25-278 and Subsequent and prior Aircraft modified per AAK 78-3, inspect buffer for excessive wear and replace if necessary.
- (9) Inspect orifice tube ends for looseness. Tube ends are swaged on; no looseness is allowed.

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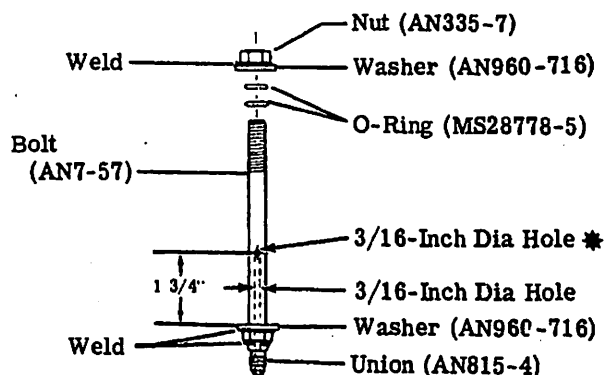
C. Assemble Main Gear Shock Strut (See figure 203.)

- (1) Immerse all GT-rings, O-rings, backup rings, seal rings, felt wiper, and rod scraper in MIL-H-5606 hydraulic fluid immediately prior to installation.
- (2) Slide retainer ring and retainer on piston.
- (3) Refer to figure 203, Details B and C, for lower bearing configurations, and install required parts.
- (4) Install lower bearing (chamfered end up) and follower (buffer end up) on piston.
- (5) Install GT-ring and backup ring on metering pin.
- (6) Install metering pin inside piston. Assure that metering pin is bottomed in piston.

CAUTION: USE CAUTION TO PREVENT DAMAGE TO PISTON ASSEMBLY'S SNUBBER ORIFICE NOZZLE OR TO BUFFER (IF INSTALLED) ON THE FOLLOWER DURING INSERTION OF PISTON ASSEMBLY INTO CYLINDER. STRIKING SNUBBER ORIFICE NOZZLE (SWAGED INTO PISTON) MAY LOOSEN IT.

- (7) The piston head consists of two matched halves and is to be replaced only as a complete unit. With piston head assembled on piston, slide piston assembly into lower end of cylinder until bottom end of cylinder contacts bottom of piston.

NOTE: During piston assembly insertion, use shim stock around follower buffer (if installed) to protect buffer from scraping against internal cylinder edges.



* Drill to meet hole through center of bolt. Do not drill completely through bolt.

**Metering Pin Removal Tool Fabrication
Figure 204**

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- (8) Secure piston in cylinder with retainer ring. Check to assure that centerline of retainer ring gap is aligned with screw hole for retainer ring lock screw and retainer ring is set at least halfway into retainer ring groove. If necessary, tap retainer ring lightly to set it into groove. Retainer ring lock screw cannot be installed if retainer ring is not properly installed.
- (9) Install retainer ring lock, with retainer ring lock screw, and secure with safety wire to adjacent cylinder screw provided for that purpose.
- (10) Assemble seal ring on orifice tube head and O-ring and backup ring on orifice tube end.
- (11) Insert orifice tube inside cylinder assuring that metering pin is aligned with metering pin hole in orifice tube head. Secure orifice end in cylinder with bolt, washer, and nut.

NOTE: Apply a fillet seal around periphery of cylinder and orifice tube head.

- (12) Replace lubricators removed during disassembly.
- (13) Replace worn or damaged bushings in pairs only. If bushings are to be replaced, proceed as follows:
 - (a) Coat outside diameter and adjacent flanges of new bushings with zinc chromate primer and press into place while primer is wet.
 - (b) After installation, coat bushing faces with zinc chromate primer. Remove excess primer from lubricator holes.
- (14) Install spacer and apex pin in torque arms and secure arms together with bolt and nut. Add AN960-916L washer if necessary to eliminate excessive clearance between torque arms.
- (15) With strut vertical, fill with MIL-H-5606 hydraulic fluid in accordance with nameplate attached to trunnion.
- (16) Install air valve and tighten securely.
- (17) Refinish all externally painted surfaces with one coat of wash primer, one coat of zinc chromate primer, and one coat of aluminized lacquer.
- (18) Perform leak test of shock strut as follows:
 - (a) Place completely assembled strut in vertical or horizontal position.
 - (b) Inflate strut using dry compressed air or nitrogen to pressure specified in Chapter 12 alternate strut inflation procedure.
 - (c) No leakage is allowed in a one-hour period.
 - (d) After completion of test, release all air from strut through air valve.
 - (e) After air pressure is released, install dust cap on air valve.
- (19) Install brakes. (Refer to 32-43-01.)
- (20) Install wheels. (Refer to 32-42-00.)

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OUTSIDE DIAMETER			
Nomenclature	Original O. D. (Inches)	Surface Finish (RMS)	Minimum O. D. (Inches)
Fwd Trunnion Pin	<u>1.2590</u> 1.2573	63	1.256
Aft Trunnion Pin	<u>1.1245</u> 1.1225	63	1.121
Actuator Pin	<u>1.122</u> 1.120	63	1.119
Axle (ends)	<u>1.4998</u> 1.4988	63	1.497
Apex Pin	<u>0.5620</u> 0.5615	32	0.558
Piston Head	<u>3.248</u> 3.244	32	3.240
Piston Cylinder	<u>2.998</u> 2.996	16	2.9945

INSIDE DIAMETER				TOTAL CLEARANCE DIAMETER	
Nomenclature	Original Hole Size I. D. (In.)	Surface Finish (RMS)	Maximum Hole Size I. D. (In.)	Original Inches	Maximum Wear (In.)
Fwd Trunnion Hole	<u>1.2608</u> 1.2598	125	1.262	<u>0.0035</u> 0.0008	0.006
Aft Trunnion Hole	<u>1.1260</u> 1.1250	125	1.128	<u>0.0035</u> 0.0005	0.007
Cylinder	<u>3.252</u> 3.250	63	3.256	<u>0.008</u> 0.002	0.016
Actuator Attach Bushings	<u>0.5640</u> 0.5625	100	0.566	<u>0.0033</u> 0.0009	0.0053
Actuator Pin Bushing	<u>1.1265</u> 1.1250	125	1.129	<u>0.0065</u> 0.0030	0.010
Apex Bushings	<u>0.5630</u> 0.5625	125	0.566	<u>0.0015</u> 0.0005	0.008
Lower Bearing	<u>3.002</u> 3.000	32	3.008	<u>0.006</u> 0.002	0.0135

Main Gear Shock Strut Wear Tolerances
Figure 205

EFFECTIVITY: ALL

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MAIN GEAR SHOCK STRUT TORQUE ARM - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Replace upper and lower torque arms in sets only.

A. Remove Torque Arms (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Remove wheels. (Refer to 32-42-00.)
- (3) Remove brakes. (Refer to 32-43-01.)
- (4) Remove torque arm pin bolts, nuts, washers, and cotter pins, and, if not already accomplished, remove apex pin bolt, nut, washers, and cotter pin.
- (5) Remove apex pin, releasing spacer and washer.
- (6) Press out torque arm pins and remove torque arms from strut.

B. Install Torque Arms (See figure 201.)

- (1) Perform Inspection/Check.
- (2) Freeze torque arm pins to -75°F.
- (3) Position upper torque arm on strut and add shims to bring clearances to 0.001 inch or less.

NOTE: Torque arm must retain freedom of movement.

- (4) Remove pin from freezer and immediately install, assuring that bolt hole in pin aligns with bolt hole in boss. Secure with bolt, washer, nut, and cotter pin.
- (5) Position lower torque arm on strut and add shims to bring clearance to 0.001 inch or less.

NOTE: Torque arm must retain freedom of movement.

- (6) Remove pin from freezer and immediately install, assuring that bolt hole in pin aligns with bolt hole in boss. Secure with bolt, washer, nut and cotter pin.
- (7) If this torque arm installation procedure is part of a main landing gear assembly procedure, proceed with remainder of gear assembly instructions. If this torque arm installation procedure is independent of other procedures, proceed with step (8).
- (8) Align upper torque arm, spacer, washer, and lower torque arm. Insert apex pin and secure with bolt, washers, and nut.
- (9) Install brakes. (Refer to 32-43-01.)
- (10) Install wheels. (Refer to 32-42-00.)
- (11) Remove aircraft from jacks.

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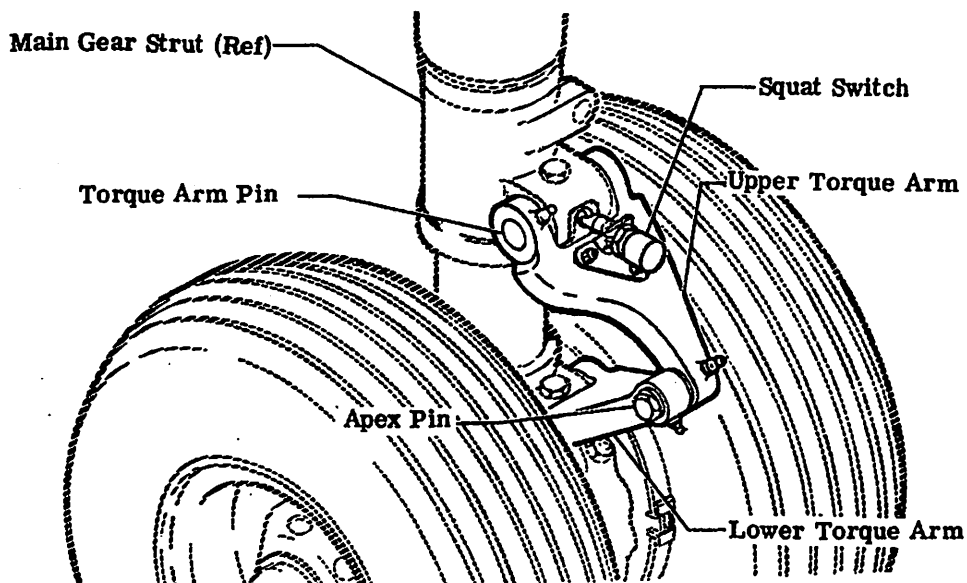
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2. Inspection/Check

- A. Inspect Torque Arm and Attaching Parts (See figures 202 and 203.)
- (1) Inspect all metal parts for evidence of structural fatigue, nicks, burrs, cracks, corrosion, worn threads, and excessive wear.
 - (2) Perform a magnetic particle inspection on all ferrous parts. Perform a fluorescent penetrant check on all aluminum parts. After ferrous parts inspection, demagnetize to remove all metal particles.
 - (3) Check all friction surfaces of bushings for out-of-round and taper using an internal micrometer, plug gages, or similar equipment.
 - (4) Replace any part which does not meet inspection requirements or tolerances.

3. Approved Repairs

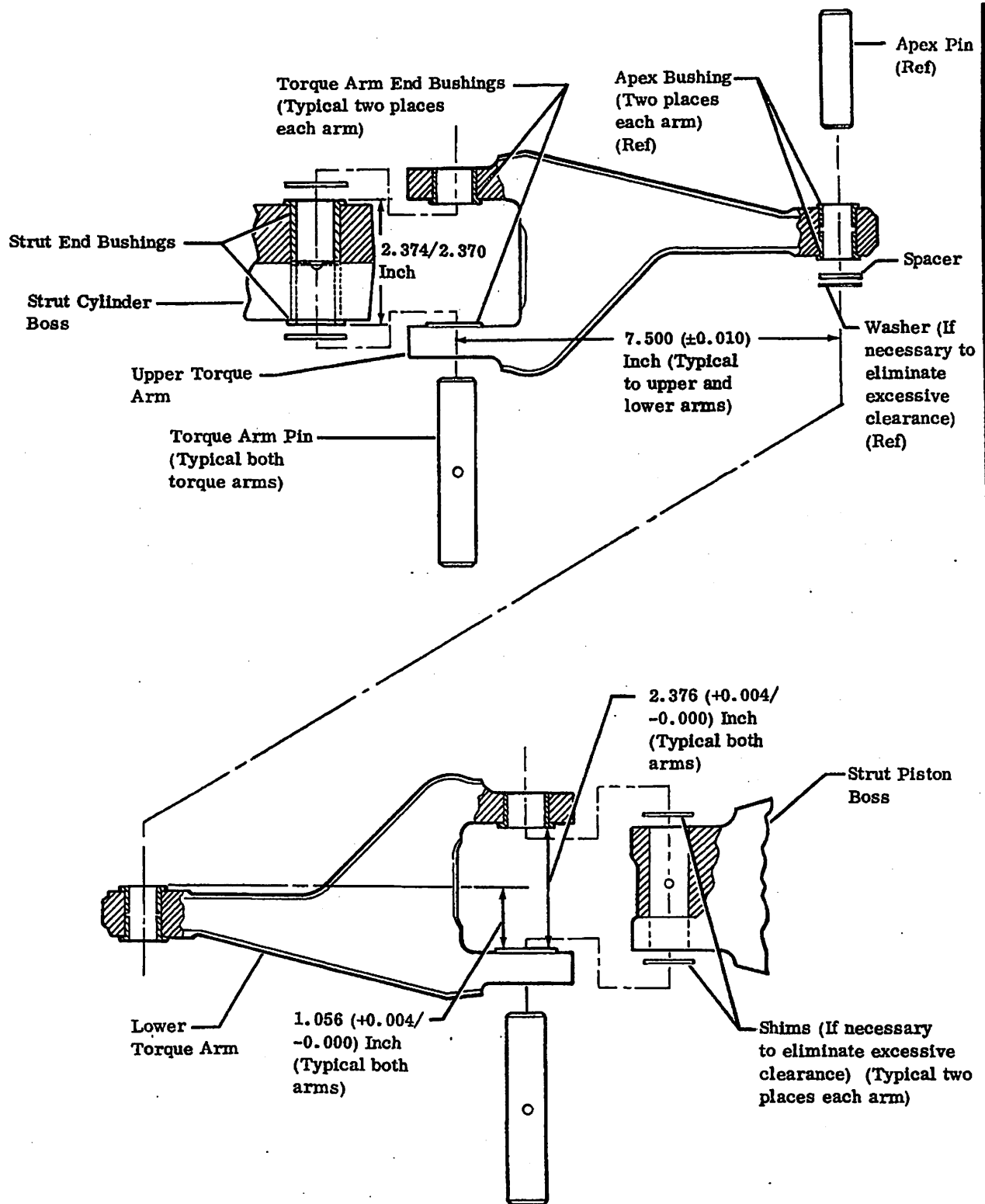
- A. Replace Torque Arm End Bushings and Strut End Bushings (See figure 202.)
- (1) Press out bushings to be replaced. Bushings must be replaced in pairs only.
 - (2) Freeze new bushings to -75°F . Remove individually from freezer.
 - (3) Apply a coat of zinc chromate primer to bushing and hole, and press bushings into hole. Care should be taken not to plug the grease hole in the bushing.
 - (4) Line ream bushings to dimensions shown in figure 202.
 - (5) Machine the flat surface of the bushings to obtain dimensions shown in figure 202.



Torque Arm Installation
Figure 201

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Torque Arm Pin and Bushing Replacement
Figure 202

6-34B

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NOMENCLATURE	INSIDE DIAMETER (INCHES)		TOTAL CLEARANCE DIAMETER (INCHES)		SURFACE FINISH
	ORIGINAL	MAXIMUM	ORIGINAL	MAXIMUM	
Torque Arm End Bushings	<u>0.7510</u> 0.7505	0.753	<u>0.0015</u> 0.0005	0.009	125 RMS
Strut End Bushings	<u>0.7505</u> 0.7500	0.7505	<u>0.0010</u> 0.0000	0.0010	125 RMS
Apex Bushings	<u>0.5630</u> 0.5625	0.566	<u>0.0015</u> 0.0005	0.008	125 RMS

NOMENCLATURE	OUTSIDE DIAMETER (INCHES)		SURFACE FINISH
	ORIGINAL	MINIMUM	
Torque Arm Pins	<u>0.7500</u> 0.7495	0.744	63 RMS
Apex Pin	<u>0.5620</u> 0.5615	0.558	32 RMS

Pin and Bushing Wear Tolerances
Figure 203

EFFECTIVITY: ALL

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MAIN GEAR DOORS - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Removal and installation procedures for the left and right doors are identical.

A. Remove Main Gear Inboard Door (See figure 201.)

- (1) Place aircraft on jacks.

WARNING: FAILURE TO DISCONNECT AIRCRAFT BATTERIES COULD RESULT IN BODILY INJURY WHILE WORKING IN WHEEL WELL AREA IF ELECTRICAL POWER IS INADVERTENTLY APPLIED TO THE LANDING GEAR SYSTEM.

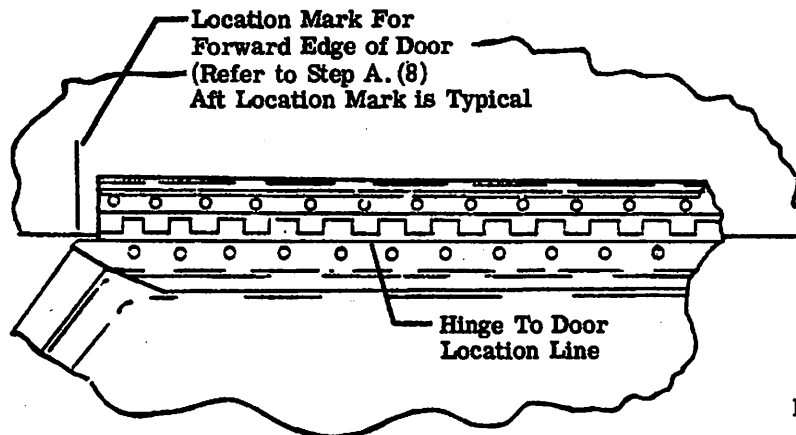
- (2) Remove safety wire and disconnect aircraft batteries.
- (3) Using a screwdriver to override uplatch actuator, release uplatch hook from roller.
- (4) To lower main gear inboard door, station a man at inboard door and a man to operate brake pedals. While pumping brakes, pull inboard door to down position.

NOTE: Brakes must be pumped during manual lowering of inboard gear door to prevent possible door actuator damage and/or nose gear vacuuming to up cycle.

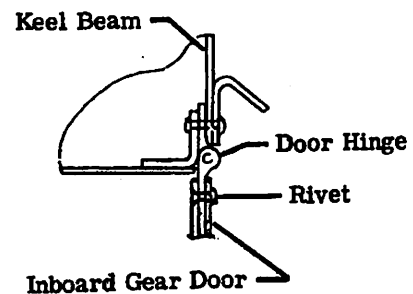
- (5) Disconnect door actuator at inboard door.
- (6) Disconnect bonding jumper at inboard door.
- (7) Remove main gear door up switch and electrical wiring and clamps from inboard door. Tie electrical wiring and switch to aircraft structure out of immediate work area.
- (8) Mark location of forward and aft inboard ends of existing door on keel beam. This will assist in locating new door.
- (9) Mark inboard edge of gear door on hinge.
- (10) Drill out rivets securing existing door to hinge and remove door from hinge.
- (11) Remove inboard door actuator attach bracket and inboard door switch bracket from door.

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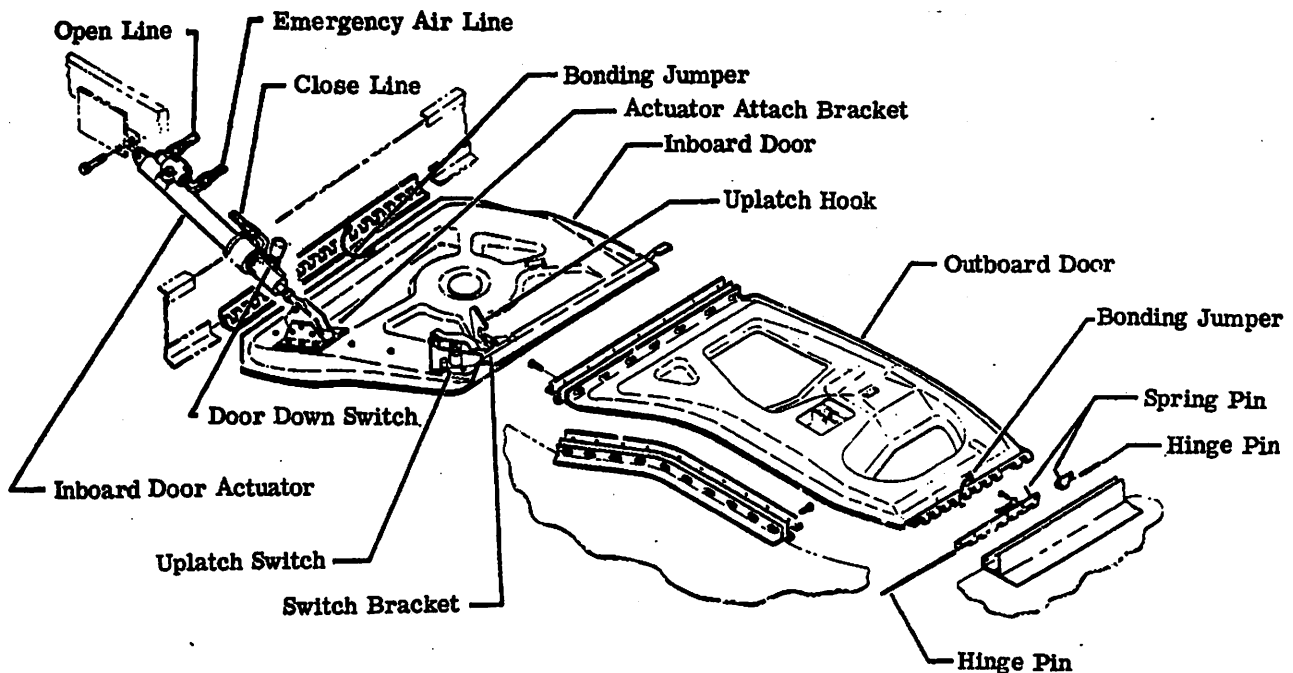
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Detail A



Detail B



**Main Gear Door Installation
Figure 201**

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- B. Install Main Gear Inboard Door (See figure 201.)**
- (1) Align inboard door and hinge with hinge on outer face of door as shown in detail A.
 - (2) Clamp door in place on hinge and assure that door is aligned per detail A.
 - (3) Drill two end holes in door using existing two end holes in hinge as a template. Cleco door to hinge.
 - (4) Drill remaining holes in door to match those in hinge.
 - (5) Mark trim contour on forward and aft sides of door. Remove door and trim to contour marks.
 - (6) Install inboard uplatch bracket and inboard door actuator attach bracket on inboard door.
 - (7) Countersink outer door skin hinge attachment holes (100° x 0.159 inch diameter).
 - (8) Position hinge on inboard door between door pan and outer door skin so that holes in hinge and door align and secure in place with rivets as shown in detail A.
 - (9) Connect bonding jumper to door.
 - (10) Final trim door to match wheel well.
 - (11) Connect aircraft batteries.
 - (12) Retract landing gear. Assure that inboard gear door actuator and uplatch acutator ends are clear of aircraft structure prior to starting retraction sequence.
 - (13) Raise inboard gear door. Mark and trim inboard gear door to match edge of outboard gear door.
 - (14) Lower landing gear and disconnect aircraft batteries.
 - (15) Connect inboard door actuator to door.
 - (16) Install main gear door up switch and clamp electrical wiring in place.
 - (17) Adjust main gear door up switch. (Refer to 32-60-02.)
 - (18) Connect aircraft batteries and safety wire.
 - (19) Remove aircraft from jacks.
- C. Remove Main Gear Outboard Door (See figure 201.)**
- (1) Place aircraft on jacks.
 - (2) Disconnect door closure link assembly from door.
 - (3) Retract landing gear to allow access to door hinge attachment screws.

WARNING: FAILURE TO DISCONNECT BATTERIES COULD RESULT IN BODILY INJURY WHILE WORKING IN WHEEL WELL AREA IF ELECTRICAL POWER IS INADVERTENTLY APPLIED TO LANDING GEAR SYSTEM.

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- (4) Remove safety wire and disconnect aircraft batteries.
 - (5) Disconnect bonding jumper from door and wing structure.
 - (6) Remove screws securing hinge to wheel well and remove outboard door from aircraft.
- D. Install Main Gear Outboard Door (See figure 201.)
- (1) Install hinge halves on aircraft structure where old hinges were installed.
 - (2) Install hinge halves together using hinge pin.
 - (3) Position and attach hinge half on outboard door.
 - (4) Temporarily install door using hinge pin and mark trim contour.
 - (5) Remove door and rough trim.
 - (6) Attach bonding jumper to door.
 - (7) Install door and final trim. Insert spring pins to secure hinge pins.
 - (8) Connect aircraft batteries and safety wire.
 - (9) Extend landing gear.
 - (10) Connect door closure link to door.
 - (11) Cycle gear and check door for proper operation and clearance.
 - (12) Remove aircraft from jacks.

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NOSE GEAR AND DOORS - DESCRIPTION AND OPERATION

1. Description

- A. The nose landing gear consists of the nose wheel and tire attached to a shock strut.
- B. The nose landing gear is attached to the fuselage structure by two trunnion pins protruding into bearing plates. Additional support is obtained from the nose gear actuator which is attached at the midpoint of the shock strut and to the fuselage structure.
- C. Normally the nose gear is extended and retracted hydraulically. A pneumatic system is installed for emergency gear blowdown.
- D. The nose landing gear doors are the split type which remain open during gear extension. The doors are opened and closed mechanically by a pushrod connected to the shock strut.
- E. The shock strut is a conventional air-hydraulic unit designed to absorb the shock of landing, takeoff, and taxiing. It is essentially a steel piston riding in a cylinder and utilizing O-rings to prevent leakage between the piston and cylinder. An air-oil filler valve is installed on the strut for servicing purposes.
- F. The nose gear is enclosed by two hinge-mounted doors. The doors are opened and closed mechanically through linkage connected to the nose gear. Adjustable pushrods are provided to bring the doors into contour with the fuselage when the gear is retracted.

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NOSE GEAR SHOCK STRUT - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Nose Gear Shock Strut (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Depressurize hydraulic system.
- (3) Remove electrical power from aircraft.
- (4) Disconnect door pushrods from nose gear strut.
- (5) Disconnect nose gear actuator from nose gear strut.
- (6) Disconnect electrical connector P33 installed on nose wheel box and remove wiring clamps as necessary to allow removal of nose gear strut.

WARNING: REMOVAL OF TRUNNION PIN RELEASES THE NOSE GEAR FROM THE AIRCRAFT. ADEQUATE SUPPORT MUST BE GIVEN THE GEAR TO PREVENT IT FROM FALLING AND CAUSING BODILY INJURY TO PERSONNEL OR STRUCTURAL DAMAGE TO THE AIRCRAFT.

- (7) Remove right and left bolts from nose gear trunnion. Insert a small punch into slot (lower side of trunnion), slide trunnion pins inboard (toward strut) until free from bearing plates, and remove strut from aircraft.

NOTE: When strut trunnion pins are slid out of bearing plates, shims (if any) will be released. Retain these shims (if serviceable) for installation of nose gear strut. To ensure proper strut and actuator alignment, install shims in original position.

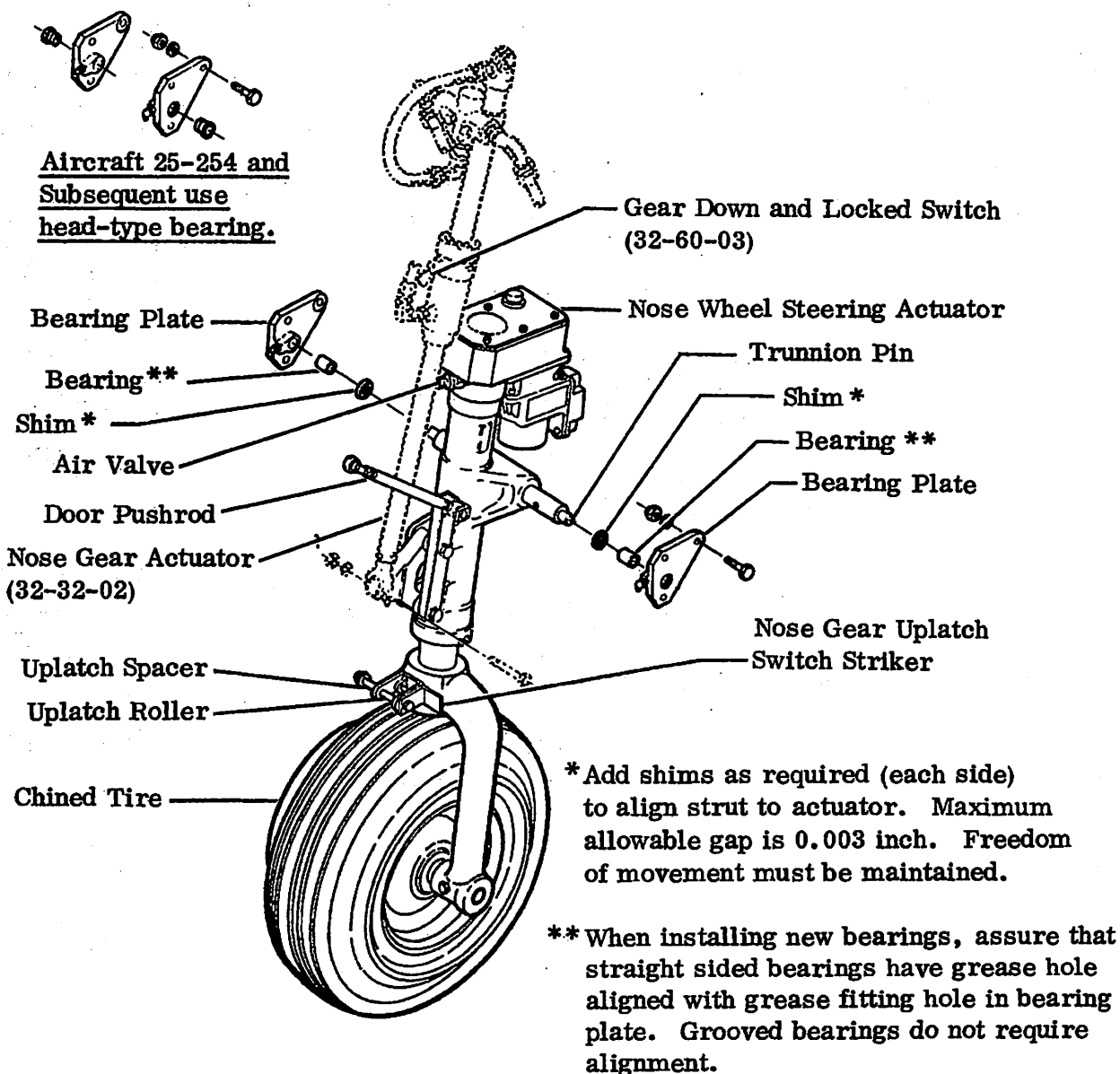
B. Install Nose Gear Shock Strut (See figure 201.)

- (1) Prior to installation of nose gear shock strut, inspect trunnion pins and bearings for general condition.
- (2) Position strut in aircraft and install previously removed shims (in original positions) between nose gear strut and bearing plates. This will properly align strut to allow attachment of actuator. Using a punch inserted into trunnion slot, slide trunnion pins into bearing plate.
- (3) Install trunnion pin bolt through trunnion and trunnion pin and secure with nut.
- (4) Connect electrical connector and replace wiring clamps.
- (5) Attach actuator to nose gear strut.

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- (6) Connect door pushrod to strut.
- (7) Restore electrical power to aircraft.
- (8) Pressurize hydraulic system. Cycle gear, checking for proper gear indication and operation.
- (9) Service nose shock strut. (Refer to Chapter 12.)
- (10) Pressurize hydraulic system. Cycle gear, checking for proper gear indication and operation.



Nose Landing Gear Installation
Figure 201

6-2C-2

EFFECTIVITY: ALL

MM-98

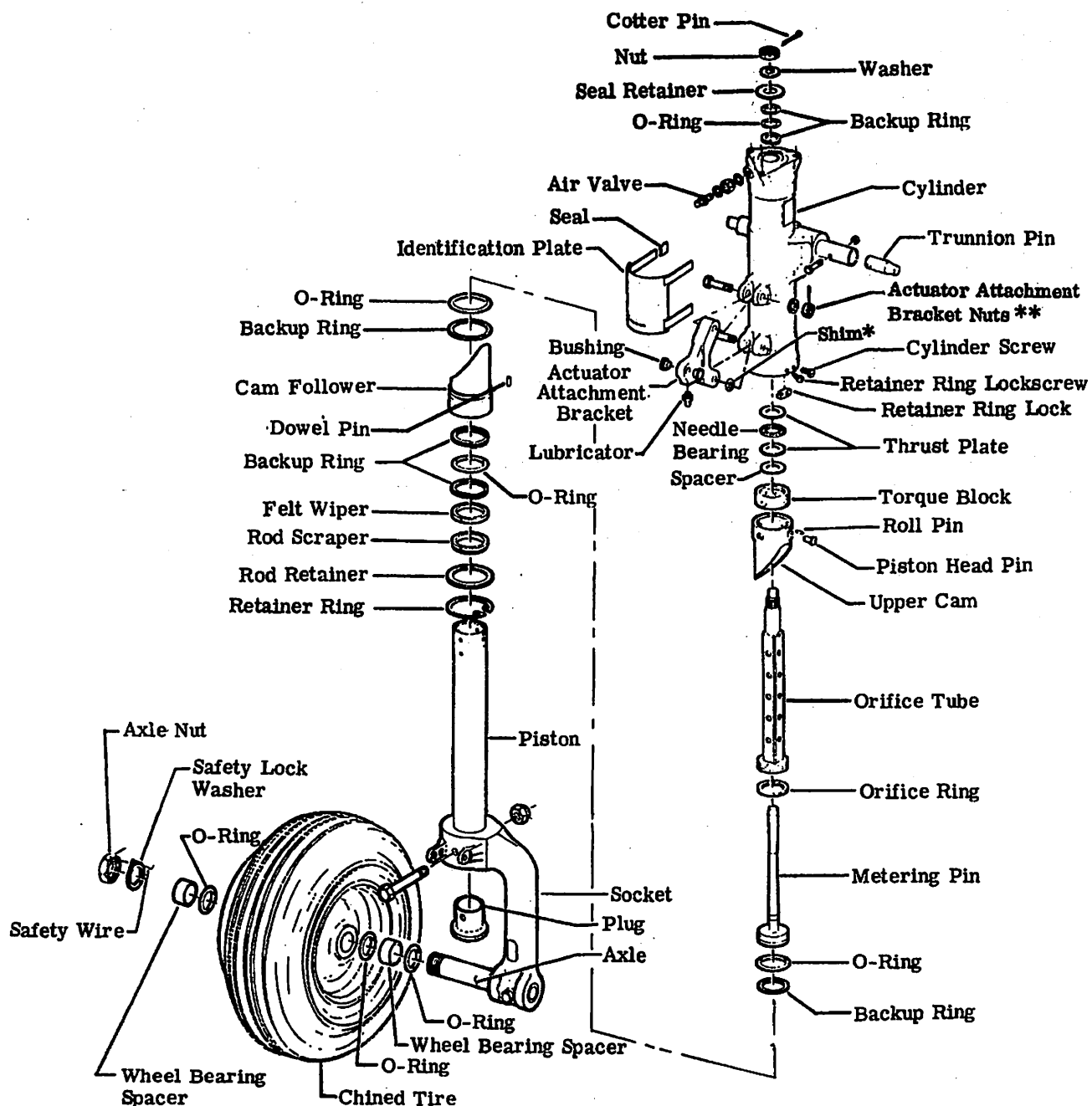
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* Peel shim as required to obtain a maximum gap of 0.002 inch (2 places).

** Torque actuator attachment bracket nuts to 190 to 290 inch-pounds.



Nose Gear Shock Strut Assembly
Figure 202

6-20C

EFFECTIVITY: ALL

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2. Approved Repairs

NOTE: During disassembly, tag and keep all parts together for assembly into the same unit. Discard all O-rings, backup rings, rod scraper and felt wiper.

A. Disassemble Nose Gear Shock Strut (See figure 202.)

WARNING: BEFORE PERFORMING ANY DISASSEMBLY OF PARTS ON THE STRUT, ASSURE THAT ALL AIR PRESSURE HAS BEEN RELEASED THROUGH THE AIR VALVE. FAILURE TO DO SO CAN CAUSE BODILY INJURY. DO NOT RELEASE AIR BY LOOSENING OR REMOVING AIR VALVE.

- (1) Remove dust cap from air valve, loosen end nut 3/4 turn, and release all air from strut.
- (2) Remove nose wheel. Refer to 32-41-00.
- (3) Remove nose wheel steering actuator. Refer to 32-50-02.
- (4) Cut safety wire and remove air valve.
- (5) Drain hydraulic fluid through air valve port. Move piston through stroke to aid draining.
- (6) Remove safety wire from retainer ring lock screw and cylinder screw. Remove retainer ring lock screw and retainer ring lock from strut.
- (7) Disengage retainer ring from groove in cylinder and slide piston straight out of cylinder. Do not twist or turn piston until dowel pin disengages from cylinder.
- (8) Remove thrust plates, needle bearing, and spacer from orifice tube.
- (9) Remove piston head pins which secure upper cam and torque block to piston as follows:

NOTE: The piston head pins incorporate internal threads (10-32UNF-3B) to aid in removal of the pins.

- (a) Fabricate piston head pin puller per figure 203.
- (b) Holding puller block against upper cam, screw puller bolt into piston head pin approximately 0.5 inch.
- (c) Tighten puller nut against washer and puller block by hand until assured a straight pull will be made on piston head pin.
- (d) Holding puller bolt steady, tighten puller nut with wrench until piston head pin is removed from upper cam.

NOTE: If piston head pin is easily removed or loose, it must be replaced. If pin is loose, it could back out and damage the cylinder wall.

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- (10) Remove roll pin from upper cam and slide upper cam off piston.
- (11) Push up against metering pin to slide metering pin, orifice tube, and torque block from piston.
- (12) Remove torque block from orifice tube and orifice tube from metering pin.
- (13) Do not remove actuator attach bracket from cylinder unless replacement is required.
- (14) Clean metal parts with Stoddard solvent or equivalent.
- (15) Remove all fingerprints from machined surfaces not plated or painted with a suitable fingerprint neutralizer. Assure that no solvent remains in internal cavities or surfaces.
- (16) Dry metal parts after cleaning with clean, dry, compressed air.
- (17) Coat internal parts with MIL-H-5606 hydraulic fluid.

OUTSIDE DIAMETER			
Nomenclature	Original O. D. (Inches)	Surface Finish (RMS)	Minimum O. D. (Inches)
Axle	$\frac{1.4998}{1.4988}$	63	1.498
Upper Cam	$\frac{2.997}{2.995}$	32	2.991
Orifice Tube	* $\frac{1.218}{1.217}$	32	* 1.218
Piston	$\frac{2.498}{2.496}$	16	2.495
Trunnion Pin	$\frac{0.6245}{0.6225}$	63	0.6215

INSIDE DIAMETER				TOTAL CLEARANCE DIAMETER	
Nomenclature	Original Hole Size I. D. (In.)	Surface Finish (RMS)	Maximum Hole Size I. D. (In.)	Original Inches	Maximum Wear (In.)
Cylinder	$\frac{3.002}{3.000}$	63	3.006	$\frac{0.007}{0.003}$	0.015
Torque Block	* $\frac{1.221}{1.220}$	32	* 1.223	$\frac{0.004}{0.002}$	* 0.007
Cam Follower	$\frac{2.502}{2.500}$	32	2.506	$\frac{0.006}{0.002}$	0.011
Bushing	$\frac{0.4385}{0.4375}$	100	0.441	$\frac{0.0027}{0.0008}$	0.008

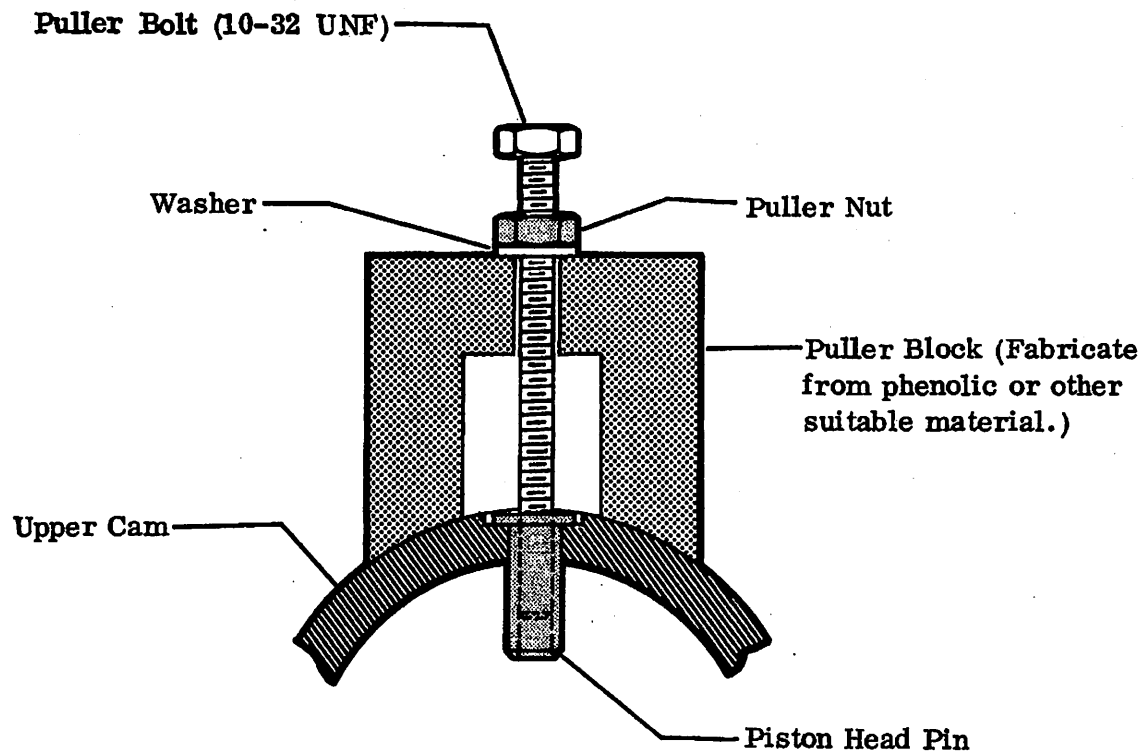
* At bearing surface of square formed parts (not diameter)

Nose Gear Shock Strut Wear Tolerances
Figure 203

EFFECTIVITY: ALL

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Piston Head Pin Puller
Figure 204

EFFECTIVITY: ALL

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- (18) Wrap cleaned parts in lint-free cloth or clean waxed paper while awaiting repair or assembly.

NOTE: When storing unplated machined parts more than a few hours, coat with MIL-C-11796 preservative. Remove preservative prior to assembly of parts.

E. Inspect Nose Gear Shock Strut for Wear (See figure 203.)

- (1) Screen drained hydraulic fluid for metal particles which may be used as an aid in determining trouble source.
- (2) Inspect interior and exterior of all metal parts for structural failure and corrosion and machined parts for excessive wear, nicks, burrs, cracks, and worn threads.
- (3) Perform a magnetic inspection on all ferrous parts. After magnetic inspection, demagnetize to remove all metal particles. Perform a fluorescent penetrant check on all aluminum parts.
- (4) Perform OD and ID tolerance inspection on piston and cylinder.
- (5) Check all friction surfaces of bushings and strut holes for out-of-round and taper using internal micrometer, plug gages, or similar equipment.
- (6) Buff out minor scratches in upper cam, trunnion pins, cam follower, piston, and cylinder. Do not create low spots by excessive buffing or polishing. Use oilstone to remove scratches and polish with crocus cloth. Check tolerance after polishing.
- (7) Replace any part that does not meet inspection requirements or tolerances.

F. Assemble Nose Gear Shock Strut (See figure 202.)

- (1) Immerse all O-rings, GT-rings, backup rings, rod scraper and felt wiper in MIL-H-5606 hydraulic fluid immediately prior to installation.

CAUTION: TO AVOID POSSIBLE DAMAGE TO NOSE GEAR PISTON, ASSURE THAT ACTUATOR ATTACHMENT BRACKET IS PROPERLY SHIMMED AND THAT ATTACHING NUTS ARE PROPERLY TORQUED.

- (2) Install nose gear actuator attachment bracket and shims (two places) if removed during disassembly. Shim to obtain 0.002 inch maximum gap between bracket and cylinder tabs. Torque nuts 190 to 290 inch-pounds.
- (3) Install GT-ring on metering pin and install metering pin in piston. Push metering pin down until it bottoms in piston.
- (4) Install orifice ring on orifice tube and install orifice tube in piston over metering pin.

EFFECTIVITY: ALL

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- (5) Insert retainer ring, rod retainer, and rod scraper over piston.
- (6) Install O-rings, backup rings, and felt wiper in cam follower and slide cam follower over piston.
- (7) Insert torque block onto orifice tube and into piston. Slide upper cam over piston. Align holes in piston, torque block, and upper cam and install piston head pin and roll pin.
- (8) Assemble spacer (chamfered edge down), needle bearing, and thrust plates on orifice tube.
- (9) Install piston into cylinder. Keep piston straight as it is being installed in cylinder to prevent jamming of needle bearing and thrust plate. Jamming will cause shimmy problems.
- (10) Extend piston approximately 5 inches and align holes in cam follower and cylinder for dowel pin.
- (11) Press dowel pin into place, assuring that pin is flush with lower end of cam follower.

CAUTION: REMOVE SHARP INSIDE EDGES OF ROD SCRAPER USING 400 GRIT ABRASIVE PAPER. SHARP EDGES MAY SCRATCH PISTON OR SCRAPE OFF ITS CHROME PLATING.

- (12) Secure piston in cylinder by installing rod scraper, rod retainer, and retainer ring. Check to assure that centerline of retainer ring opening is aligned with screw hole for retainer ring lock and retainer ring is set at least halfway into retainer ring groove. If necessary, tap retainer ring lightly to set it into groove. Retainer ring lock cannot be installed if retainer ring is not properly installed.

NOTE: Retainer ring has one side beveled and bevel must be on down side.

- (13) Install retainer ring lock and secure with retainer ring lock screw. Install cylinder screw.
- (14) Safety wire cylinder screw to retainer ring lock screw.
- (15) After complete assembly of strut, except for air valve, fully compress piston.
- (16) Place strut in vertical position and fill with clean MIL-H-5606 hydraulic fluid through air valve port until overflow exists. Pour fluid slowly to avoid foaming and accumulation of air pockets.
- (17) After servicing, install packing, reducer, seal, and air valve.
- (18) Perform leak test of shock strut as follows:
 - (a) Place completely assembled strut in either vertical or horizontal position.

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- (b) With piston fully extended, inflate strut with dry compressed air or nitrogen to pressure specified in Chapter 12 alternate strut inflation procedure.
 - (c) There shall be no leakage for a one-hour period.
 - (d) After completion of test, remove dust cap from air valve and loosen end nut 3/4 turn to release all air from strut.
 - (e) Install dust cap on air valve.
- (19) Install nose steering actuator. (Refer to 32-50-02.)
 - (20) Install nose wheel. (Refer to 32-41-00.)

EFFECTIVITY: ALL

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NOSE GEAR DOORS - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Removal and installation of both gear doors is identical.

A. Remove Nose Gear Door (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Lower tailcone access door and disconnect aircraft batteries.
- (3) Disconnect and remove applicable nose gear door pushrod.
- (4) Remove nose compartment access doors and remove equipment and equipment racks necessary to gain access to nose gear door hinge attach rivets.
- (5) Scribe a line on the aircraft structure around hinge. This line will be used in locating new door and hinge.
- (6) Drill out rivets and remove door from aircraft.

B. Install Nose Gear Door (See figure 201.)

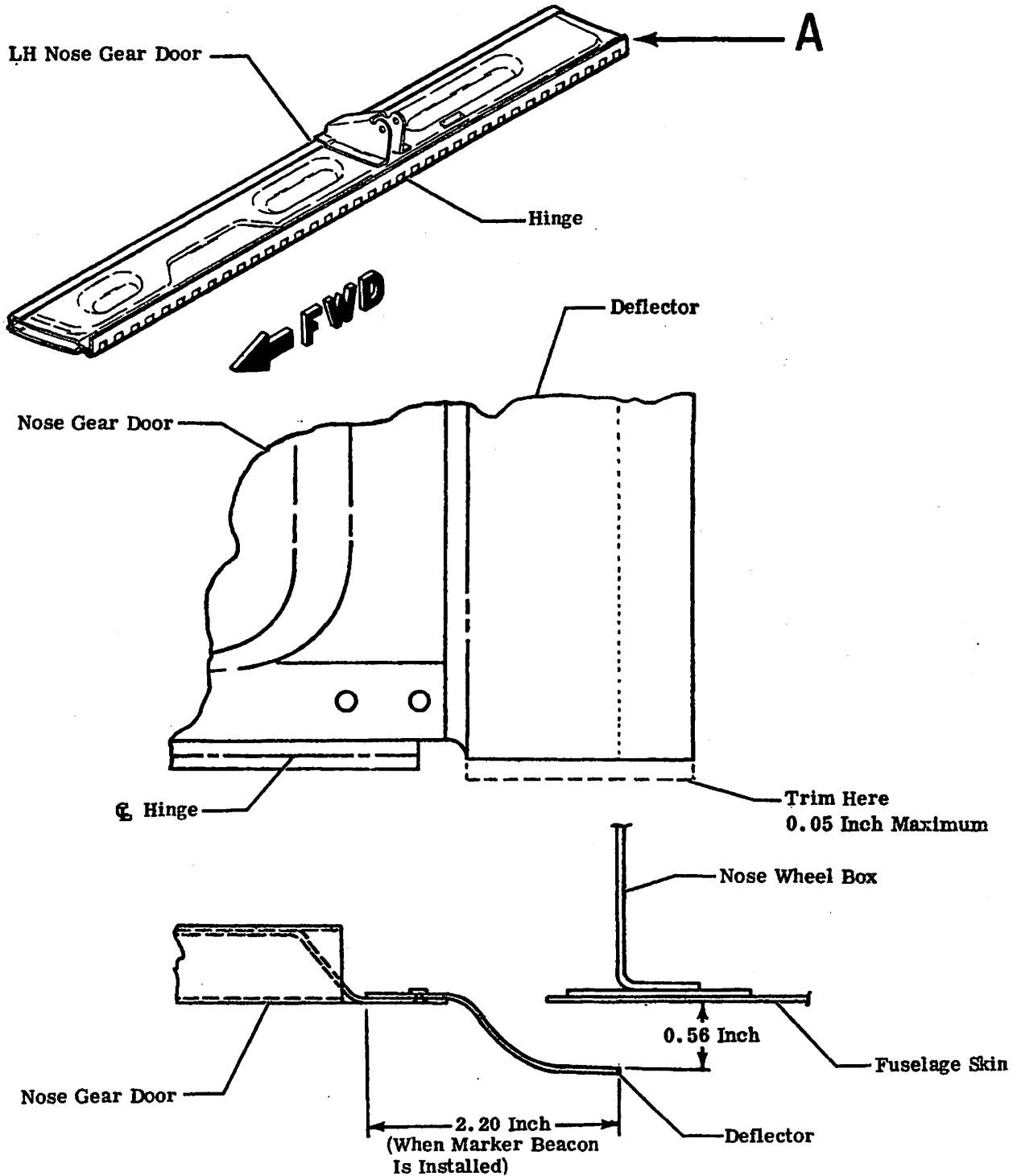
- (1) Position new door-hinge assembly on aircraft and align hinge with scribe mark which was made during removal. Clamp in place.
- (2) Using existing holes in aircraft structure as a template, drill a hole in each end of hinge and cleco in place.
- (3) Remove clamps and drill remainder of holes in hinge to match aircraft structure.
- (4) Secure hinge to structure with rivets.
- (5) Temporarily attach door pushrod to door and, with door lowered, check for clearance between outboard end of deflector and fuselage skin. If necessary, outboard end of deflector may be trimmed a maximum of 0.05 inch to obtain clearance.

NOTE: If a marker beacon is installed and it interferes with deflector, deflector must be trimmed to a length of 2.20 inches as shown in figure 201.

- (6) Remove door pushrod and close door until it is faired with the fuselage skin. Check for 0.56 inch clearance between deflector and fuselage skin.
- (7) Install nose compartment equipment and access doors.
- (8) Rig nose gear doors. (Refer to Adjustment/Test.)
- (9) Remove aircraft from jacks and connect aircraft batteries.

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Detail A
Nose Gear Door Installation
Figure 201

EFFECTIVITY: ALL

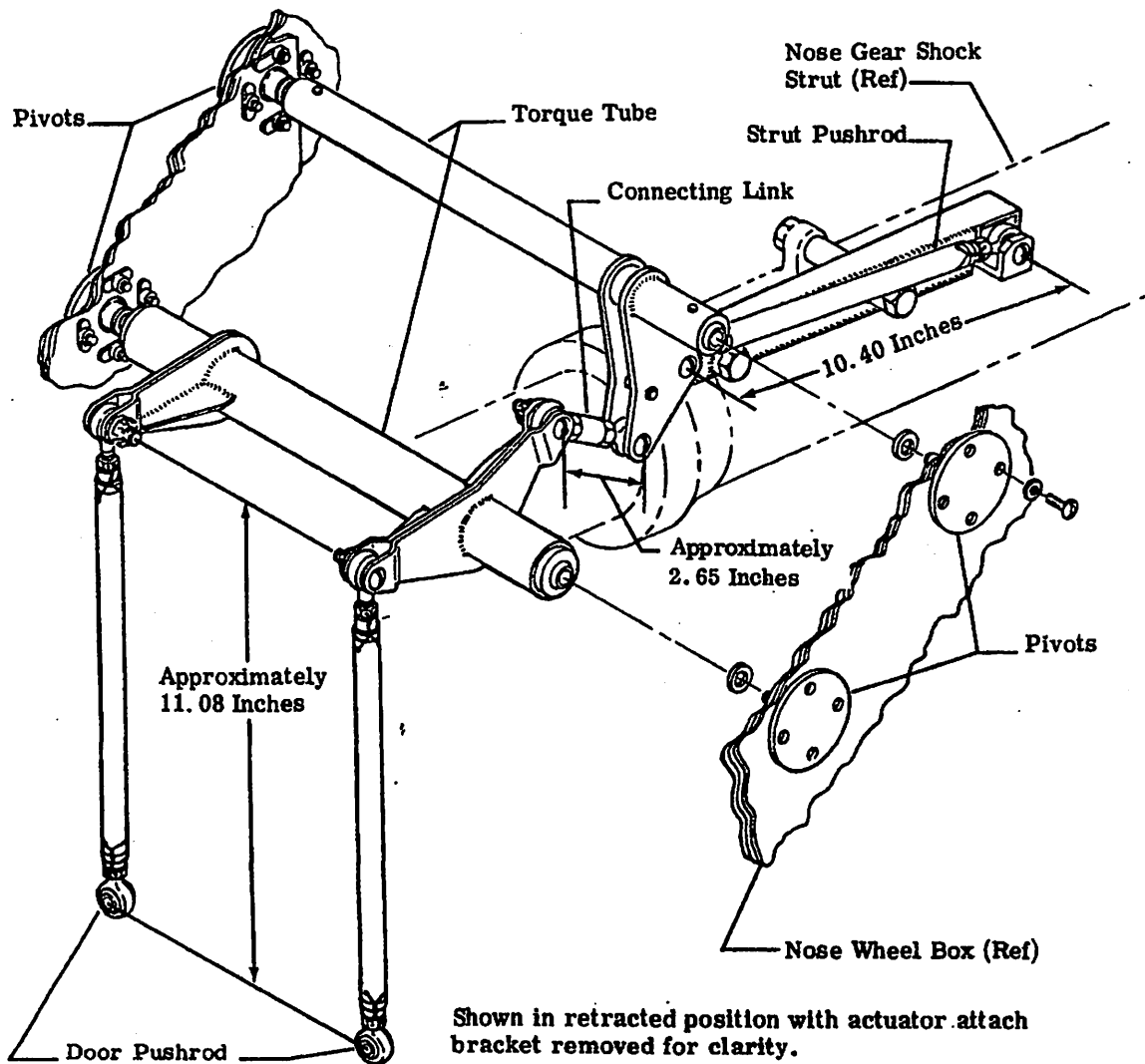
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2. Adjustment/Test

A. Rig Nose Gear Doors (See figure 202.)

- (1) Place aircraft on jacks and disconnect aircraft batteries.
- (2) With nose gear down, adjust strut pushrod to obtain 10.40 inches between rod end bearing centers.
- (3) Adjust connecting link to obtain 2.65 inches between rod end bearing centers.
- (4) Disconnect right and left door pushrods and retract gear.
- (5) Adjust door pushrods as required to bring the doors into contour with the fuselage. Approximate door pushrod length between rod end bearing centers is 11.08 inches.
- (6) Extend gear and connect door pushrods.
- (7) Remove aircraft from jacks and connect aircraft batteries.



Nose Gear Door Linkage Rigging
Figure 202

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EXTENSION AND RETRACTION - DESCRIPTION AND OPERATION

1. Description

- A. The landing gear extension and retraction system consists of the components necessary to extend and retract the landing gear, retain the gear in the up and locked position, and open and close the gear doors.
- B. The landing gear is extended by hydraulic actuators which contain internal locking devices to lock the gear in the down position. Hydraulic pressure is required to unlock and retract the landing gear.
- C. The main gear doors and the nose gear shock strut are held in the up position by uplatch hooks. The uplatch hooks are operated by uplatch actuators.
- D. The nose gear doors and the outboard main gear doors are opened and closed mechanically through linkage connected to the shock strut. The inboard main gear doors are opened and closed by a hydraulic actuator.
- E. In addition to normal gear extension, an emergency gear extension system is provided. The emergency gear extension system is controlled by an emergency gear control valve through shuttle valves.
- F. Component Description
 - (1) The gear up switches are actuated by strikers mounted on top of the left and right landing gear trunnions. These switches energize the gear door control valve to the up position to close the main gear inboard doors. If these switches are not correctly adjusted, the main gear inboard doors will remain closed and the landing gear will not retract or extend.
 - (2) The right gear down safety switch is actuated by a striker mounted on top of the right gear trunnion. This switch supplies continuous voltage to the landing gear selector valve which maintains the gear in its selected position.
 - (3) The left and right door down switches, one located on each main gear inboard door actuator, energize the gear selector valve to the up or down position after the main gear inboard doors are open. This allows the landing gear to extend or retract. Remove and discard washers and one jamnut supplied with each new switch.
 - (4) The left and right main gear squat switches are located on each strut torque arm. When the aircraft is on the ground, the squat switches:
 - (a) Complete a circuit preventing the landing gear selector valve from energizing to the up position.
 - (b) Break a circuit that prevents aircraft landing with brakes applied.
 - (c) Complete a circuit that energizes the squat switch relay panel. (Refer to Chapter 39 for squat switch relay panel.)

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2. Operation

A. Landing Gear Down Cycle Operation (See figure 1.)

- (1) Setting the Landing Gear Selector Switch to DN energizes the gear door control valve to the down position. System pressure enters the valve and is directed to the door uplatch actuators and door actuators. Pressure unlatches the main gear door uplatches and opens the main gear inboard doors. When the uplatches open, the door up switches cause the gear indication red UNSAFE lights to illuminate. When the main gear doors are open, the door down switches are actuated and energize the landing gear selector valve to the down position. Pressure flows through the valve to release the nose gear uplatch actuator and extend the nose gear. The nose gear doors and main gear outboard doors are opened by mechanical linkage between the struts and the doors. Restrictors, installed in the main and nose gear actuator return lines, dampen main gear extension. The restrictors have a free-flow bypass feature in the opposite direction for rapid gear retraction. The nose gear down and locked switch illuminates the green LOCKED DN light when the nose gear is fully extended and causes the red UNSAFE light to extinguish. When the main gear is extended, the main gear down and locked switches energize the gear door control valve to the up position and illuminate the green LOCKED DN lights. Pressure is routed to the door actuators, and the main gear inboard doors close. The main gear inboard door uplocks are engaged by actuator spring tension. When the uplock mechanism is engaged, the door up switches are actuated and the left and right red UNSAFE lights extinguish. This completes the landing gear down cycle.

B. Landing Gear Up Cycle Operation (See figure 2.)

- (1) When the landing gear clears the ground on takeoff or when the aircraft is on jacks, the squat switches are actuated to allow the landing gear selector valve to be energized to the up position. Setting the landing gear selector switch to GEAR UP energizes the gear door control valve to the down position. System pressure enters the valve and is directed to the door uplock actuators and door actuators. Pressure unlatches the gear door uplocks and opens the main gear inboard doors. When the main gear inboard doors are opened, the door down switches are actuated and the landing gear selector valve is energized to the up position. Pressure flows through the valve to the nose gear and main gear actuators causing the landing gear to retract. The nose gear doors and main gear outboard doors are closed by mechanical linkage. When the gear is retracted, the gear up switches are actuated, energizing the gear door control valve to the up position. Pressure is routed to the door actuators and the main gear inboard doors close. The nose gear and main gear inboard door uplatches are engaged by spring tension. When the gear is fully retracted (all doors closed), the door up switches and nose gear up and locked switch are actuated, causing the red UNSAFE lights to extinguish. This completes the landing gear up cycle.

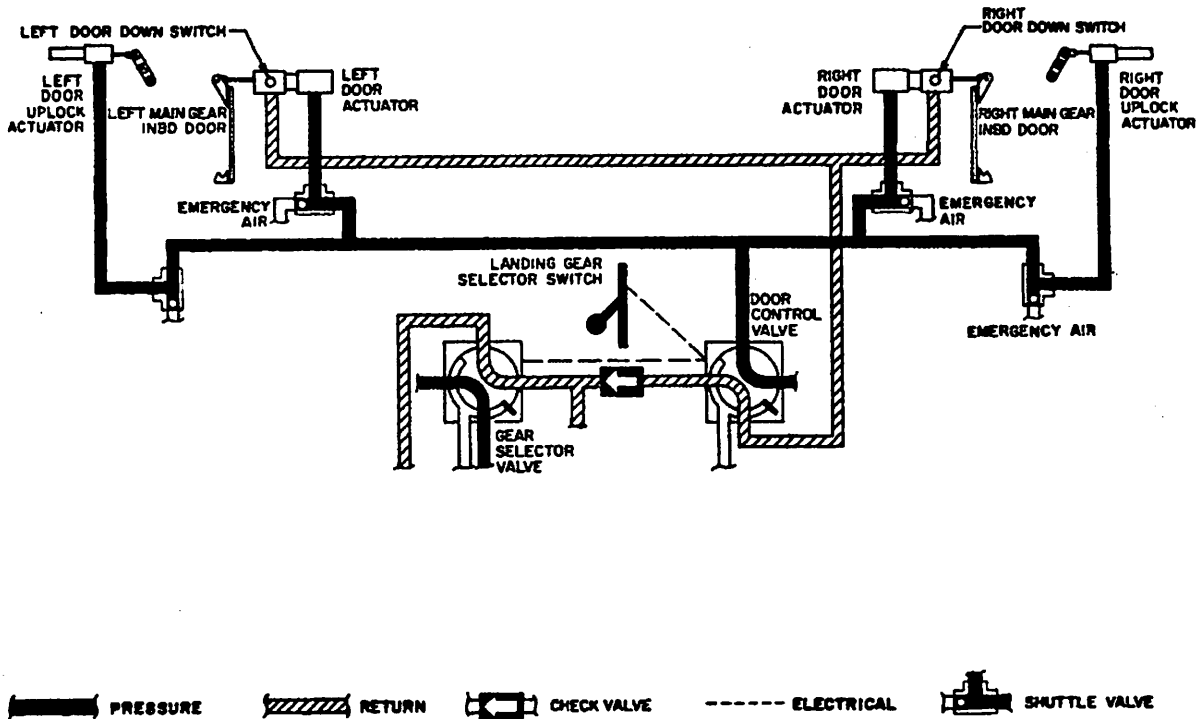
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- C. Landing Gear Down Cycle, Emergency Extension (See figure 3.)
- (1) The emergency air system to the landing gear and inboard main gear doors is operated by the emergency gear extension handle, located on the left side of the cockpit center pedestal. When the handle is depressed, air from the emergency air bottle located in the nose cone is directed to the nose gear uplatch actuator, the main gear inboard door uplatch actuators, the nose gear actuators, the main gear inboard door actuators, and both main gear actuators. Shuttle valves, installed in the system isolate the pneumatic system from the hydraulic system. After pneumatic extension of the landing gear, air pressure may be bled from the system through an overboard vent by pulling the ratchet spring release. This allows the spring-loaded handle to return to the off position.

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GEAR DOWN OPERATION SEQUENCE - HYDRAULIC SYSTEM

Note

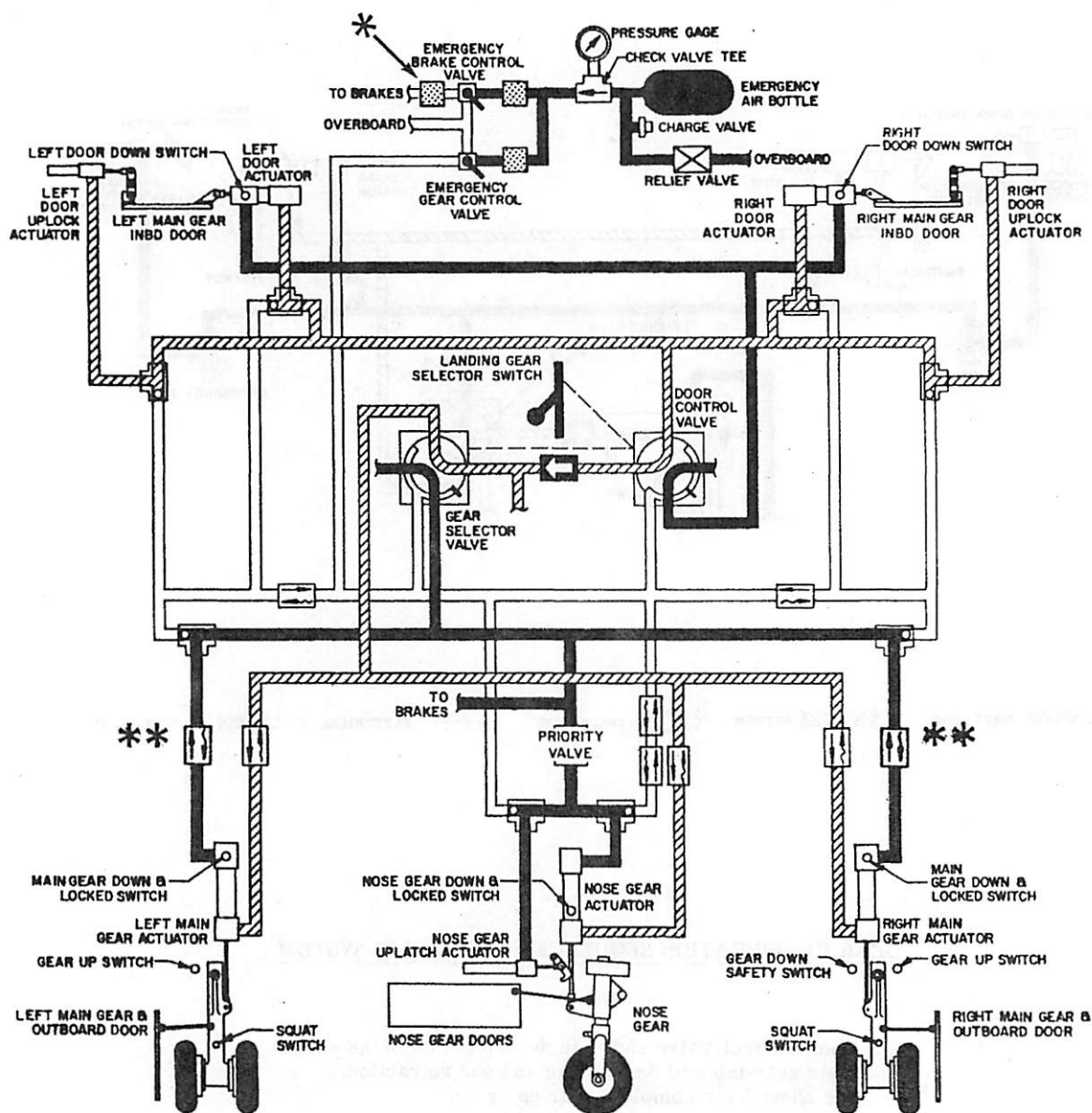
Door Control Valve shown in down position to open main gear inboard doors prior to gear extension.
See Sheet 2 for complete gear down cycle.

1. Landing Gear Selector Switch to GEAR DOWN.
2. Door Control Valve energized to down position.
3. Hydraulic pressure releases door uplock actuators and main gear inboard doors open.
4. Door down switches energize Gear Selector Valve to down position (See Sheet 2).
5. Hydraulic pressure releases nose gear uplatch actuator and extends landing gear.
6. Gear down and locked switches energize Door Control Valve to up position and main gear inboard doors close.
7. Main gear inboard doors lock in position by spring tension of door uplock actuators.

Landing Gear Hydraulic System Schematic - Gear Down Cycle
Figure 1 (Sheet 1 of 2)

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** EFFECTIVE AIRCRAFT 25-181 AND SUBSEQUENT
 * EFFECTIVE AIRCRAFT 25-195 AND SUBSEQUENT

———— PRESSURE
 - - - - - RETURN

[Symbol] CHECK VALVE
 [Symbol] RESTRICTOR
 [Symbol] FILTER

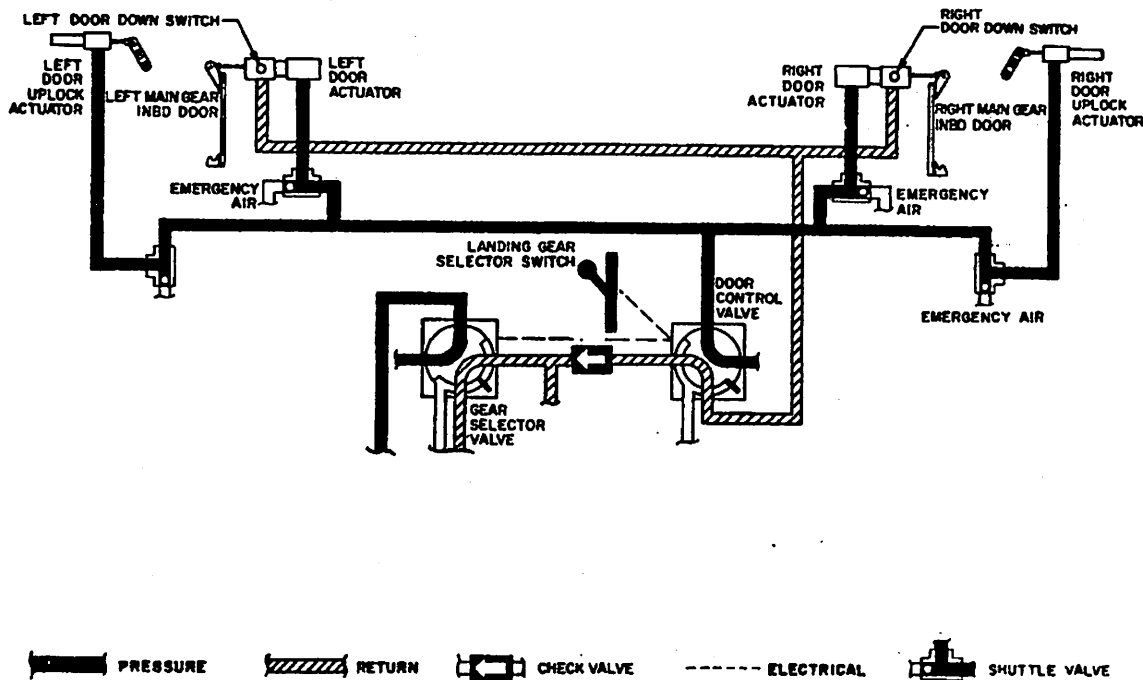
- - - - - ELECTRICAL
 [Symbol] SHUTTLE VALVE

Landing Gear Hydraulic System Schematic - Gear Down Cycle
 Figure 1 (Sheet 2 of 2)

EFFECTIVITY: ALL

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GEAR UP OPERATION SEQUENCE - HYDRAULIC SYSTEM

Note

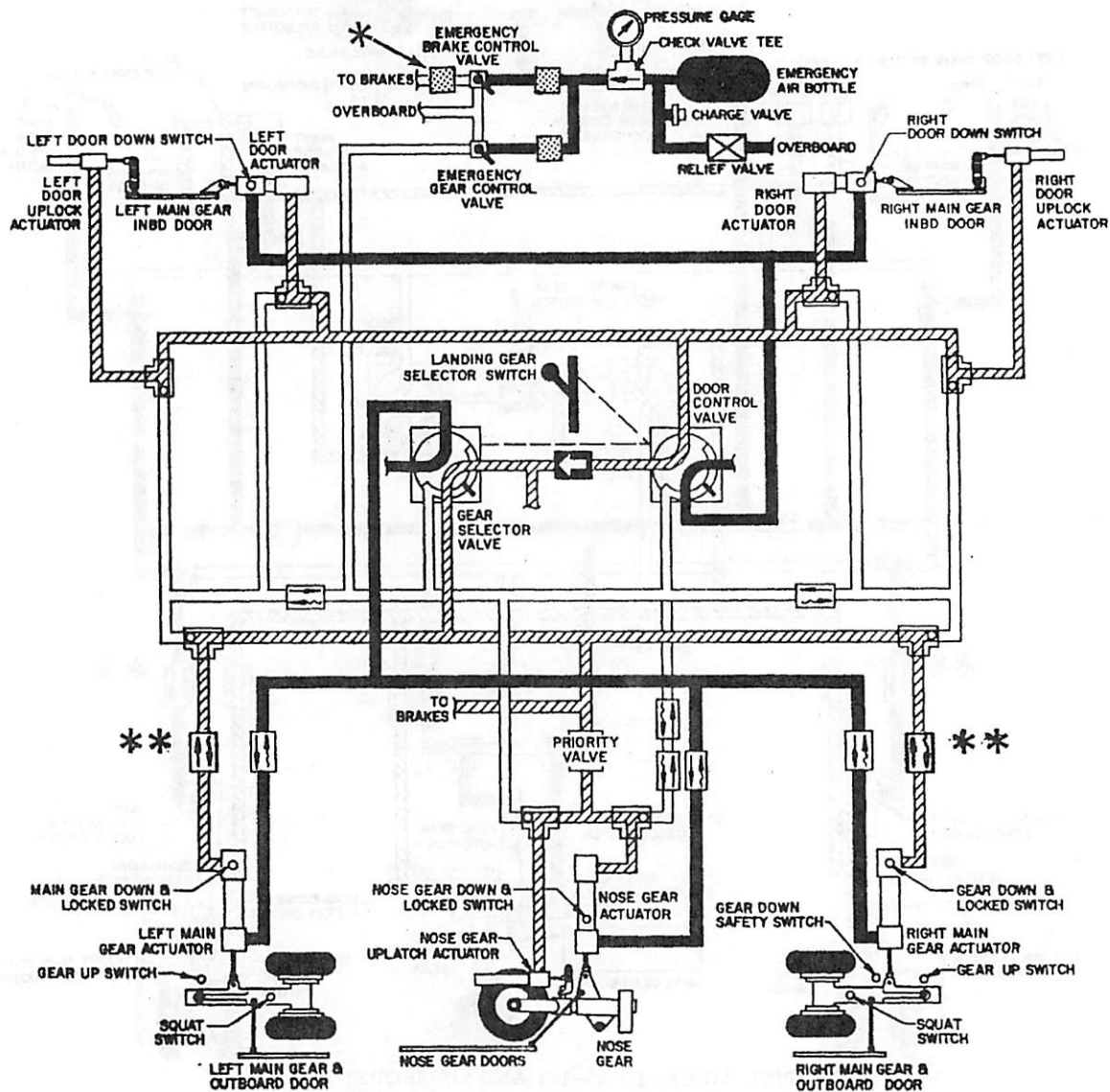
Door Control Valve shown in down position to open main gear inboard doors prior to gear retraction.
See Sheet 2 for complete gear up cycle.

1. Landing Gear Selector Switch to GEAR UP.
2. Door Control Valve energized to down position.
3. Hydraulic pressure releases door uplock actuators and main gear inboard doors open.
4. Door down switches energize Gear Selector Valve to up position (See Sheet 2).
5. Hydraulic pressure retracts landing gear.
6. Gear up switches energize Door Control Valve to up position and main gear inboard doors close.
7. Main gear inboard doors lock in position by spring tension of door uplock actuators.

Landing Gear Hydraulic System Schematic - Gear Up Cycle
Figure 2 (Sheet 1 of 2)

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** EFFECTIVE AIRCRAFT 25-181 AND SUBSEQUENT
 * EFFECTIVE AIRCRAFT 25-195 AND SUBSEQUENT

— PRESSURE
 — RETURN

— CHECK VALVE
 — RESTRICTOR
 — FILTER

--- ELECTRICAL
 — SHUTTLE VALVE

Landing Gear Hydraulic System Schematic - Gear Up Cycle
 Figure 2 (Sheet 2 of 2)

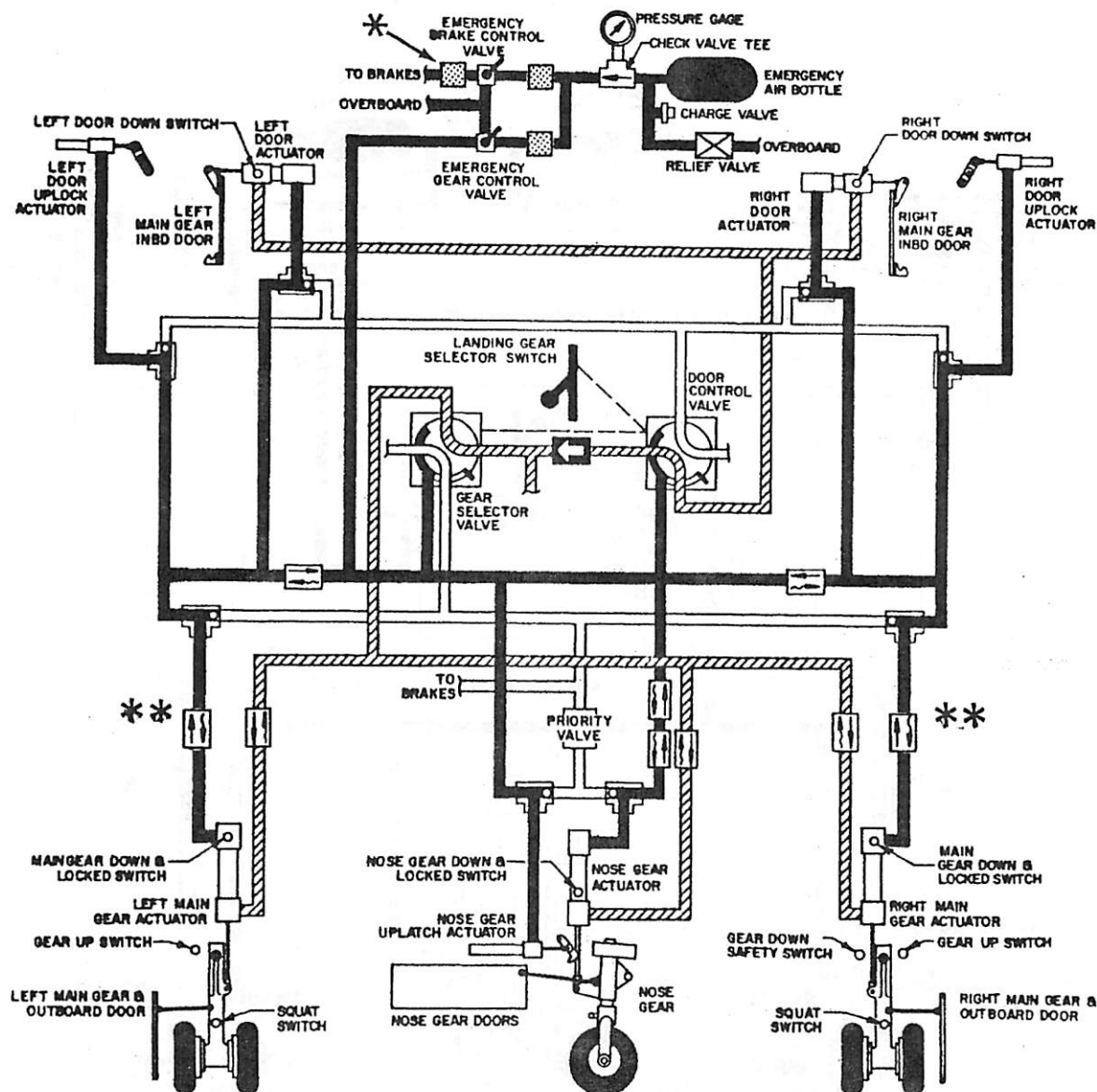
EFFECTIVITY: ALL

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maintenance manual



** EFFECTIVE AIRCRAFT 25-181 AND SUBSEQUENT

** EFFECTIVE AIRCRAFT 25-195 AND SUBSEQUENT

— PRESSURE

/// RETURN

⊢ CHECK VALVE

⊢ RESTRICTOR

⊢ FILTER

--- ELECTRICAL

⊢ SHUTTLE VALVE

GEAR DOWN OPERATION SEQUENCE - EMERGENCY AIR

1. Emergency Gear Control Valve to down position.

Note

Air pressure moves all system shuttle valves to positions shown.

2. Emergency air pressure moves door control valve and gear selector valve to down position.
3. Door uplock actuators release and main gear inboard doors open.
4. Nose gear uplatch actuator releases and gear extends. Inboard doors will remain open.

Landing Gear Emergency Air System Schematic

Figure 3

EFFECTIVITY: ALL

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GEAR UP SWITCH - MAINTENANCE PRACTICES

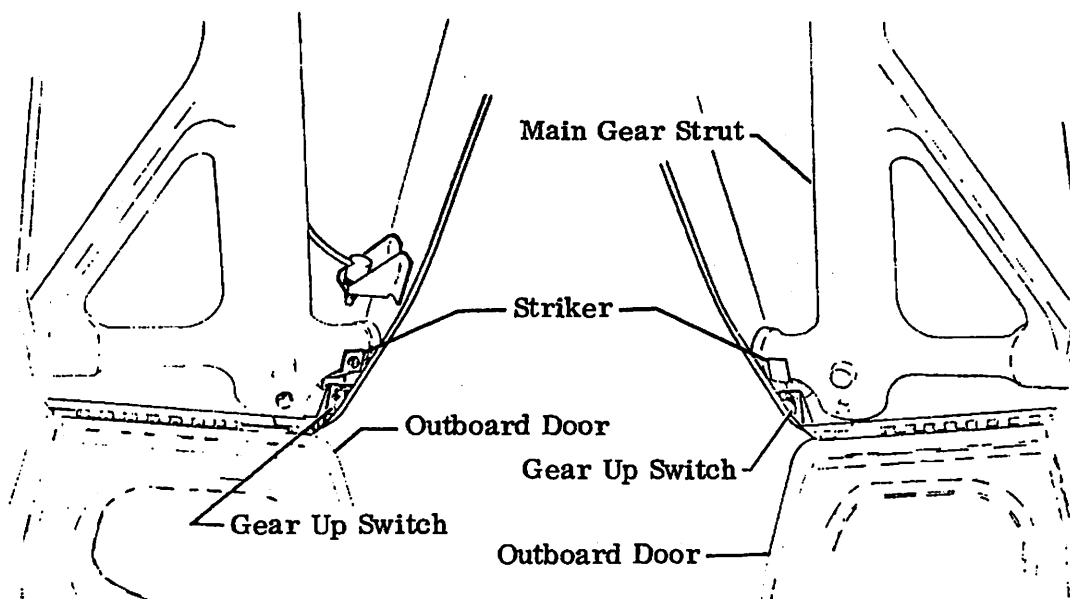
1. Adjustment/Test

A. Adjust Gear Up Switch (See figure 201.)

- (1) Place aircraft on jacks and disconnect main gear outboard door(s).
- (2) Retract landing gear.
- (3) Depressurize hydraulic system, allowing gear to drop down on door(s).
- (4) Attach an ohmmeter to switch wires No. 1 and 3.
- (5) Loosen and adjust jamnuts until ohmmeter actuation occurs, then adjust jamnuts one revolution beyond point of actuation.
- (6) Secure jamnuts and safety wire.
- (7) Connect main gear outboard door(s).
- (8) Pressurize the hydraulic system and cycle gear while checking proper operation and gear indication.
- (9) Remove aircraft from jacks.

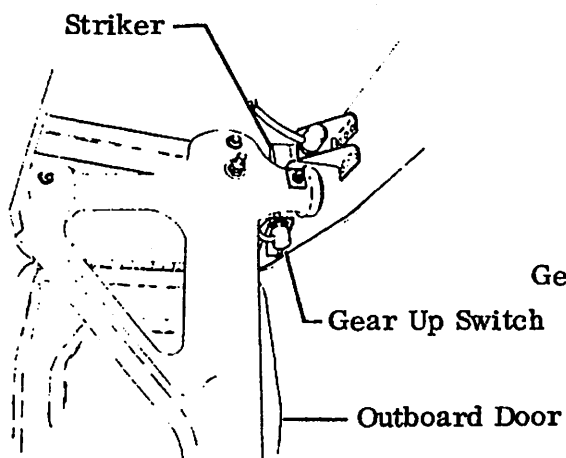
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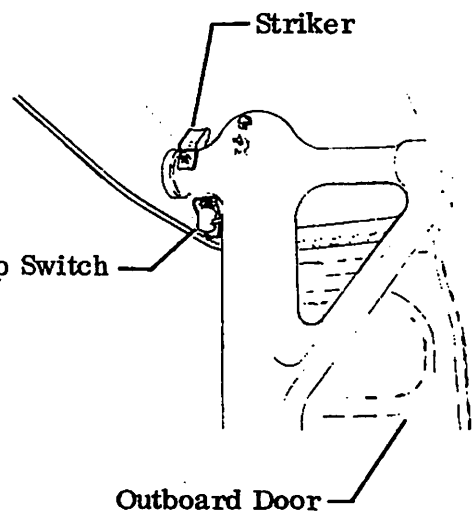


RIGHT GEAR UP

LEFT GEAR UP



RIGHT GEAR DOWN



LEFT GEAR DOWN

**Gear Up Switch Adjustment
Figure 201**

EFFECTIVITY: ALL

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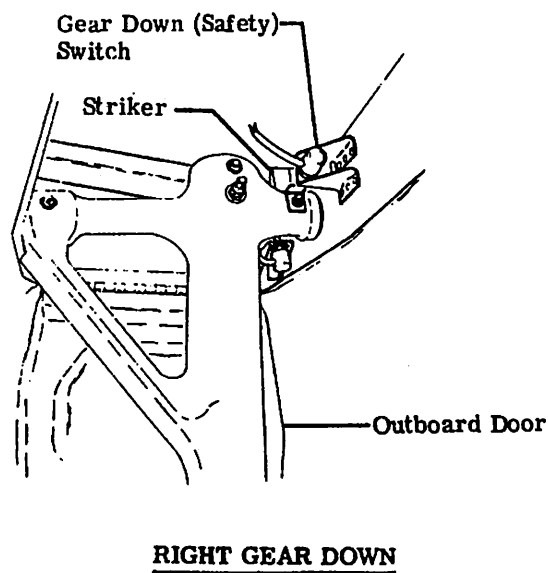
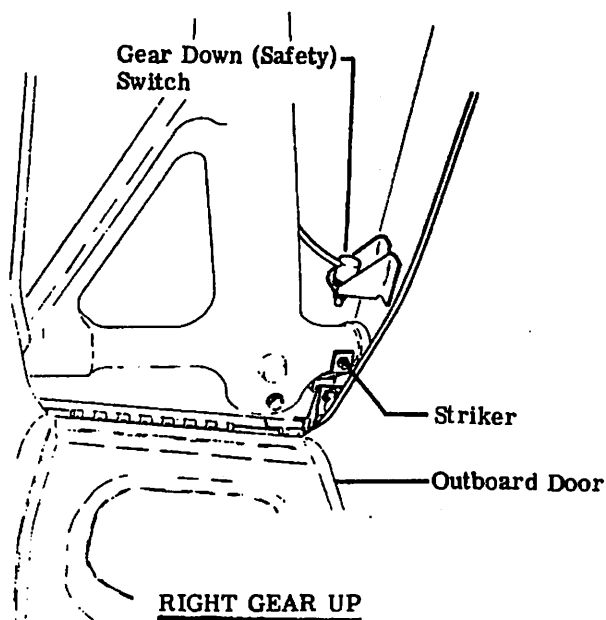
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RIGHT GEAR DOWN SAFETY SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Gear Down Safety Switch (See figure 201.)

- (1) Place aircraft on jacks.
- (2) With landing gear extended, attach an ohmmeter to switch wires No. 1 and 3.
- (3) Loosen jamnuts and adjust until ohmmeter actuation occurs, then adjust jamnuts one revolution beyond point of actuation.
- (4) Secure jamnuts and safety wire.
- (5) Cycle gear while checking for proper operation and gear indication.
- (6) Remove aircraft from jacks.



Right Gear Down Safety Switch Adjustment
Figure 201

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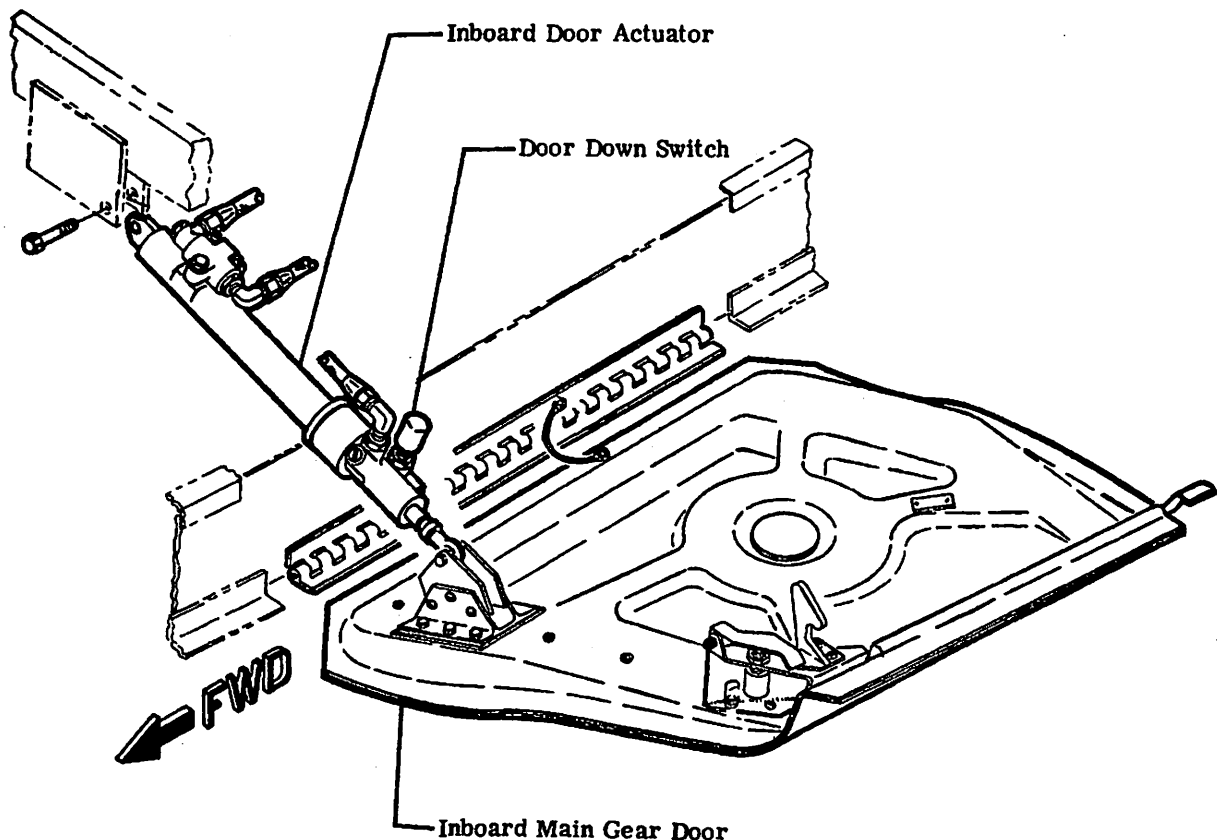
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DOOR DOWN SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Door Down Switch (See figure 201.)

- (1) Place aircraft on jacks and disconnect main gear inboard doors.
- (2) Retract landing gear.
- (3) Attach an ohmmeter to switch wires No. 4 and 6.
- (4) Loosen jamnut and screw switch into actuator until ohmmeter actuation occurs.
- (5) Back switch off from one-half to one turn until switch contact breaks and wiring is routed in line with actuator.
- (6) Secure jamnut and safety wire.
- (7) Cycle gear while checking for proper operation and indication.
- (8) Remove aircraft from jacks.



Door Down Switch Adjustment
Figure 201

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MAIN GEAR SQUAT SWITCHES - MAINTENANCE PRACTICES

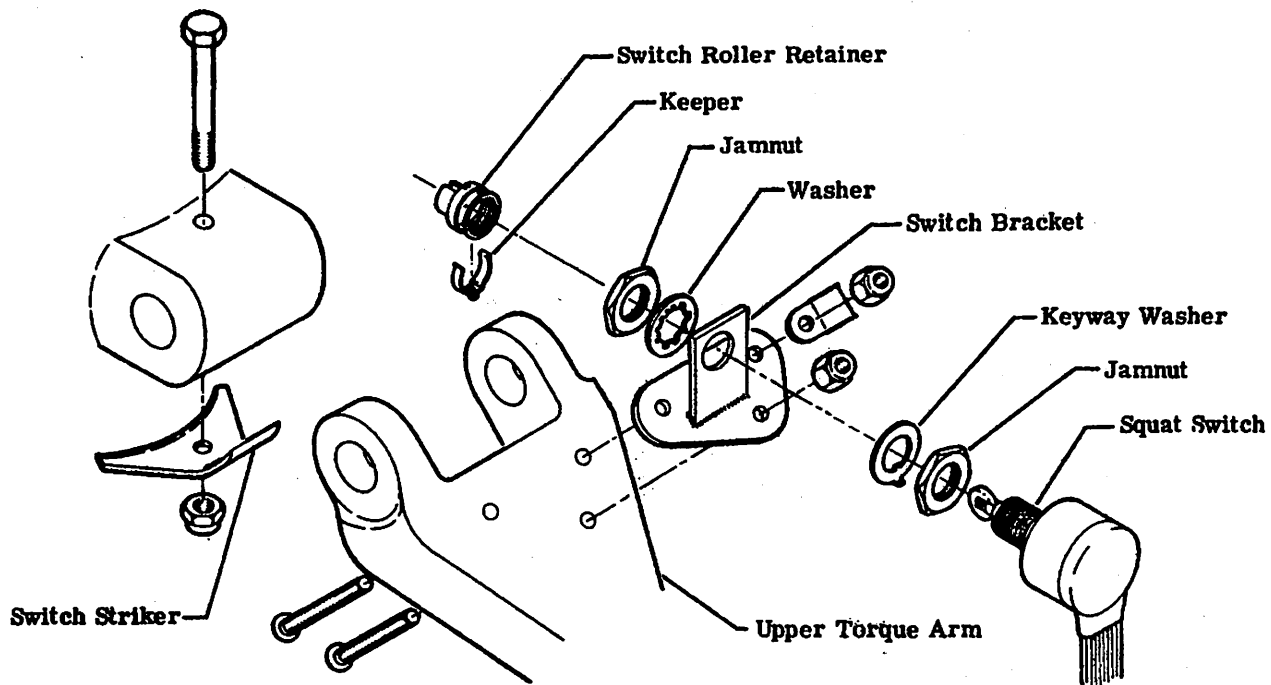
1. Removal/Installation

- A. Remove Squat Switch (See figure 201.)
 - (1) Remove attaching parts and switch bracket and switch from strut torque arm.
 - (2) Remove keeper from switch roller retainer.
 - (3) Remove switch roller retainer. Count number of turns required to remove switch roller retainer from switch.
 - (4) Remove safety wire from switch jamnuts.
 - (5) Remove jamnuts and squat switch from switch bracket.
- B. Install Squat Switch (See figure 201.)
 - (1) Install jamnut, keyway washer, and washer on switch.
 - (2) Position switch through hole in switch bracket and install remaining jamnut.
 - (3) Install switch roller retainer on switch same number of turns as required for its removal in step A. (3).
 - (4) Install keeper and assure that keeper is properly seated through hole in roller retainer and switch keyway.
 - (5) Install switch bracket on strut torque arm and secure with attaching parts.
 - (6) Adjust squat switch. (Refer to Adjustment/Test.)

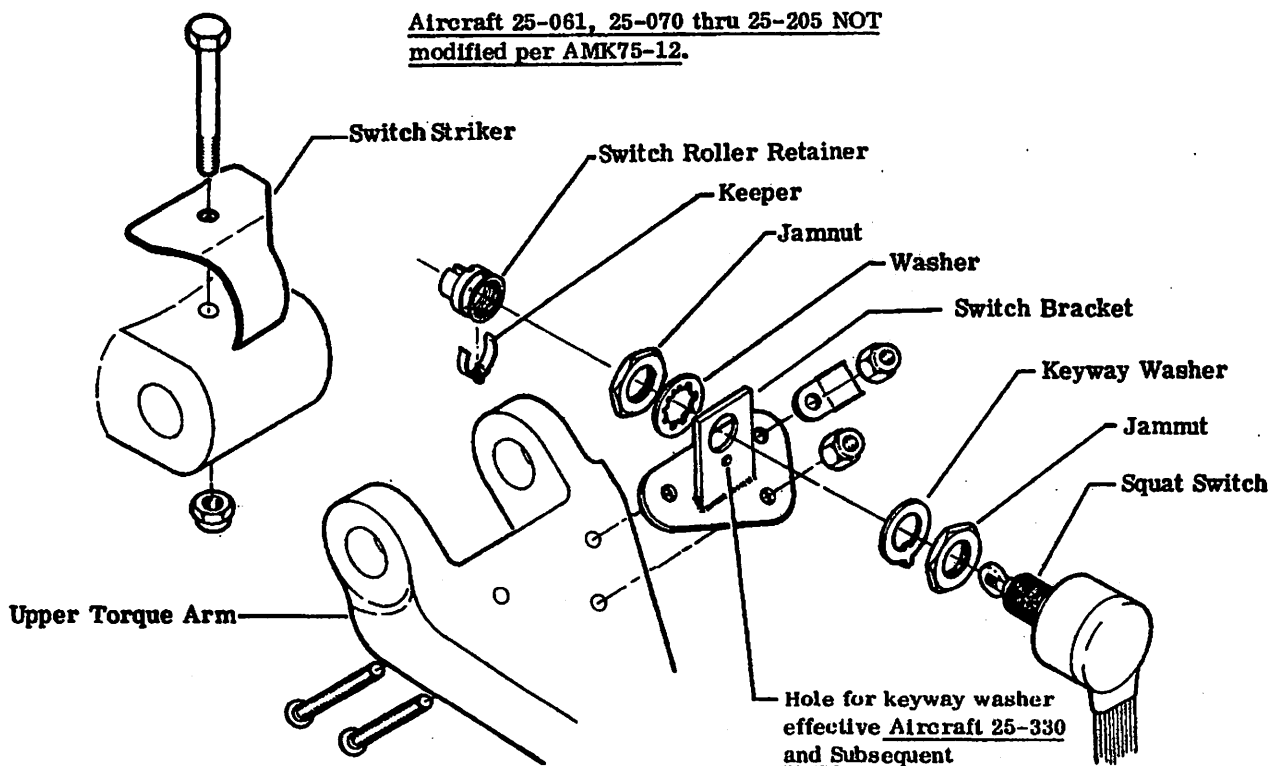
2. Adjustment/Test

- A. Adjust Main Gear Squat Switches (Aircraft 25-061, 25-070 thru 25-205 not modified per AMK 75-12) (See figure 201.)
 - (1) Place aircraft on jacks.
 - (2) Attach an ohmmeter to switch wires No. 1 and 3.
 - (3) Loosen jamnuts and adjust until ohmmeter actuation occurs, with strut within 0.50 (± 0.10) inch from fully extended position. After extension, the gear may be manually positioned to make necessary switch adjustments. Do not rotate switches after final adjustment is made.
 - (4) Secure jamnuts and safety wire.
 - (5) Cycle gear while checking for proper operation and indication.
 - (6) Remove aircraft from jacks.
- B. Adjust Main Gear Squat Switches (Aircraft 25-206 and Subsequent and prior Aircraft modified per AMK 75-12) (See figure 201.)
 - (1) Attach an ohmmeter to switch wires No. 1 and 3.
 - (2) Loosen jamnuts and adjust until ohmmeter actuation occurs.
 - (3) Secure jamnuts and safety wire.
 - (4) Place aircraft on jacks.
 - (5) Cycle gear while checking for proper operation and indication.
 - (6) Remove aircraft from jacks.

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Aircraft 25-061, 25-070 thru 25-205 NOT
 modified per AMK75-12.



Aircraft 25-206 and Subsequent and
 Aircraft modified per AMK 75-12.

**Main Gear Squat Switch Adjustment
 Figure 201**

EFFECTIVITY: NOTED

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MAIN GEAR EXTENSION AND RETRACTION - DESCRIPTION AND OPERATION

1. **Description**

- A. The main gear extension and retraction system consists of two main gear actuators, two main gear inboard door actuators, and two main gear inboard door uplatches.

2. **Operation**

- A. Each main gear is retracted and extended by an actuator which is attached to the shock strut and to a pillar in the wing structure. A spring load, built into the actuator, drives a downlock ball housing which entraps steel balls in a notch in the piston rod causing a positive mechanical lock when the main gear is in the down position. An integral switch in the actuator transmits a gear down and locked indication to the position indicator lights when the downlock ball housing has extended over the mechanical lock. Hydraulic pressure in excess of 250 psi is required to retract the housing against the spring to release the downlock mechanism and permit gear retraction. During an emergency, the acutator is extended by emergency air pressure to the down position. Refer to 32-33-00 for emergency extension system.

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MAIN GEAR ACTUATOR - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Main Gear Actuator (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Lower tailcone access door and disconnect batteries.
- (3) Depressurize hydraulic system. (Refer to Chapter 29.)
- (4) Note position of hydraulic hoses at retract port and mark clocking of fitting on actuator.
- (5) Disconnect hydraulic hoses. Cap hoses and plug actuator ports.
- (6) Remove actuator down and locked switch and electrical wiring clamps.
- (7) Cut safety wire (noting manner in which safety wire is installed) and remove cotter pin, nut, retaining washer, and washer from end of actuator attached to pillar in wing.
- (8) Remove cotter pin, nut, washer, and bolt securing actuator to shock strut.
- (9) Remove actuator from aircraft. As actuator is being removed from aircraft, rotate actuator to dislocate and remove down and locked switch ball.

B. Install Main Gear Actuator (See figure 201.)

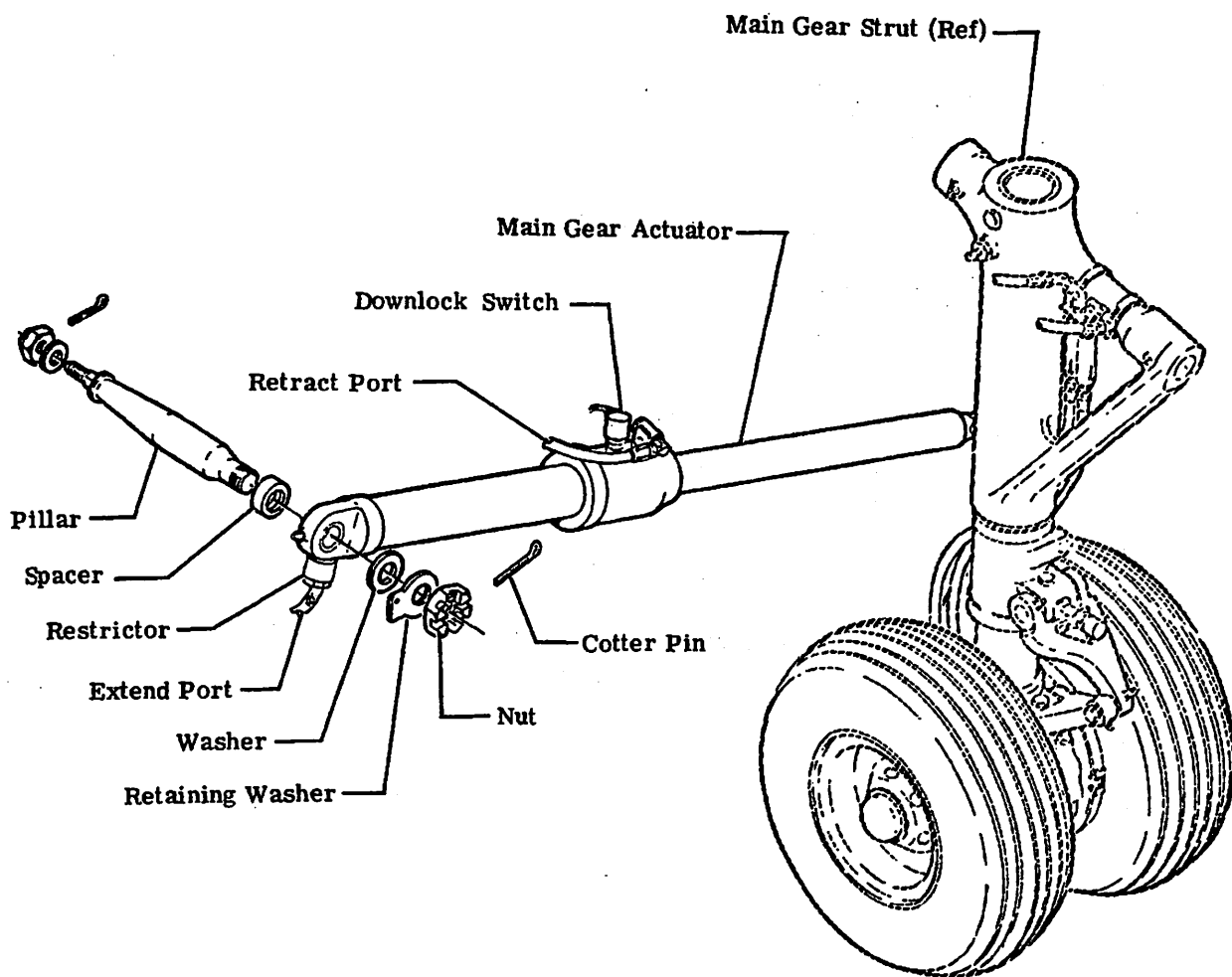
NOTE: A gap of 0.001 to 0.036 inch must exist between bearing surfaces of actuator pin bushing and actuator pin to assure pin's freedom of movement.

- (1) Position actuator on pillar and strut and secure with attaching parts. Safety wire.
- (2) Install switch actuator ball, down and locked switch, and electrical wiring clamps. Adjust down and locked switch. (Refer to 32-60-01.)

CAUTION: ASSURE THAT ACTUATOR RETRACT HOSE IS ROUTED AND CONNECTED THE SAME AS IT WAS PRIOR TO REMOVAL. THIS WILL PREVENT HOSE FROM BEING TWISTED. AN IMPROPERLY INSTALLED HOSE WILL RESULT IN HOSE RUPTURE AND HYDRAULIC SYSTEM FAILURE.

- (3) Remove plugs and caps and connect hydraulic hoses to actuators.
- (4) Connect main gear inboard door actuator to door and secure with attaching parts.

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Main Gear Actuator Installation
Figure 201

EFFECTIVITY: ALL

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- (5) Pressurize hydraulic system, cycle landing gear system, and check for proper gear indication and operation.
- (6) Remove aircraft from jacks.
- (7) With the aircraft fully fueled and resting on landing gear, main gear struts shall be perpendicular to ground line within $0^{\circ} 15'$.

2. Adjustment/Test

A. Functionally Test Main Gear Actuator

- (1) After assembly, place actuator in a horizontal position with extend port up.
- (2) Fill through extend port with clean MIL-H-5606 hydraulic fluid, bleeding unit of all air.
- (3) Place actuator in a horizontal fully extended position.
- (4) Connect a hydraulic power source to actuator and cycle actuator twenty-five times at 1500 psi. Leakage shall not exceed one drop.
- (5) Connect a hydraulic test bench, capable of at least 3000 psi, to extend port with retract port open.
- (6) Pressurize to 2250 psi for 2 minutes, then drop pressure to 5 psi for an additional 2 minutes. There shall be no leakage from the open retract port during this period.
- (7) Disconnect extend port and connect hydraulic test bench to retract port with extend port open.
- (8) Pressurize to 3000 psi for 2 minutes, then drop pressure to 5 psi for an additional 2 minutes. There shall be no leakage from the open extend port during this period.

3. Approved Repairs

A. Disassemble Main Gear Actuator (See figure 202.)

- (1) During disassembly, tag and keep all parts together for assembly into same unit. Do not remove nameplate unless it is damaged.

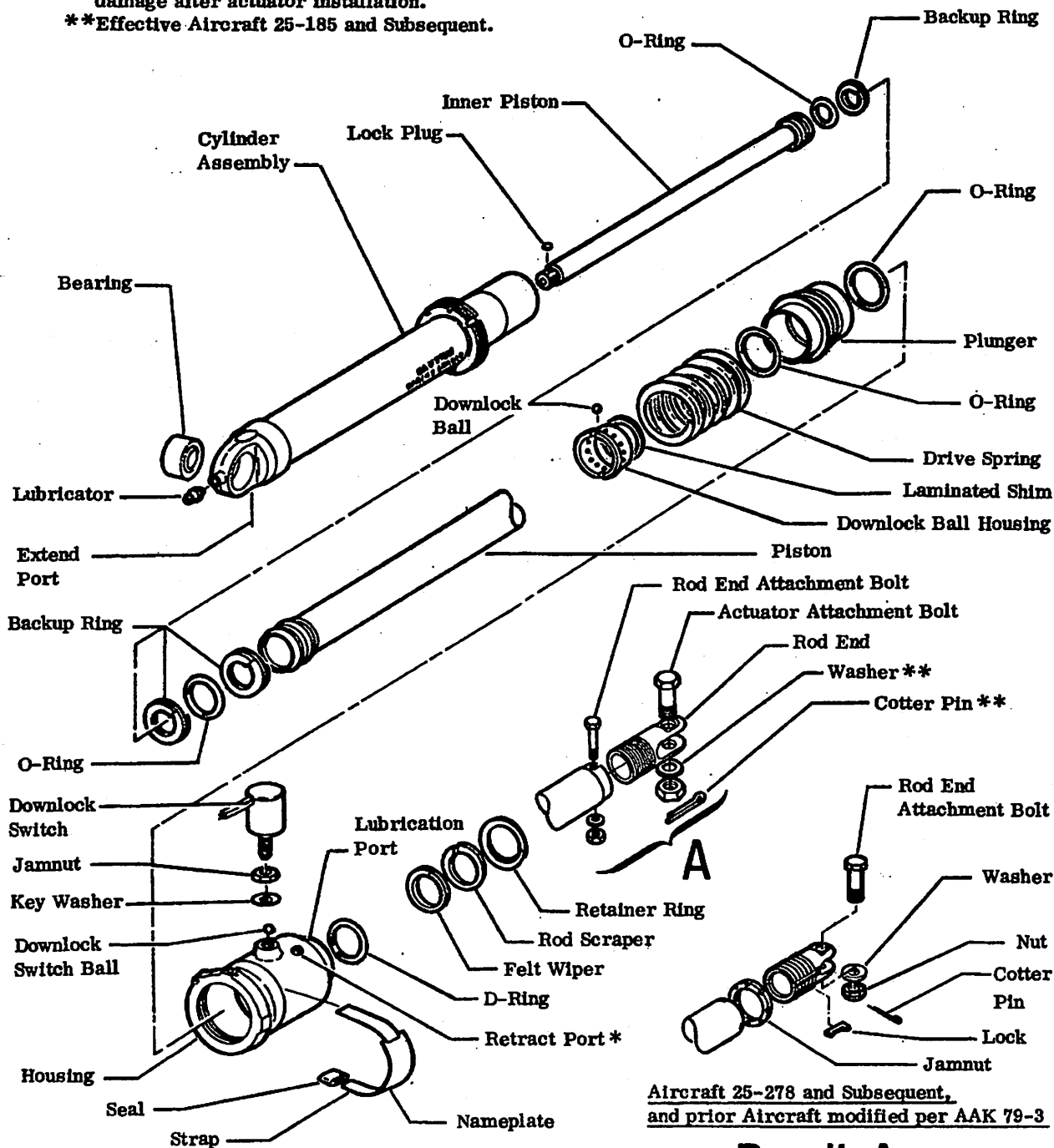
CAUTION: THE ACTUATOR IS SPRING-LOADED TO 310 POUNDS. USE SPECIAL PRECAUTION WHEN UNSCREWING CYLINDER ASSEMBLY FROM HOUSING TO PREVENT EITHER PART FROM SPRINGING OUT OF CONTROL. OTHERWISE BODILY INJURY MAY RESULT.

- (2) Remove lockscrew and, while taking care to control actuator spring load, slowly unscrew cylinder assembly from housing.
- (3) Remove nut and bolt attaching rod end to piston and remove rod end from piston.

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*Clock hose swivel fitting in same position as it was prior to removal to avoid hose damage after actuator installation.
 **Effective Aircraft 25-185 and Subsequent.



Detail A

Main Gear Actuator Assembly
Figure 202

11-36B

EFFECTIVITY: ALL

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- (4) Slide piston, inner piston, laminated shim, drive spring, plunger, and downlock ball housing from actuator housing. Retrieve downlock balls which will fall free as piston is removed.
- (5) Remove retainer ring and rod scraper from actuator housing. Remove and discard all O-rings, D-ring, backup rings, and felt wiper.
- (6) Bearing and lubricators are pressed into place. Bearings are staked when installed. Remove only if replacement is required.
- (7) Clean metal parts with Stoddard solvent or equivalent. Assure that no solvent is allowed to remain in internal cavities or surfaces.
- (8) Remove all fingerprints from machined surfaces not plated or painted using a suitable fingerprint neutralizer.
- (9) Dry metal parts after cleaning with clean, dry, compressed air.
- (10) Coat internal surfaces with MIL-H-5606 hydraulic fluid.
- (11) Wrap cleaned parts in lint-free cloth or clean waxed paper while awaiting repair and assembly.
- (12) When storing unplated, machine surfaced parts more than a few hours, coat with MIL-C-11796 preservative. Remove preservative prior to assembly of parts.

B. Inspect Main Gear Actuator (See figure 203.)

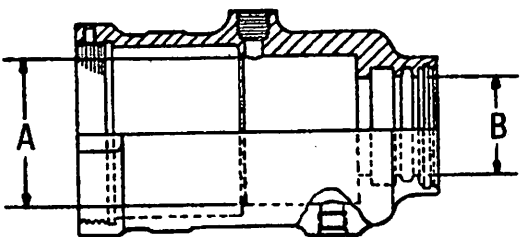
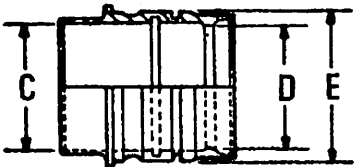
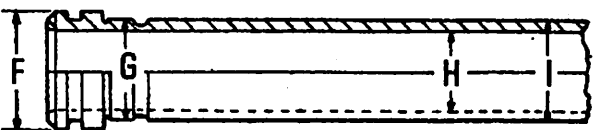
- (1) Screen drained hydraulic fluid for metal particles which may be used as an aid in determining trouble source.
- (2) Inspect all metal parts including welds and machined and forged parts for structural failure, nicks, burrs, corrosion, worn threads, and excessive wear.
- (3) Perform a magnetic-type inspection on all ferrous parts. Perform fluorescent penetrant check on all aluminum parts. After ferrous parts inspection, demagnetize to remove all metal particles.
- (4) Inspect OD and ID surfaces of piston, cylinder assembly, and ID surfaces of housing assembly for galls, scratches, scoring, corrosion, or worn plating.
- (5) Check to assure that spring height is at least 2.44 inches when supporting a 310-pound load.

NOTE: Aircraft 25-278 and Subsequent and prior Aircraft modified per AAK 79-3, are equipped with a jamnut and lock-type rod end which eliminates possibility of looseness. Aircraft with jamnut type installation should skip steps (6) and (8).

- (6) With rod end threaded into piston, check rod end for looseness. If looseness exists between actuator piston and rod end, remove rod end.
- (7) Inspect threads of rod end and piston. Replace if damaged.

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Nomenclature	Outside Diameter (Inches)		Inside Diameter (Inches)		Total Clearance Diameter (Inches)		Surface Finish
	Original	Minimum	Original	Maximum	Original	Maximum	
Cylinder	<u>2.436</u> 2.434	2.433	<u>2.247</u> 2.245	2.252	<u>0.006</u> 0.002	0.016	16 RMS
Inner Piston	<u>1.489</u> 1.487	1.484					63 RMS
Piston Head (F)	<u>2.243</u> 2.241	2.236					63 RMS
Piston Rod (G)	<u>1.873</u> 1.871	1.870					32 RMS
Piston (I)	<u>1.873</u> 1.871	1.870					16 RMS
Piston (H)			<u>1.493</u> 1.491	1.496	<u>0.006</u> 0.002	0.012	16 RMS
Ball Housing	<u>2.217</u> 2.215	2.213	<u>1.877</u> 1.875	1.880	<u>0.006</u> 0.002	0.010	125 RMS
Plunger (E)	<u>2.811</u> 2.809	2.807					32 RMS
Plunger (C)			<u>2.440</u> 2.438	2.442	<u>0.006</u> 0.002	0.009	32 RMS
Plunger (D)			<u>2.230</u> 2.228	2.235	<u>0.015</u> 0.011	0.022	125 RMS
Housing (B)			<u>1.878</u> 1.876	1.885	<u>0.007</u> 0.003	0.015	32 RMS
Housing (A)			<u>2.815</u> 2.813	2.819	<u>0.006</u> 0.002	0.012	32 RMS

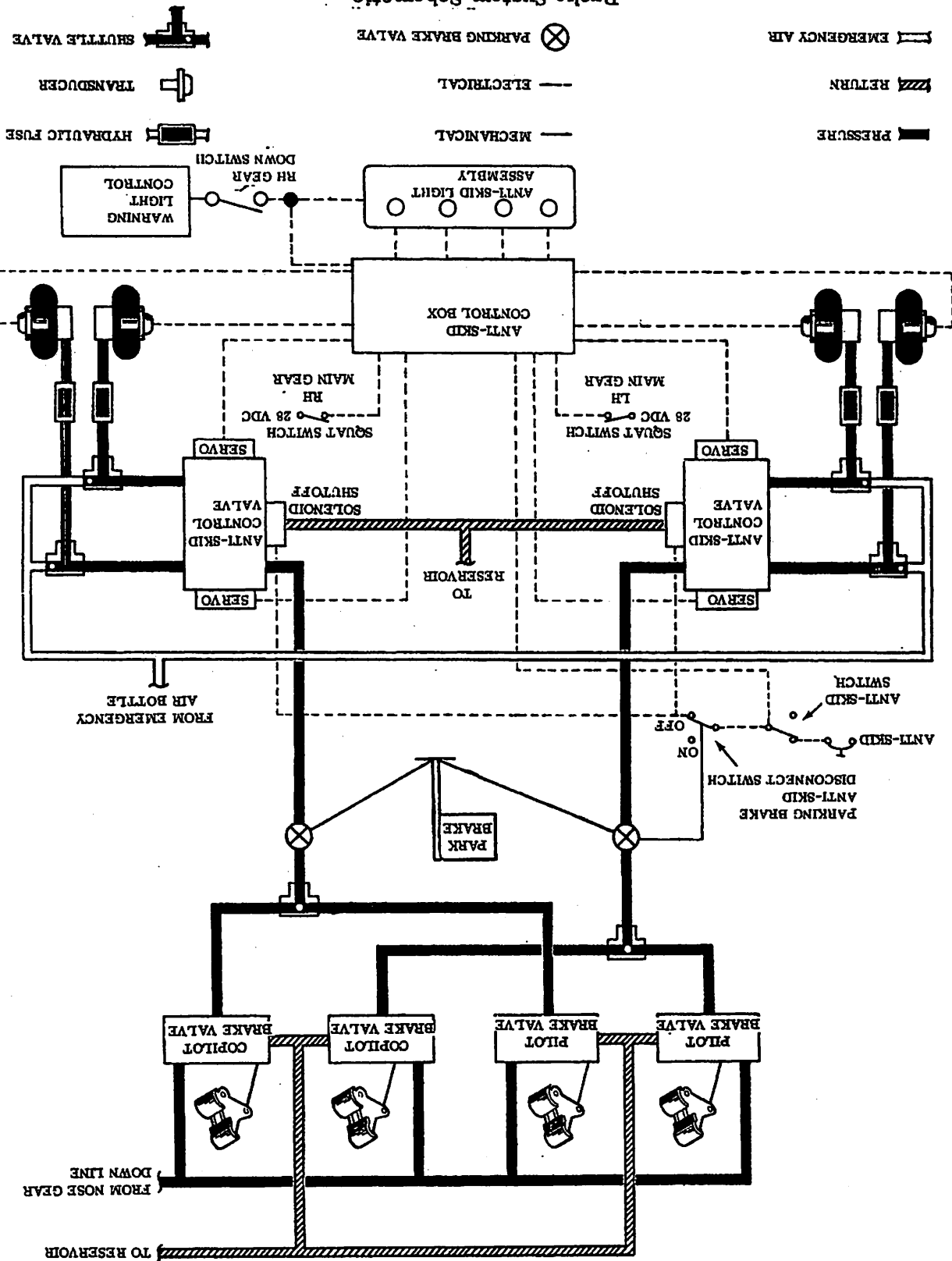
Main Gear Actuator Wear Tolerances
Figure 203

EFFECTIVITY: ALL

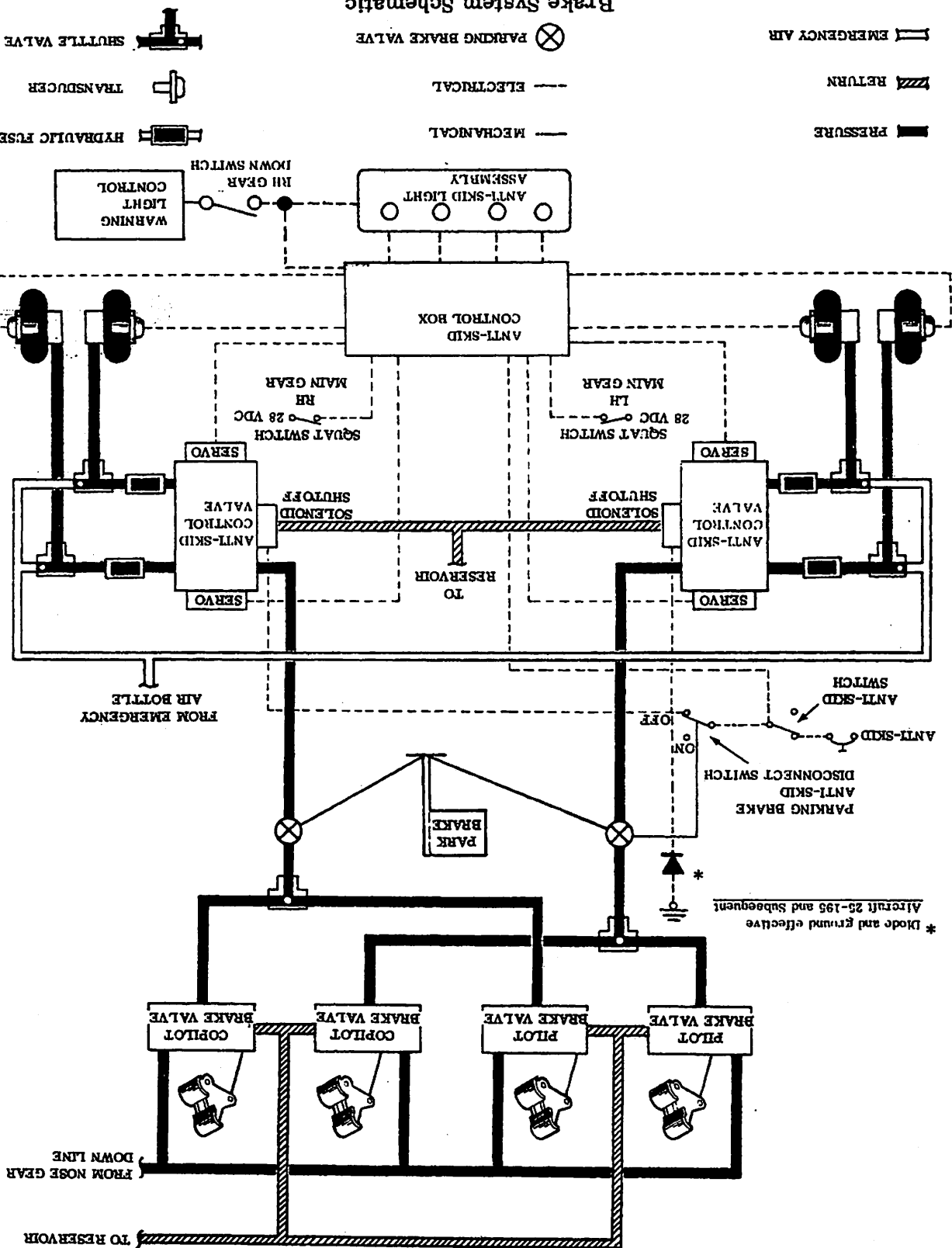
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11-8A-1

Brake System Schematic
 Figure 1 (Sheet 1 of 2)



EFFECTIVITY: 25-061, 25-070 THRU 25-184



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BRAKE SYSTEM - MAINTENANCE PRACTICES

1. Servicing

- NOTE:
- Brake system bleeding is required whenever the system is opened at any point between the power brake valves and the wheel brake assemblies, whenever the brakes become spongy in service or whenever the parking brake will no longer hold. In case of parking brake malfunction, the system should be further checked for leakage.
 - Normally the head pressure in the hydraulic reservoir will be sufficient to bleed the brakes; however, if the head pressure is not sufficient, the hydraulic system should be pressurized using a ground hydraulic power cart. System pressure should be maintained at approximately 600 psi during bleeding. (Refer to Chapter 29 for hydraulic reservoir.)
 - When bleeding the brakes, use care to depress brake pedal slowly or excessive fluid flow rate through the bleed port could cause the hydraulic fuses to close.
 - Hydraulic fuses which are accidentally closed should automatically open after approximately one minute. Fuses which do not open must be removed, placed on a hydraulic test bench, and subjected to reverse flow.

A. Bleed the Brake System

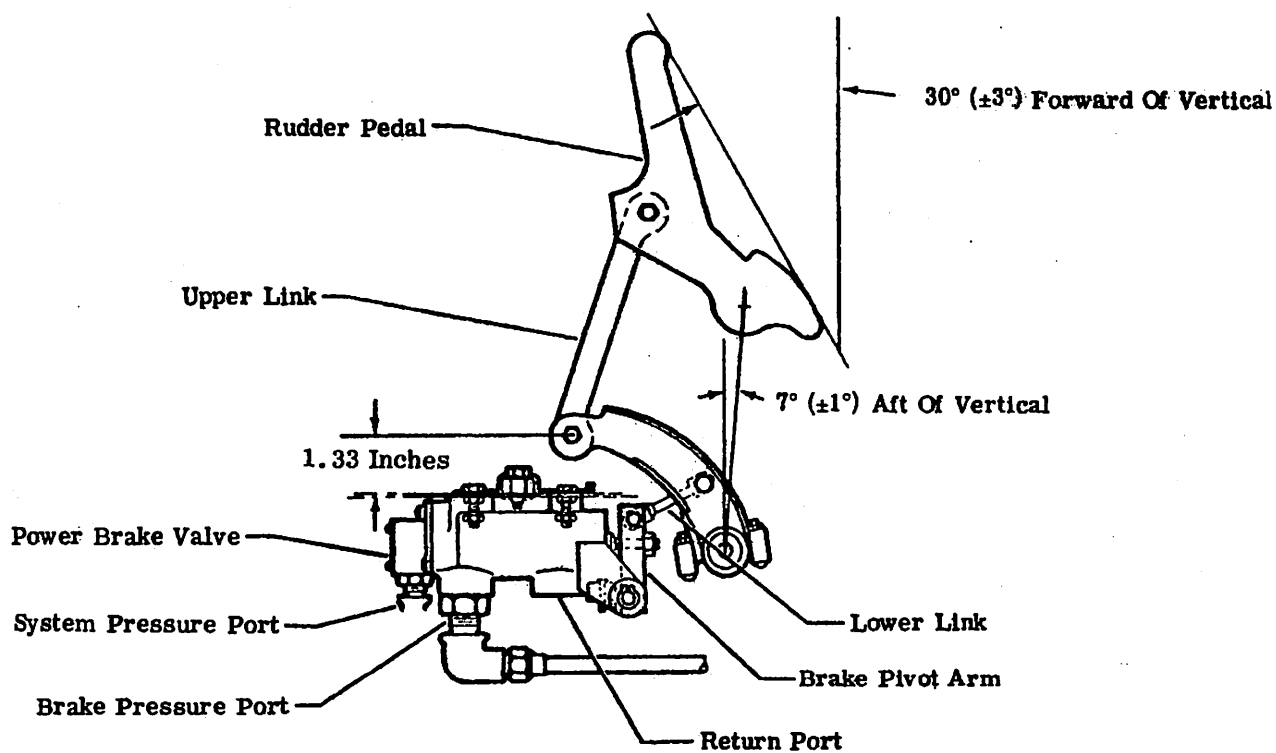
- (1) Remove safety wire and bleeder screw from brake housing. Install a suitable adapter with a length of transparent hose attached.
- (2) Slowly depress applicable brake pedal and allow hydraulic fluid to flow until there are no air bubbles evident.
- (3) Remove adapter and hose and install bleeder screw prior to releasing brake pedal. Safety wire bleeder screw.
- (4) Repeat steps 1 through 3 for remaining brake assemblies.
- (5) Service hydraulic reservoir. (Refer to Chapter 12.)

2. Adjustment/Test

A. Rig the Brakes (See figure 201.)

- (1) Rig rudder pedals. (Refer to Chapter 27.)
- (2) Adjust lower link to obtain brake pedal angle of 30° ($\pm 3^{\circ}$) forward of vertical (1.33 inches above floorboard as shown).
- (3) Install a pressure gage (0 to 2000 psi) in brake pressure line.
- (4) With pedals in neutral, apply an 80-pound load with a spring scale to pedal toe position. Pressure gage should indicate 400 psi.
- (5) With either pedal fully forward and brakes applied, pressure gage should indicate approximately 1500 psi.
- (6) With maximum braking force applied to both pedals, gage should indicate approximately 1500 psi normal system pressure.

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Brake Rigging
Figure 201

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MULTIPLE DISC BRAKE - MAINTENANCE PRACTICES

1. Removal/Installation

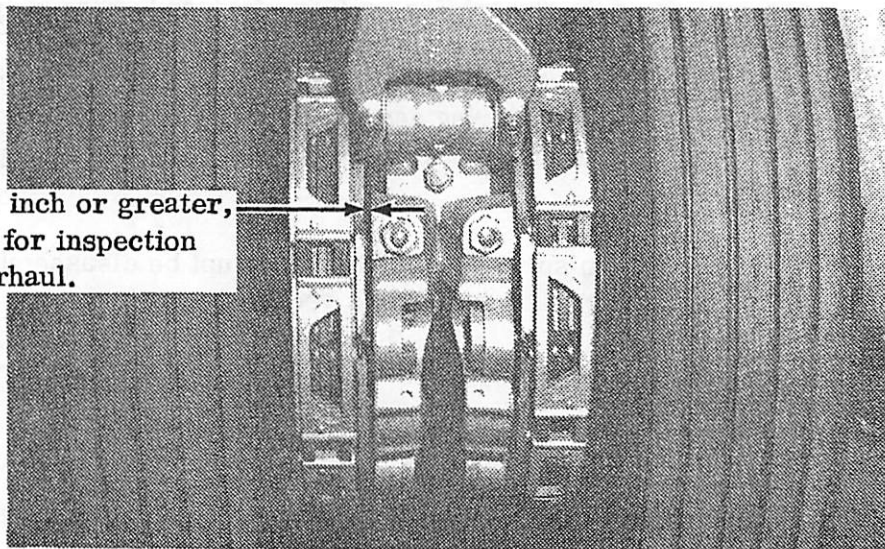
A. Remove Multiple Disc Brake

- (1) Place aircraft on jacks.
- (2) Depressurize hydraulic system.
- (3) Remove main landing gear wheels. Refer to 32-42-00.
- (4) Disconnect hydraulic hoses at brake housings. Cap or plug all exposed hydraulic fittings.
- (5) Remove the two self-locking nuts, washers, and bolts attaching brake housings together.
- (6) Remove brake assembly from aircraft.

B. Install Multiple Disc Brake

- (1) Install brake assembly on main gear axle.
- (2) Secure brake assemblies together using the two housing attachment bolts, washers and self-locking nuts.
- (3) Install main landing gear wheels. Refer to 32-42-00.
- (4) Connect hydraulic brake hoses to brake housing ports.
- (5) Pressurize hydraulic system and bleed brakes. Refer to 32-43-00.

If 0.330 inch or greater,
remove for inspection
and overhaul.



Multiple Disc Brake Check
Figure 201

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2. Inspection/Check

A. Inspect Brakes for Excessive Wear (See figure 201.)

- (1) Pressurize hydraulic system.
- (2) Depress brake pedals and set parking brake.
- (3) Measure distance between brake housing and pressure plate as shown.
- (4) If distance between brake housing and pressure plate is 0.330 inch or greater, remove brake assemblies for inspection and overhaul.

3. Adjustment/Test

A. Functionally Test Multiple Disc Brake

- (1) Apply normal braking pressure and hold several times while checking for leakage.
- (2) Release brake pressure and observe brake release.
- (3) Rotate main landing gear wheels and check for freedom of movement.

4. Approved Repairs

A. Disassemble Multiple Disc Brake (See figure 202.)

- (1) During disassembly, tag and keep all parts together for assembly into same unit from which removed.

CAUTION: DISASSEMBLE BRAKE ASSEMBLY WITH CARE. DO NOT USE IMPACT OR POWER WRENCHES ON THREADED PARTS.

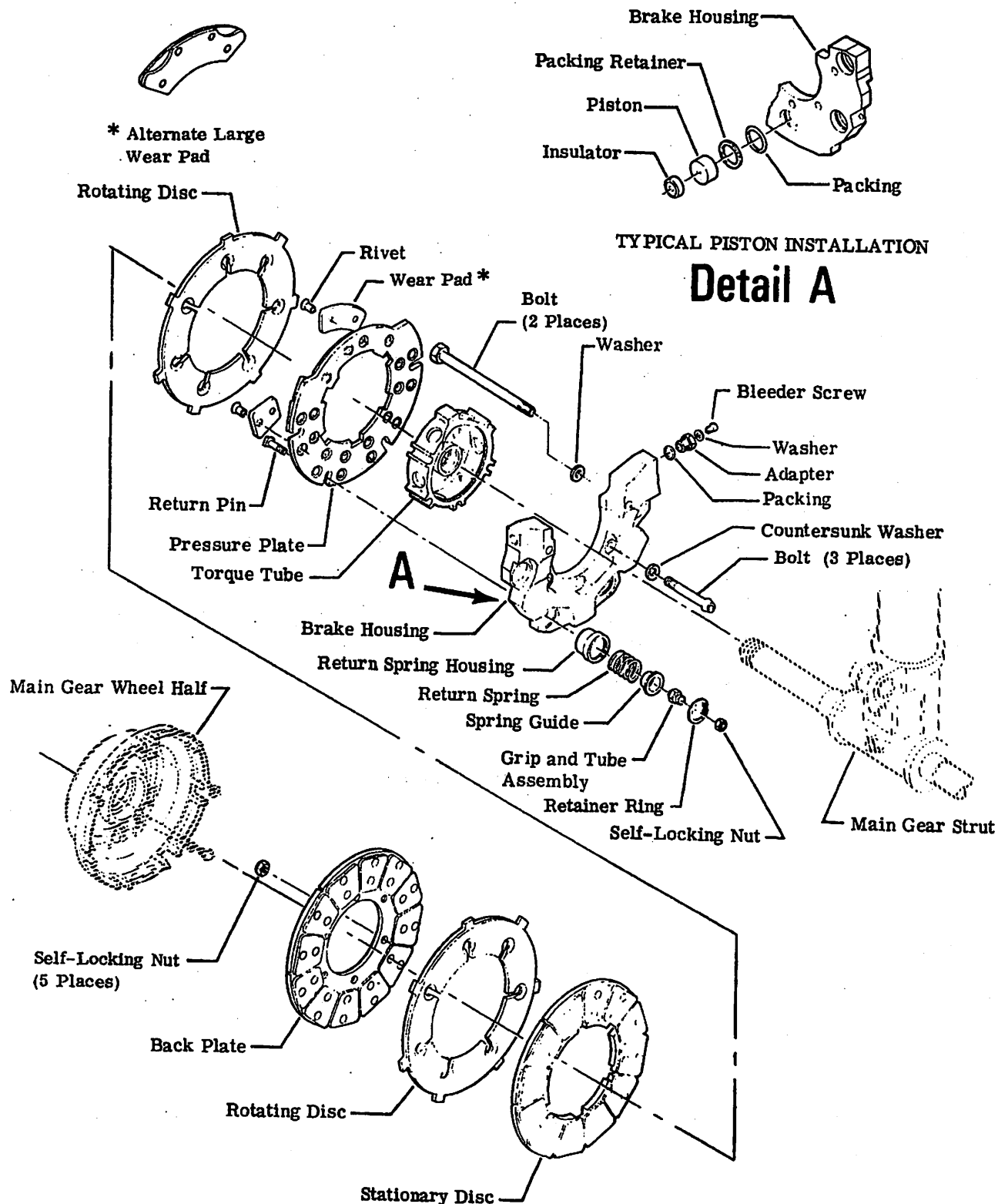
- (2) Remove the three self-locking nuts, bolts, and washers to release back plate and torque tube from brake housing.
- (3) Remove back plate, rotating discs, stationary discs, and torque tube.
- (4) Remove self-locking nuts from brake return pins to release spring assemblies, grip assemblies, return pins, and pressure plate.
- (5) Carefully remove insulators and pistons from housing.
- (6) Remove packing retainer and packing from piston cavities.
- (7) Disassemble return spring assembly by depressing spring guide, removing retainer ring, and removing spring guide and springs from spring housing.

NOTE: Return spring assemblies need not be disassembled unless replacement of parts is required.

- (8) Remove bleeder screw, washer, adapter, and packing from housing.

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6-18C-1
6-40A
6-17C

Multiple Disc Brake Assembly
Figure 202

EFFECTIVITY: ALL

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B. Inspect Multiple Disc Brake Components (See figure 203.)

- (1) Prior to inspection, clean all dirt and grease from brake components as follows:

WARNING: DRY-CLEANING SOLVENTS ARE TOXIC AND VOLATILE. USE IN A WELL-VENTILATED ROOM. DO NOT INHALE SOLVENT VAPORS OR ALLOW SOLVENT TO CONTACT SKIN OR CLOTHING.

- (a) Clean all metal parts except rotating discs, stationary discs, backplate, pressure plate, and tube subassemblies with P-D-680 dry-cleaning solution. Use soft bristle brush when necessary.
- (b) Clean rotating discs, stationary discs, backplate, and pressure plate with compressed air and a soft wire or bristle brush.

WARNING: RUST VETO IS A HIGHLY FLAMMABLE MATERIAL. ASSURE THAT PROPER VENTILATION IS PROVIDED AND EXTREME CAUTION IS EXERCISED.

- (c) Clean grip and tube subassembly tubes with a soft bristle brush or by rubbing with a cloth dipped in trichloroethane, or equivalent, to remove contaminants and buildup of brass on tube. After cleaning, dip subassembly in Rust Veto 377 and cycle grips several times.

NOTE: Grip-tubes must not be cleaned by tumbling, polishing compounds, or with wire brushes. This type cleaning polishes or grinds away the finish requirement that is necessary to give consistent optimum results with respect to load.

- (d) Clean all insulators, packings, and packing retainers with isopropyl alcohol. Dry with a clean, soft cloth.

NOTE: It is recommended that new packings and packing retainers be installed at each brake overhaul.

- (2) Inspect rotating discs as follows:

- (a) Inspect disc relief slot terminal for cracks. Replace cracked discs.

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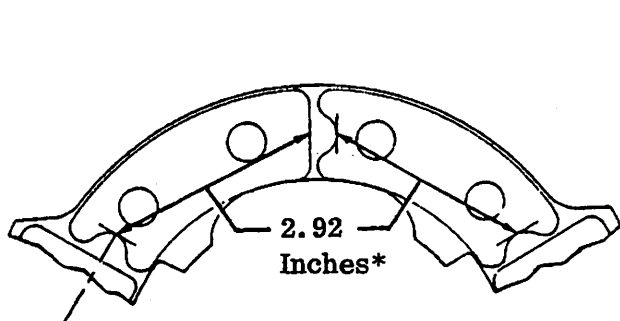
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- (b) Inspect disc for wear. Replace disc when minimum thickness across unchipped, bonded surface reaches 0.300 inch, or if bonded surface is entirely worn off one side in any segment. Up to 25% of the bonded surface may be lost by chipping, provided chipping is scattered.
 - (c) Inspect discs for dishing. Discs may be dished a maximum of 0.030 inch when used with other dished discs.
 - (d) Replace discs that are out-of-round 0.125 inch or more.
 - (e) Inspect discs for shrinkage. Replace discs that have shrunk to an outside diameter of 9.120 inches or less.
 - (f) Check disc tangs for battering. Replace discs if tangs have worn smaller than 0.615 inch in width.
- (3) Inspect stationary disc as follows:
- (a) Inspect disc relief slot terminal holes for cracks. Replace cracked discs.
 - (b) Check disc thickness. Replace wear pads when total thickness measured at a point midway between wear pad rivets is less than 0.431 inch.
 - (c) Inspect discs for dishing. Discs can be dished a maximum of 0.030 inch when used with other dished discs.
 - (d) Check the inside diameter of the discs for size. Replace disc if the measurement is greater than 4.68 inches or less than 4.25 inches.
 - (e) Inspect discs for condition of rivets and wear pads. Whenever one or more rivets is sheared or when the dimension shown is less than 1.43 inches on three or more wear pads per side, replace all wear pads. Wear pads have a tendency to curl away from base plate and wear to a sharp edge. Slight curling is not reason for replacement.
- (4) Inspect back plate assembly as follows:
- (a) Inspect back plate assembly for cracks. Replace cracked back plate assemblies.
 - (b) Check back plate assembly thickness. If worn to less than 0.486 inch at a point midway between rivets, replace wear pad segments.
 - (c) Inspect back plate for condition of rivets and wear pads. Whenever one or more rivets is sheared, or when wear pads measure less than 1.43 inches on three or more segments, replace all wear pad segments. Wear pads have a tendency to curl away from base plate and wear to sharp edges. Slight curling is not reason for replacement.
 - (d) Inspect back plate for dishing. Back plate can be dished a maximum of 0.040 inch (unbolted). Replace excessively dished back plate.

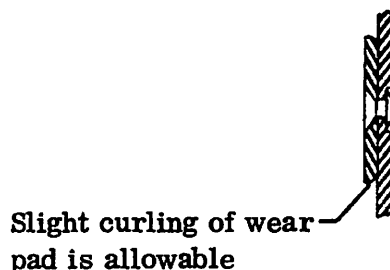
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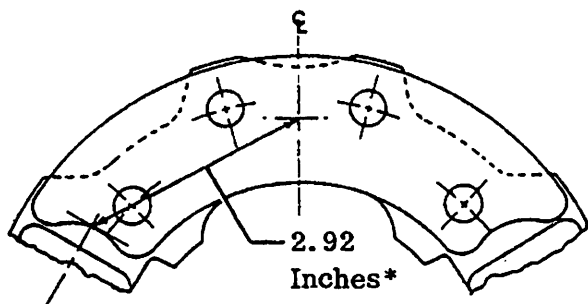
- (5) Inspect pressure plate assembly as follows:
- (a) Inspect pressure plate for cracks. Replace cracked pressure plate.
 - (b) Check pressure plate assembly thickness. If worn to less than 0.321 inch at a point midway between rivets, replace wear pad segments.
 - (c) Inspect pressure plate assembly for condition of rivets and wear pad segments. Whenever one or more rivets is sheared, or when the wear pads measure less than 1.43 inches on two or more small wear pads, or less than 2.92 inches on the large wear pads as shown, replace all wear pads. Wear pads have a tendency to curl away from base plate and wear to sharp edges. Slight curling is not reason for replacement.
 - (d) Inspect pressure plate assembly for flatness. If pressure plate is more than 0.030 inch out-of-flat, replace.



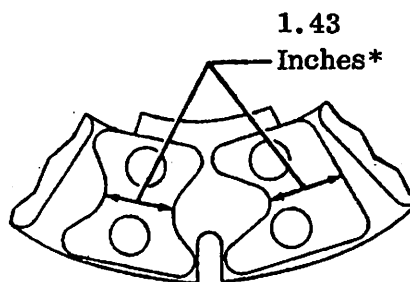
Large Wear Pad



Cross Section of Installed Wear Pad



Alternate Large Wear Pad



Small Wear Pad

* Dimension to be measured from center of wear pad width (typical)

Brake Pad Inspection
Figure 203

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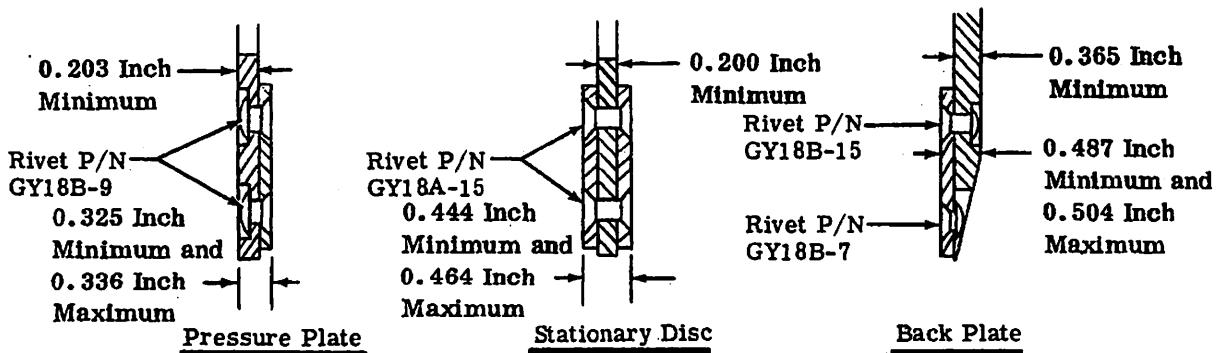
- (6) Inspect brake housing as follows:
 - (a) Check for cracks, using Zyglol or dye-penetrant method. Replace cracked castings.
 - (b) Inspect spring cavities and mating surfaces for corrosion, wear, and other damage.
 - (c) Inspect piston cavities for scoring and wear.
 - (d) Inspect threads in bleed and pressure ports for fit, damage, or cross-threading.
 - (7) Inspect brake pistons as follows:
 - (a) Check pistons for wear. Measure diameter of pistons at three places around circumference. Replace pistons if diameter measures 1.120 inches or less at any point.
 - (b) Check pistons for burrs, scratches, or nicks. Replace any pistons that are damaged more than 0.003 inch deep on seal contacting surfaces. Blend out and polish scratches, nicks, and burrs on seal contacting surfaces up to 0.003 inch deep. Avoid creating local indentations.
 - (c) Ascertain that pistons are free to move in their cavities.
 - (8) Magnetic-particle inspect brake bolts. Carefully check for cracks under bolt head and in thread adjacent to bolt shank. Replace bolts having damaged threads.
 - (9) Inspect piston insulators for blisters.
 - (10) Inspect brake return pins for damaged threads, cracks under the head, and bending. Replace damaged pins.
 - (11) Inspect all self-locking nuts for locking feature. Replace nuts after 10 applications or at any time nut can be turned with the fingers.
 - (12) Inspect brake return springs. Replace springs that require less than 67 pounds to compress to a height of 0.635 inch.
 - (13) Inspect torque tube for cracks or damage. Replace if cracks are found.
 - (14) Inspect grip and tube assembly for bending, damage, or corrosion of tubes. Replace damaged or corroded grip and tube assemblies.
 - (15) Check grip and tube assembly to assure that 150 pounds (minimum) of force is required to move grip in either direction over length of tube up to within 0.11 inch of either end.
 - (16) Reset tubes of grip and tube assemblies as shown in figure 205.
- C. Replace Multiple Disc Brake Wear Pads (See figure 204.)
- (1) Using a 17/32 drill, remove existing wear pads from back plate, pressure plate, and stationary discs by drilling out shop head of rivets and punching out remaining portion of rivets. Use care when performing drilling operation to avoid damaging or enlarging rivet holes.

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- (2) Any raised metal areas on faces of base plate should be removed by grinding before installation of wear pads in order to ensure proper seating of wear pads. Grinding can be accomplished as follows:
- (a) Using a grinder with a magnetic chuck, float base plate on shims until original surface is level. Grind raised areas or high spots until parallel to, and almost even with, surrounding surfaces. Turn plate over and repeat grinding operation on opposite side without use of shims. Turn plate over and grind surface flat. Repeat on opposite side until both sides are parallel and flat within 0.010 inch. Minimum thickness of stationary disc base plate after grinding is 0.200 inch.
 - (b) Grind pad seating surfaces until flat within 0.010 inch. Minimum thickness of back plate base plate after grinding is 0.365 inch measured on a centerline between rivet holes. Minimum thickness of pressure plate base plate after grinding is 0.203 inch measured on a centerline between rivet holes.

NOTE: Make all thickness measurements on the centerline between rivets at the outer edge of the wear pad. If total thickness exceeds the maximum thickness listed above, rivets must be removed and wear pads reinstalled correctly before brake is assembled.



Installation Tolerances of New Wear Pads
Figure 204

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- (3) Install new wear pads using rivets as shown. Install rivets by compression method so that formed head is flush with or slightly below surface of wear pad. If rivet head protrusion above wear pad surface occurs, grinding will be required to provide flush rivets. Maximum allowable head grind should not exceed 35% of total rivet head area. Care must be exercised to produce the proper rivet squeeze. Excessive squeeze pressure will result in wear pad corner lift-up, while insufficient squeeze pressure will result in loose pads and improperly formed heads. Rivet installation shall be such that a minimum force of 2 pounds and maximum force of 100 pounds shall be required to move pad.
- (4) Assembled disc shall be flat within 0.010 inch. Check thickness of stationary discs, back plate, and pressure plate. Refer to figure 204.
- (5) No more than one crack is allowable in each rivet head and it cannot extend into the rivet shank.
- (6) If reworked discs are to be stored, apply protective coating as follows:
 - (a) Degrease disc in a vapor degreaser or other means of assuring that all grease and grinding materials are removed.

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ASSURE THAT PROPER VENTILATION IS PROVIDED
AND MAINTENANCE PERSONNEL EXERCISE
EXTREME CAUTION.**

- (b) Immerse discs in Rust Veto 377 or equivalent. Do not dilute.
 - (c) Let disc drip and air dry for 5 minutes.
- D. Repair Multiple Disc Brake Housing
- (1) Polish away burrs in spring cavities and elsewhere on brake housing with 280 grit wet-or-dry sandpaper. Polish piston cavities with fine (400 grit) wet-or-dry aluminum oxide cloth.
 - (2) Repair damaged housing inlet or bleeder threads as follows:
 - (a) Tap drill inlet or bleeder with a 9/16 tap drill to a depth of 0.48 (+0.02/-0.00) inch.
 - (b) Counterbore inlet or bleeder to 0.94 (+0.015/-0.00 inch depth using a 0.64 (+0.005/-0.000) inch counterbore.
 - (c) Countersink hole, using a 120° x 0.750 (+0.015/-0.000) inch diameter countersink.
 - (d) Tap 5/8-18 UNF-3B full threads to a depth of 0.46 (±0.02) inch.
 - (e) Rinse reworked area with hot water and thoroughly dry the part with compressed air.
 - (f) Treat newly machined surfaces with MIL-C-5541 chemical film solution.

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- (g) Repaint reworked area on brake housing (except in piston cavities and surfaces of bushings) with one coat of MIL-P-8585A zinc chromate primer and one coat of TT-L-32 aluminum lacquer.
 - (h) Lubricate an MS 28778-6 preformed packing with MIL-G-81322 grease.
 - (i) Install lubricated packing on a 9535617 repair bushing. Install bushing in housing.
 - (j) Stamp the letter "R" adjacent to repaired inlet hole.
- E. Repair Multiple Disc Brake Insulator
- (1) Remove surface blisters from insulators with a file or by touching lightly to an emery wheel.
- F. Assemble Multiple Disc Brake (See figure 202.)
- (1) Place brake housing assembly on a clean, flat surface with piston cavities up.
 - (2) Lubricate preformed packings, retainer, housing cylinder walls, and seal grooves and contacting surfaces of pistons with a light coat of MIL-H-5606 hydraulic fluid.
 - (3) Carefully install one preformed packing, then one packing retainer in groove of each cylinder wall.
 - (4) Install a piston in each cylinder (with recessed end for insulator out) carefully, so as not to displace packing or retainer. This procedure can be facilitated by turning piston slightly during installation.
 - (5) Place one piston insulator in each piston recess with impressed part numbers toward piston.
 - (6) Insert heads of return pins into return pin slots of pressure plate assembly and position pressure plate against piston insulators with return pins through return spring assembly holes.

NOTE: Pressure plate assembly must be installed with two large wear pads (or alternate single large wear pad) at top of brake housing over open section between housing ends.

- (7) Grasp assembled unit firmly, turn over, and place on working surface.
- (8) Assemble four return spring assemblies as follows:
 - (a) Place return spring in return spring housing.
 - (b) Place spring guide in return spring.
 - (c) Depress spring guide and install retaining ring in groove of return spring housing.
- (9) Install assembled spring assemblies over return pins and into brake housing.

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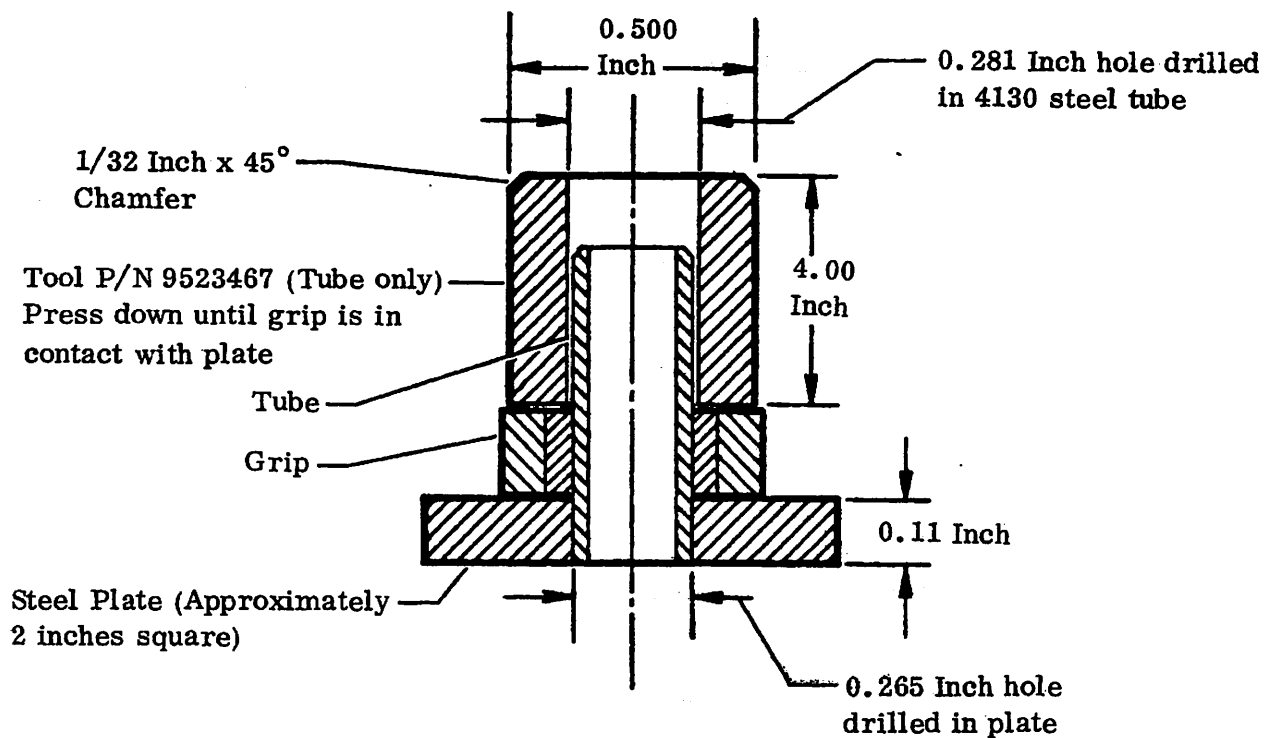
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- (10) Reset grip and tube assembly to 0.11 inch position. (See figure 205.) Force required to move grip on tube can not be less than 150 pounds.
- (11) Install grip and tube assemblies on return pins. Assure that short end of tube (0.11 inch) is facing pressure plate. Install self-locking nuts and torque to 25 inch-pounds.
- (12) Lubricate bolt and nut threads and bearing surface of bolts, washers, and nuts with MIL-T-5544 anti-seize compound.
- (13) Insert bolts and washers in three holes in bottom of housing opposite open section. Countersunk washer must be placed on bolt with countersink facing bolt head.
- (14) Hold bolts in position and turn brake over.
- (15) Position torque tube over pressure plate with axle flange away from pressure plate and solid key facing open section of brake housing. Slide torque tube over brake bolts and into pressure plate.

CAUTION: ASSURE THAT PROPER ROTATING DISCS P/N 5003147 ARE USED WITH BRAKE ASSEMBLY 5003096, AND ROTATING DISCS P/N 9543992 ARE USED WITH BRAKE ASSEMBLY 9550608. SUBSTITUTION OR MIXING OF ROTATING DISC P/N'S IN EITHER BRAKE ASSEMBLY CANNOT BE PERMITTED UNDER ANY CIRCUMSTANCES BECAUSE OF THE DIFFERENCES IN FRICTION MATERIAL AND RELATED BRAKE OPERATIONAL CHARACTERISTICS. ALSO BRAKE ASSEMBLY NUMBERS MUST NOT BE MIXED ACROSS THE AIRCRAFT.

- (16) Place stationary disc between two rotating discs and slide entire stack onto torque tube.
- (17) Position back plate assembly on three bolts with back plate wear pads facing rotating disc. Secure back plate assembly with self-locking nuts. Tighten nuts evenly and Lubtork to 120 inch-pounds.
- (18) Lubricate bleed adapter packing with a light coat of MIL-H-5606 hydraulic fluid and install on bleeder adapter. New packings are recommended at each overhaul.
- (19) Screw adapter into brake housing, then place bleeder washer on bleeder screw and screw into bleeder adapter. Torque adapter to 120 inch-pounds.

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Grip and tube assembly resetting tool P/N 9523467 is available from Goodyear Tire and Rubber Co. Aviation Products Division, Akron, Ohio 44316 or Los Angeles, California 90054; or may be fabricated locally.

Grip and Tube Assembly Resetting
Figure 205

6-41A

EFFECTIVITY: ALL

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POWER BRAKE VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Power Brake Valve (See figure 201.)

- (1) Depressurize hydraulic system by alternately extending and retracting flaps.
- (2) Remove pilot's and copilot's rudder pedals. (Refer to Chapter 27.)
- (3) Disconnect power brake valve hydraulic lines at frame 6. Cap or plug all exposed hydraulic fittings.
- (4) Remove power brake valve support channel attaching parts and remove power brake valves, lines, and support channel.
- (5) Remove attaching parts securing power brake valve to power brake valve support channel.

B. Install Power Brake Valve (See figure 201.)

- (1) Install power brake valve on power brake valve support channel using attaching parts.
- (2) Position power brake valve support channel on aircraft structure and connect power brake valve hydraulic lines to hydraulic fittings at frame 6.
- (3) Secure power brake valve support channel to aircraft structure using attaching parts.
- (4) Pressurize hydraulic system and check power brake valve hydraulic lines for leaks.
- (5) Install rudder pedals. (Refer to Chapter 27.)

2. Adjustment/Test

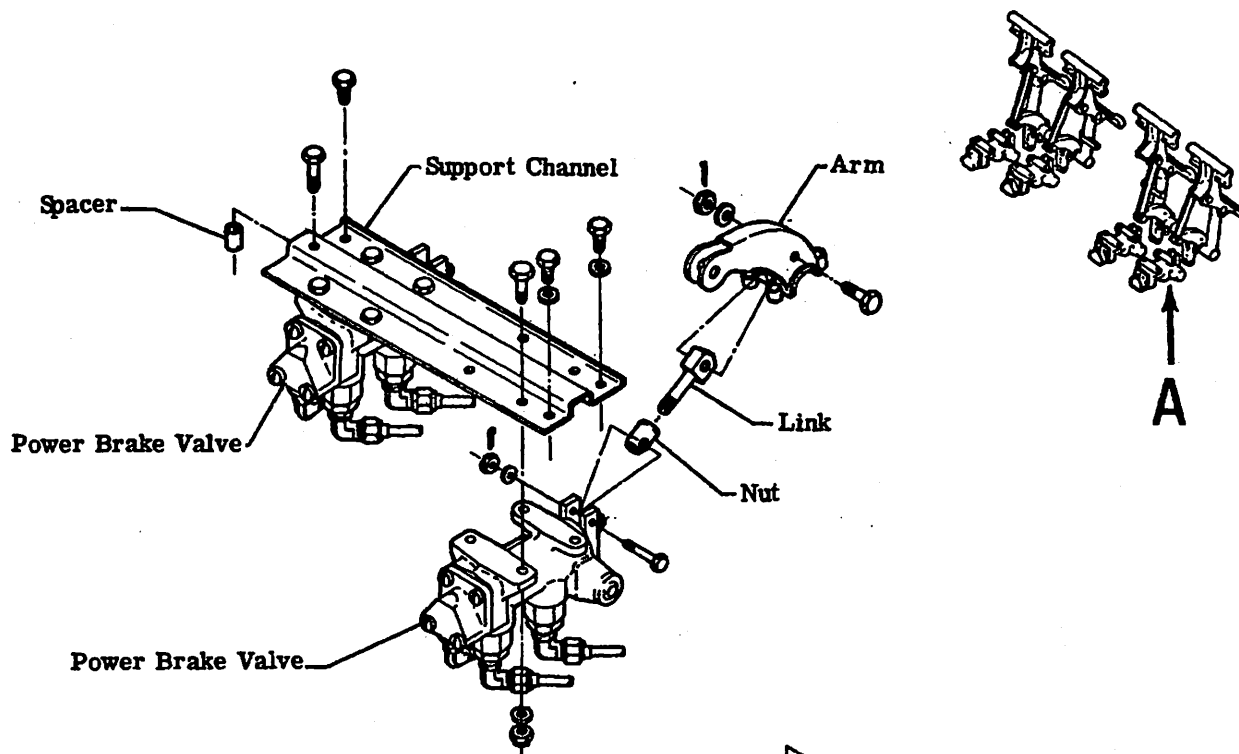
A. Adjust Power Brake Valve

- (1) Turn adjustment screw until valve lever is 90° to the centerline of the valve assembly.
- (2) Connect valve to a hydraulic test bench capable of 1500 psi.
- (3) With 1500 psi applied at a system pressure port, loosen jamnut and turn lever adjustment screw until valve poppet cracks, then back off adjustment screw until poppet seats plus 3/4 additional turn. Tighten jamnut.

NOTE: On Aircraft 25-205 and Subsequent and aircraft equipped with power brake valves incorporating key washers, bend tabs to free jamnut. After adjustments are made and jamnut is tightened, bend tabs back down to lock jamnut in place.

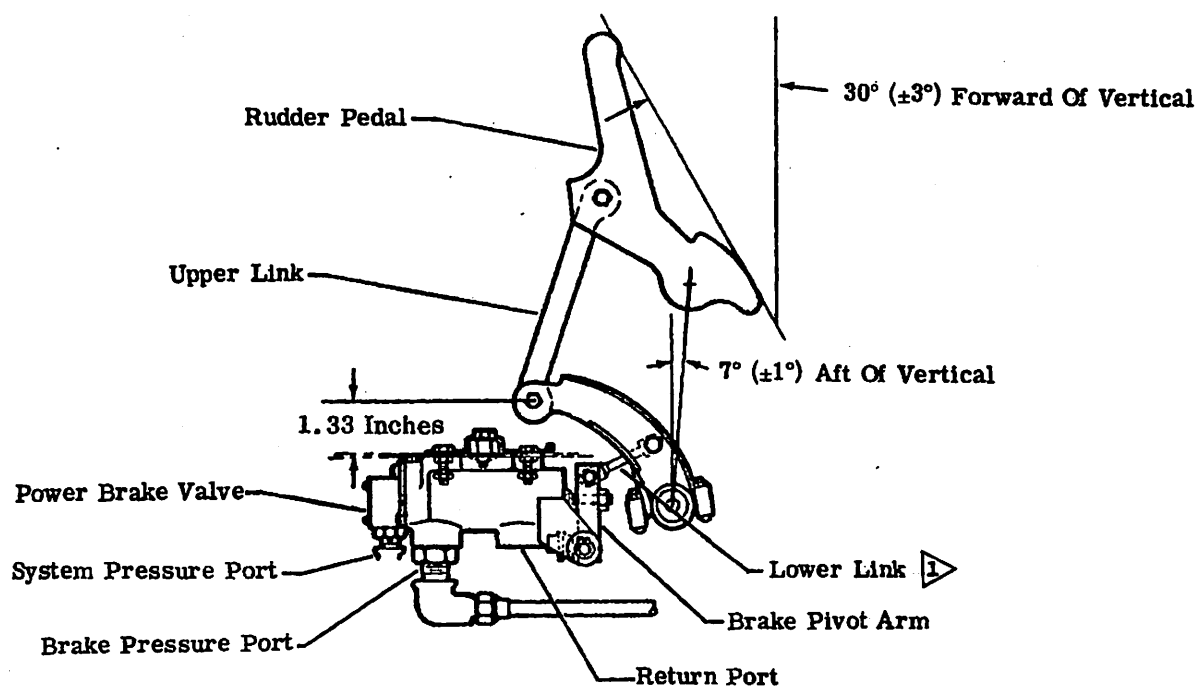
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Detail A

1 Adjust lower link to obtain 1.33 inch dimension as shown.

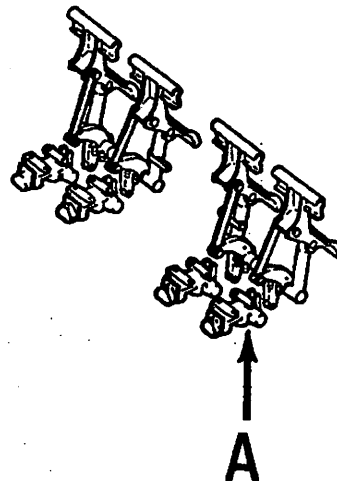


**Power Brake Valve Installation
Figure 201**

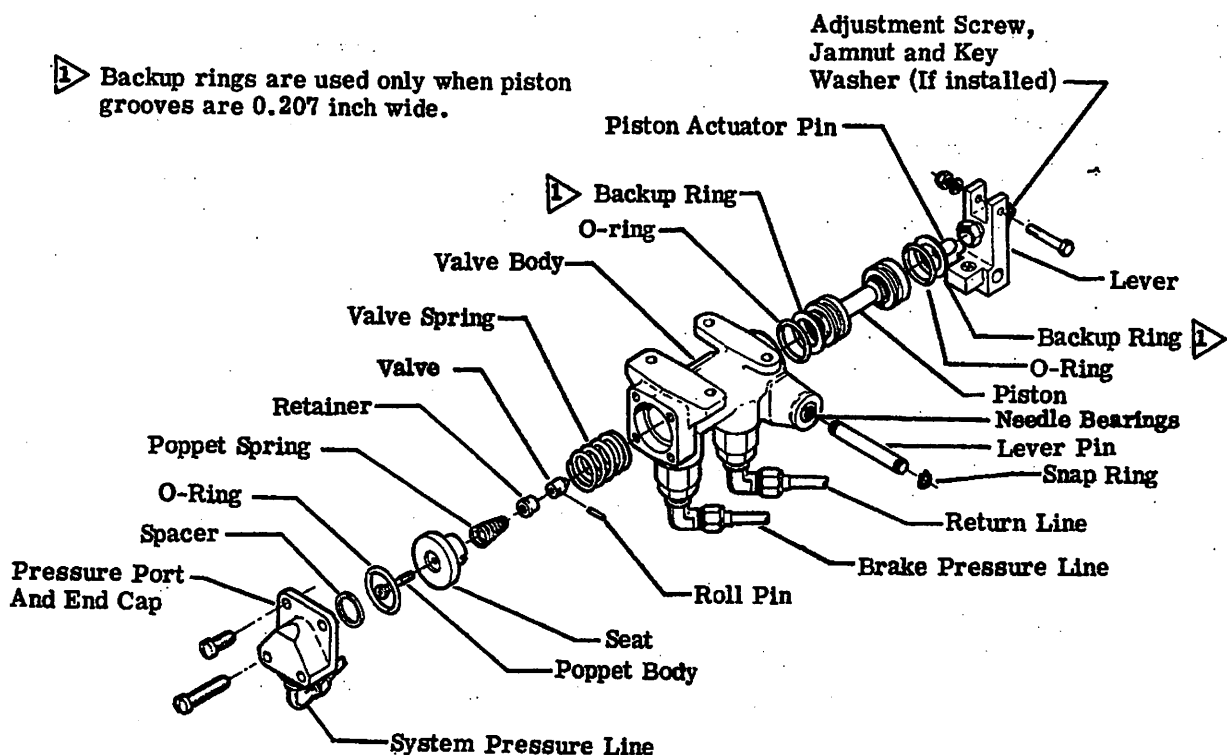
EFFECTIVITY: ALL

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1 Backup rings are used only when piston grooves are 0.207 inch wide.



Detail A

Power Brake Valve Assembly
Figure 202

EFFECTIVITY: ALL

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3. Approved Repairs

A. Disassemble Power Brake Valve (See figure 202.)

- (1) Drain hydraulic fluid from valve.
- (2) Remove snap rings and lever pin which retain power brake lever, and remove lever and piston actuator pin from valve. Do not disassemble power brake valve lever.
- (3) Remove screws from system pressure port and end cap assembly. Remove end cap assembly from valve.
- (4) Remove spacer, O-ring, seat with poppet assembly attached, and valve spring from valve body.
- (5) Remove piston from valve body and remove O-ring and backup rings (if installed) from piston.
- (6) Clean all metal parts with Stoddard solvent or equivalent. Visually inspect valve bore, piston, seat, and poppet assembly for scoring, scratches, and corrosion.
- (7) Seat and poppet assembly may be disassembled for thorough cleaning as follows:
 - (a) Push retainer back and remove roll pin from valve.
 - (b) Remove valve, retainer, and poppet spring from poppet and remove poppet from seat.

CAUTION: DO NOT CREATE LOW SPOTS BY EXCESSIVE BUFFING OR POLISHING. USE OIL STONE TO REMOVE SCRATCHES AND POLISH WITH CROCUS CLOTH.

- (8) Buff out minor scratches and corrosion.
- (9) Coat internal parts with MIL-H-5606 hydraulic fluid.
- (10) Wrap cleaned parts in lint-free cloth or clean waxed paper while awaiting repair or assembly.

B. Assemble Power Brake Valve (See figure 202.)

- (1) Immerse all internal parts, O-rings and backup rings in MIL-H-5606 hydraulic fluid prior to installation.
- (2) Assemble seat, valve, retainer, poppet, and poppet spring as follows:
 - (a) Install poppet in seat and assemble poppet spring and retainer on poppet.
 - (b) Install valve on poppet and secure with roll pin.
- (3) Install seat and poppet assembly in valve body.
- (4) Install O-ring and spacer in pressure port and end cap assembly and secure to valve body with screws.

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- (5) Install O-rings and backup rings (if required) on piston and install valve spring and piston in valve body.
- (6) Lubricate needle bearings with MIL-G-3278 grease, install power brake control lever, and secure lever with lever pin and snap rings.

NOTE: Assure that piston actuator pin is installed between piston and adjustment screw as power brake lever is being installed.

- (7) Adjust power brake valve as instructed in Adjustment/Test.
- (8) Connect valve system pressure port line to hydraulic source containing MIL-H-5606 hydraulic fluid.
- (9) With valve control lever in static position, apply 2250 psi pressure to valve; hold for 5 minutes. There shall be no leakage.
- (10) Disconnect power brake valve line from hydraulic source and cap lines.

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HYDRAULIC BRAKE FUSE - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Hydraulic fuses which have closed should automatically open after approximately one minute. Fuses which do not open automatically must be removed, placed on a hydraulic test bench, and subjected to reverse flow.

A. Remove Hydraulic Brake Fuse (See figure 201.)

- (1) Depressurize hydraulic system.
- (2) Disconnect and cap brake pressure line from brake fuse.
- (3) Disconnect and remove brake fuse from line at shuttle valve. Cap exposed fittings.

B. Install Hydraulic Brake Fuse (See figure 201.)

- (1) Install brake fuse with closing flow arrow pointing away from shuttle valve.
- (2) Pressurize hydraulic system.
- (3) Bleed brake system. (Refer to 32-43-00.)

2. Adjustment/Test

NOTE: Hydraulic fuses must be functionally tested in accordance with current inspection intervals in Chapter 5.

A. Functionally Test Hydraulic Brake Fuse

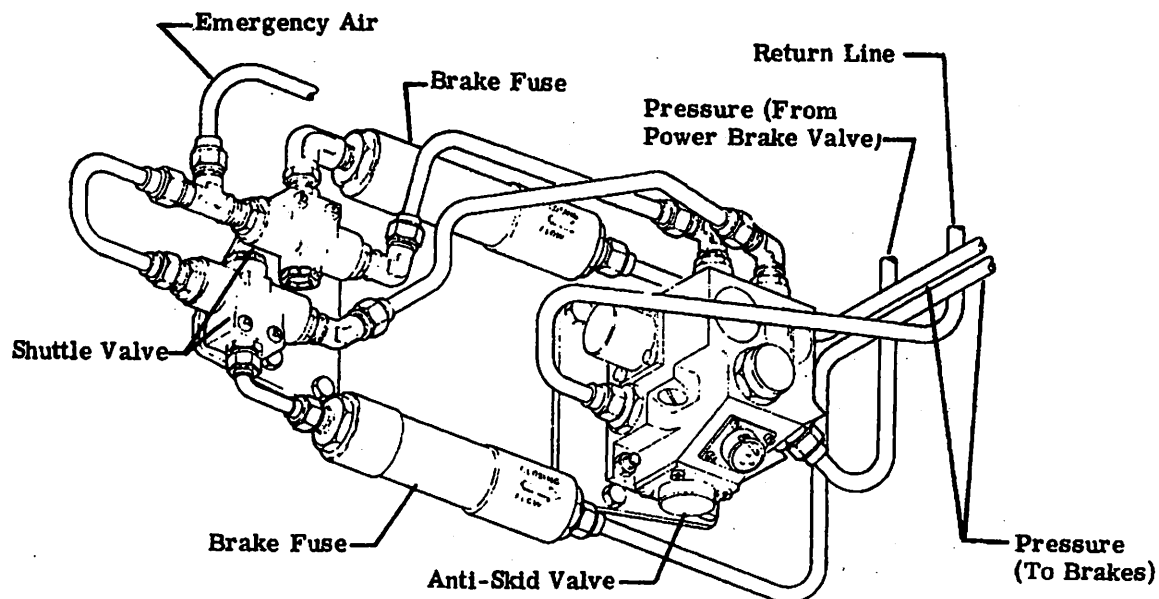
- (1) Attach a supply line to the inlet port.
- (2) Apply fluid to brake fuse and measure amount of fluid passed through fuse before it closes. Amount of fluid passed shall be 5.5 (± 1.1) fluid ounces. Check amount of fluid three times.
- (3) Apply a proof pressure of 2250 psi for a period of 3 minutes. There shall be no leakage.
- (4) Remove proof pressure and recheck fluid volume as described in step 2.
- (5) Remove brake fuse from supply line.

3. Approved Repairs

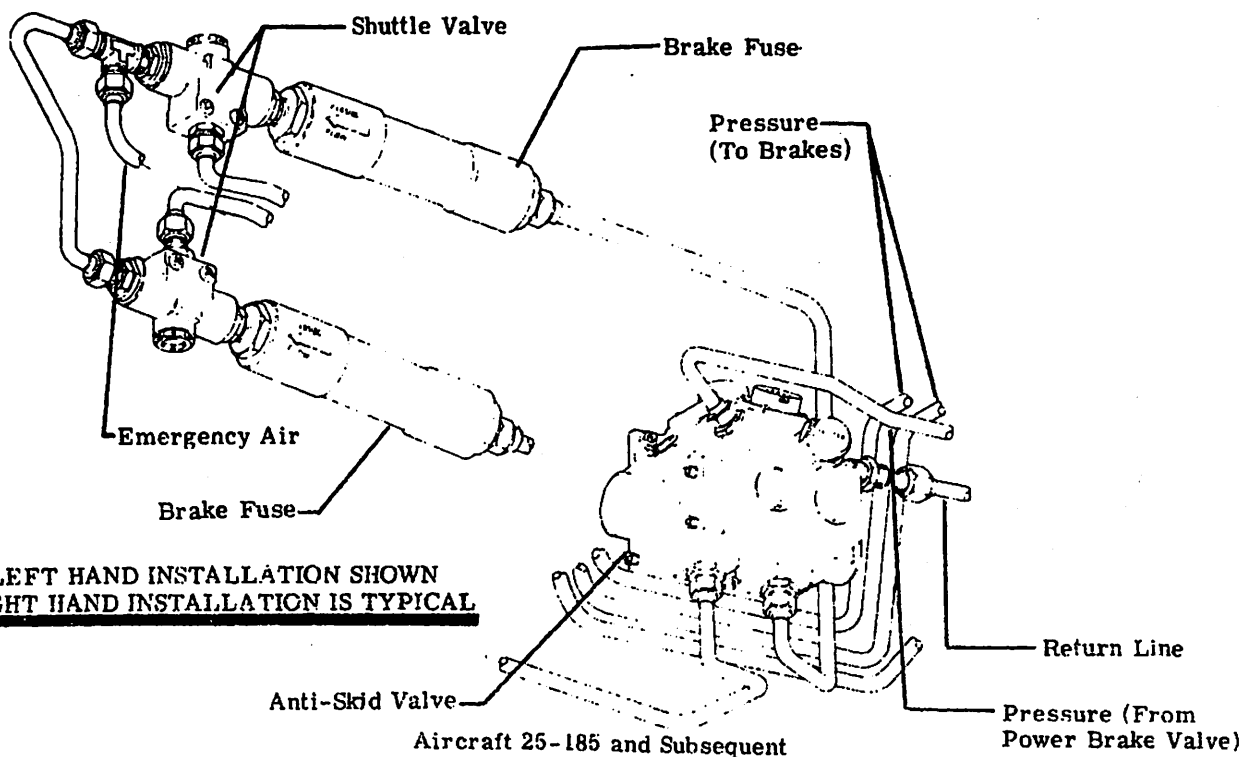
A. Disassemble Hydraulic Brake Fuse (See figure 202.)

- (1) Remove safety wire and end cap from valve housing.
- (2) Remove flow deflector, measuring float, diffuser, diffuser spring, flow barrel collector ring, and spring washer from valve housing.
- (3) Remove snap ring from flow deflector. Remove orifice support, O-ring, orifice disc, orifice housing, and filler ring from flow deflector.

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Aircraft 25-061, 25-070 thru 25-184



LEFT HAND INSTALLATION SHOWN
RIGHT HAND INSTALLATION IS TYPICAL

Aircraft 25-185 and Subsequent

**Hydraulic Brake Fuse Installation
Figure 201**

EFFECTIVITY: ALL

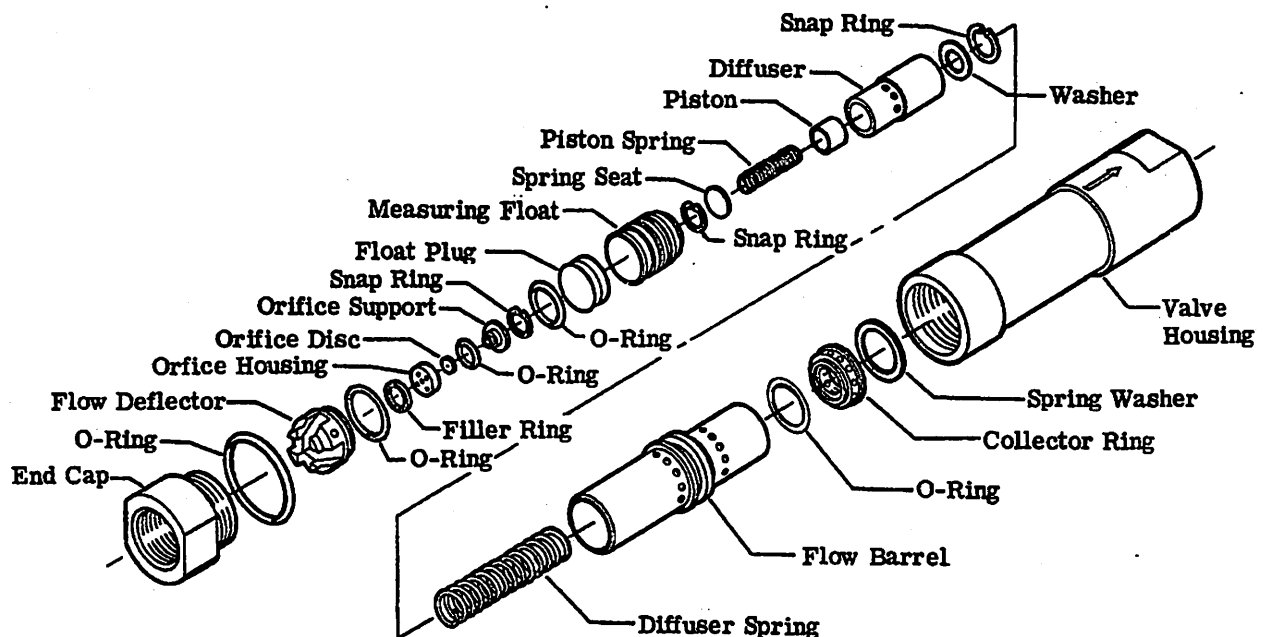
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- (4) Remove float plug from measuring float.
 - (5) Remove snap ring from diffuser. Remove spring seat, piston spring, and piston from diffuser. Remove snap ring and washer from opposite end of diffuser.
 - (6) Remove and discard all old O-rings.
 - (7) Clean all metal parts with Stoddard solvent or equivalent and dry with compressed air.
 - (8) Clean orifice disc hole.
- B. Assemble Hydraulic Brake Fuse (See figure 202.)
- (1) Install new O-rings on end cap, flow deflector, orifice support, and flow barrel.
 - (2) Assemble filler ring, orifice housing, orifice disc, and orifice support into flow deflector and secure with snap ring.
 - (3) Install float plug in measuring float.
 - (4) Assemble piston, piston spring, and spring seat into diffuser and secure with snap ring. Install washer and snap ring in opposite end of diffuser.
 - (5) Install spring washer and collector ring in valve housing. Insert flow barrel in valve housing until it is firmly seated against collector ring and spring washer.
 - (6) Insert diffuser spring and diffuser assembly in flow barrel.
 - (7) Insert measuring float into flow barrel.
 - (8) Install flow deflector assembly on end of flow barrel and install end cap. Safety wire end cap to valve housing.



Hydraulic Brake Fuse Assembly
Figure 202

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PARKING BRAKE VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Parking Brake Valve (See figure 201.)

- (1) Depressurize hydraulic system.
- (2) Remove RH floorboards between frames 10 and 13 to gain access to parking brake valves.
- (3) Disconnect hydraulic lines from valve and cap or plug hydraulic line and valve ports.
- (4) Disconnect link assembly from valve control levers.
- (5) Disconnect anti-skid disconnect switch (on left hand forward valve assembly) electrical wiring.
- (6) Remove attaching parts and valves from the aircraft.

B. Install Parking Brake Valve (See figure 201.)

- (1) Install valve assembly on valve support assembly using attaching parts.
- (2) Connect link assembly to control lever.
- (3) Connect electrical wiring to anti-skid disconnect switch.
- (4) Connect hydraulic lines to valve ports.
- (5) Pressurize hydraulic system and bleed brake system. (Refer to 32-43-00.)
- (6) Rig parking brake valve. (Refer to Adjustment/Test.)

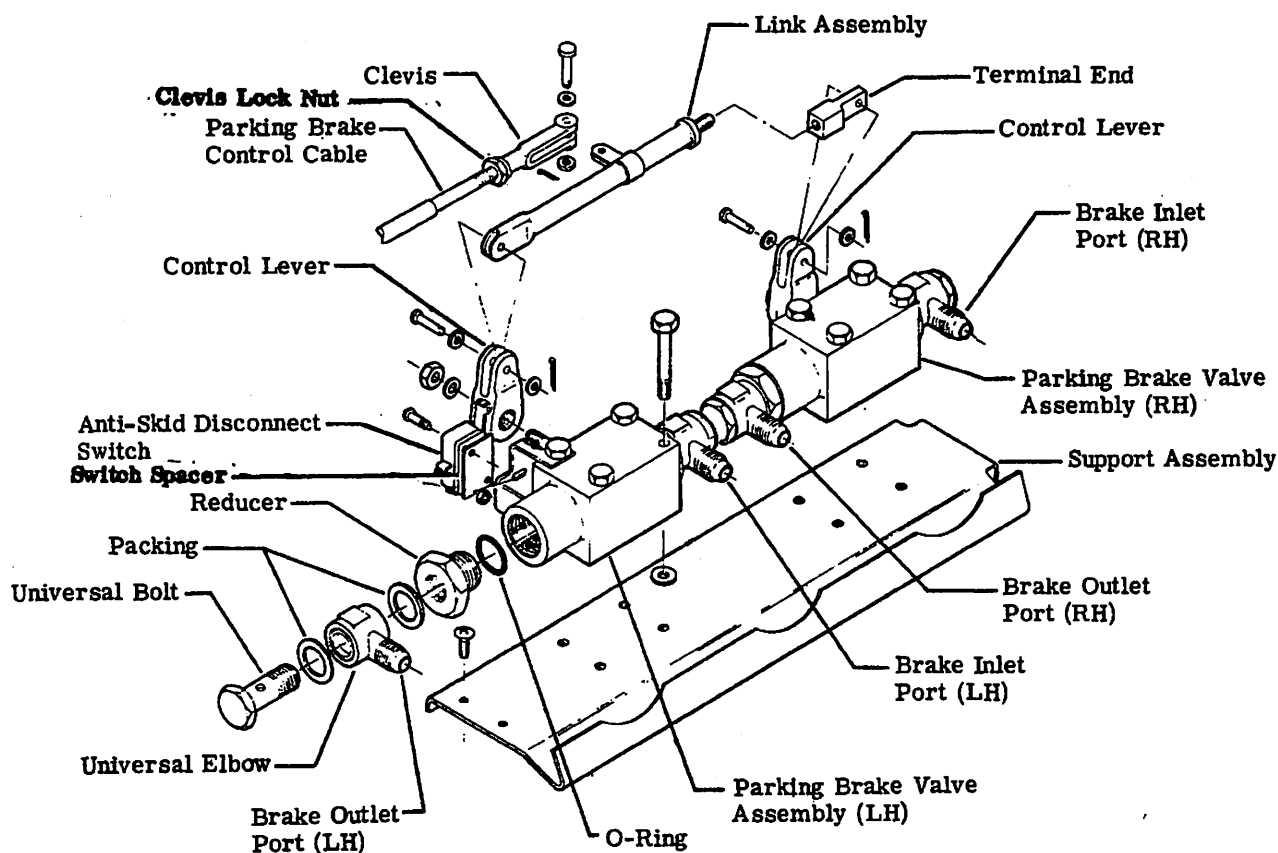
2. Adjustment/Test

A. Rig Parking Brake Valve (See figure 201 and 202.)

- (1) Connect an ohmmeter between C and NC terminals of anti-skid disconnect switch installed on forward parking brake valve.
- (2) Disconnect link assembly from control cable clevis end and aft control lever.
- (3) Check that anti-skid disconnect switch actuates (continuity) when valve control lever is at 15° to 18° forward of vertical. If adjustment is required, loosen switch attaching screws and position switch as required to obtain actuation.
- (4) Rotate control levers on both valves to aft position.
- (5) Adjust terminal end of link assembly so that attaching holes align with both control levers, and install link assembly between control levers.
- (6) Check that parking brake lever is fully down.
- (7) If control cable clevis does not align with hole in link assembly, loosen lock nut and adjust clevis as required.
- (8) Tighten clevis lock nut, and secure cable clevis to link assembly.
- (9) With handle fully down, ohmmeter will indicate continuity.

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TYPICAL PARKING BRAKE VALVE INSTALLATION

**Parking Brake Valve Installation
Figure 201**

11-10C-1

EFFECTIVITY: ALL

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- (10) Slowly pull handle up. Ohmmeter will show switch actuation to no continuity at a point 1.00 to 1.30 inches from bottom of handle travel. Travel is measured along quadrant surface.
- (11) If adjustment is required, loosen clevis lock nut and adjust clevis as required to obtain actuation.
- (12) Tighten lock nut, disconnect ohmmeter from switch, and install floorboards removed to gain access to parking brake valve.

B. Functionally Test Parking Brake Valve

- (1) Install pressure cap on outlet port.
- (2) Connect valve to hydraulic source containing MIL-H-5606 hydraulic fluid.
- (3) Apply 2250 psi pressure for 3 minutes; there shall be no external leakage or rupture.
- (4) Disconnect hydraulic source and remove outlet cap from valve.
- (5) Connect valve to hydraulic source using outlet port.
- (6) With valve in closed position, apply 1550 psi pressure to outlet port. After 3 minutes there shall be no external leakage and internal leakage shall not exceed 10 drops per minute from inlet port.
- (7) Reduce pressure to 5 psi, open and close valve. After 3 minutes there shall be no external leakage and internal leakage shall not exceed 10 drops per minute from inlet port.
- (8) Disconnect hydraulic source, cap ports, and clean external surface.

3. Approved Repairs

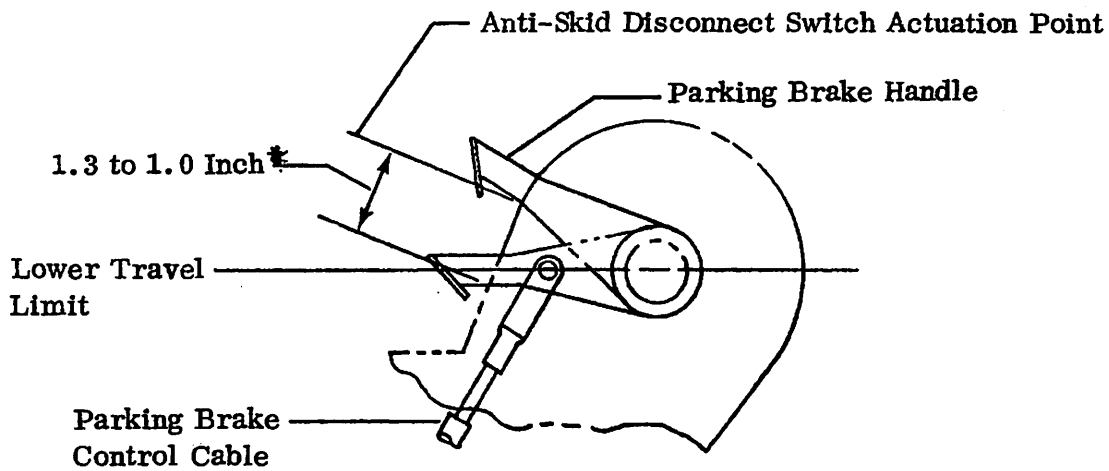
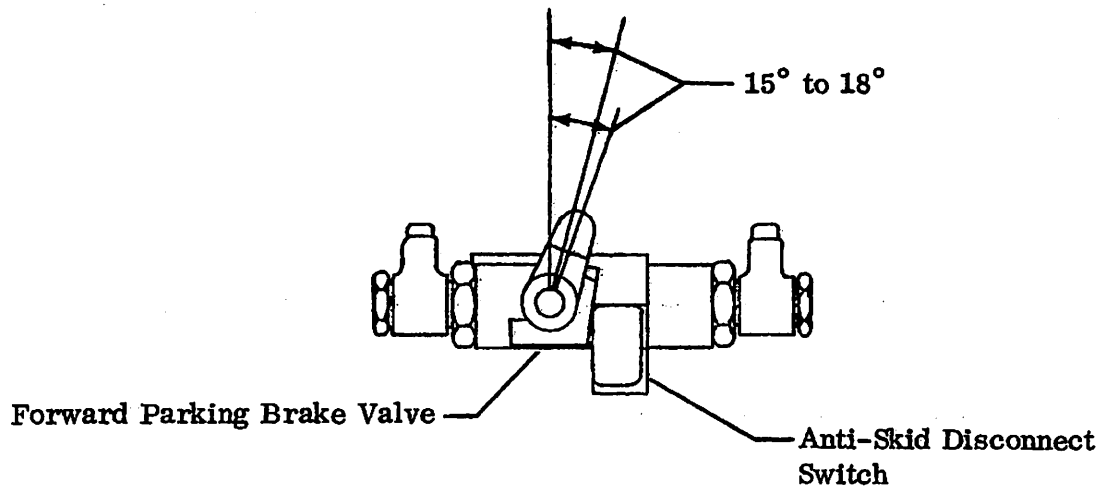
A. Disassemble Parking Brake Valve (See figures 201 and 202.)

- (1) Remove universal bolt, universal elbow, packings, reducer, and O-ring from inlet and outlet ports of valve body. Discard packings and O-rings.
- (2) If applicable, remove anti-skid disconnect switch bracket with switch attached from valve body.
- (3) Unscrew spring retainer from valve body and remove spring, spring seat, and poppet.

CAUTION: THE CONTROL LEVER AND SHAFT ARE CLOCKED AND BONDED TOGETHER FOR PROPER VALVE CONTROL. REMOVING CONTROL LEVER FROM SHAFT MAY CAUSE DAMAGE TO SHAFT.

- (4) Remove screws securing shaft retainer to valve body and remove shaft retainer.
- (5) Pull shaft from valve body. Remove and discard O-rings.

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* Measure distance along surface of quadrant.

Anti-Skid Disconnect Switch Rigging
Figure 202

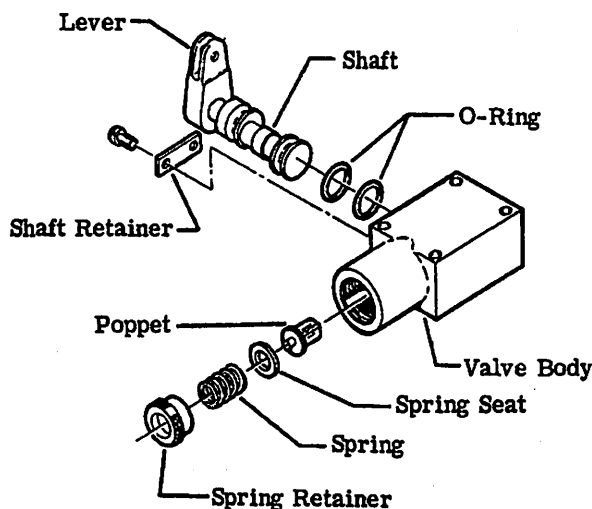
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- (6) Clean metal parts in Stoddard solvent or equivalent.
 - (7) Visually inspect interior of valve body and all removed parts. Replace all parts which show signs of excessive wear, nicks, or scoring.
- B. Assemble Parking Brake Valve (See figures 201 and 202.)

NOTE: Use new O-rings at each assembly of a valve.

- (1) Immerse O-rings in MIL-H-5606 hydraulic fluid and install on shaft. Insert shaft into valve body.
- (2) Position shaft retainer on valve body so that screw holes align. Apply Loctite Screw Lock to screw threads and install screws.
- (3) Insert poppet, spring seat, and spring in valve body. Screw in spring retainer until it bottoms out, then back out 1-1/2 turns to assure that valve is working properly.
- (4) If applicable, install anti-skid disconnect switch bracket with switch attached. Apply Loctite Screw Lock to screw threads and secure bracket.
- (5) Immerse O-rings in MIL-H-5606 hydraulic fluid and install on reducers. Screw reducers into valve body.
- (6) Install universal elbows using new packings and secure with universal bolt.
- (7) Perform functional test of parking brake valve. (Refer to Adjustment/Test.)



Parking Brake Valve Assembly
Figure 203

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SHUTTLE VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Shuttle Valve

- (1) Depressurize hydraulic system, check that parking brake is released, and assure that emergency brake control valve handle is in off position.
- (2) Disconnect hydraulic and pneumatic lines at shuttle valve. Discard pneumatic line B-nut fitting seal.
- (3) Remove attaching parts and shuttle valve.

B. Install Shuttle Valve

- (1) Install shuttle valve in its original position and secure with attaching hardware.
- (2) Connect hydraulic and pneumatic lines to their original fittings. Use a new fitting seal at pneumatic B-nut connection.

2. Adjustment/Test

A. Functionally Test Shuttle Valve

- (1) With outlet port capped and valve connected to a hydraulic test bench, apply a proof pressure of 2250 psi for 3 minutes to each inlet port. There shall be no sign of leakage.

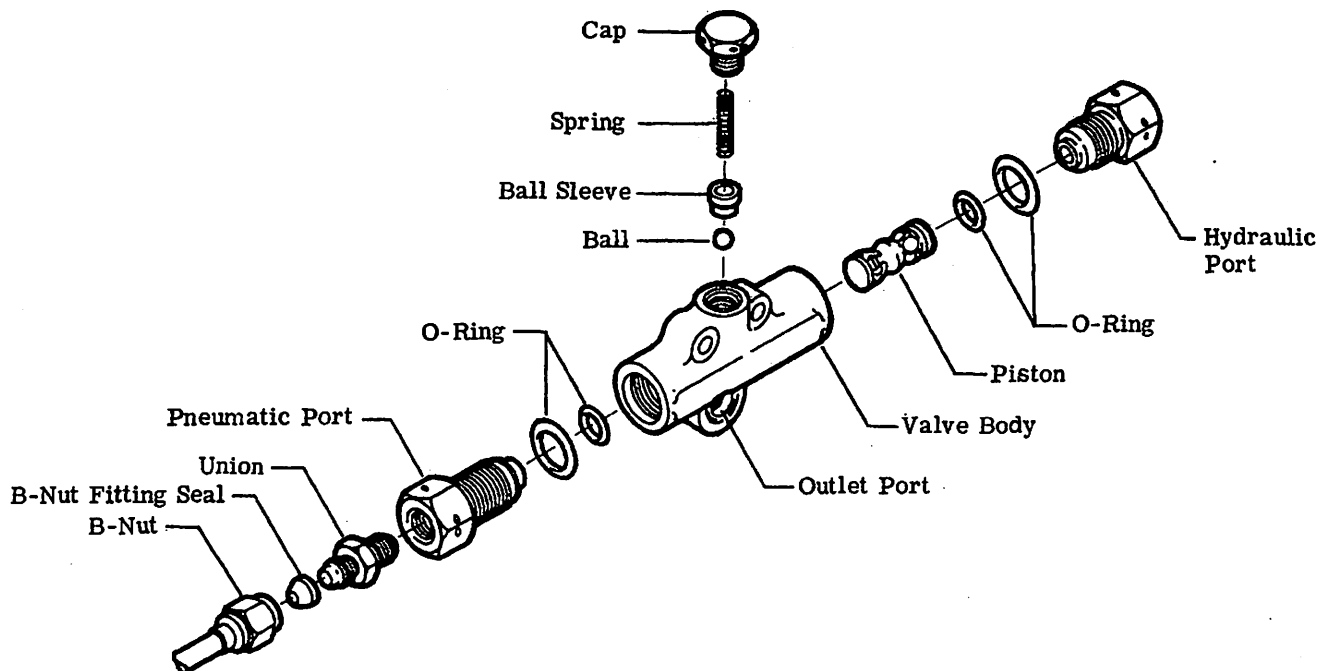
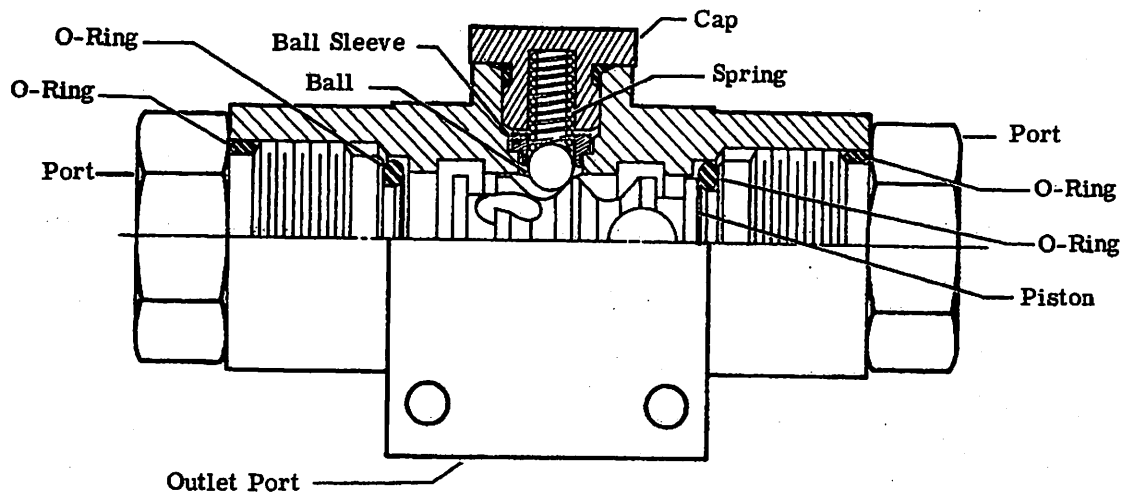
3. Approved Repairs

A. Due to the many configurations of shuttle valves, only general repair instructions can be given.

- (1) Disassemble shuttle valve beginning with cap bolt and/or port. Note sequence of parts removal.
- (2) Discard O-rings, packing, and backup rings (if installed).
- (3) Inspect for scoring, scratches, or corrosion. Buff out minor damage with crocus cloth using caution not to create any low spots.
- (4) Clean thoroughly with Stoddard solvent and blow dry to remove foreign particles.
- (5) Reassemble in reverse order of disassembly using new O-rings, packing, and backup rings (if required). Immerse O-rings in MIL-H-5606 hydraulic fluid prior to assembly.

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Typical Shuttle Valve Assembly
Figure 201

EFFECTIVITY: ALL

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EMERGENCY AIR BRAKE SYSTEM - DESCRIPTION AND OPERATION

1. Description

- A. The emergency air brake system consists of an emergency brake control valve and air pressure line plumbing from the control valve to the shuttle valves in each main gear wheel well.
- (1) The emergency brake control valve is installed on the RH side of the center pedestal and is manually controlled by a lever.

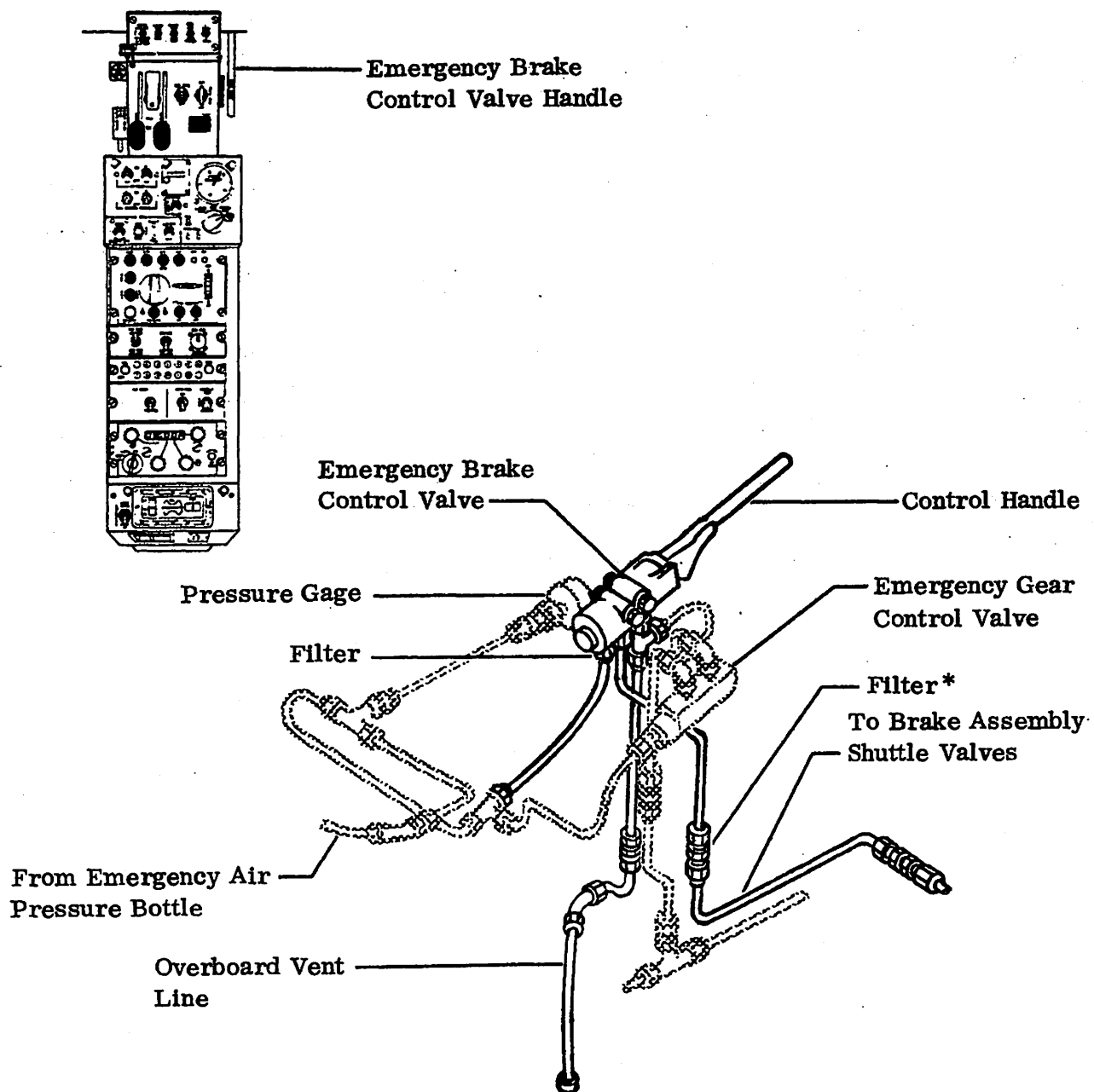
2. Operation

- A. Depressing the emergency brake control handle releases emergency air pressure through the emergency brake control valve, to the shuttle valves, and the brake assemblies.
- B. A pressure regulator within the valve limits brake pressure to approximately 600 psi. To release the brakes, raise the control handle; brake air pressure will be vented overboard, and emergency air will be shut off. The emergency air bottle must be recharged with dry air or nitrogen when the indicated pressure falls below the green arc on the EMERGENCY AIR pressure gage. To recharge the emergency air bottle, refer to Chapter 12. To return brake system to normal, refer to 32-44-00 Servicing.

NOTE: Brake anti-skid system will be inoperative during emergency braking because the emergency air pressure bypasses the anti-skid control valves.

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* On Aircraft 25-195 and Subsequent Only.

Emergency Air Brake System
Figure 1

EFFECTIVITY: ALL

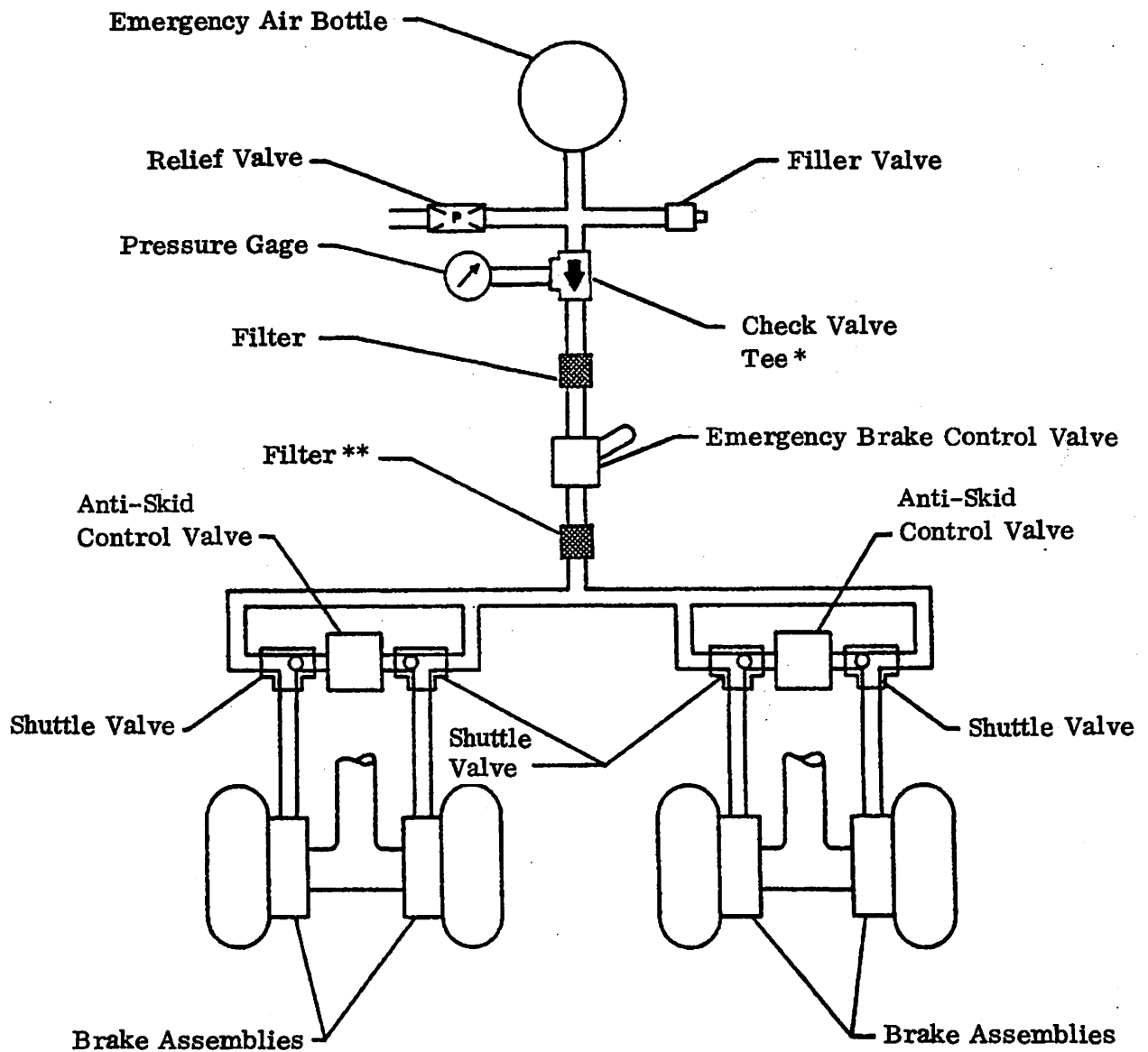
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NOTE: For clarity, emergency gear extension system is not shown.

*Effective 25-270, 25-272 and Subsequent

**Effective 25-195 and Subsequent



Emergency Air Brake System Schematic
Figure 2

EFFECTIVITY: ALL

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EMERGENCY AIR BRAKE SYSTEM - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Return system to normal operation

- (1) Assure that emergency brake control valve handle is up and in locked position.
- (2) Repair hydraulic brake system as required.
- (3) Bleed hydraulic brake system as instructed under Servicing 32-43-00.
- (4) Service hydraulic reservoir and emergency air bottle as required.
Refer to Chapter 12.

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SHUTTLE VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Shuttle Valve

- (1) Depressurize hydraulic system. Assure that emergency brake control valve handle is in off position.
- (2) Disconnect hydraulic and pneumatic lines at shuttle valve. Discard pneumatic line B-nut fitting seal.
- (3) Remove attaching parts and shuttle valve from aircraft.

B. Install Shuttle Valve

- (1) Install shuttle valve into its original position and secure with its attaching hardware.
- (2) Connect hydraulic and pneumatic lines to their original fittings. Use a new fitting seal at pneumatic B-nut connection.

2. Adjustment/Test

A. Functionally Test Shuttle Valve

- (1) With outlet port capped and valve connected to a hydraulic test bench, apply a proof pressure of 2250 psi for 3 minutes to each inlet port. There shall be no sign of leakage.

3. Approved Repairs

A. Due to the many configurations of shuttle valves, only general repair instructions can be given.

- (1) Disassemble shuttle valve beginning with cap bolt and/or port. Note sequence of parts removal.
- (2) Discard O-rings, packing, and backup rings (if installed).
- (3) Inspect for scoring, scratches, or corrosion. Buff out minor damage with crocus cloth using caution not to create any low spots.
- (4) Clean thoroughly with Stoddard solvent and blow dry to remove foreign particles.
- (5) Reassemble in reverse order of disassembly using new O-rings, packing, and backup rings (if required). Immerse O-rings in MIL-H-5606 hydraulic fluid prior to assembly.

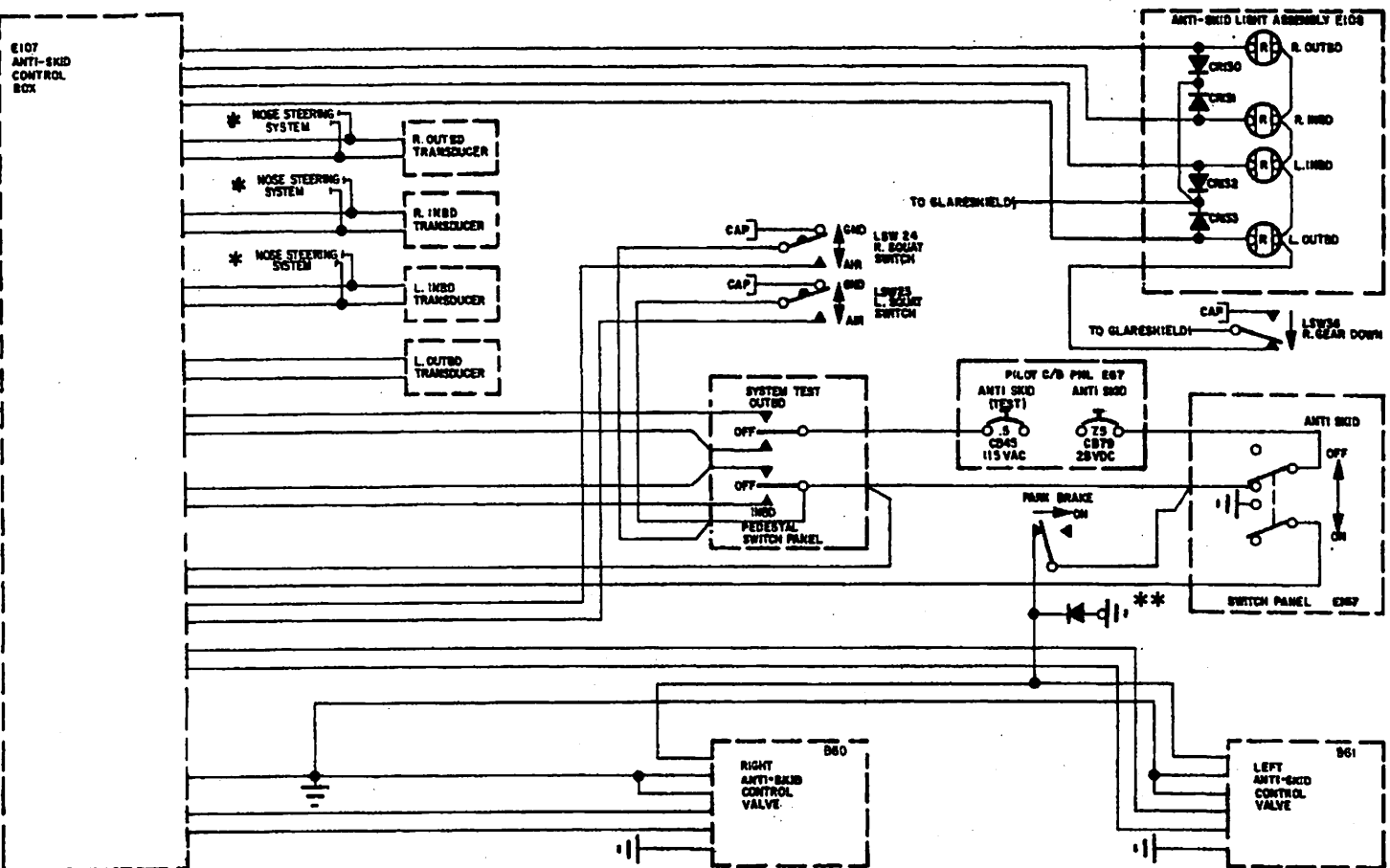
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ANTI-SKID SYSTEM - DESCRIPTION AND OPERATION

1. Description

- A. The Anti-Skid system is an integrated electro-mechanical and hydraulic system incorporated in the main hydraulic brake system to allow even braking on all types and conditions of runways without skidding the landing gear tires.
- B. The brake anti-skid system consists of an anti-skid control valve in each brake pressure line, two wheel speed transducers in each main gear axle, and an anti-skid control box located in the tailcone. Four shuttle valves, one located in each brake pressure line between the anti-skid valve and the brakes, are provided for emergency braking. The Anti-Skid Switch controls system electrical power. On Aircraft 25-061, 25-070 thru 25-196 and 25-198 thru 25-204, the Anti-Skid Test Switch is used to test the anti-skid electrical system and will cause the anti-skid lights to illuminate. On Aircraft 25-197, 25-205 and Subsequent, there is no test switch. Four individual anti-skid lights are installed and provide an indication of system malfunction when illuminated.
- C. Component Description
 - (1) The wheel transducers transmit sinusoidal frequency-modulated signals to the control box. The signals are in direct proportion to each wheel's speed.
 - (2) The anti-skid control valve is electro-hydraulically operated. The control valve relieves applied hydraulic pressure in the event of an anticipated wheel skid.
 - (3) The control box receives signals from the wheel transducers and determines the signal to be sent to the pressure control valves to regulate the brake pressure. The control box also monitors the operating condition of the system components and will cause the respective indicator to illuminate on the anti-skid panel, indicating what section of the system has a malfunction.
 - (4) The anti-skid panel assembly consists of four indicator lights: two lights for the left landing gear brake system and two lights for the right landing gear brake system. The indicator lights will be extinguished with the landing gear retracted. With the landing gear extended and the Anti-Skid Switch in the OFF position, the indicator lights will be illuminated. With the landing gear extended and the Anti-Skid Switch in the ANTI-SKID position, the indicator lights will be extinguished unless there is a malfunction in the anti-skid system.
 - (5) The Anti-Skid Switch controls the electrical power used during normal operation, and the Anti-Skid Test Switch tests the circuits in the system, causing the four indicator lights to illuminate.

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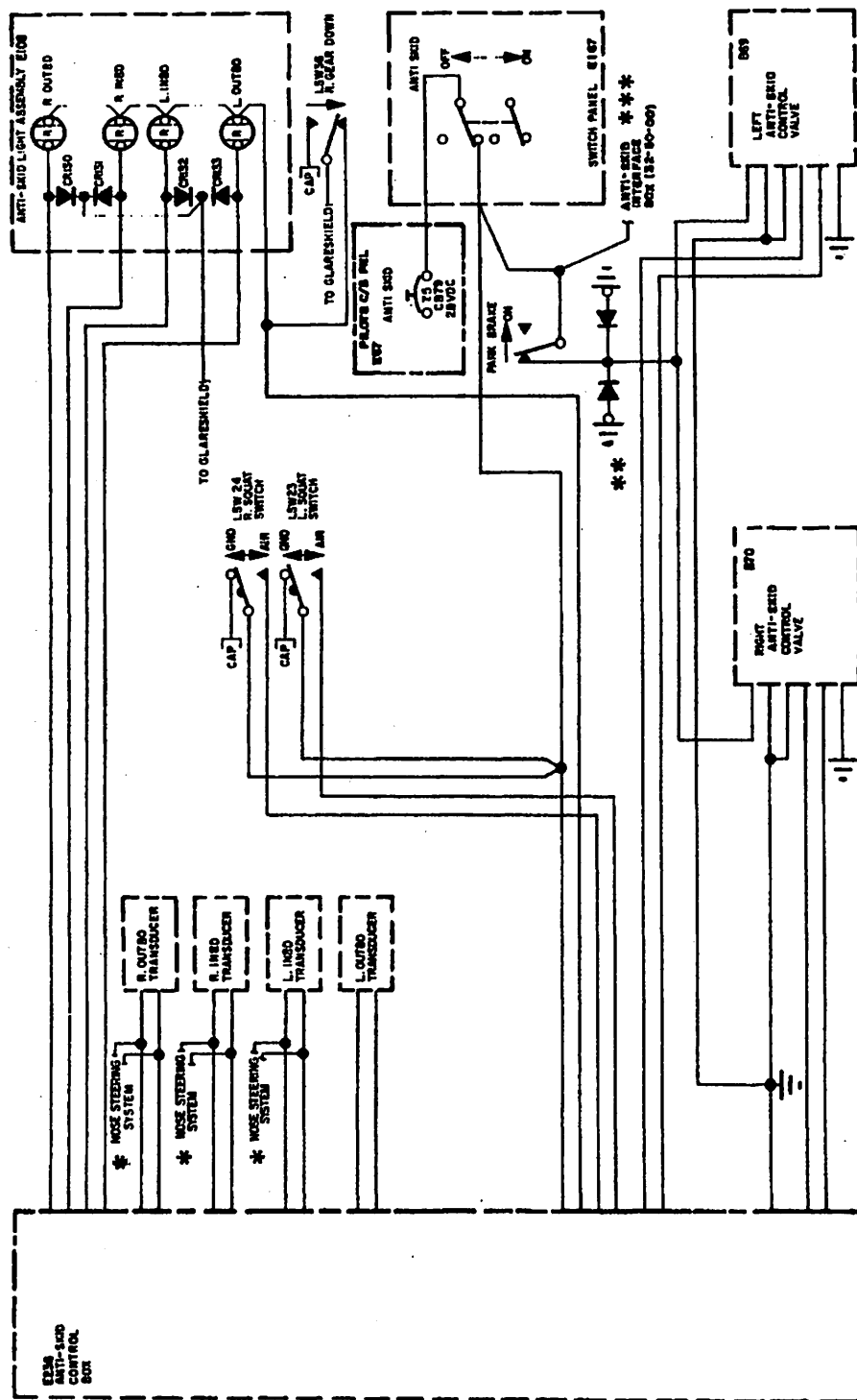
- * Effective Aircraft 25-104 and Subsequent.
- ** Diode and Ground Effective 25-195 and Subsequent.

Brake Anti-Skid System Electrical Schematic
Figure 1 (Sheet 1 of 2)

EFFECTIVITY: 25-061, 25-070 thru 25-196, and 25-198 thru 25-204
Not Equipped with Reduced Approach Speed System

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Brake Anti-Skid System Electrical Schematic
Figure 1 (Sheet 2 of 2)

- * Effective on Aircraft 25-104 and Subsequent.
- ** Second Diode and Ground Effective 25-195, 25-196 and 25-198 thru 25-204.
- *** Not used on Aircraft 25-197 (only) without AMK 76-3 (Anti-Skid Interface Box); and not used on Aircraft 25-230 and Subsequent.

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2. Operation

- A. With the Anti-Skid Switch in the ANTI-SKID position, power is applied to the control box and excitation voltage to the wheel transducers.
- B. Wheel rotation causes each wheel transducer to generate a signal to the control box. The control box compares each transducer's signals with the others' and determines if any one wheel is rotating slower than its mate or if one gear's wheels are rotating slower than the opposite side. It then commands the appropriate control valve to relieve some brake pressure from the appropriate brake assembly or assemblies to preclude wheel skidding.
- C. The anti-skid system is inoperative with the main landing gear in the retracted position. The monitoring circuits in the system monitor the hydraulic pressure within the system and will cause the respective anti-skid warning light to illuminate should there be a loss of hydraulic pressure. The anti-skid system is inoperative when the parking brake valves are not fully opened. (Refer to 32-43-04.)

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ANTI-SKID SYSTEM - MAINTENANCE PRACTICES

1. Adjustment/Test

- NOTE:**
- Erratic braking, blown tires, or flat spots on tires are indicative of possible anti-skid system malfunction. A complete functional test of the system should be accomplished to isolate hydraulic or electrical fault.
 - Cleanliness during maintenance procedures cannot be overemphasized due to the extremely close tolerances used during manufacturing of hydraulic components.
 - Entrapped air in the system is the most common cause of malfunction and is usually the result of allowing the fluid level to drop below the sight gage on the hydraulic reservoir.
 - Erratic braking can occur during brake application if the refrigeration system is being utilized. The electrical system voltage drop caused by the initial high current draw of the compressor motor creates a false signal to the anti-skid system. As a result, the anti-skid will dump the brake pressure for approximately 2 to 3 seconds if the compressor is turned on while the brakes are applied.
 - When using the Anti-Skid Test Switch (Aircraft 25-061, 25-070 thru 25-196, 25-198 thru 25-204 Not Equipped with Reduced Approach Speed System), do not hold switch in one position for more than 20 seconds.

A. Tools and Equipment

NAME	NUMBER	MANUFACTURER	USE
Spin Adapter	2571003-1* 2571004-1**	Gates Learjet	To adapt drill to transducer.
Reversible Drill Motor (capable of 700 to 1000 rpm)			To spin transducer.
Hydraulic Cart (capable of 6 gpm @ 1500 psi)			To pressurize brake system.
Voltmeter	260	Simpson	Check voltages.

*2571003-1 - Used on Aircraft 25-061, 25-070 thru 25-193 not incorporating AAK 75-1 (Anti-Skid Transducer Drive Assembly Improvement).

**2571004-1 - Used on Aircraft 25-194 and Subsequent and previous Aircraft incorporating AAK 75-1 (Anti-Skid Transducer Drive Assembly Improvement).

EFFECTIVITY: SEE TEXT

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- B. Prepare Aircraft for Anti-Skid Functional Test (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)
- (1) Place aircraft on jacks.
 - (2) Pull all circuit breakers except anti-skid test, anti-skid power, warning lights, left low-oil pressure, and inverters.
 - (3) Disconnect anti-skid control box electrical connector (P551).
 - (4) Figure 201 is for use when performing trouble shooting procedures. Pin numbers are those of P551 disconnected from anti-skid control box.
- C. Perform Anti-Skid Functional Test - Part I. (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)
- (1) Apply electrical power to aircraft. With squat switches in air mode, check for 28 vdc at P551, pins 25 and 26. With squat switches in ground mode, check for open circuit to pins 25 and 26.
 - (2) Verify that pins 27, 28, and 29 are at ground potential and measured resistance of pins to ground post is no greater than 0.5 ohm.
 - (3) Set Anti-Skid Switch to OFF and check for open circuit to pins 23 and 24. Verify that pin 30 is grounded.
 - (4) Set Anti-Skid Switch to ANTI-SKID and check for 28 vdc at pins 23 and 24. Verify open circuit at pin 30.
 - (5) Actuate Landing Gear Switch to UP; pins 17, 18, 19, and 20 shall lose 28 vdc power (go to open circuit). Actuate Landing Gear Switch to DN position; pins 17, 18, 19, and 20 shall have 28 vdc.
- CAUTION: WHEN USING ANTI-SKID TEST SWITCH, DO NOT HOLD SWITCH IN ONE POSITION FOR MORE THAN 20 SECONDS.
- (6) Place and hold Anti-Skid Test Switch to INBD; pin 3 (system enable INBD) shall go from open to 28 vdc. Place and hold Anti-Skid Test Switch to OUTBD; pin 4 (system enable OUTBD) shall go from open to 28 vdc. With the Anti-Skid Test Switch in its neutral position, pins 3 and 5 shall be open.
 - (7) Place and hold Anti-Skid Test Switch to INBD position; pin 1 (system enable INBD) shall go from open to 26 vac, 400 Hz. Place and hold Anti-Skid Test Switch to OUTBD; pin 2 (system enable OUTBD) shall go from open to 26 vac, 400 Hz.
 - (8) Check for proper indicator light actuation as follows:
 - (a) Ground pin 17; LI light shall illuminate.
 - (b) Ground pin 18; LO light shall illuminate.
 - (c) Ground pin 19; RI light shall illuminate.
 - (d) Ground pin 20; RO light shall illuminate.

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P551	
PIN NO.	RELATED SYSTEM
1	- System Test Inboard
2	- System Test Outboard
3	- System Test Enable Inboard
4	- System Test Enable Outboard
5	- Left Anti-Skid Control Valve
6	- Right Anti-Skid Control Valve
7	- Left Anti-Skid Control Valve
8	- Right Anti-Skid Control Valve
9	- Left Wheel Inboard Transducer
10	- Left Wheel Outboard Transducer
11	- Right Wheel Inboard Transducer
12	- Right Wheel Outboard Transducer
13	- Left Wheel Inboard Transducer Return
14	- Left Wheel Outboard Transducer Return
15	- Right Wheel Inboard Transducer Return
16	- Right Wheel Outboard Transducer Return
17	- Left Wheel Inboard Indicator Lamp
18	- Left Wheel Outboard Indicator Lamp
19	- Right Wheel Inboard Indicator Lamp
20	- Right Wheel Outboard Indicator Lamp
21	- Not Used
22	- Not Used
23	- System Power
24	- System Power
25	- Left MLG Squat Switch
26	- Right MLG Squat Switch
27	- Case Ground
28	- System Ground
29	- System Ground
30	- Reset
31	- Spare

P551 Pins and Related Systems
Figure 201

EFFECTIVITY: 25-061, 25-070 THRU 25-196, AND 25-198
THRU 25-204 NOT equipped with Reduced
Approach Speed System

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- (9) Perform voltage and resistance test of anti-skid control valve solenoid as follows:
 - (a) Disconnect electrical connector (P552) from right anti-skid control valve.
 - (b) Connect volt-ohmmeter (set to volts dc scale) across pins d and g.
 - (c) Set parking brake handle to OFF. Voltmeter shall register 28 vdc.
 - (d) Set parking brake handle to ON. Voltmeter shall register 0 vdc.
 - (e) Set volt-ohmmeter to ohms scale. Resistance between pins d and g shall be 30 to 50 ohms.
 - (f) Connect electrical connector (P552) to the right anti-skid control valve.
 - (g) Disconnect electrical connector (P553) from left anti-skid control valve.
 - (h) Perform steps (b) through (e) for the left anti-skid control valve.
 - (i) Connect electrical connector (P553) to the left anti-skid control valve.
 - (10) Set Anti-Skid Switch to OFF.
 - (11) Assure that anti-skid control valve electrical connectors (P552 and P553) are installed properly by performing the following resistance check at P551.
 - (a) Resistance between pin 5 and ground shall be 180 to 200 ohms.
 - (b) Resistance between pin 6 and ground shall be 180 to 200 ohms.
 - (c) Resistance between pin 7 and ground shall be 180 to 200 ohms.
 - (d) Resistance between pin 8 and ground shall be 180 to 200 ohms.
 - (12) Assure that wheel transducer connectors are properly installed by performing a resistance check at P551 as follows:
 - (a) Check resistance from pin 10 to pin 14 (LO transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (b) Check resistance from pin 13 to pin 9 (LI transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (c) Check resistance from pin 16 to pin 12 (RO transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (d) Check resistance from pin 11 to pin 15 (RI transducer) for 400 (± 25) ohms at 70°F (21.1°C).
- D. Prepare Aircraft for Functional Test - Part II. (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)
- (1) Connect anti-skid control box electrical connector.
 - (2) Verify that parking brake is released (handle set to OFF).
 - (3) Place aircraft on jacks.
 - (4) Verify that ANTI-SKID and WRN LTS circuit breakers are closed.

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- E. Perform Anti-Skid Functional Test - Part II (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)

NOTE: The anti-skid indicator lights are reset by momentarily setting the Anti-Skid Switch to OFF and back to ANTI-SKID. If a malfunction is present, the affected indicator will illuminate after resetting switch to ON.

- (1) Apply 28 vdc to aircraft.
- (2) Set Anti-Skid Switch to OFF.
- (3) Disconnect electrical connector from the left anti-skid control valve.
- (4) Block squat switches to ground mode.
- (5) With Anti-Skid Switch set to OFF, all four anti-skid indicator lights shall illuminate.
- (6) Set Anti-Skid Switch to ANTI-SKID. Right inboard and right outboard lights shall not illuminate. Left inboard and left outboard lights shall illuminate after an 8-second delay (maximum).
- (7) Set Anti-Skid Switch to OFF and position squat switches to flight mode. All four anti-skid indicator lights shall illuminate.
- (8) Set Anti-Skid Switch to ANTI-SKID. Right inboard and right outboard lights shall not illuminate. Left outboard and left inboard lights shall illuminate after an 8-second delay (maximum).
- (9) Return squat switches to ground mode.
- (10) Connect electrical connector to left anti-skid control valve.
- (11) Set Anti-Skid Switch to OFF and disconnect electrical connector from right anti-skid control valve.
- (12) All four anti-skid indicator lights shall illuminate.
- (13) Set Anti-Skid Switch to ANTI-SKID. Left outboard and left inboard lights shall not illuminate. Right inboard and right outboard lights shall illuminate after an 8-second delay (maximum).
- (14) Set Anti-Skid Switch to OFF and position squat switches to flight mode. All four anti-skid indicator lights shall illuminate.
- (15) Set Anti-Skid Switch to ANTI-SKID. Left inboard and right outboard lights shall not illuminate. Right inboard and right outboard lights shall illuminate after an 8-second delay (maximum).
- (16) Connect electrical connector to right anti-skid control valve.
- (17) Set Anti-Skid Switch to OFF. All four indicator lights will illuminate.
- (18) Return squat switches to ground mode. All four indicator lights shall illuminate.
- (19) Set Anti-Skid Switch to ANTI-SKID and set parking brake. Check that no wheels will rotate.
- (20) Release parking brake. All wheels shall be free to rotate.
- (21) Remove external power from aircraft, blocks from squat switches, and aircraft from jacks.

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F. Perform Anti-Skid Functional Check - Part III as follows: (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)

- (1) Place aircraft on jacks.
- (2) Connect auxiliary hydraulic and electrical power to aircraft.
- (3) Remove transducer drive cap from each wheel, exposing transducer.
- (4) Block squat switches to ground mode.
- (5) Block both brake pedals so that braking pressure is applied. All four wheels shall be locked and shall not rotate.
- (6) Set Anti-Skid Control Switch to ANTI-SKID.
- (7) Set parking brake. Check to assure that no wheel will rotate.
- (8) Unblock both brake pedals and release parking brake. Assure that wheels will now rotate.
- (9) Block both brake pedals so that braking pressure is applied.
- (10) Perform operations 1 through 4 in Table I.

NOTE:

- When spinning transducers, maintain a constant speed of 700 to 1000 rpm.
- After each transducer spin, reset the system by setting the Anti-Skid Control Switch to OFF and back to ANTI-SKID.

- (11) Place squat switches in air mode.
- (12) Perform operations 1 through 4 in Table II.

NOTE:

- When spinning transducers, maintain a constant speed of 700 to 1000 rpm.
- After each transducer spin, reset the system by setting the Anti-Skid Control Switch to OFF and back to ANTI-SKID.

- (13) Remove blocks from brake pedals.
- (14) Assure that squat switches are in ground mode.
- (15) Remove hydraulic and electrical power.
- (16) Replace transducer drive caps.
- (17) Remove aircraft from jacks.

G. Perform Anti-Skid Functional Check - Part IV as follows: (Aircraft 25-061, 25-070 thru 25-196, and 25-198 thru 25-204 NOT equipped with Reduced Approach Speed System.)

- (1) Set Battery and Primary Inverter Switches to ON and allow 30-second warmup period. Set Anti-Skid Switch to OFF. This will simulate an electrical power failure and all four anti-skid indicator lights shall illuminate.
- (2) Set Anti-Skid Switch to ANTI-SKID, indicator lights shall extinguish.
- (3) Hold Anti-Skid Test Switch to OUTBD (minimum 4 seconds, maximum 20 seconds); indicator lights shall illuminate. Release test switch; lights shall extinguish.

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TABLE 1 (GROUND MODE)

Operation	Transducer Spin	Wheels Free to Rotate	Wheel With Brake Pressure	Brake Pressure Within 5 Sec.	Indicator Light Within 5 Sec.
1	R-OUTBD*	L-OUTBD	R-OUTBD L-INBD R-INBD	L-OUTBD	L-OUTBD
2	R-INBD*	L-INBD	R-INBD L-OUTBD R-OUTBD	L-INBD	L-INBD
3	L-INBD*	R-INBD	L-INBD L-OUTBD R-OUTBD	R-INBD	R-INBD
4	L-OUTBD*	R-OUTBD	L-OUTBD L-INBD R-INBD	R-OUTBD	R-OUTBD

*When spinning force is removed, that wheel should be free to rotate for a period of 2 to 10 seconds.

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TABLE II (AIR MODE)

Operation	Transducer Spin	Wheels Free To Rotate	Wheel With Brake Pressure
1	R-OUTBD*	L-INBD L-OUTBD R-INBD	R-OUTBD
2	R-INBD*	L-INBD L-OUTBD R-OUTBD	R-INBD
3	L-INBD*	R-OUTBD R-INBD L-OUTBD	L-INBD
4	L-OUTBD*	R-INBD R-OUTBD L-INBD	L-OUTBD

*When spinning force is removed, that wheel should be free to rotate for a period of 2 to 10 seconds.

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- (4) Hold Anti-Skid Test Switch to INBD (minimum 4 seconds, maximum 20 seconds); indicator lights shall illuminate. Release test switch; lights shall extinguish.

NOTE: This procedure is used for normal ground check operation or if a tire has blown during brake application.

- H. Prepare Aircraft for Anti-Skid Functional Test - Part I as follows: (Aircraft 25-197, 25-205 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
- (1) Place aircraft on jacks.
 - (2) Pull all circuit breakers except ANTI-SKID, GEAR, WRN LTS, bus ties and bus breakers.
 - (3) Disconnect anti-skid control box electrical connector (P551).
 - (4) Figure 202 has been provided for use as an aid when performing trouble shooting procedures. Pin numbers are those of P551 disconnected from anti-skid control box.
- I. Perform Anti-Skid Functional Test - Part I as follows: (Aircraft 25-197, 25-205 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
- (1) Apply electrical power to aircraft. With squat switches in air mode, check for 28 vdc at P551, pins 25 and 26. With squat switches in ground mode, check for open circuit to pins 25 and 26.
 - (2) Verify that pins 27, 28, and 29 are at ground potential and that measured resistance of pins to ground post is no greater than 0.5 ohm.
 - (3) Set Anti-Skid Switch to OFF and check for open circuit to pins 23 and 24. Verify voltage at pin 30.
 - (4) Set Anti-Skid Switch to ANTI-SKID and check for 28 vdc at pins 23 and 24.
 - (5) Raise landing gear; pins 17, 18, 19, and 20 shall lose power (to open circuit). Lower landing gear; pins 17, 18, 19, and 20 shall have voltage (12 - 18 vdc).
 - (6) Check for proper indicator light actuation as follows:
 - (a) Ground pin 17; LI light shall illuminate.
 - (b) Ground pin 18; LO light shall illuminate.
 - (c) Ground pin 19; RI light shall illuminate.
 - (c) Ground pin 20; RO light shall illuminate.
 - (7) Perform voltage and resistance test of anti-skid control valve solenoid as follows:
 - (a) Disconnect electrical connector (P552) from right anti-skid control valve.
 - (b) Connect volt-ohmmeter (set to volts dc scale) across pins d and g.
 - (c) Set parking brake handle to OFF. Voltmeter shall register 28 vdc.
 - (d) Set parking brake handle to ON. Voltmeter shall register 0 vdc.

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P551 PIN NO.	RELATED SYSTEM
1	- Not used
2	- Not used
3	- Not used
4	- Not used
5	- Left Anti-Skid Control Valve
6	- Right Anti-Skid Control Valve
7	- Left Anti-Skid Control Valve
8	- Right Anti-Skid Control Valve
9	- Left Wheel Inboard Transducer
10	- Left Wheel Outboard Transducer
11	- Right Wheel Inboard Transducer
12	- Right Wheel Outboard Transducer
13	- Left Wheel Inboard Transducer Return
14	- Left Wheel Outboard Transducer Return
15	- Right Wheel Inboard Transducer Return
16	- Right Wheel Outboard Transducer Return
17	- Left Wheel Inboard Indicator Lamp
18	- Left Wheel Outboard Indicator Lamp
19	- Right Wheel Inboard Indicator Lamp
20	- Right Wheel Outboard Indicator Lamp
21	- Not used
22	- Not used
23	- System Power
24	- System Power
25	- Left MLG Squat Switch
26	- Right MLG Squat Switch
27	- Case Ground
28	- System Ground
29	- System Ground
30	- Warning Lamp Voltage
31	- Spare

P551 Pins and Related Systems
Figure 202

EFFECTIVITY: 25-197, 25-205 AND SUBSEQUENT AND
Earlier Aircraft Equipped with Reduced
Approach Speed System

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- (e) Set voltmeter to ohms scale. Resistance between pins d and g shall be 30 to 50 ohms.
- (f) Connect electrical connector (P552) to right anti-skid control valve.
- (g) Disconnect electrical connector (P553) from left anti-skid control valve.
- (h) Perform steps (b) through (e) for left anti-skid control valve.
- (i) Connect electrical connector (P553) to left anti-skid control valve.
- (8) Set Anti-Skid Switch to OFF.
- (9) Assure that anti-skid control valve electrical connectors (P552 and P553) are installed properly by performing following resistance check at P551.
 - (a) Resistance between pin 5 and ground shall be 180 to 200 ohms.
 - (b) Resistance between pin 6 and ground shall be 180 to 200 ohms.
 - (c) Resistance between pin 7 and ground shall be 180 to 200 ohms.
 - (d) Resistance between pin 8 and ground shall be 180 to 200 ohms.
- (10) Assure that wheel transducer connectors are properly installed by performing a resistance check at P551 as follows:
 - (a) Check resistance from pin 10 to pin 14 (LO transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (b) Check resistance from pin 13 to pin 9 (LI transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (c) Check resistance from pin 16 to pin 12 (RO transducer) for 400 (± 25) ohms at 70°F (21.1°C).
 - (d) Check resistance from pin 11 to pin 15 (RI transducer) for 400 (± 25) ohms at 70°F (21.1°C).
- J. Prepare Aircraft for Functional Test - Part II as follows: (Aircraft 25-197, 25-205 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
 - (1) Connect anti-skid control box electrical connector.
 - (2) If aircraft is on jacks, block squat switch to ground mode.
 - (3) Verify that parking brake is released.
 - (4) Set Anti-Skid Switch to OFF.
 - (5) Assure that Anti-Skid circuit breaker is depressed.
- K. Perform Anti-Skid Functional Test - Part II as follows: (Aircraft 25-197, 25-205 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
 - (1) Apply 28 vdc to aircraft. All four anti-skid indicator lights shall illuminate.
 - (2) Set Anti-Skid Switch to ANTI-SKID. All four anti-skid indicator lights must extinguish.
 - (3) Disconnect electrical connector from left anti-skid control valve. Both left anti-skid indicator lights must illuminate.

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- (4) Connect electrical connector on left anti-skid control valve.
 - (5) Reset system by setting Anti-Skid Switch to OFF and then to ANTI-SKID.
 - (6) Disconnect electrical connector from right anti-skid control valve.
Both right anti-skid indicator lights must illuminate.
 - (7) Connect electrical connector on right anti-skid control valve.
 - (8) Reset system by setting Anti-Skid Switch to OFF and then to ANTI-SKID.
 - (9) Remove transducer drive cap and transducer and disconnect electrical connector from left outboard wheel transducer. Left outboard anti-skid indicator light must illuminate.
 - (10) Connect electrical connector on left outboard wheel transducer.
 - (11) Reset system by setting Anti-Skid Switch to OFF and then to ANTI-SKID.
 - (12) Repeat steps (9), (10), and (11) with each remaining wheel transducer and its respective anti-skid light.
 - (13) Remove power from aircraft.
- L. Perform Anti-Skid Functional Test - Part III (Aircraft 25-197, 25-205 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
- (1) Place aircraft on jacks.
 - (2) Connect auxiliary hydraulic and electrical power to aircraft.
 - (3) Remove transducer drive cap from each wheel, exposing transducer.
 - (4) Block squat switches to ground mode.
 - (5) Set Anti-Skid Switch to ANTI-SKID.
 - (6) Set parking brake. Check each wheel to assure that no wheel will rotate.
 - (7) Release parking brake. Assure that all wheels are free to rotate.
 - (8) Place squat switches in air mode.
 - (9) Block both brake pedals so that braking pressure is applied. All four wheels shall be unlocked and free to rotate.
 - (10) Perform operations 1 through 4 in Table III.
- NOTE:
- When spinning transducers, maintain a constant speed of 700 to 1000 rpm.
 - After each transducer spin, reset the system by setting the Anti-Skid Switch to OFF and back to ANTI-SKID.
- (11) Place squat switches in ground mode.
 - (12) Perform operations 1 through 4 in Table IV.
- NOTE:
- When spinning transducers, maintain a constant speed of 700 to 1000 rpm.
 - After each transducer spin, reset the system by setting the Anti-Skid Switch to OFF and back to ANTI-SKID.
- (13) Remove blocks from brake pedals.

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- (14) Remove aircraft from jacks.
- (15) Replace transducer drive caps.
- (16) Disconnect auxiliary hydraulic and electrical power supply.
- (17) Assure that squat switches are in ground mode.
- M. Perform Anti-Skid Functional Check - Part IV (Aircraft 25-197, 25-204 and Subsequent and Earlier Aircraft Equipped with Reduced Approach Speed System.)
 - (1) Set Battery Switches to BAT 1 and BAT 2 and allow 30-second warmup period. Set Anti-Skid Switch to OFF. This will simulate an electrical power failure and all four anti-skid indicator lights shall illuminate.
 - (2) Set Anti-Skid Switch to ANTI-SKID; indicator lights shall extinguish.

NOTE: This procedure is used for normal ground check operation or if a tire has blown during brake application.

2. Approved Repairs

- A. Repairs are allowed on the anti-skid control box, wheel speed transducers, and the anti-skid valves provided the repair facility obtains the applicable vendor repair manuals and test equipment specified by the vendor.

NOTE: Components of the anti-skid system which are still under factory warranty must be returned to the factory for repair.

- B. Vendor manuals are listed under "Supplementary Publications" in the Introduction section of the maintenance manual.

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TABLE III (AIR MODE)

Operation	Transducer Spin *	Wheels Free To Rotate	Wheel With Brake Pressure
1	R-OUTBD	L-INBD L-OUTBD R-INBD	R-OUTBD
2	R-INBD	L-INBD L-OUTBD R-OUTBD	R-INBD
3	L-INBD	R-OUTBD R-INBD L-OUTBD	L-INBD
4	L-OUTBD	R-INBD R-OUTBD L-INBD	L-OUTBD

*When spinning force is removed, that wheel should be free to rotate after a period of 2 to 10 seconds.

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TABLE IV (GROUND MODE)

Operation	Transducer Spin	Wheels With Brake Pressure	Wheel With Brake Release Then Reapplication of Pressure**
1	R-OUTBD*	R-OUTBD L-INBD R-INBD	L-OUTBD
2	R-INBD*	R-INBD L-OUTBD R-OUTBD	L-INBD
3	L-INBD*	L-INBD L-OUTBD R-OUTBD	R-INBD
4	L-OUTBD*	L-OUTBD L-INBD R-INBD	R-OUTBD

*When spinning force is removed, that wheel should be free to rotate for a period of 2 to 10 seconds.

**When spinning force is removed, the opposite brake will reapply within 2 to 10 seconds.

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WHEEL SPEED TRANSDUCER - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Wheel Speed Transducer (See figure 201.)

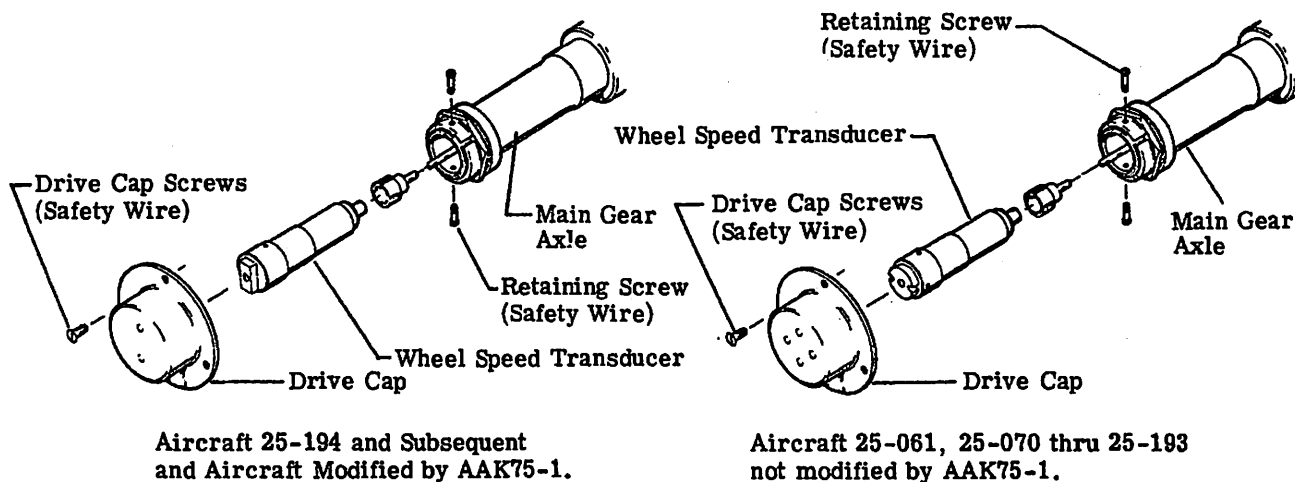
- (1) Remove safety wire and screws securing drive cap to main gear wheel and remove drive cap.

CAUTION: WHEN REMOVING TRANSDUCER, PULL TRANSDUCER JUST FAR ENOUGH TO GAIN ACCESS TO CONNECTOR PLUG; OTHERWISE WIRING SEAL THROUGH AXLE MAY BE DAMAGED.

- (2) Remove safety wire, loosen retaining screws in axle, remove transducer, and disconnect electrical connector.

B. Install Wheel Speed Transducer (See figure 201.)

- (1) Connect electrical connector, insert wiring and transducer into axle and secure with retaining screws. Safety wire retaining screws.
- (2) Position drive cap on main gear wheel and secure with screws. Safety wire screws.
- (3) Check anti-skid system. (Refer to 32-45-00, Adjustment/Test.)



TYPICAL TRANSDUCER INSTALLATION

Wheel Speed Transducer Installation
Figure 201

EFFECTIVITY: ALL

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ANTI-SKID VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Anti-Skid Valve (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Remove electrical power from aircraft.
- (3) Remove hydraulic pressure from brake system.
- (4) Disconnect plumbing connections from anti-skid valve. Cap or plug exposed fittings.
- (5) Disconnect electrical connector from anti-skid valve.
- (6) Remove attaching parts securing anti-skid valve to wing spar and remove valve from aircraft.

B. Install Anti-Skid Valve (See figure 201.)

- (1) Secure anti-skid valve to wing spar with attaching parts.
- (2) Remove plumbing caps and plugs, then connect plumbing to anti-skid valve.
- (3) Connect electrical connector to anti-skid valve.
- (4) Apply electrical power to aircraft.
- (5) Depress brake pedals to apply pressure to anti-skid system and check for leaks.
- (6) Release brakes.
- (7) Remove aircraft from jacks.

2. Cleaning/Painting

NOTE: Each anti-skid valve contains a filter which must be cleaned in accordance with current inspection intervals in Chapter 5.

A. Clean Filter Element

- (1) Unscrew filter cover from anti-skid valve. Remove and discard packing and backup ring.
- (2) Extract filter element from anti-skid valve. Remove packing from filter element and discard.
- (3) Fill filter element cavity with full strength liquid detergent (biodegradable).
- (4) Insert a rubber plug with a hole through center into element neck and, using clean, dry air, force detergent out through element.
- (5) Remove rubber plug and reverse flush element with tap water to remove soap.
- (6) Reinsert rubber plug and blow water out of element.
- (7) Remove rubber plug and reverse flush element with tap water until water runs clear from element.
- (8) Repeat steps (3) through (7) until element is clean.

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- (9) Dry element with clean, dry air or nitrogen.
- (10) Install new MS 28775-013 packing on filter element and insert element into anti-skid valve.
- (11) Install new MS 28774-022 backup ring and MS 28775-022 packing on filter cover and screw filter cover into anti-skid valve. Torque filter cover to 5 to 10 inch-pounds and safety wire.

EFFECTIVITY: ALL

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STEERING - DESCRIPTION AND OPERATION

1. Description

- A. On Aircraft 25-061, 25-070 thru 25-103, the hi-low nose wheel steering system is operable only when the aircraft is on the ground and one of the Wheel Master or Primary Steering Switches is depressed.
- B. On Aircraft 25-104 and Subsequent, the variable authority nose wheel steering system is operable only when the aircraft is on the ground and either the pilot's or copilot's Wheel Master Switch is depressed or the Steer Lock Switch is engaged.
- C. On Aircraft 25-061, 25-070 thru 25-103, the Steering Lock Switch (momentary type) energizes the steering lock circuit which applies continuous 28 vdc to energize the nose steering and stall warning computer for steering travel of 6° to 10°. Depressing the Wheel Master Switch automatically removes the steering lock from the circuit. With a Primary Steering Switch depressed, the nose wheel is steerable through 40° to 50° either side of neutral for low-speed taxiing and parking purposes.
- D. On Aircraft 25-104 and Subsequent, nose steering turn angle varies with ground speed. At ground speeds of approximately 45 knots, steering travel is 8°. Steering travel tapers inversely with ground speed to approximately 50° at ground speeds below 10 knots. A momentary Steering Lock Switch energizes the steering lock circuit which applies continuous 28 vdc to energize the nose steering computer. Depressing a Wheel Master Switch automatically removes the steering lock from the circuit.
- E. On Aircraft 25-061, 25-070 thru 25-103, system components consist of an electrically powered steering actuator and followup mounted on top of the nose gear strut, a steering computer-amplifier mounted in the left side of the nose section, a followup mechanically connected to the rudder pedals, a steering relay located in the nose section, a Wheel Master Switch on each control wheel, two Primary Steering Switches, and a Steering Lock Switch.
- F. On Aircraft 25-104 and Subsequent, system components consist of an electrically powered steering actuator and followup mounted on top of the nose gear strut, a steering computer-amplifier mounted in the right side of the nose section, a followup mechanically connected to the rudder pedals, a steering relay located in the nose section, a Wheel Master Switch on each control wheel, and a Steering Lock Switch. The system also utilizes the left inboard, right outboard, and right inboard wheel speed transducers to provide an input signal to the computer-amplifier.
- G. System power is 28 vdc for the actuator servo motor and 115 vac for the steering computer-amplifier.
- H. When the nose gear steering is not being used, the steering actuator acts as a shimmy damper.

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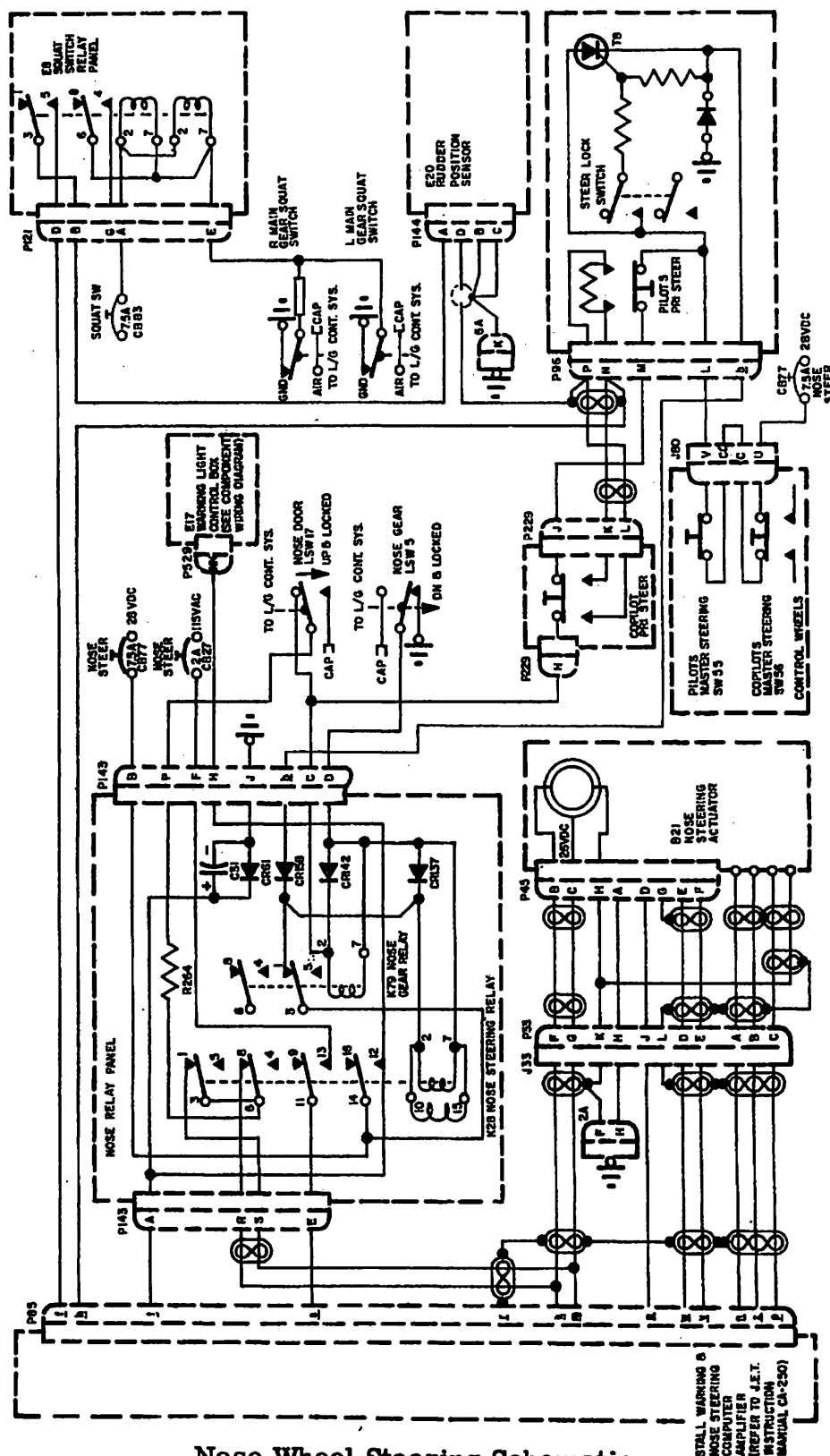
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2. Operation

- A. On Aircraft 25-061, 25-070 thru 25-103, rudder pedal movement drives the rudder pedal followup which applies a voltage displacement signal to the input of the computer-amplifier. The computer-amplifier applies a signal to a "clockwise" or "counterclockwise" magnetic particle clutch in the steering actuator of a magnitude corresponding to the degree of rudder pedal displacement. The signal application causes the magnetic clutch particles to solidify and engage the clutch with the DC motor of the steering actuator. The clutch output torque drives the actuator gear train and positions the nose wheel to the selected position. Two additional signals are also applied to the input of the computer-amplifier: a nose steering followup signal and a DC velocity generator feedback signal from the steering actuator. The nose steering followup signal and the rudder pedal followup signal are algebraically summed in the computer-amplifier with the steering actuator feedback signal tending to incrementally cancel the rudder pedal followup signal. The nose steering followup signal eventually cancels the rudder pedal followup signal when the nose gear has attained its selected position.
- B. On Aircraft 25-104 and Subsequent, rudder pedal movement drives the rudder pedal followup which applies a voltage displacement signal to the input of the nose steering computer-amplifier. The computer-amplifier applies a clockwise or counterclockwise signal to the steering actuator. Signal application causes the actuator clutch to engage the actuator motor. The clutch output torque drives the actuator gear train and positions the nose wheel to the selected position. Two additional signals are also applied to the input of the computer-amplifier: a nose steering followup signal and a DC velocity generator feedback signal from the steering actuator. The nose steering followup signal and rudder pedal followup signals are algebraically summed in the computer-amplifier with the steering followup signal tending to incrementally cancel the rudder pedal followup signal. Nose steering followup signal eventually cancels the rudder pedal followup signal when the gear has attained its selected position. A signal from the left inboard, right inboard and right outboard wheel speed transducer modifies the rudder pedal followup signal to accomplish the variable authority nose wheel steering.

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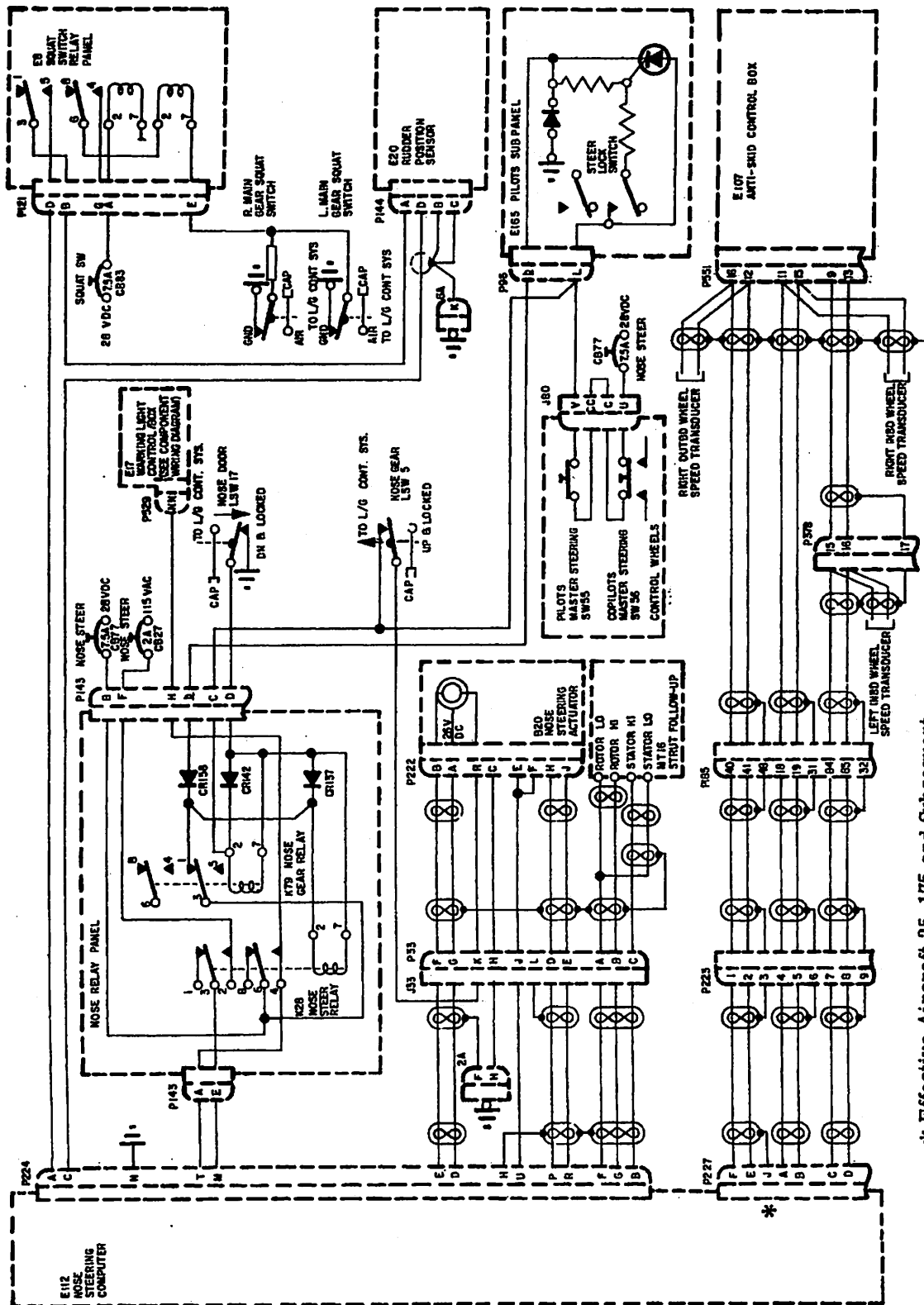
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Nose Wheel Steering Schematic
Figure 1 (Sheet 1 of 8)

EFFECTIVITY: Aircraft 25-061, 25-070 thru 25-103
not equipped with Reduced Approach Speed System

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* Effective Aircraft 25-175 and Subsequent,
Delete P227-J connection to shield.

Nose Wheel Steering Schematic
Figure 1 (Sheet 2 of 8)

EFFECTIVITY: Aircraft 25-104 thru 25-179
not equipped with Reduced Approach Speed System

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Nose Wheel Steering System Schematic

EFFECTIVITY: Aircraft 25-180 thru 25-195

not equipped with Reduced Approach Speed System

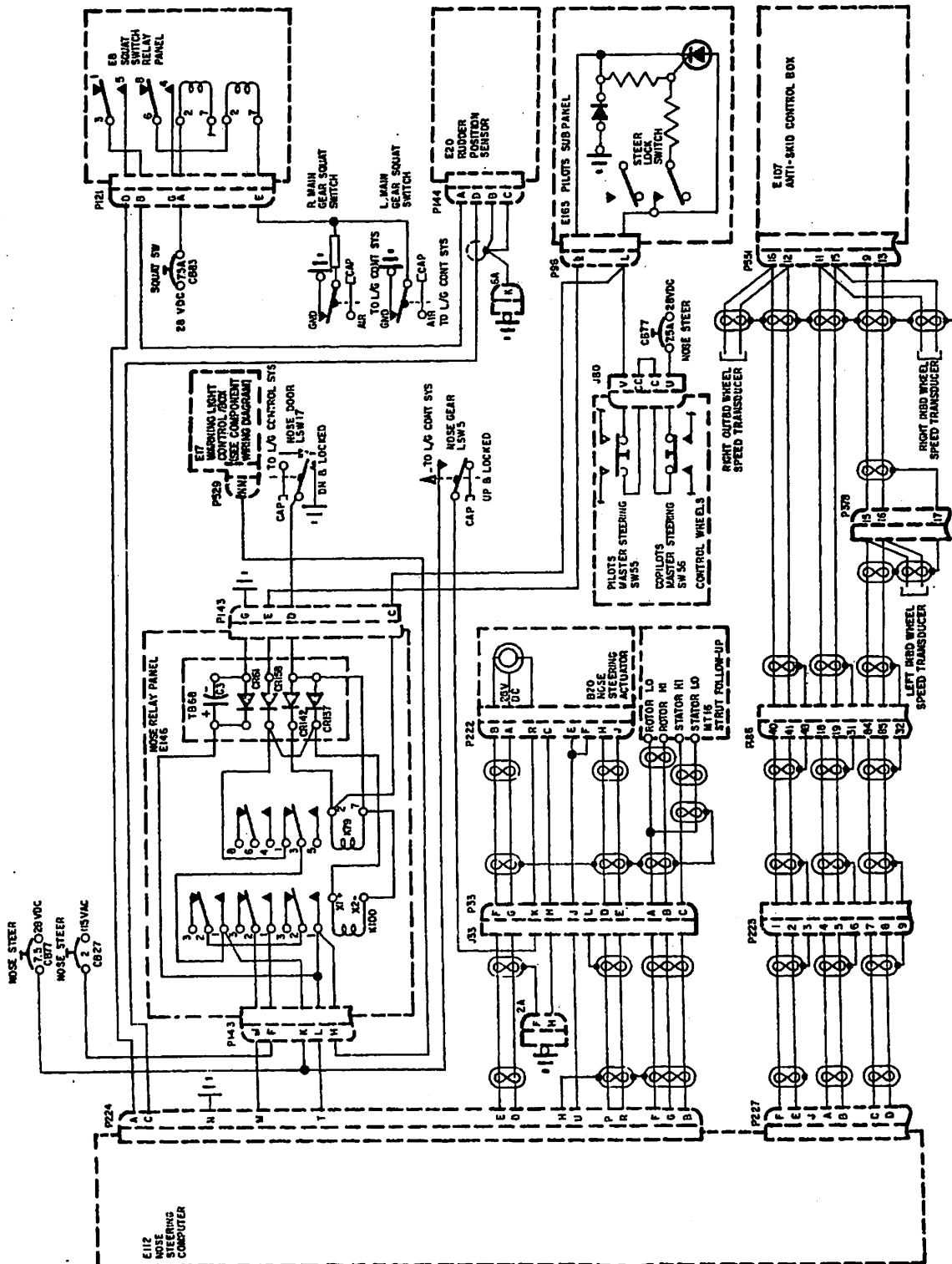
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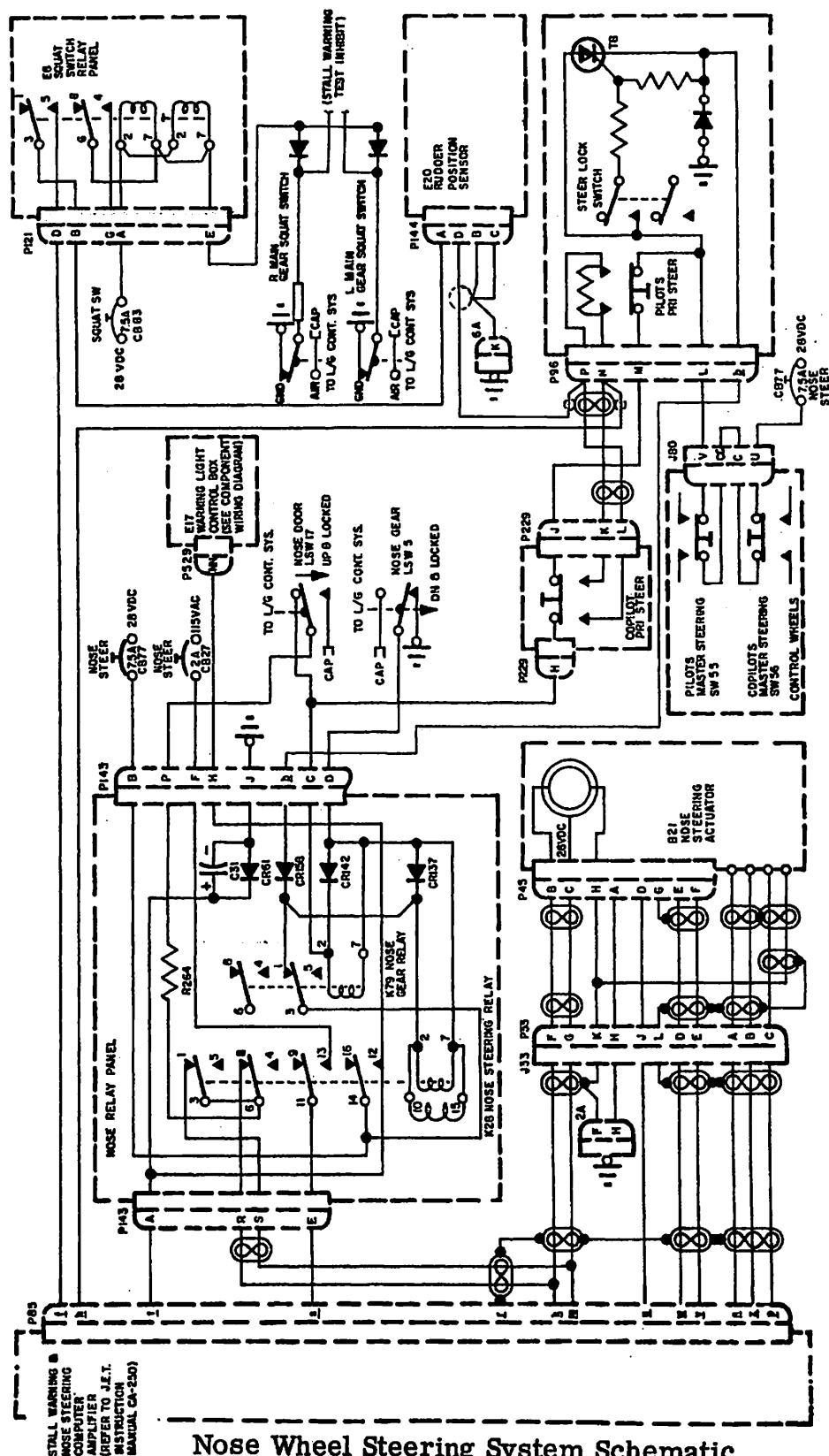
Nose Wheel Steering System Schematic
Figure 1 (Sheet 4 of 8)

EFFECTIVITY: Aircraft 25-196 thru 25-205
not equipped with Reduced Approach Speed System

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Nose Wheel Steering System Schematic
Figure 1 (Sheet 5 of 8)

EFFECTIVITY: Aircraft 25-061, 25-070 thru 25-103
Equipped with Reduced Approach Speed System

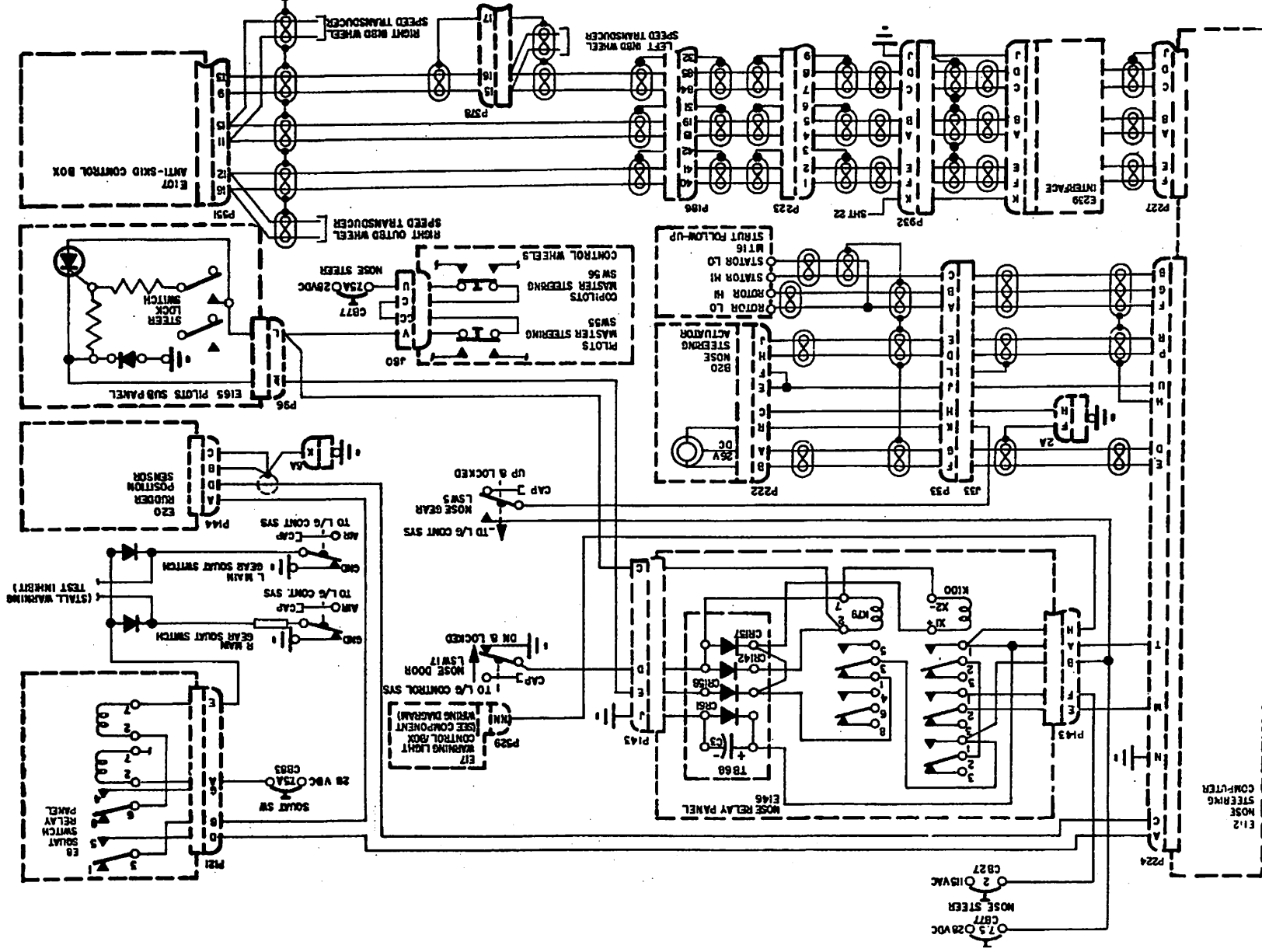
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Nose Wheel Steering System Schematic

Figure 1 (Sheet 7 of 8)

EFFECTIVITY: Aircraft 25-180 thru 25-195
Equipped with Reduced Approach Speed System

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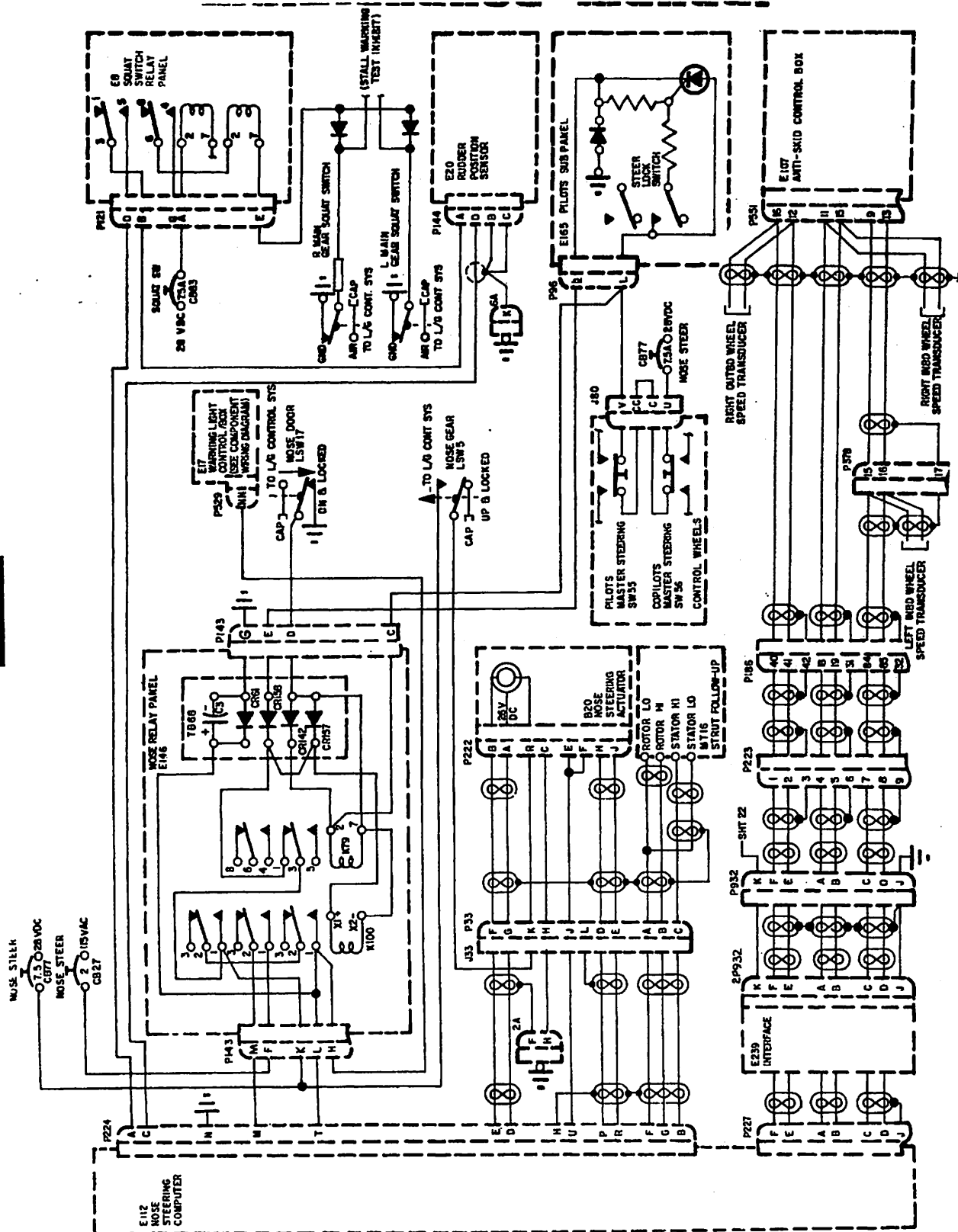
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Nose Wheel Steering System Schematic
Figure 1 (Sheet 8 of 8)

EFFECTIVITY: Aircraft 25-196 thru 25-205 Equipped with Reduced Approach Speed System and 25-206 and Subsequent

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STEERING - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Tools and Equipment

NAME	NUMBER	MANUFACTURER	USE
Nose Wheel Protractor	2471001	Gates Learjet Corp.	Measure deflection.
Test Set*	800585 (or fabricate per Figure 202)	LearAvia	To allow separate test for each transducer circuit.
Transducer Spin Adapter*	2571003-1 or 2571004-1	Gates Learjet Corp.	To simulate aircraft ground speed for vari- able authority check.

Drill Motor (Capable of 1700 rpm)*

Voltmeter* Mod 8000A-01 Fluke
Digital Multimeter
or equivalent.

Auxiliary Power Unit* Provide constant 28
vdc to aircraft.

*Used with Variable Authority Nose Wheel Steering Systems.

B. Rig Hi-Low Nose Wheel Steering System (See figure 201)

- (1) Place jack under nose of aircraft and remove nose gear shock strut.
(Refer to 32-21-00.)

NOTE: Main gear squat switches must be in ground mode to apply
signal from rudder pedal followup.

WARNING: BEFORE PERFORMING ANY MAINTENANCE ON THE
STRUT, ASSURE THAT ALL AIR PRESSURE HAS BEEN
RELEASED THROUGH THE AIR VALVE. DO NOT RE-
LEASE AIR BY LOOSENING OR REMOVING AIR VALVE.

- (2) Place the nose gear shock strut in a stand. The stand must be high
enough to allow clearance of wheel when strut is fully extended.

EFFECTIVITY: ALL

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- (3) Fabricate a jumper cable using same type and size wires as those which were disconnected, a Bendix P/N PTO6SE-14-12P plug and a Bendix PTOOSE-14-12S receptacle. Wire length should be approximately 6 feet. Connect jumper cable between aircraft wiring and nose gear.
- (4) Actuate cable between aircraft wiring and nose gear downlock switch by extending nose gear actuator as far as possible. Nose gear downlock switch must be actuated to apply 28 vdc to nose steering actuator.
- (5) Connect auxiliary power unit 28 vdc to aircraft.
- (6) Set Battery and Primary Inverter Switches on.
- (7) Center rudder pedals and depress Wheel Master Steering Switch on pilot's control wheel. If followup is off null, nose wheel will move out of mechanical detent and position itself with respect to followup.
- (8) Loosen followup attachment screws and adjust followup until nose wheel moves into mechanical detent. Tighten screws.
- (9) Move rudder pedals to apply either right or left rudder. Nose wheel should follow direction of rudder pedal. If nose wheel follows direction of rudder pedals, torque followup attachment screws 20 to 25 inch-pounds, install safety wire, and disconnect wire bundle. If nose wheel turns 180° or moves in opposite direction of rudder pedals, followup shaft is turned 180°; proceed as follows:
 - (a) Disconnect wire bundle from nose gear and place nose wheel in mechanical detent.
 - (b) Remove lid attachment screws from lid and carefully raise lid. Do not remove lid completely from case.
 - (c) Noting position of gear in relation to lid, rotate followup gear 180°. Install lid on case and repeat steps (4) through (8). Torque lid attachment screws 20 to 25 inch-pounds. Safety wire screws.
- (10) Set Battery and Primary Inverter Switches to OFF and inflate nose gear strut. (Refer to Chapter 12.)
- (11) Install nose gear strut. (Refer to 32-21-00.)
- (12) Place nose gear on nose wheel protractor (P/N 2471001) and remove jack from aircraft.
- (13) Turn pedal control potentiometer completely counterclockwise.
- (14) Turn velocity generator potentiometer completely counterclockwise.
- (15) Turn followup potentiometer approximately one-fourth its full clockwise range.
- (16) Turn balance potentiometer approximately one-half its full clockwise range.
- (17) Set Battery and Primary Inverter Switches on.
- (18) Move rudder pedals to full right rudder position and hold.
- (19) Turn pedal control potentiometer clockwise until nose wheel is turned 40° to 50° right of center.
- (20) Center rudder pedals and depress Primary Steering Switch of either instrument subpanel; nose wheel shall move to center and oscillate.

EFFECTIVITY: SEE TEXT

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- (21) Turn velocity generator potentiometer clockwise until oscillation stops plus one-eighth additional turn to assure stability.

NOTE: If nose wheel moves sluggishly, a higher setting of the followup potentiometer may be necessary. If so, turn followup potentiometer $\frac{3}{8}$ of its full clockwise range, return to step (16) and proceed again.

- (22) With rudder pedals centered, turn balance potentiometer until nose wheel is centered.
- (23) Move rudder pedals to full left rudder position and hold; nose wheel shall turn 40° to 50° left of center.
- (24) Center rudder pedals and nose wheel shall return to center and stop.

NOTE: If wheel tends to be sluggish during full right or full left command, velocity generator is adjusted too far clockwise. If nose wheel moves quickly and has a tendency to oscillate, velocity generator is adjusted too far counterclockwise.

- (25) With rudder pedals and nose wheel centered, depress and release Primary Steering Switch while observing nose wheel. If wheel shifts direction when Primary Steering Switch is depressed and released, adjustment of rudder pedal synchro housing is required. This is to assure that synchro is nulled when rudder is exactly centered relative to aircraft centerline.
- (26) Depress Wheel Master Switch on either control wheel and move rudder pedals to full left and full right rudder positions while observing nose wheel travel. Nose wheel travel shall be 6° to 20° both left and right of center.
- (27) Actuate Primary Steering Switch and move nose wheel to approximately 40° right.

C. Rig LearAvia Variable Authority Nose Wheel Steering System (Aircraft 25-104 thru 25-229) (See figure 201.)

- (1) Place jack under nose of aircraft and remove nose gear shock strut. (Refer to 32-21-00.)

WARNING: ASSURE THAT ALL AIR PRESSURE HAS BEEN RELEASED THROUGH THE AIR VALVE. DO NOT RELEASE AIR PRESSURE BY LOOSENING OR REMOVING AIR VALVE.

- (2) Place nose gear shock strut in a stand. Stand must be high enough to allow clearance of wheel when strut is fully extended.

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- (3) Fabricate a jumper cable to reconnect nose steering actuator and follow-up electrically with aircraft. Use same type and size wires as those which were disconnected, a Bendix P/N PTOSE-14-12P plug and a Bendix P/N PTOOSE-14-12S receptacle. Wire lengths should be approximately six feet. Connect jumper cable between aircraft wiring and nose gear.
- (4) Remove floorboard that is between and aft of copilot's rudder pedals. Insert rigging pin through rigging pin hole in rudder outboard bellcrank assembly. This will assure that pedals are neutral when mulling pedal followup.
- (5) Remove RH nose compartment door to gain access to nose wheel steering computer-amplifier. Using a small screwdriver, turn BAL potentiometer and FU potentiometer clockwise approximately ten complete turns.
- (6) Actuate nose gear down and locked switch by assuring nose gear actuator is extended as far as possible. Nose gear down and locked switch must be actuated to apply 28 vdc to nose steering actuator clutch (P222, pin B).
- (7) Assure that squat switches are in ground mode. Squat switches must be in ground mode to apply 1 vac excitation to rudder pedal followup (P144, pin A).
- (8) Set Battery, Primary Inverter, and Anti-Skid Switches on.
- (9) Insert one voltmeter lead into back of connector P224, pin C and ground other voltmeter lead.
- (10) Loosen rudder pedal followup retaining screws (located on frame 5 forward of copilot's rudder pedals) and adjust followup for a low voltage null (as close as zero vac as possible). Retighten retaining screws.
- (11) Remove voltmeter from ground and pin C of P224.
- (12) Insert one voltmeter lead into back of connector P224, pin G and insert other voltmeter lead (gnd) into back of connector P224, pin F.
- (13) Remove safety wire from steering followup retaining screws (steering followup located on top of nose wheel steering actuator), loosen steering followup retaining screws, and adjust followup for a voltage null (as close to 0 vac as possible). Retighten retaining screws.
- (14) Move rudder pedals to apply either right or left rudder. Nose wheel should follow direction of rudder pedals. If nose wheel follows direction of rudder pedals, torque followup attachment screws 20 to 25 inch-pounds and safety wire. If nose wheel turns 180° or moves in opposite direction of rudder pedals, followup shaft is turned 180°; proceed as follows:
 - (a) Disconnect wire bundle from nose gear and place nose wheel in mechanical detent.
 - (b) Remove lid attachment screws from lid and carefully raise lid. Do not remove lid completely from case.

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- (c) Noting position of gear in relation to lid, rotate followup gear 180°. Install lid on case and repeat steps (6) through (10). Torque lid attachment screws 20 to 25 inch-pounds. Safety wire screws.
- (15) Set Battery and Primary Inverter Switches to OFF. Disconnect wire bundle and inflate nose gear shock strut. (Refer to Chapter 12.)
 - (16) Install nose gear shock strut. (Refer to 32-21-00.)
 - (17) Place aircraft on nose wheel protractor and adjust so that, with nose gear centered, protractor is positioned at zero degrees.
 - (18) Set Battery, Primary Inverter, and Anti-Skid Switches on.
 - (19) Remove rigging pin from rudder bellcrank and, with rudder pedals centered, energize nose wheel steering system.
 - (20) Adjust followup potentiometer (on computer) until nose wheel travel has a total of 90 degrees (± 5 degrees) when pushing followup potentiometer rudder pedals left and right.
 - (21) Adjust balance potentiometer (on computer) to obtain nose wheel deflection of 45 degrees (± 5 degrees) both left and right of center when pushing rudder pedals.
 - (22) Disconnect P227 at computer-amplifier and connect test set (P/N 800585, or fabricated per figure 202) between computer-amplifier and P227.
 - (23) Remove wheel speed transducer drive caps from left inboard, right inboard, and right outboard wheels.
 - (24) Set test switch to position 1. Attach spin adapter to drill motor and drive right outboard transducer at 1700 rpm.
 - (25) Depress control wheel Master Switch and fully depress right and left rudder pedals. Nose wheel deflection shall be 8 (± 2) degrees in both directions.
 - (26) Set test switch to position 2.
 - (27) Drive left inboard transducer at 1700 rpm.
 - (28) Depress Master Switch and fully depress right and left rudder pedals. Nose wheel deflection shall be 8 (± 2) degrees in both directions.
 - (29) Set test switch to position 3.
 - (30) Drive right inboard transducer at 1700 rpm.
 - (31) Depress Master Switch and fully depress right and then left rudder pedals. Nose wheel deflection shall be 8 (± 2) degrees in both directions.
 - (32) Set Battery and Primary Inverter Switches to OFF.
 - (33) Disconnect test set and connect P227 to computer-amplifier.
 - (34) Remove aircraft from nose wheel protractor.
 - (35) Install wheel speed transducer drive caps.
 - (36) Install floorboard and RH nose compartment access door.

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D. Rig J.E.T. Variable Authority Nose Wheel Steering System. (Aircraft 25-230 and Subsequent.) (See figure 201.)

- (1) Place jack under nose of aircraft and remove nose gear shock strut. (Refer to 32-21-00.)

WARNING: ASSURE THAT ALL AIR PRESSURE HAS BEEN RELEASED THROUGH THE AIR VALVE. DO NOT RELEASE AIR BY LOOSENING OR REMOVING AIR VALVE.

- (2) Place nose gear shock strut in a stand high enough to allow wheel clearance when strut is fully extended.
- (3) Fabricate a jumper cable to electrically reconnect nose steering actuator and followup with aircraft. Use same type and size wires as those which were disconnected, a Bendix P/N PTOSE-14-12P plug, and a Bendix P/N PTOOSE-14-12S receptacle. Wire lengths should be approximately six feet. Connect jumper cable between aircraft wiring and nose gear.
- (4) Remove floorboard between and aft of copilot's rudder pedals. Insert rigging pin through rigging pin hole in rudder bellcrank. This will assure that pedals are neutral when mulling pedal followup.
- (5) Remove RH nose compartment door to gain access to the nose wheel steering computer-amplifier. CMD potentiometer and BAL potentiometer are located beneath a cover on computer end.
- (6) Actuate nose gear down and locked switch by extending nose gear actuator as far as possible. Nose gear down and locked switch must be actuated to apply 28 vdc to nose steering actuator clutch (P222, pin R).
- (7) Assure that squat switches are in ground mode. Squat switches must be in ground mode to apply 1 vac excitation to rudder pedal followup (P144, pin A).
- (8) Set Battery and Primary Inverter and Anti-Skid Switches on.
- (9) Insert one voltmeter lead into back of connector P224, pin C and ground other lead.
- (10) Loosen rudder pedal followup retaining ring screws and adjust followup for a low voltage null (as close to zero vac as possible). Tighten retaining ring screws. Followup is located on frame 5 forward of copilot's rudder pedals.
- (11) Remove voltmeter from ground connection and connector P224, pin C.
- (12) Insert one voltmeter lead into back of connector P224, pin G and other lead into P224, pin F.
- (13) Remove safety wire from steering followup retaining ring screws. Loosen retaining ring screws and adjust followup for a voltage null (as close to 0 vac as possible). Tighten retaining ring screws. Remove voltmeter from connector P224.

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- (14) Move rudder pedals to apply either right or left rudder. Nose wheel should follow direction of rudder pedal. If nose wheel follows direction of rudder pedals, torque followup attachment screws 20 to 25 inch-pounds and safety wire. If nose wheel turns 180° or moves in opposite direction of rudder pedals, followup shaft is turned 180°, proceed as follows:
- (a) Disconnect wire bundle from nose gear and place nose wheel in mechanical detent.
 - (b) Remove lid attachment screws from lid and carefully raise lid. Do not remove lid completely from case.
 - (c) Noting position of gear in relation to lid, rotate followup gear 180°. Install lid on case and repeat steps (6) through (10). Torque lid attachment screws 20 to 25 inch-pounds. Safety wire screws.
- (15) Set Battery, Inverter, and Anti-Skid Switches to OFF.
- (16) Install nose gear shock strut. (Refer to 32-21-00.)
- (17) Place aircraft on nose wheel protractor and adjust so that, with nose gear centered, protractor is positioned at zero degrees.
- (18) Set Battery, Inverter, and Anti-Skid Switches to on.
- (19) Remove rigging pin from rudder bellcrank and, with rudder pedals centered, energize nose wheel steering system.
- (20) Adjust CMD potentiometer (on computer) for 90 (±5) degrees total nose wheel travel with full left to full right rudder applied.
- (21) Adjust BAL potentiometer (on computer) to obtain nose wheel deflection of 45 (±5) degrees both right and left of center.
- (22) Remove wheel speed transducer drive caps from left inboard, right inboard, and right outboard wheels.
- (23) Attach spin adapter to drill motor and spin right outboard transducer at 1700 rpm.
- (24) Depress Master Switch on control wheel and fully depress right and then left rudder pedals. Nose wheel deflection shall be 8 (±2) degrees in both directions.
- (25) Spin right inboard transducer at 1700 rpm.
- (26) Depress Master Switch and fully depress right and then left rudder pedals. Nose wheel deflection shall be 8 (±2) degrees in both directions.
- (27) Spin left inboard transducer at 1700 rpm. Depress Master Switch and fully depress right and then left rudder pedals. Nose wheel deflection shall be 8 (±2) degrees in both directions.
- (28) Repeat steps (21) through (26) with Anti-Skid Switch set to OFF.

NOTE: When Anti-Skid Switch is set to OFF, nose wheel will momentarily deflect up to 10°, then return.

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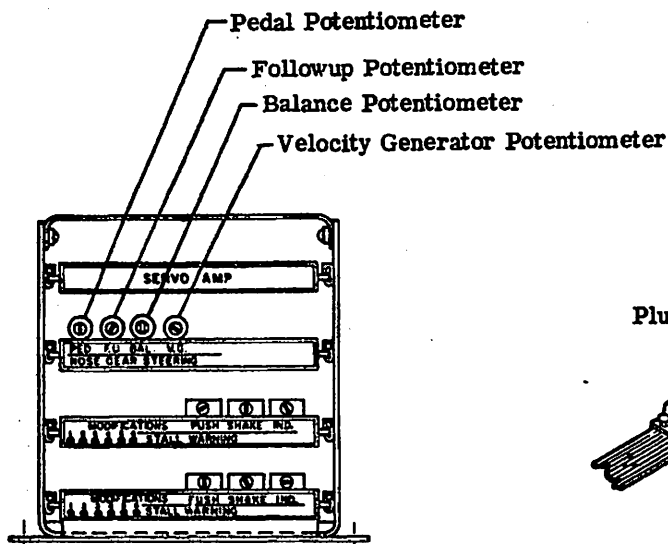
- (29) Check steering computer test feature as follows:
- (a) Fully depress rudder pedals right, depress Master Switch, and depress TEST switch on computer. Nose wheel will return to 8 (± 2) degrees deflection.
 - (b) Fully depress rudder pedals left, depress Master Switch, and depress TEST switch on computer. Nose wheel will return to 8 (± 2) degrees deflection.

NOTE: Depressing computer TEST switch performs an internal computer circuit test, not a nose wheel steering system test.

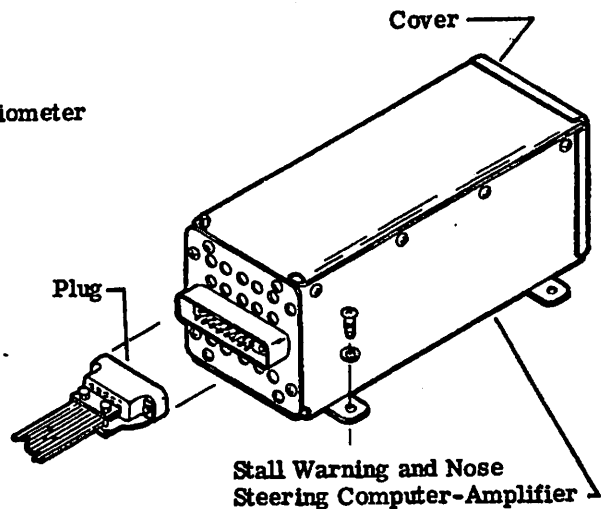
- (30) Center rudder and set squat switches in flight mode. Check that nose wheel remains centered.
- (31) Remove aircraft from nose wheel protractor.
- (32) Install wheel speed transducer drive caps.
- (33) Secure steering computer potentiometer protective cover.
- (34) Install floorboard nose compartment access doors.
- (35) Set Battery and Primary Inverter Switches to OFF.

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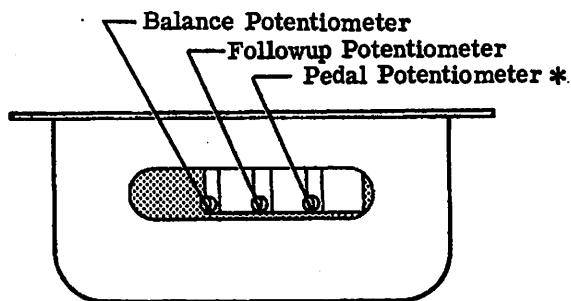
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VIEW OF END OPPOSITE
CONNECTORS WITH COVER REMOVED



HI-LOW COMPUTER-AMPLIFIER
AIRCRAFT 25-061 AND 25-070 THRU 25-103



VIEW OF END OPPOSITE
CONNECTORS WITH
COVER REMOVED

* On some later model Nose Steering Com-
puters this potentiometer is not installed.

J227 (Transducer
Input)

J224

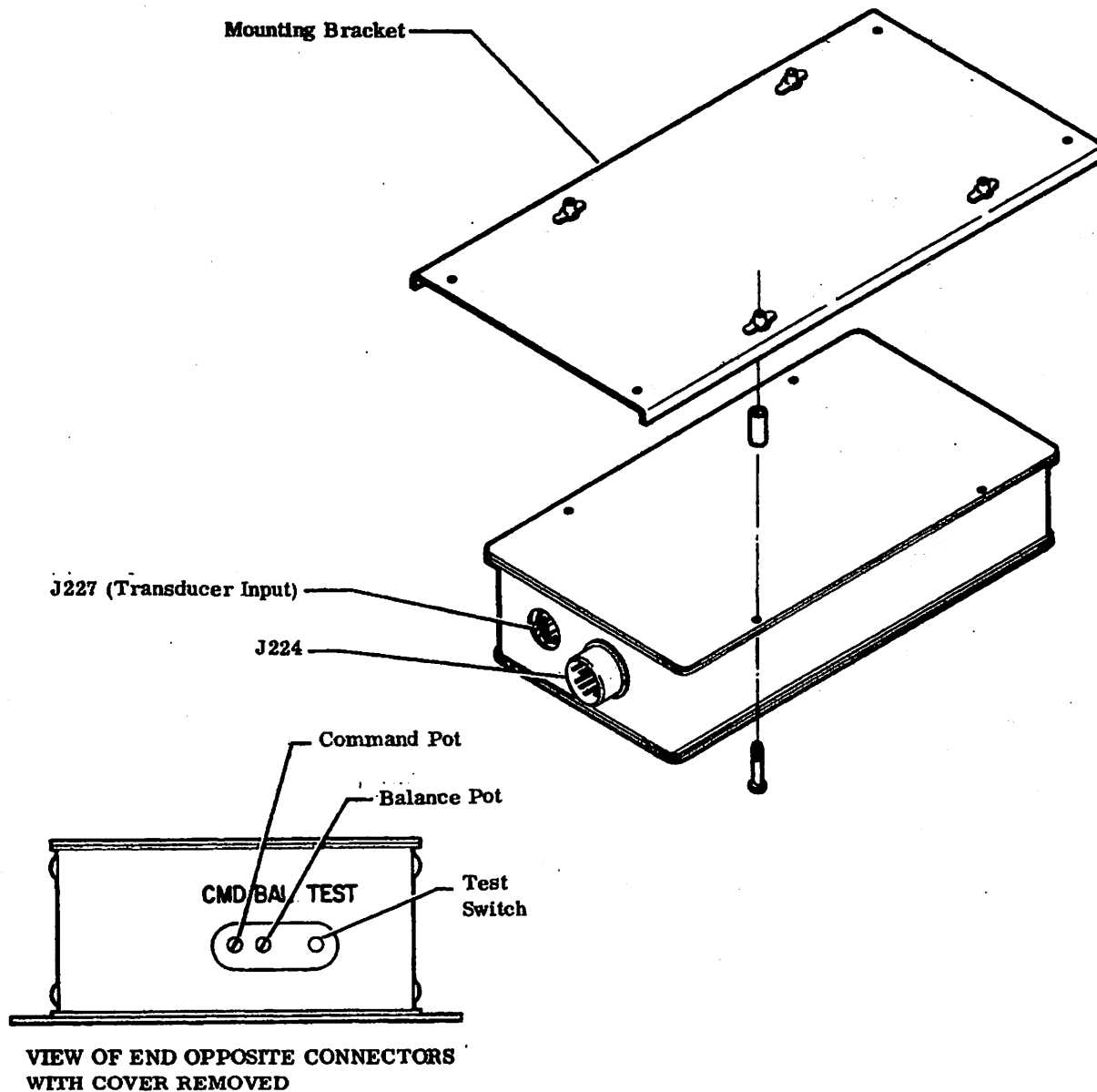
Steering Computer-
Amplifier Assembly

LEAR AVIA COMPUTER-AMPLIFIER
AIRCRAFT 25-104 THRU 25-229

Nose Steering Computer-Amplifier
Figure 201 (Sheet 1 of 2)

EFFECTIVITY: NOTED

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**J. E. T. COMPUTER-AMPLIFIER
AIRCRAFT 25-230 AND SUBSEQUENT**

**Nose Steering Computer-Amplifier
Figure 201 (Sheet 2 of 2)**

10-26A
10-27A

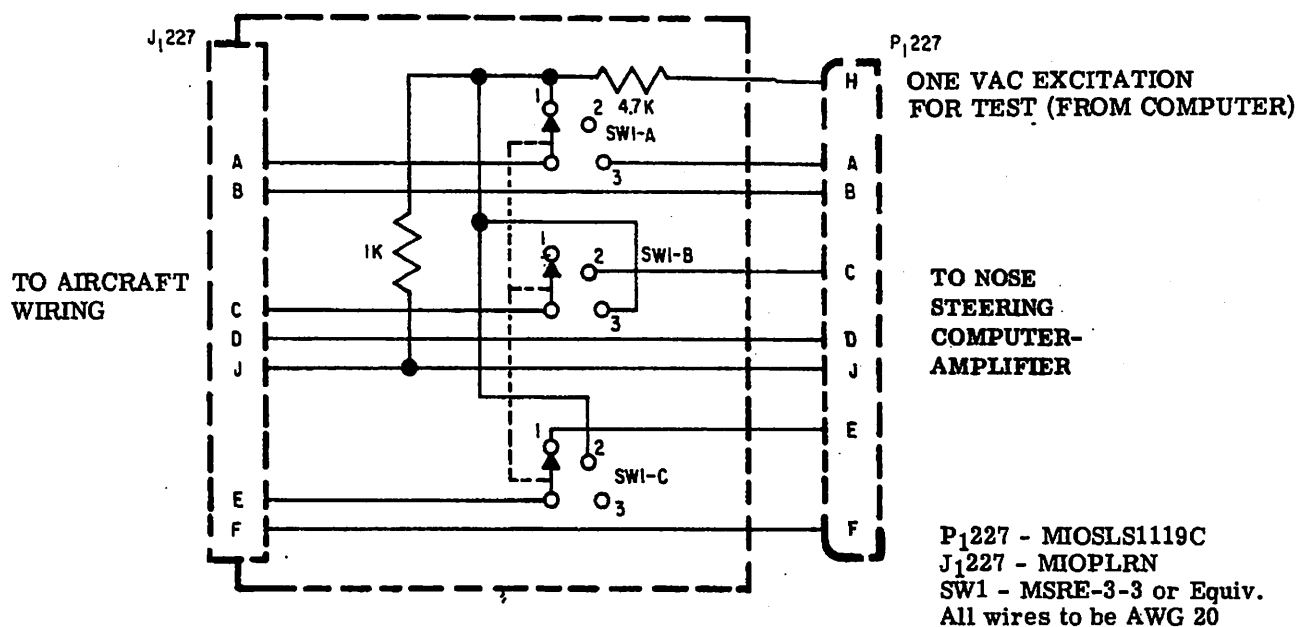
EFFECTIVITY: NOTED

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AIRCRAFT 25-104 THRU 25-229

Nose Steering Computer Test Set
 Figure 202

EFFECTIVITY: NOTED

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STEERING FOLLOWUP - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Steering Followup (See figure 201)

- (1) Place aircraft on jacks and remove nose gear shock strut. (Refer to 32-21-00.)

WARNING: BEFORE PERFORMING ANY DISASSEMBLY OF PARTS ON THE STRUT, ASSURE THAT ALL AIR PRESSURE HAS BEEN RELEASED THROUGH THE AIR VALVE. FAILURE TO DO SO CAN CAUSE BODILY INJURY. DO NOT RELEASE AIR BY LOOSENING OR REMOVING AIR VALVE.

- (2) Place nose gear strut in a stand high enough to allow wheel clearance when strut is fully extended.
- (3) Assure that nose gear shock strut is fully extended and in mechanical detent position. If nose gear wheel cannot physically be turned, it is in mechanical detent.
- (4) Remove safety wire from lid attachment screws.
- (5) Remove screws and washers from lid.

CAUTION: USE EXTREME CARE TO PREVENT DAMAGE TO LID OR CASE WHEN REMOVING LID FROM CASE. BOTH LID AND CASE HAVE MACHINED SURFACES AND ANY DAMAGE MAY CAUSE AN OIL LEAK.

- (6) Remove lid from case.
- (7) Loosen setscrews and remove followup gear from followup shaft.
- (8) Remove followup attachment screws, ring, and followup from lid.

B. Install Steering Followup (See figure 201)

- (1) Install followup and ring on lid and secure with screws.
- (2) Apply Loctite retaining compound to followup gear shaft hole and position followup gear on followup shaft approximately 1/16 inch from bottom surface of lid.
- (3) Secure followup gear to followup shaft with setscrews.

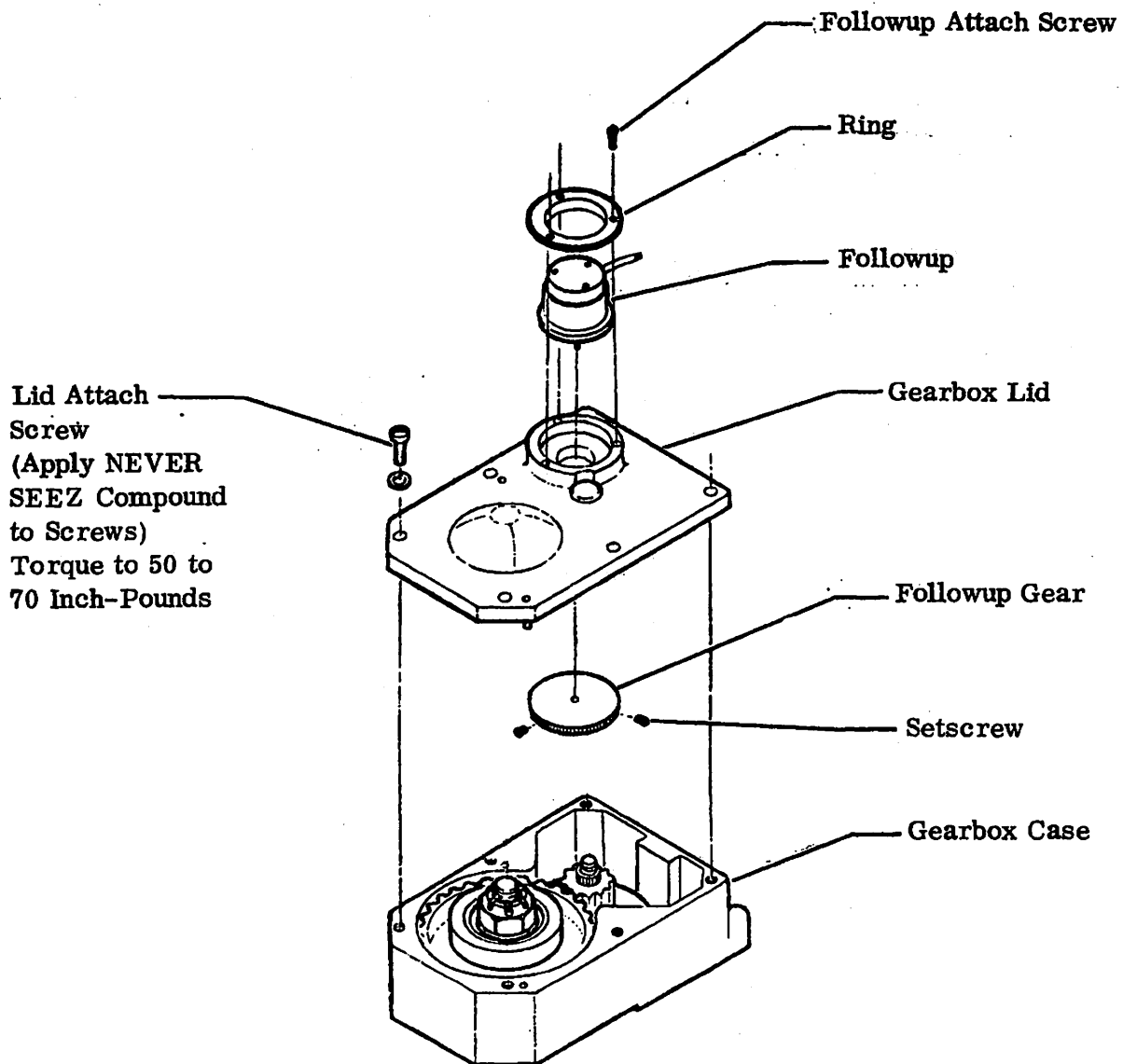
CAUTION: WHEN INSTALLING LID, THE GEARS MUST MESH PERFECTLY OR GEAR DAMAGE WILL OCCUR WHEN LID IS IN PLACE.

- (4) Insert a small metal strip between lid and case to move followup gear slightly until it meshes with gear cluster.
- (5) Install lid and secure with washers and lid attachment screws.
- (6) Adjust nose wheel steering system. (Refer to 32-50-00.)

EFFECTIVITY: ALL

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Nose Steering Followup Installation
Figure 201

6-11C

EFFECTIVITY: ALL

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STEERING ACTUATOR ASSEMBLY - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: The steering actuator may be removed without disassembly or removal of the steering actuator gearbox from the strut. The nose gear will have to be removed from the aircraft to gain access to the steering actuator attachment bolts.

A. Remove Steering Actuator Gearbox (See figure 201.)

- (1) Remove nose gear from aircraft. (Refer to 32-21-00.)
- (2) Remove steering servo attachment bolts and remove steering servo from gearbox case.

WARNING: BEFORE PERFORMING GEARBOX REMOVAL, ASSURE THAT ALL AIR PRESSURE HAS BEEN RELEASED FROM STRUT. FAILURE TO DO SO CAN CAUSE BODILY INJURY. DO NOT RELEASE AIR BY LOOSENING OR REMOVING AIR VALVE.

- (3) Remove safety wire from lid attachment screws.
- (4) Remove screws and washers from lid.

CAUTION: USE EXTREME CARE NOT TO DAMAGE THE LID AND CASE MATING SURFACES AS THEY ARE MACHINE FINISHED AND ANY DAMAGE WILL RESULT IN AN OIL LEAK.

- (5) Carefully remove lid from case.
- (6) Remove cotter pin, strut attachment nut, washer, and ball bearing from strut.
- (7) Fabricate a main drive gear removal tool as shown in figure 201.
- (8) Screw main drive gear removal tool fully onto strut shaft. Strike tool with a series of sharp blows until main drive gear is loosened on strut shaft.
- (9) Unscrew main drive gear removal tool from strut shaft, assuring that shaft does not fall into strut.
- (10) Remove main drive gear and cluster gear from gearbox.

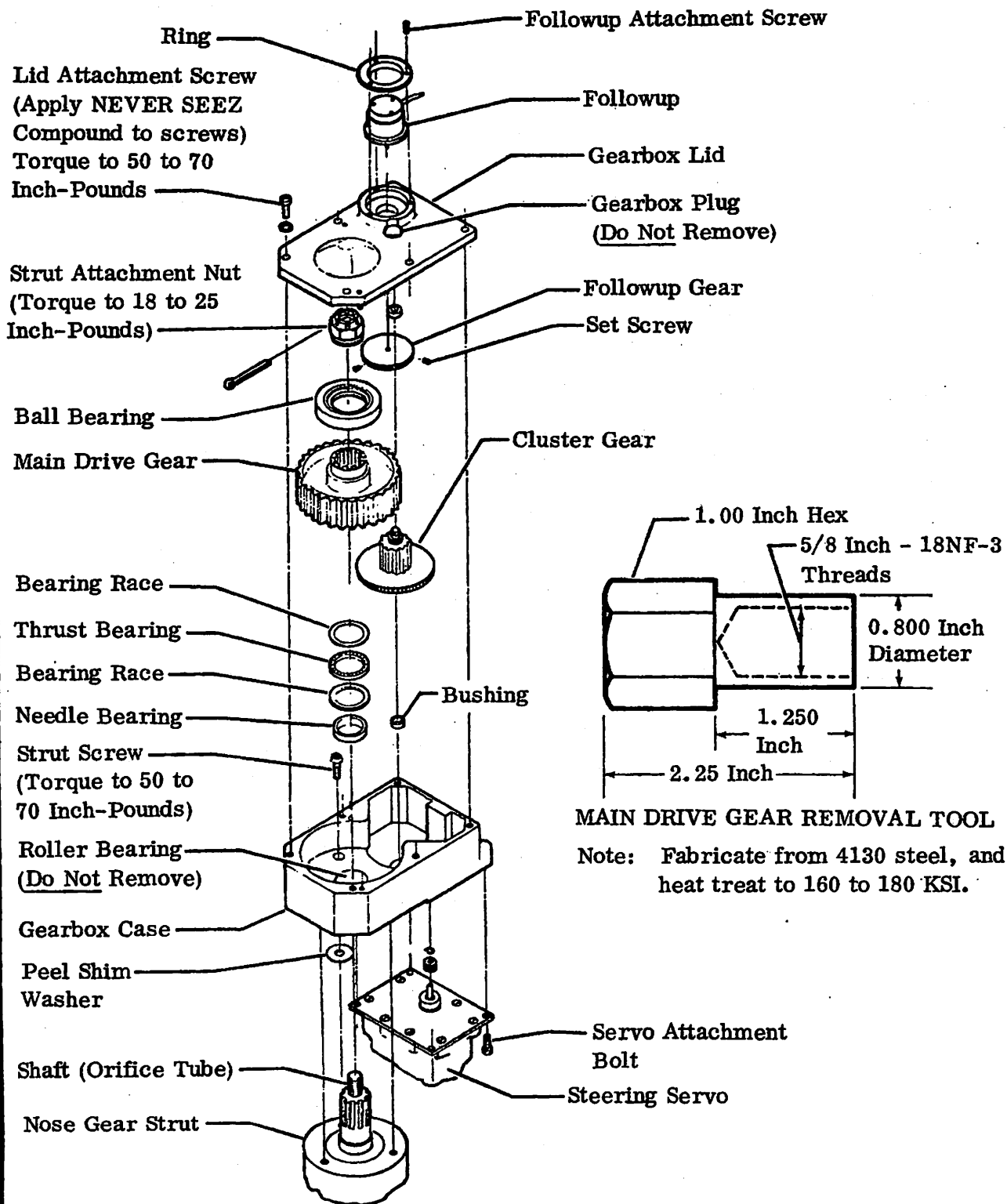
NOTE: • Removal of the roller bearing pressed into case is not required.

• Removal of the bushing pressed into the lower portion of lid is not required.

EFFECTIVITY: ALL

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Nose Steering Actuator Installation
Figure 201

6-11C

EFFECTIVITY: ALL

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- (11) Remove the two bearing races and thrust bearing from gearbox case and inspect for damage caused by driving shaft out of main drive gear.
- (12) Remove strut screws and case from nose gear strut.

NOTE: When removing the case, spacers will be released from in between the strut and case. Retain these spacers for re-installation.

- (13) Remove setscrews and followup gear from followup shaft.
- (14) Remove followup attachment screws, ring, and followup from lid.
- B. Install Steering Actuator Gearbox (See figure 201.)
 - (1) Apply Grade B Loctite to threads of all screws immediately prior to installation.
 - (2) Attach steering servo to gear box case using steering servo attachment screws. Tighten screw and safety wire.
 - (3) Apply MIL-G-23827A or MIL-G-81322 grease to all gears as required prior to installation.
 - (4) Install followup on actuator lid using ring and followup attachment screws.
 - (5) Apply Loctite retaining compound to followup gear shaft hole and position followup gear on followup shaft approximately 1/16 inch from bottom surface of lid. Secure with setscrews.
 - (6) Position case on strut and measure gap between case and strut. If old washers are unusable, fabricate peel shim washers to same thickness of gap and insert between case and strut and align with screw holes. Secure case to strut and torque screws 50 to 70 inch-pounds.
 - (7) Install bearing races and needle bearing on strut.
 - (8) Install cluster gear and bushing.
 - (9) Apply Loctite Retaining Compound No. 680 to main drive gear internal spline and shaft external spline according to manufacturers instructions and install main drive gear on shaft.

NOTE: Allow Loctite to cure for a minimum of 3 hours before proceeding with assembly of nose gear steering actuator.

- (10) Install ball bearing, washer, and strut attachment nut, torque nut to 18 to 25 inch-pounds. Install cotter pin.

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CAUTION: WHEN REPLACING LID ON CASE, IT IS EXTREMELY IMPORTANT THAT GEARS MESH PERFECTLY OR DAMAGE COULD OCCUR WHEN LID IS IN PLACE.

- (11) Insert a small metal strip between lid and case. Move followup gear slightly until it meshes with gear cluster.
- (12) Secure lid with washers and screws. Torque screws 50 to 70 inch-pounds and safety wire.

NOTE: Apply NEVER SEEZ compound to gearbox lid screws prior to installation.

- (13) Perform nose wheel steering system Adjustment/Test. (Refer to 32-50-00.)

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RUDDER PEDAL FOLLOWUP - MAINTENANCE PRACTICES

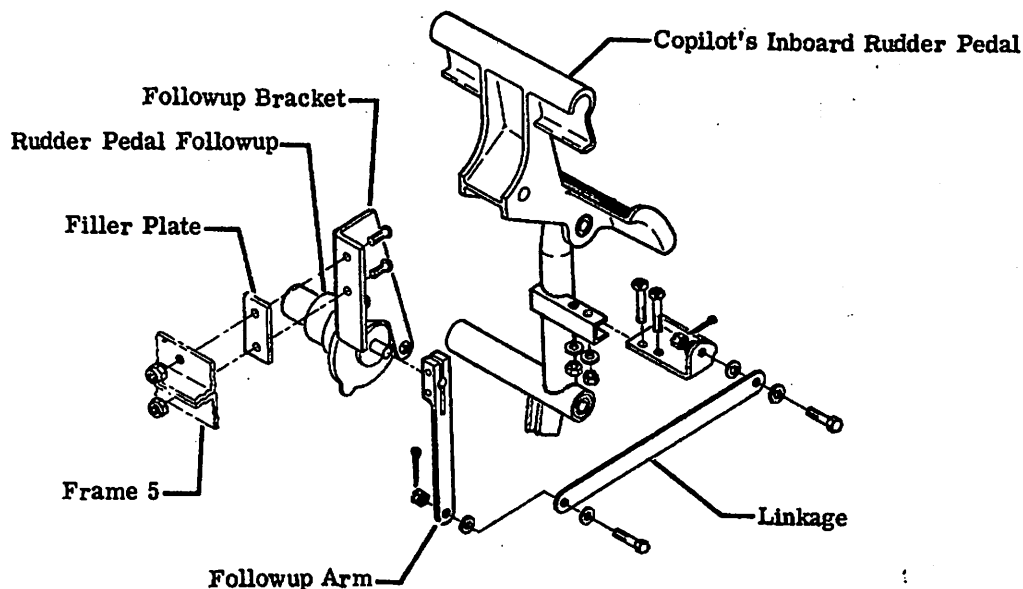
1. Removal/Installation

A. Remove Rudder Pedal Followup (See figure 201)

- (1) Set Battery Switches to OFF.
- (2) Remove floorboards immediately aft of rudder pedals. Position rudder pedals to neutral and insert rigging pin through rigging pin hole in rudder bellcrank. This will maintain rudder pedals in neutral.
- (3) Locate rudder pedal followup on frame 5 and forward of copilot's inboard rudder pedal. Disconnect followup arm from followup shaft.
- (4) Disconnect followup electrical connector.
- (5) Remove attaching parts and followup from aircraft.

B. Install Rudder Pedal Followup (See figure 201)

- (1) Position followup on frame 5 and secure with attaching parts.
- (2) Connect followup electrical connector.
- (3) If red dots are present on followup shaft and followup housing, assure that they are aligned. Connect followup arm to followup shaft. Do not secure followup arm to shaft until after adjustment.
- (4) Adjust computer-amplifier and followup per nose wheel steering system Adjustment/Test. Refer to 32-50-00.



Rudder Pedal Followup Installation
Figure 201

EFFECTIVITY: ALL

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POSITION AND WARNING - DESCRIPTION AND OPERATION

1. Description

- A. The position and warning system provides an indication of landing gear position and warns of a malfunction during normal landing gear extension and retraction.
- B. A malfunction of any of the position and warning system electrical components will not affect ultimate landing gear operation as the two systems are on different circuits.
- C. Integral switches in the nose gear actuator and both main gear actuators provide a down and locked position indication. Uplatch switches located on the nose gear uplatch mechanism and on both main gear doors provide an up and locked position indication.
- D. Whenever the landing gear is neither up and locked nor down and locked, the red UNSAFE lights will be illuminated.
- E. An aural warning horn sounds when the gear is retracted and the thrust levers are retarded below 70% or the flaps are extended more than 25°. Refer to Chapter 76 for gear aural warning switch adjustment.
- F. Each main gear has a gear down and locked switch installed on its actuator. Each main gear down and locked switch energizes its main gear green DN light. The left and right switches together energize the main gear door control valve to close the inboard gear doors.
- G. A main gear door up switch is installed adjacent to each door uplatch hook and energizes the main gear red UNSAFE light while the inboard door is not up and locked.
- H. The nose gear down and locked switch is installed on the nose gear actuator and energizes the nose gear green DN light.
- I. The nose gear up and locked switch is installed adjacent to the uplatch mechanism and energizes the nose gear red UNSAFE light while the nose gear is not down and locked or up and locked.

2. Operation

- A. When the landing gear selector switch is set to DN, the uplatch mechanisms disengage and the red UNSAFE lights illuminate. As each gear reaches down and locked position, its corresponding green LOCKED DN light will illuminate. If a gear fails to lock in the extended position, its corresponding red UNSAFE light will remain illuminated.
- B. When the landing gear selector switch is set to UP, the internal down-lock in the actuators will disengage and the red UNSAFE lights will illuminate. As each gear reaches the up and locked position, its corresponding UNSAFE light will go out. If a gear fails to lock in the up position, its corresponding UNSAFE light will remain illuminated.

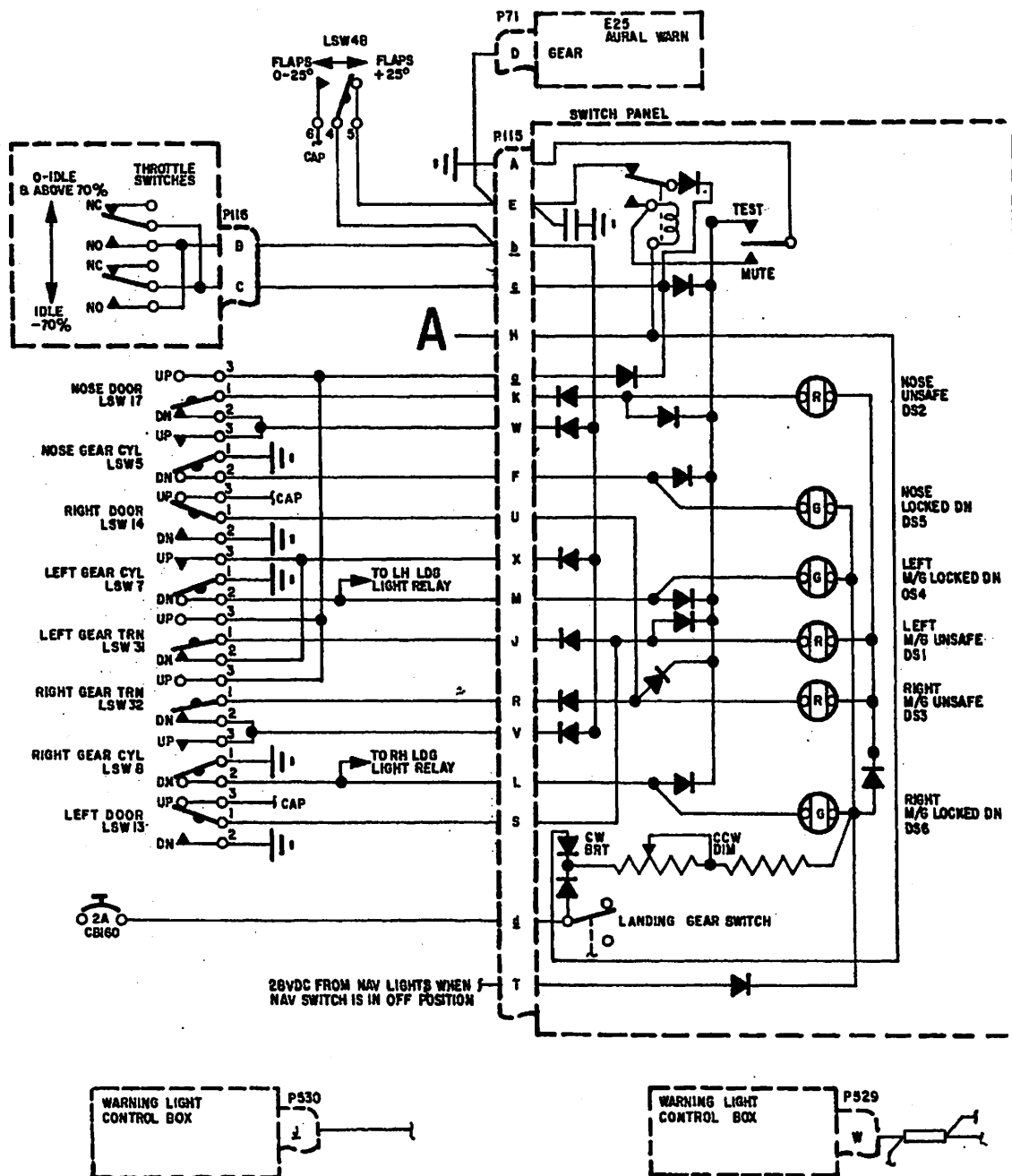
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- I**
- C. When the thrust levers are retarded below 67.5% (± 2.5) rpm with the gear retracted, an aural warning horn will sound until the gear is down and locked. The horn may be silenced by using the MUTE/TEST Switch. When the flaps are extended beyond 25°, the aural warning horn cannot be silenced using the MUTE/TEST Switch. The MUTE/TEST Switch can also be used to test the gear indication lights.

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Aircraft 25-081 and Subsequent

Aircraft 25-061, 25-070 thru 25-080

Detail A

Landing Gear Warning and Monitor Schematic
Figure 1 (Sheet 1 of 3)

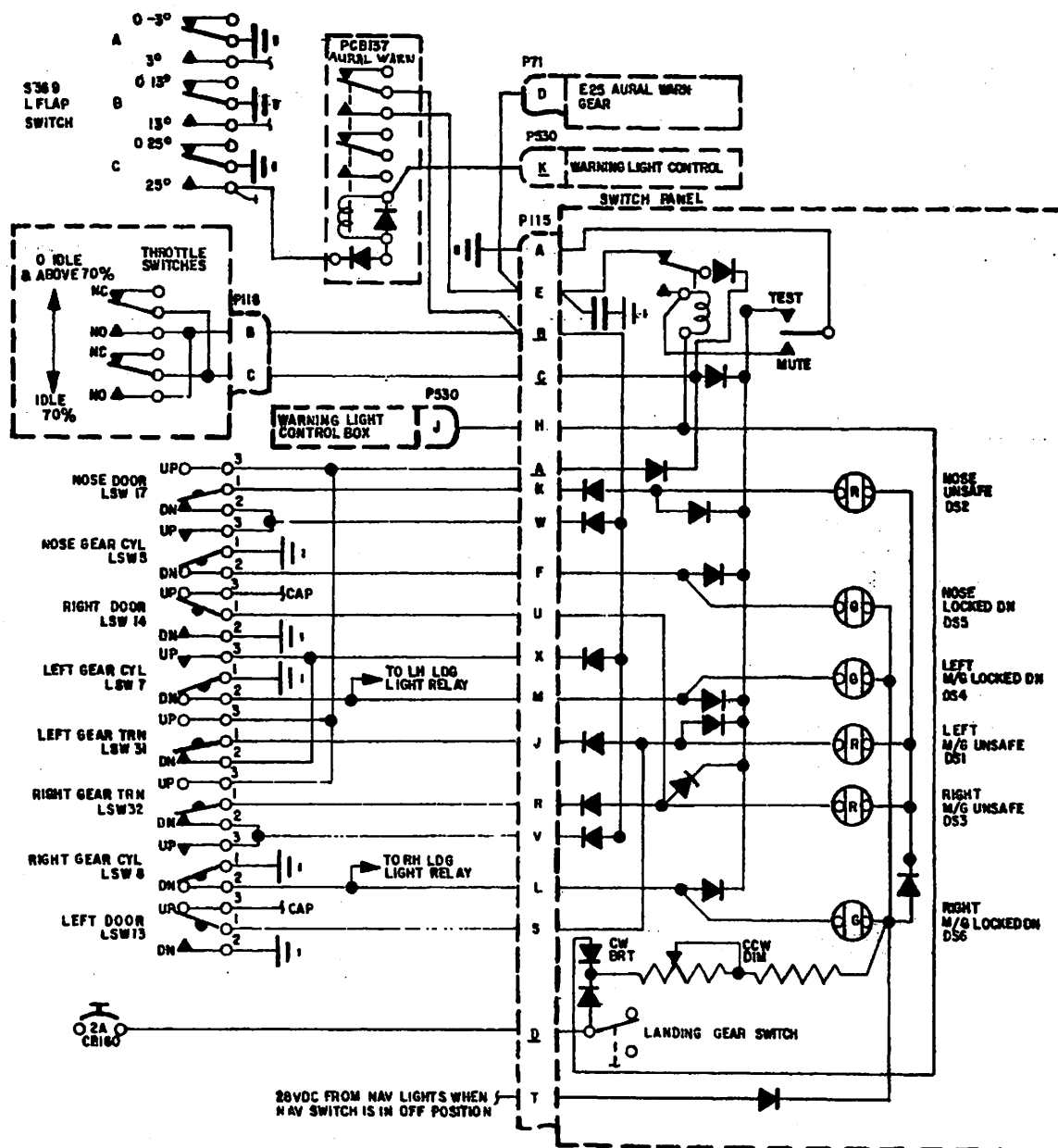
EFFECTIVITY: Aircraft 25-061, 25-070 thru 25-205
Not Equipped with Reduced Approach Speed System

32-60-00

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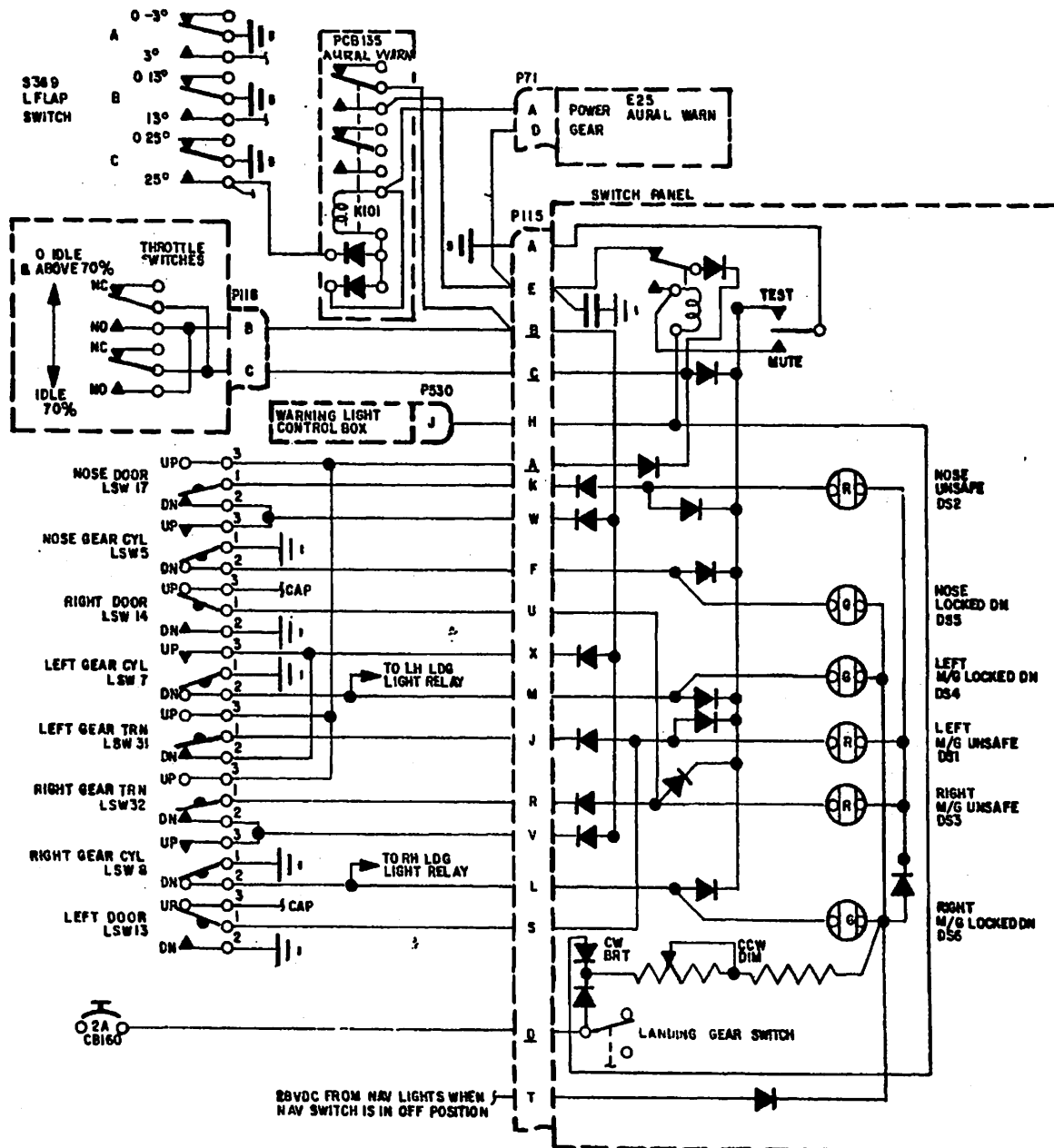


Landing Gear Warning and Monitor Schematic
Figure 1 (Sheet 2 of 3)

EFFECTIVITY: Aircraft 25-061, 25-070 thru 25-205
 Equipped with Reduced Approach Speed System

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Landing Gear Warning and Monitor Schematic
Figure 1 (Sheet 3 of 3)

EFFECTIVITY: Aircraft 25-206 and Subsequent

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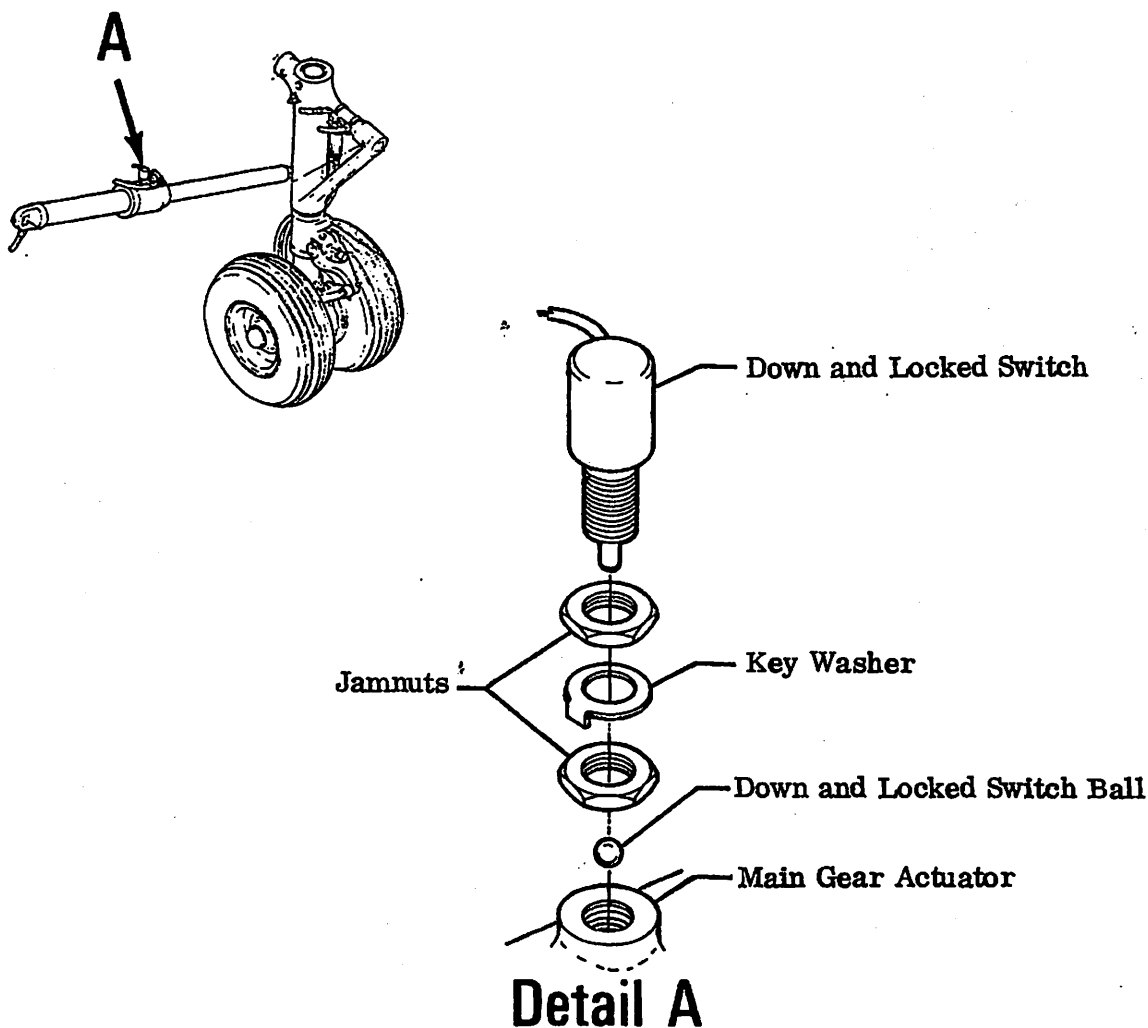
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MAIN GEAR DOWN AND LOCKED SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Main Gear Down and Locked Switch (See figure 201)

- (1) Place aircraft on jacks.
- (2) Attach an ohmmeter to switch wires No. 1 and 3 at splices.
- (3) Loosen jamnuts and screw switch into actuator until ohmmeter actuation occurs. Then back off a minimum of 1/2 to a maximum of 1-1/2 revolutions until wiring is routed along actuator.
- (4) Secure jamnuts and safety wire.
- (5) Cycle gear and check for proper operation and indication.
- (6) Remove aircraft from jacks.



Main Gear Down and Locked Switch Installation
Figure 201

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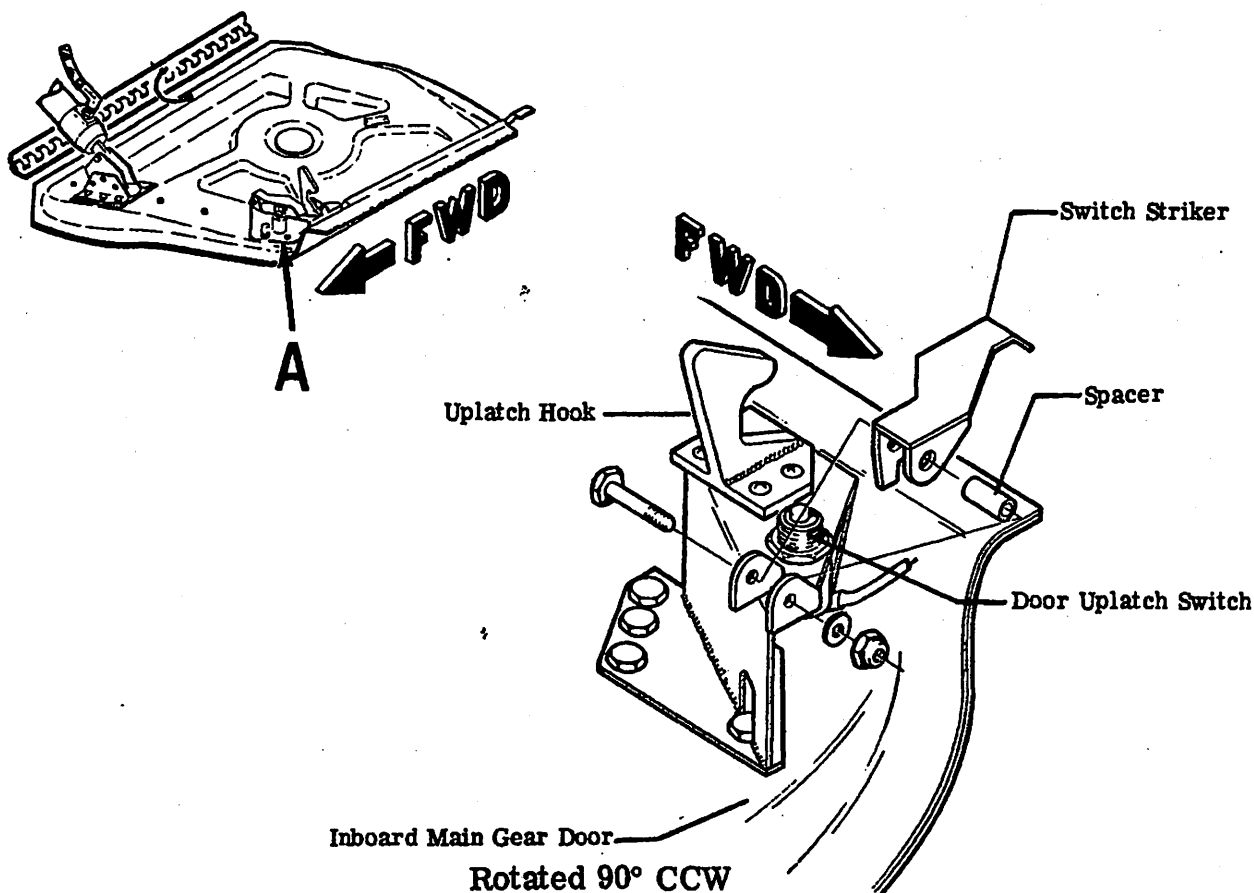
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MAIN GEAR DOOR UPLATCH SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Main Gear Door Uplatch Switch (See figure 201)

- (1) Place aircraft on jacks and pressurize hydraulic system.
- (2) Disconnect outboard gear door and fully retract gear.
- (3) Release hydraulic pressure so that uplatch hook hangs on roller.
- (4) Attach an ohmmeter to No. 1 and 3 wires of uplatch switch.
- (5) Loosen jamnuts and adjust switch until ohmmeter actuation occurs.
- (6) Adjust switch one revolution beyond point where ohmmeter actuation occurs. Secure jamnuts and safety wire.
- (7) Extend landing gear and connect outboard door linkage.
- (8) Remove aircraft from jacks.



Detail A

Main Gear Door Uplatch Switch Installation
Figure 201

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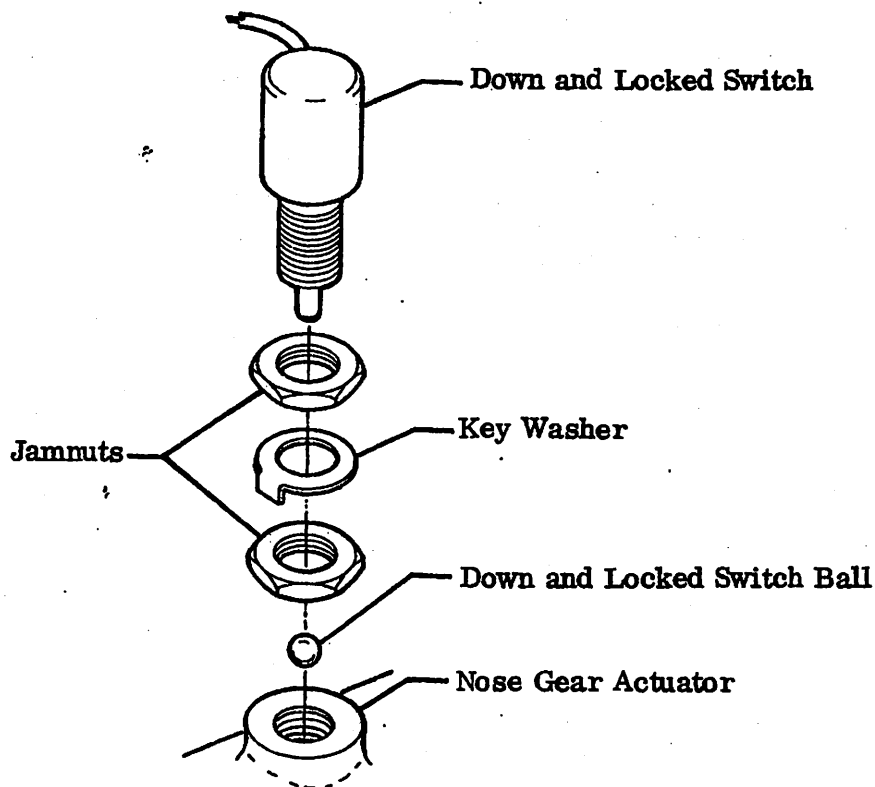
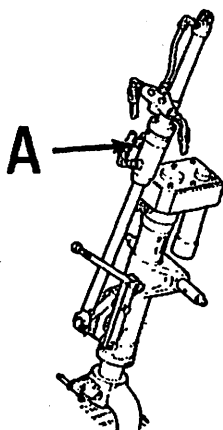
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NOSE GEAR DOWN AND LOCKED SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Nose Gear Down and Locked Switch (See figure 201)

- (1) Remove nose compartment access doors and gain access to nose gear down and locked switch.
- (2) Attach an ohmmeter to No. 1 and 3 wires of switch.
- (3) Loosen jamnut and adjust switch until ohmmeter actuation occurs.
- (4) Back off switch one revolution, secure jamnut and safety wire.



Detail A

Nose Gear Down and Locked Switch Installation
Figure 201

EFFECTIVITY: ALL

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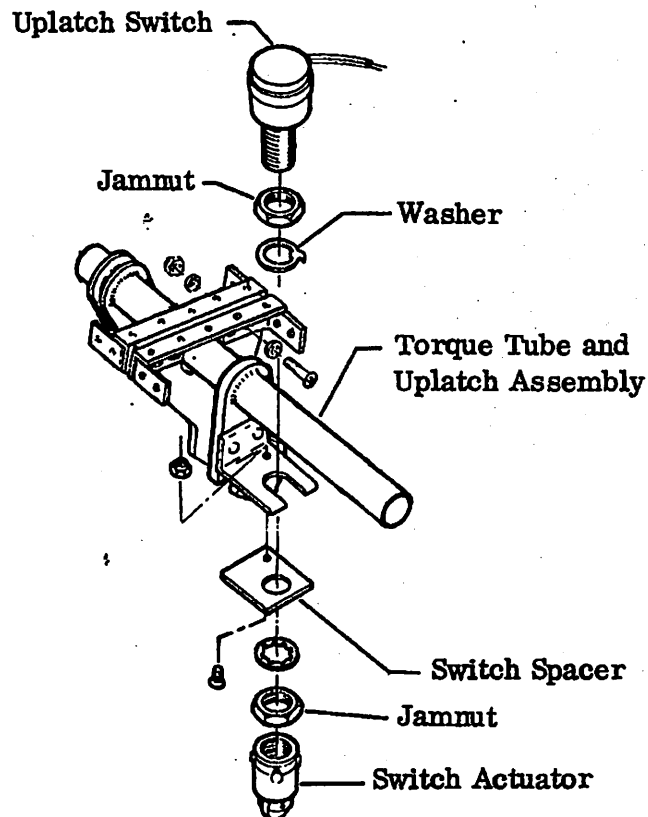
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NOSE GEAR UPLATCH SWITCH - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Adjust Nose Gear Uplatch Switch (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Disconnect nose gear door linkage.
- (3) Retract nose gear.
- (4) Adjust nose gear uplatch mechanism if required. (Refer to 32-32-03.)
- (5) If red UNSAFE lights illuminate, extend gear.
- (6) Adjust switch and jammuts in 1/2 turn increments, retracting gear each time, until red UNSAFE lights do not illuminate.
- (7) Adjust switch and jammuts one more full turn to further depress switch plunger.
- (8) Secure and safety wire jammuts using double-twist method.
- (9) Connect nose gear door linkage.



Nose Gear Uplatch Switch Installation
Figure 201

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- (8) If threads are not damaged and end play does not exceed 0.004 inch, apply Loctite Retaining Compound No. 680 evenly to threads and install rod end. If looseness exceeds 0.004 inch, replace defective part.
- (9) Buff out minor scratches in piston, cylinder assembly, inner piston, downlock ball housing, plunger, or housing assembly. Do not create low spots by excessive buffing or polishing. Use oil stone to remove scratches and polish with crocus cloth. Check tolerances after polishing.
- (10) Replace any parts which are damaged, cracked, excessively worn, scored, corroded, or do not meet inspection requirements or tolerances.

C. Assemble Main Gear Actuator (See figure 202.)

- (1) Assure that all items are thoroughly clean and immerse new O-rings, D-rings, backup rings, and felt wiper in MIL-H-5606 hydraulic fluid immediately prior to installation.
- (2) Install new lubricator and/or bearing if removed during disassembly. If new bearing is to be installed, install as follows:

NOTE: Maximum tolerance between new bearing and cylinder assembly is 0.004. If dimension is greater, replace cylinder assembly.

- (a) Clean bearing and cylinder assembly using trichlorethane solvent. Bearing and bearing housing must be clean of all traces of oil, dust, condensation, or other contaminants.
 - (b) Apply Locquic primer (Grade T) to mating surfaces of bearing outer race and to actuator bearing housing outer race and to actuator bearing housing and wipe off while still wet. Reapply primer to bearing and bearing housing and allow to dry until primer odor is no longer present.
 - (c) Apply an even coat of Loctite compound No. 680 evenly to mating surfaces of bearing and cylinder assembly. Excessive compound may be removed using a rag moistened in trichlorethane.
 - (d) Press bearing into cylinder assembly and keep aligned until Loctite cures. Cure time is 20 to 30 minutes. Do not stake bearing.
- (3) Install D-ring and felt wiper inside actuator housing assembly.
 - (4) Install O-ring inside plunger and O-ring on outside of plunger.
 - (5) Install O-ring and backup ring on piston and inner piston.
 - (6) Assemble downlock ball housing, shoulder first, on piston.
 - (7) Install plunger in actuator housing, assuring that plunger is bottomed.

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- (8) Assemble laminated shim on piston.
- (9) Install balls in downlock ball housing.
- (10) With actuator housing assembly held vertically, slide piston with laminated shim, balls, and downlock ball housing into actuator housing assembly and plunger.
- (11) With actuator housing assembly held securely, pull piston to extended position. Pulling piston to extended position will seat balls inside downlock ball housing into groove just below piston head and will hold piston in extended position.
- (12) Place spring inside housing assembly and bottom against shoulder on actuator plunger.
- (13) Screw inner piston into cylinder assembly.
- (14) Align and insert inner piston through housing assembly into piston and compress spring enough to screw cylinder into housing assembly and bottom together. With cylinder bottomed in housing assembly, extend and retract ports must be aligned on same plane, within 5°, and lock-screw slot in cylinder lined up with lockscrew hole in housing assembly.
- (15) Add to or remove laminations from shim as required to obtain firm seating of spacer shim and downlock ball housing inside of housing.
- (16) Install lockscrew in housing assembly, securing housing assembly and cylinder as a unit.
- (17) Install rod scraper on piston and secure in housing assembly with retainer ring.
- (18) Install rod end in piston and adjust so that dimension from center of rod end bolt hole to center of cylinder assembly bolt hole is 45.79 inches extended and locked (26.76 inches compressed). Secure rod end with attaching parts.
- (19) Refinish all exterior painted surfaces, as required, with one coat of wash primer, one coat of MIL-L-8585 zinc chromate primer, and one coat of MIL-L-7178 aluminized lacquer.
- (20) Lubricate main gear actuator piston rod end felt wiper with MIL-H-5606 hydraulic fluid. Lubrication is accomplished through lubrication hole on rod end of actuator.
- (21) Perform Adjustment/Test of main gear actuator.

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MAIN GEAR INBOARD DOOR UPLATCH - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Uplatch Actuator (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Lower tailcone access door and disconnect aircraft batteries.
- (3) Depressurize hydraulic system.
- (4) Disconnect hydraulic hose from uplatch actuator. Cap and plug exposed hydraulic fittings.
- (5) Remove cotter pins, nuts, washers and bolts securing actuator and remove actuator from aircraft.

B. Install Uplatch Actuator (See figure 201.)

- (1) Install actuator and secure with bolts, washers, nuts and cotter pins.
- (2) Connect hydraulic hose to uplatch actuator.
- (3) Adjust uplatch mechanism. Refer to Adjustment/Test.
- (4) Connect aircraft batteries and secure tailcone access door.
- (5) Remove aircraft from jacks.

2. Adjustment/Test

A. Adjust Uplatch Mechanism (See figure 202.)

- (1) Place aircraft on jacks and pressurize hydraulic system.
- (2) Disconnect outboard door linkage and fully retract gear.
- (3) Adjust uplatch cam bracket to obtain 0.10 (± 0.03) inch between uplatch roller and top of uplatch hook.
- (4) Adjust uplatch actuator rod end until roller contacts hook, then back rod end 1/2 turn to next terminal index and safety wire.
- (5) Release hydraulic pressure so that door hook hangs on uplatch roller.
- (6) Attach an ohmmeter to No. 1 and 3 wires of door up switch.
- (7) Adjust door up switch one revolution beyond point where ohmmeter actuation occurs.
- (8) Secure jamnuts and safety wire.
- (9) Extend landing gear and connect outboard door linkage.
- (10) Remove aircraft from jacks.

B. Functionally Test Uplatch Actuator

- (1) Connect actuator to a hydraulic source and pressure check at 2250 psi for 3 minutes. Leakage rate shall not exceed one drop in the 3-minute period.
- (2) While actuator is in the extended position, adjust for 9.34 (± 0.015) inch distance between centerlines of rod end and actuator bearing.
- (3) Remove hydraulic pressure and check retracted length of 8.44 (± 0.045) inches. Safety wire rod end lock in place.

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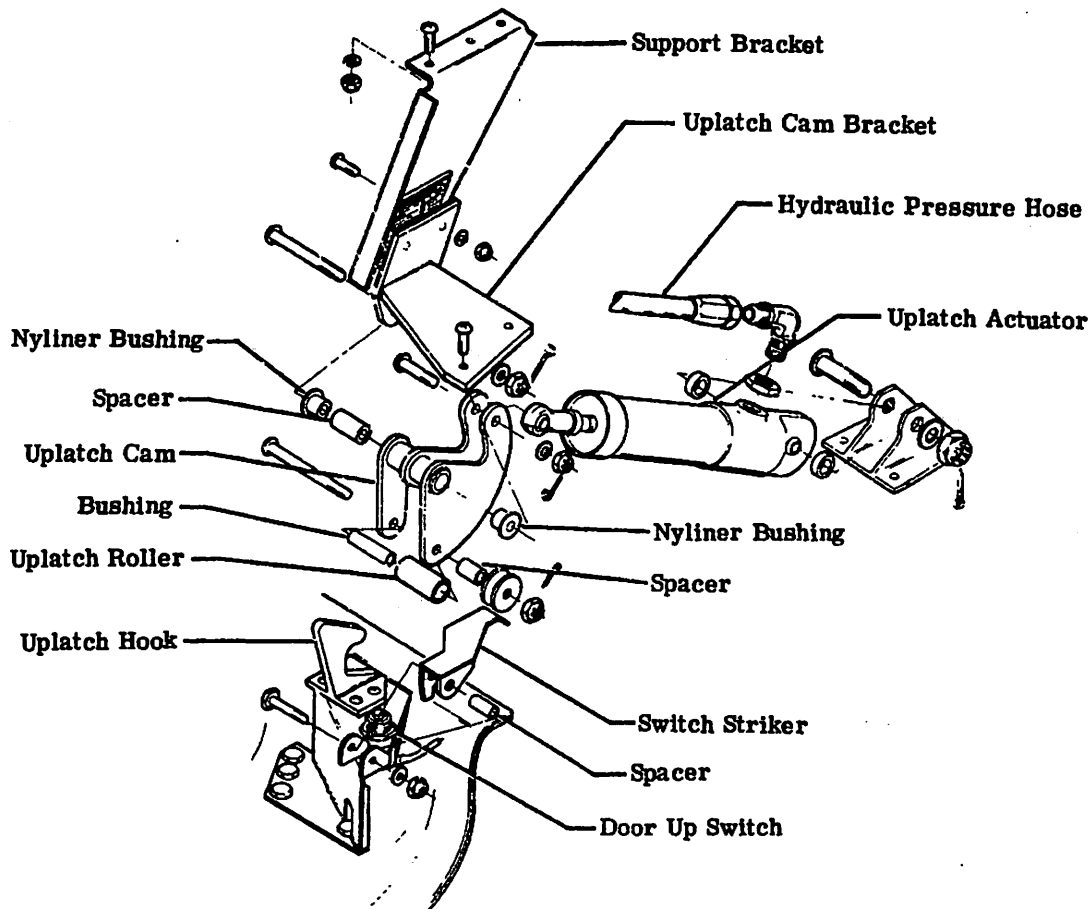
3. Approved Repairs

A. Disassemble Uplatch Actuator (See figure 203)

- (1) Cut safety wire and loosen locknut sufficiently to release rod end lock from piston rod. Remove rod end, rod end lock, and locknut from piston rod.

WARNING: USE EXTREME CARE WHEN REMOVING LOCK RING AND PLUG FROM BARREL. THE ACTUATOR IS SPRING LOADED TO 40 POUNDS IN THE RETRACTED POSITION AND COULD SPRING OUT OF CONTROL, CAUSING BODILY INJURY.

- (2) Press plug inboard to allow removal of lock ring. Remove lock ring and slowly release plug.
- (3) When spring tension is released, remove plug and spring from barrel.
- (4) Remove piston rod from barrel. Remove O-ring and backup rings from piston rod. Remove felt wiper from plug.
- (5) Clean all parts with Stoddard solvent or equivalent. Dry parts with clean, dry, compressed air.

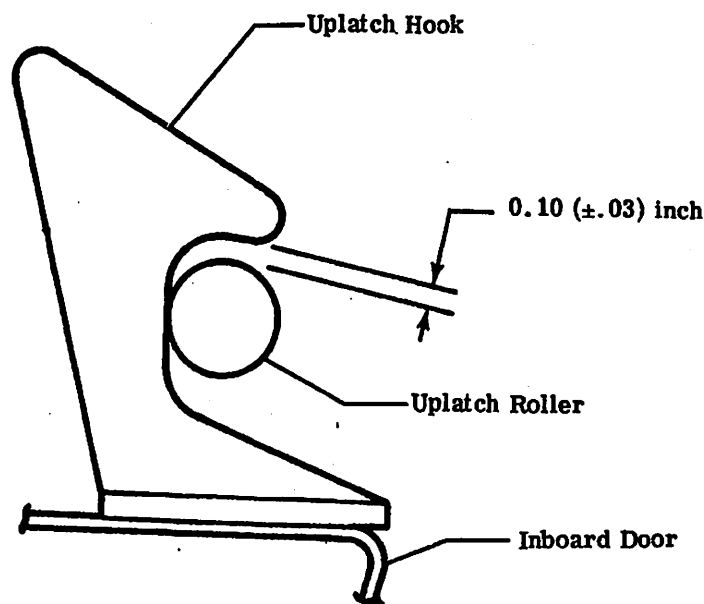


Main Gear Inboard Door Uplatch Installation
Figure 201

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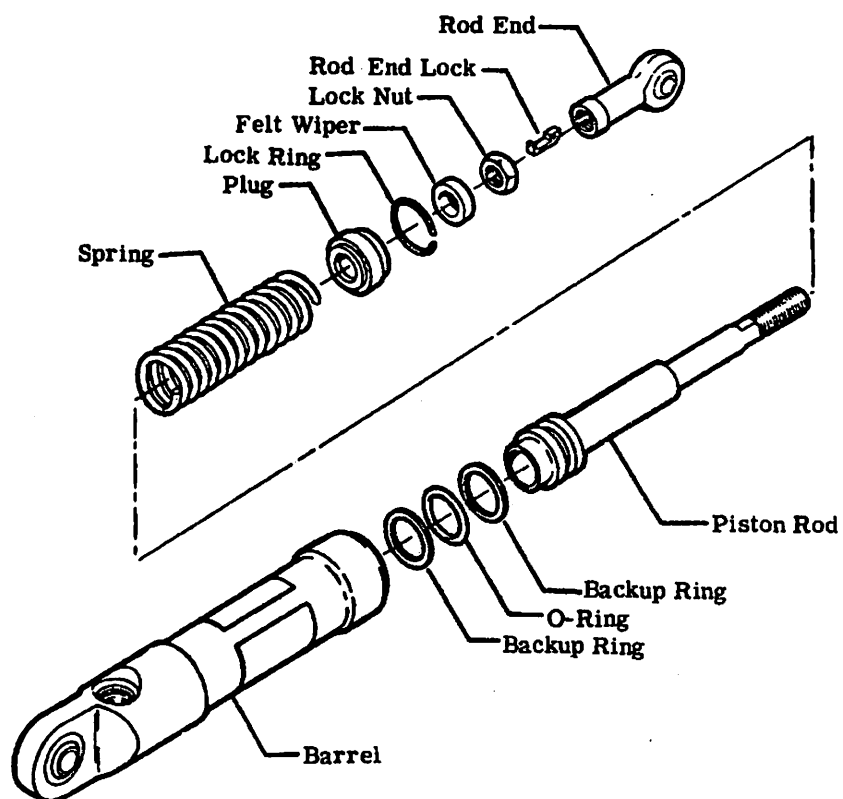
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- (6) Inspect all parts for nicks, scratches, or excessive wear. Replace defective parts.
 - (7) Buff out minor scratches using an oil stone and polish with crocus cloth.
- B. Assemble Uplatch Actuator (See figure 203.)
- (1) Use new O-rings, backup rings, and felt wiper at each assembly. Immerse these items in MIL-H-5606 hydraulic fluid immediately prior to installation.
 - (2) Assemble O-ring and backup rings on piston rod and insert felt wiper in plug.
 - (3) Position piston rod and spring in barrel. Install plug on piston rod and push in, compressing spring sufficiently to allow installation of lock ring.
 - (4) Install lock ring and slowly release plug until firmly seated against lock ring.
 - (5) Install rod end lock, locknut, and rod end on piston rod.
 - (6) Functionally test actuator. (Refer to Adjustment/Test.)



Main Gear Inboard Door Uplatch Adjustment
Figure 202

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Uplatch Actuator Assembly
Figure 203

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MAIN GEAR INBOARD DOOR ACTUATOR - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Main Gear Inboard Door Actuator (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Lower tailcone access door and disconnect aircraft batteries.
- (3) Depressurize hydraulic system.
- (4) Lower inboard gear door as follows:
 - (a) Station a man at inboard door and a man to operate brake pedals.
 - (b) While pumping brakes, pull inboard door to down position.
- (5) Remove door down switch from actuator.
- (6) Remove screws and universal bolt securing shuttle valve to actuator and cap exposed fittings.
- (7) Disconnect hydraulic hose from door actuator and cap exposed fittings.
- (8) Remove bolts securing each end of actuator and remove actuator from aircraft.

B. Install Main Gear Inboard Door Actuator (See figure 201.)

- (1) Position actuator on aircraft structure and secure with bolt.
- (2) Attach shuttle valve to actuator using universal bolt and screws.
- (3) Connect hydraulic hose to actuator.
- (4) Install door down switch and adjust switch. (Refer to 32-30-03.)
- (5) Attach actuator to inboard door and secure with attaching parts.
- (6) Cycle gear and check for proper operation.
- (7) Connect aircraft batteries, secure tailcone access door and remove aircraft from jacks.

2. Approved Repairs

A. Disassemble Main Gear Inboard Door Actuator (See figure 202.)

- (1) Cut safety wire and remove rod end lock, locknut, and rod end from piston rod.
- (2) Loosen spanner nut and unscrew cap from body. Remove piston rod from body.
- (3) Remove retainer ring, washer, scraper ring, felt wiper, spacer, and all O-rings and backup rings from cylinder body, piston head, and cap.
- (4) Unscrew piston head from piston rod. Remove O-rings and backup rings from piston rod. Retain spacer for reinstallation.
- (5) Remove plug from cap to gain access to actuating pin. Insert a small screwdriver through hole in cap and remove actuating pin from piston.
- (6) Insert screwdriver through hole into slot in sleeve. Remove sleeve, stem, and actuating ball from cap.

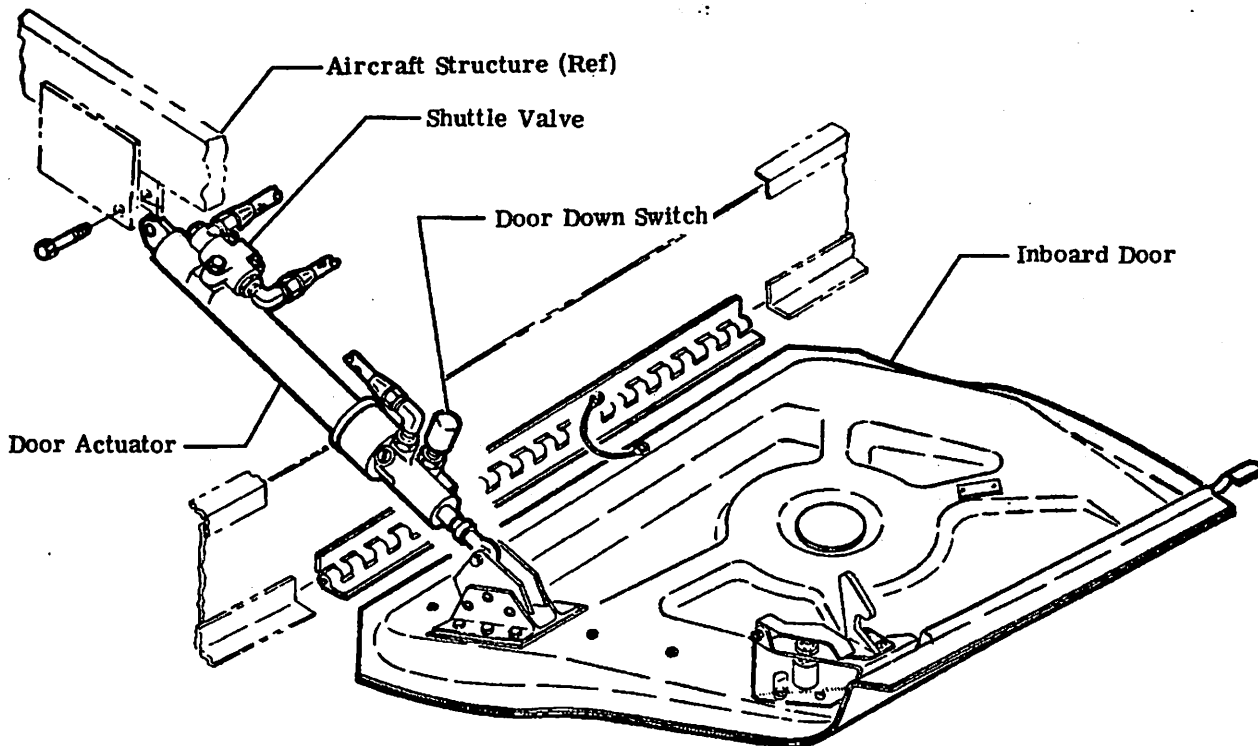
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- (7) Slowly remove plug from cap, releasing tension on spring. Remove spring and piston from cap. Remove O-rings and backup rings from piston.
- (8) Clean all parts in Stoddard solvent or equivalent and dry with clean, dry, compressed air.
- (9) Inspect all parts for nicks, scratches, or excessive wear. Replace defective parts.

B. Assemble Main Gear Inboard Door Actuator (See figure 202.)

- (1) Immerse all O-rings and backup rings in MIL-H-5606 hydraulic fluid prior to installation.
- (2) Install O-rings and backup rings on piston and insert piston in cap as far as possible. Install actuating pin in piston. Install O-ring on plug and install plug in cap.
- (3) Install O-ring on plug and install spring and plug in cap.
- (4) Insert actuating ball, stem, and sleeve in cap. Insert screwdriver into slot in sleeve and tighten.
- (5) Assemble O-ring, spacer, and backup rings on piston rod and screw piston head on piston rod.
- (6) Assemble O-ring and backup rings in body. Assemble O-ring and backup rings on piston head. Insert piston rod (with piston head attached) in body.

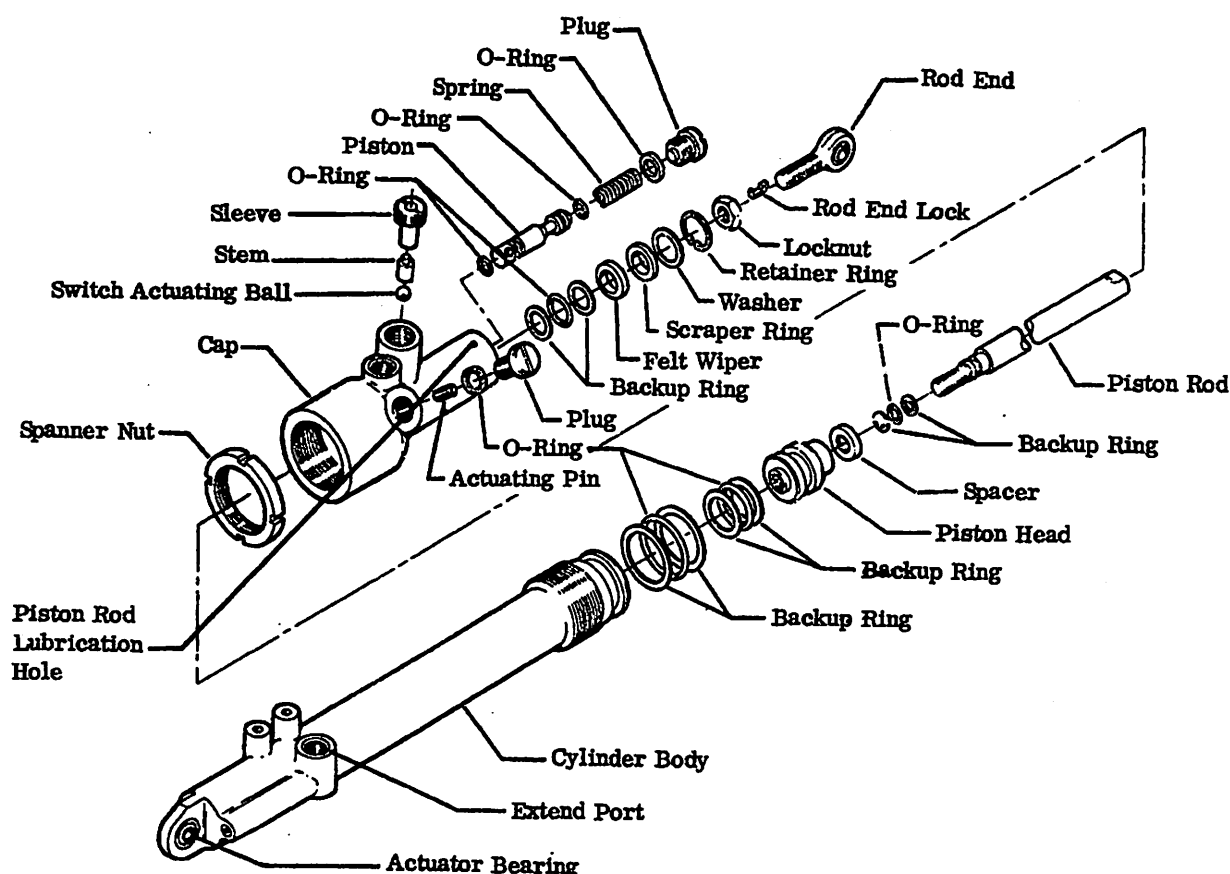


Main Gear Inboard Door Actuator Installation
Figure 201

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- (7) Assemble O-ring, backup rings, and felt wiper in cap. Slip cap over piston rod and screw cap on body as far as possible. Tighten spanner nut against cap and safety wire.
- (8) Assemble scraper ring, washer, and retainer ring in end of cap. Install rod end, locknuts, and rod end lock on piston rod.
- (9) With piston rod fully bottomed, adjust rod end for 17.55 inches between centerlines of actuator bearing and rod end bearing and safety wire.
- (10) Lubricate main inboard gear door actuator piston rod end felt wiper. Lubrication is accomplished through lubrication hole on rod end of actuator. Lubricate only with MIL-H-5606 hydraulic fluid.
- (11) Connect a hydraulic pressure source to extend and retract ports of actuator. Pressure check at 2250 psi for 3 minutes in both extend and retract positions. Leakage rate should not exceed one drop in 3-minute period.



Main Gear Inboard Door Actuator Assembly
Figure 202

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NOSE GEAR EXTENSION AND RETRACTION - DESCRIPTION AND OPERATION

1. Operation

- A.** The nose gear is retracted and extended by an actuator which is attached to the shock strut and to the fuselage structure. A spring load built into the actuator drives a downlock mechanism (a sleeve arrangement which entraps 10 steel balls that engage a notch in the piston rod) for a positive mechanical locking action when the nose gear is in the down position. An integral switch in the actuator transmits a gear down and locked indication (green LOCKED DN light) when the sleeve has extended to the locked position. To retract the gear, hydraulic pressure in excess of 250 pounds is required to retract the sleeve assembly against the drive spring and release the downlock mechanism to permit gear retraction. During an emergency, the actuator is extended pneumatically to the down position.
- B.** The priority valve is a 500 to 550 psi valve located in the nose gear extend line. During gear extension, the priority valve offers no restriction to fluid flow; but during gear retraction, the extend line functions as a return line and the priority valve stops fluid flow required for nose gear retraction. This directs full hydraulic pump flow to main gear retraction and avoids overloading the hydraulic pump. When system pressure exceeds 500 to 550 psi, the main gear will have been retracted and full pump flow is admitted through the nose gear retract system.

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PRIORITY VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Priority Valve (See figure 201.)

- (1) Depressurize hydraulic system.
- (2) Remove LH nose access door.
- (3) Disconnect and cap hydraulic lines at priority valve.

NOTE: The priority valve is located left of the nose wheel box on frame 5.

- (4) Remove clamp and priority valve from aircraft.

B. Install Priority Valve (See figure 201.)

- (1) Install priority valve and secure with clamp.
- (2) Connect hydraulic lines to priority valve.
- (3) Pressurize and bleed hydraulic system.
- (4) Place aircraft on jacks and cycle gear to check for proper operation.
- (5) Remove aircraft from jacks.

2. Approved Repairs

A. Disassemble Priority Valve (See figure 202.)

WARNING: PRIORITY VALVE IS SPRING LOADED; THEREFORE, CAREFULLY REMOVE END CAP TO RELEASE SPRING TENSION OR BODILY INJURY MAY OCCUR.

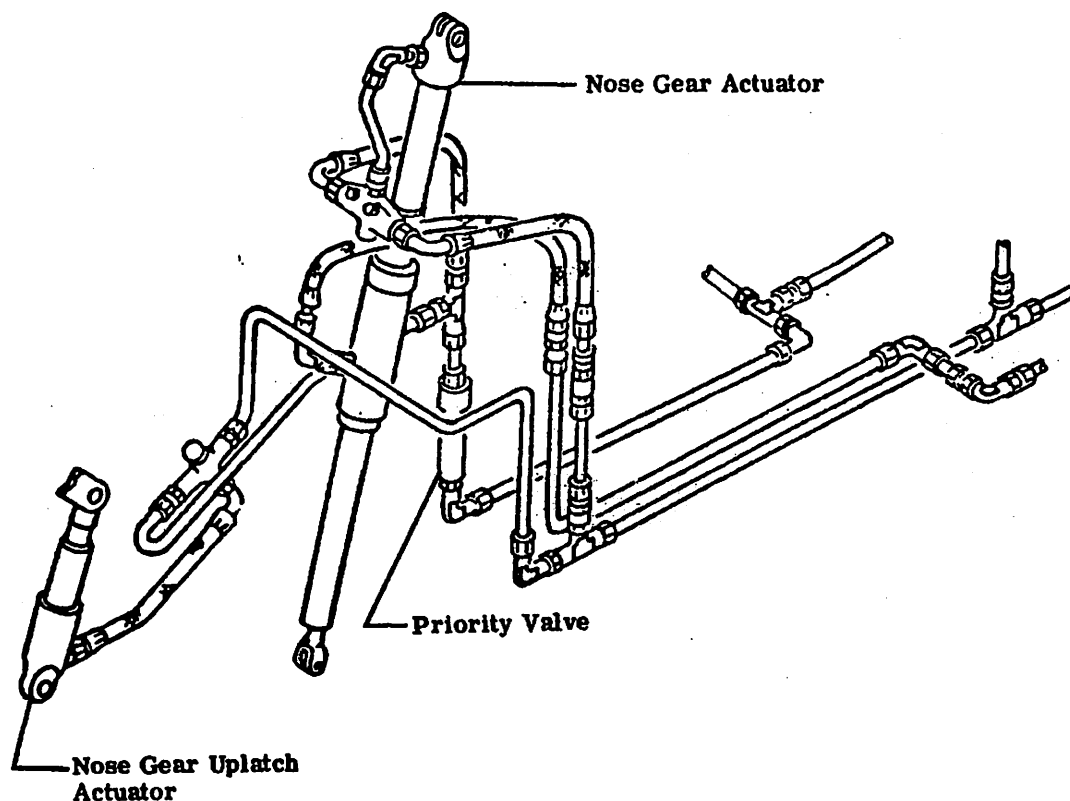
- (1) Remove end cap from valve housing.
- (2) Remove poppet valve, cylinder, and valve spring from valve housing. The adjustment screw, located in the valve housing, is factory set and no adjustment is allowed.
- (3) Remove cotter pin and washer from poppet and unscrew collar from poppet.
- (4) Remove poppet and poppet spring from poppet valve.
- (5) Remove and discard O-rings and backup rings from cylinder and end cap.
- (6) Clean all parts in Stoddard solvent and air dry.
- (7) Inspect all parts for nicks, scratches, or badly worn areas, paying particular attention to poppet valve and cylinder areas.
- (8) If parts are badly worn or nicked, replace the valve.

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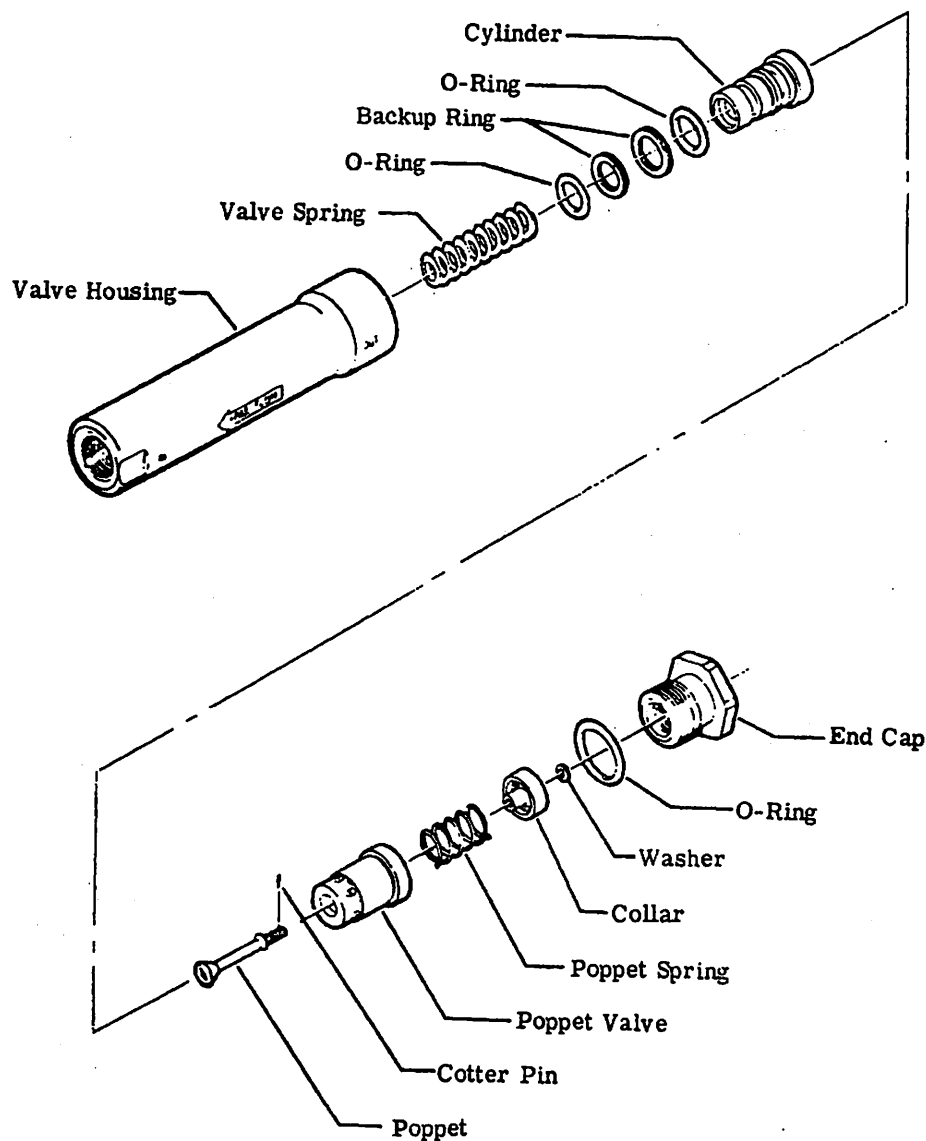
B. Assemble Priority Valve (See figure 202.)

- (1) Install all O-rings and backup rings on cylinder. Install O-ring on end cap.
- (2) Install poppet and spring in poppet valve. Screw collar on poppet and install washer and cotter pin.
- (3) Insert valve spring, cylinder, and poppet valve in valve housing.
- (4) Install end cap.
- (5) Connect a pressure source to valve port marked IN. With pressure set at 5 psi, leakage rate shall not exceed 6 cc's in 3 minutes.
- (6) Connect pressure source to valve port marked OUT. With opposite end of valve plugged, apply pressure of 2250 psi; no leakage is allowed.



Priority Valve Installation
Figure 201

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**Priority Valve Assembly
Figure 202**

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NOSE GEAR ACTUATOR - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Nose Gear Actuator (See figure 201.)

- (1) Place aircraft on jacks.
- (2) Disconnect aircraft batteries.
- (3) Depressurize hydraulic system.
- (4) Remove nose compartment doors and electronic equipment from rack.
- (5) Remove clamp securing shuttle valve to actuator. Disconnect and remove hydraulic line, hose, and elbows from actuator. Cap or plug all exposed hydraulic fittings.
- (6) Remove safety wire and gear down and locked switch from actuator.
- (7) Remove clamp securing actuator boot to actuator.
- (8) Remove attaching parts from upper and lower end of actuator and remove actuator through nose compartment. Switch ball will be released by removal of gear down and locked switch. Before actuator is removed from aircraft, rotate actuator to remove switch ball.

B. Install Nose Gear Actuator (See figure 201.)

- (1) Install actuator in aircraft and secure with attaching parts.
- (2) Secure actuator boot to actuator with clamp.
- (3) Secure shuttle valve to actuator and connect elbow and tube to extend port. Connect elbow and hose to retract port.
- (4) Install switch ball in actuator, install gear down and locked switch in actuator, and adjust switch. (Refer to 32-60-03.)
- (5) Connect aircraft batteries and bleed the hydraulic system. When bleeding system, check for proper operation of actuator.
- (6) Install electronic equipment and nose compartment doors.
- (7) Remove aircraft from jacks.

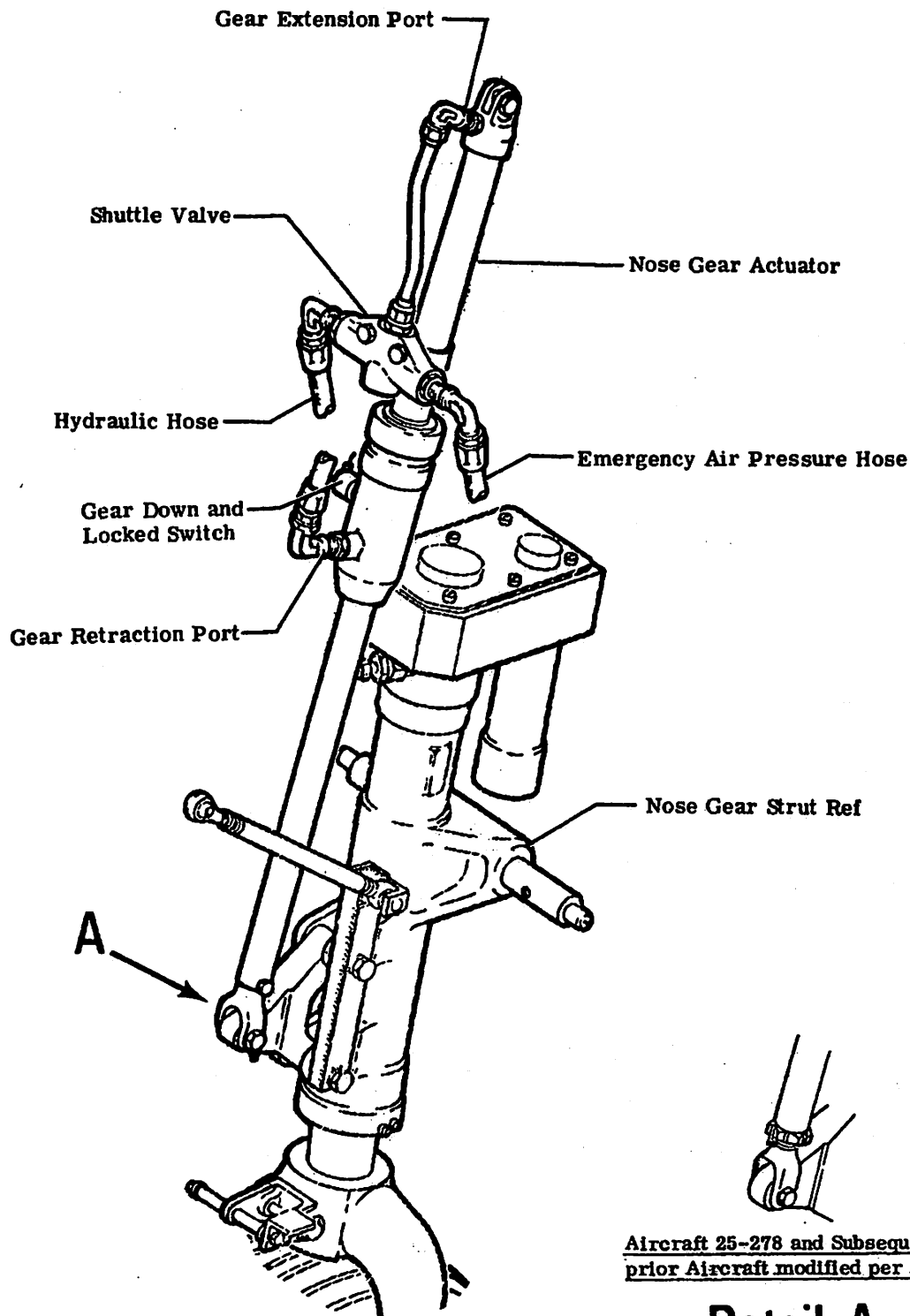
2. Adjustment/Test

A. Functionally Test Nose Gear Actuator

- (1) Fill actuator through extend port with clean MIL-H-5606 hydraulic fluid.
- (2) Bleed all air from actuator.
- (3) Place actuator in a horizontal, fully extended position.
- (4) Connect a hydraulic power source to actuator and cycle actuator twenty-five times at 1500 psi. Leakage shall not exceed one drop.
- (5) Connect a hydraulic source, capable of at least 3000 psi, to extend port with retract port open.
- (6) Pressurize to 2250 psi for 2 minutes, then drop pressure to 5 psi for an additional 2 minutes. There shall be no leakage from open retract port during this period.

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Aircraft 25-278 and Subsequent and
prior Aircraft modified per AAK 79-3.

Detail A

Nose Gear Actuator Installation
Figure 201

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- (7) Disconnect hydraulic source from extend port and connect to retract port with extend port open.
- (8) Bleed all air from actuator.
- (9) Pressurize to 2250 psi for 2 minutes, then drop pressure to 5 psi for an additional 2 minutes. There shall be no leakage from open extend port during this period.

3. Approved Repairs

A. Disassemble Nose Gear Actuator (See figure 202.)

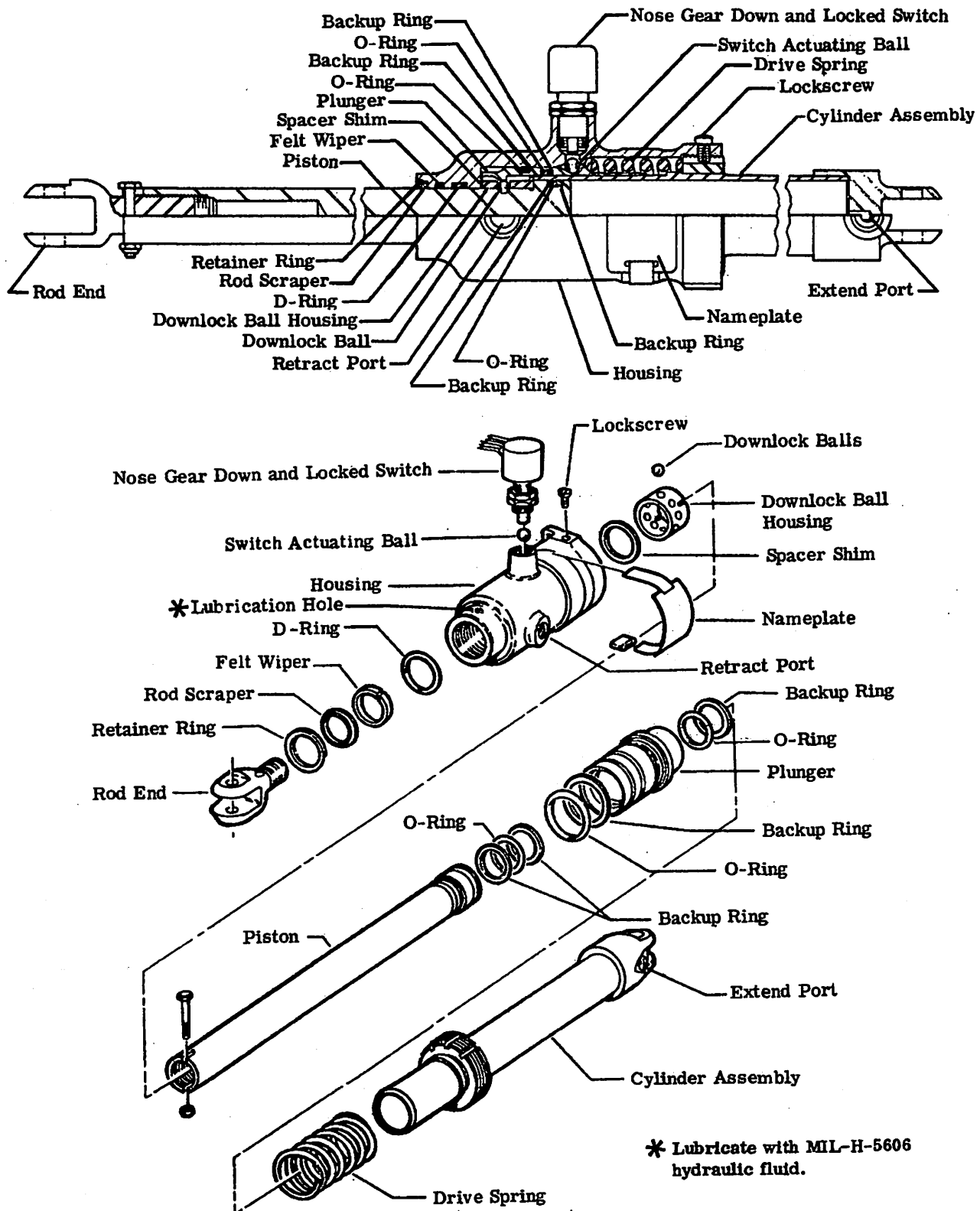
- (1) During disassembly tag and keep all parts together for assembly into the same unit. Do not remove actuator nameplate unless it is damaged.

WARNING: THE ACTUATOR IS SPRING LOADED TO 161 POUNDS. USE SPECIAL PRECAUTION WHEN UNSCREWING CYLINDER ASSEMBLY FROM HOUSING TO PREVENT EITHER PART FROM SPRINGING OUT OF CONTROL. OTHERWISE BODILY INJURY MAY RESULT.

- (2) Remove lockscrew and, while taking care to control actuator spring load, slowly unscrew cylinder assembly and remove from housing.
- (3) Aircraft 25-061, 25-070 thru 25-277 NOT modified per AAK 79-3; remove nut and bolt attaching rod end to piston and remove rod end from piston. Aircraft 25-278 and Subsequent and prior Aircraft modified per AAK 79-3; loosen rod end lock nut so that lock key can be extracted from piston. Remove rod end, with lock key and lock nut installed, from piston.
- (4) Slide piston, spacer shim, drive spring, plunger, and downlock ball housing from housing. Retrieve downlock balls which fall free as piston is removed.
- (5) Remove retainer ring and rod scraper from housing. Remove and discard D-ring and felt wiper.
- (6) Clean metal parts with Stoddard solvent or equivalent. Assure that no solvent remains in internal cavities or surfaces.
- (7) Remove all fingerprints from machined surfaces not plated or painted using a suitable fingerprint neutralizer.
- (8) Dry metal parts after cleaning with clean, dry, compressed air.
- (9) Coat internal surfaces with MIL-H-5606 hydraulic fluid.
- (10) Wrap cleaned parts in lint-free cloth or clean waxed paper while awaiting repair and assembly.
- (11) When storing unplated, machine surfaced parts more than a few hours, coat with MIL-C-11796 preservative. Remove preservative prior to assembly of parts.

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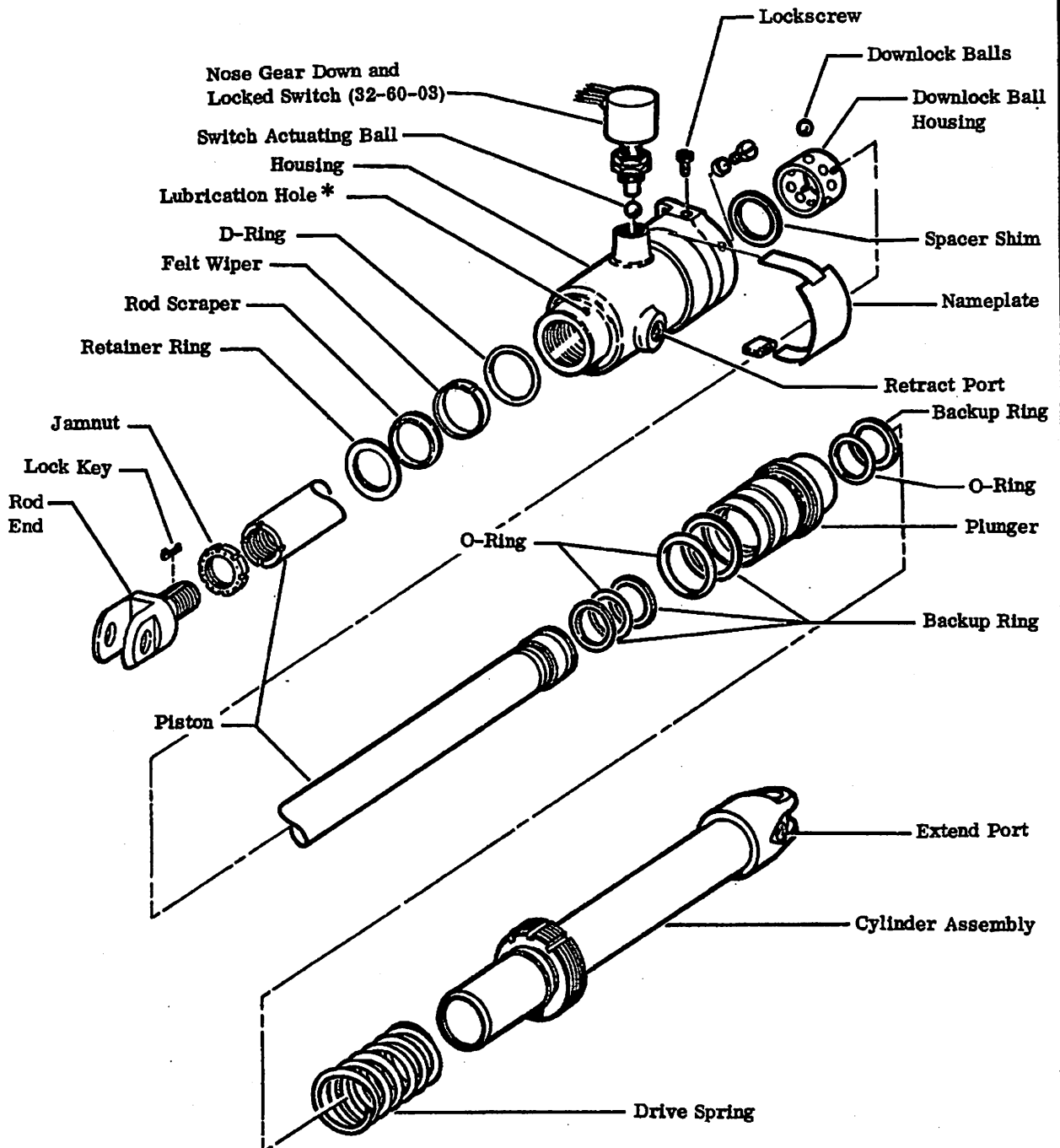
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Nose Gear Actuator Assembly
Figure 202 (Sheet 1 of 2)

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* Lubricate with MIL-H-5606 hydraulic fluid only.

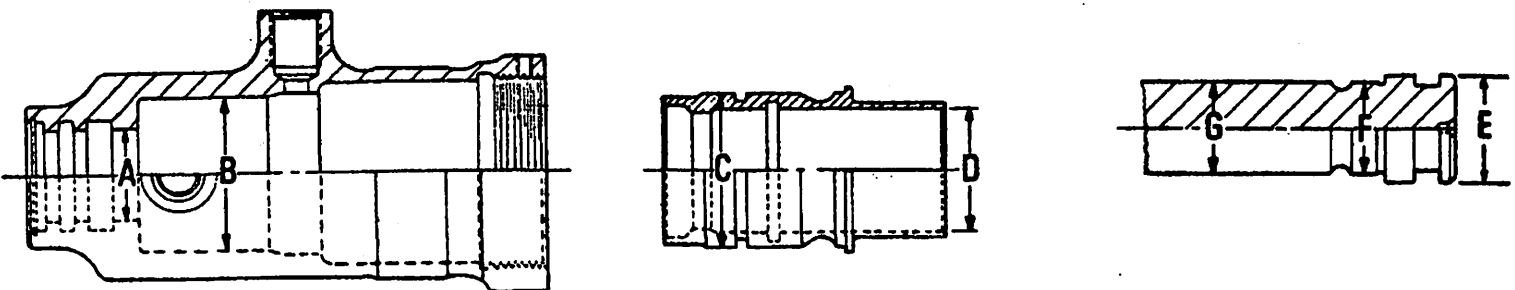
Nose Gear Actuator Assembly
Figure 202 (Sheet 2 of 2)

11-32B

EFFECTIVITY: 25-278 AND SUBSEQUENT AND PRIOR
AIRCRAFT MODIFIED PER AAK 79-3

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Nomenclature	Outside Diameter (Inches)		Inside Dieameter (Inches)		Total Clearance Diameter (Inches)		Surface Finish
	Original	Minimum	Original	Maximum	Original	Maximum	
Upper and Lower Attach Bolts	0.4367 to 0.4358	0.433					32 RMS
Cylinder Assembly	1.627 to 1.625	1.624	1.438 to 1.436	1.440	0.006 to 0.002	0.010	16 RMS
Plunger (C)	1.990 to 1.988	1.986					32 RMS
Plunger (D)			1.631 to 1.629	1.633	0.006 to 0.002	0.009	32 RMS
Piston (E)	1.434 to 1.432	1.430					32 RMS
Piston (F)	1.124 to 1.122	1.120					32 RMS
Piston (G)	1.124 to 1.122	1.121					16 RMS
Downlock Ball			1.128 to 1.126	1.130	0.006 to 0.002	0.010	125 RMS
Housing (B)			1.994 to 1.992	1.997	0.006 to 0.002	0.011	16 RMS
Housing (A)			1.127 to 1.125	1.131	0.005 to 0.001	0.010	32 RMS

Nose Gear Actuator Wear Tolerances
Figure 203

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B. Inspect Nose Gear Actuator (See figure 203.)

- (1) Screen drained hydraulic fluid for metal particles which may be used as an aid in determining trouble source.
- (2) Inspect all metal parts including welds and machined and forged parts for structural failure, nicks, burrs, cracks, corrosion, worn threads, and excessive wear.
- (3) Perform a magnetic-type inspection on all ferrous parts. Perform a fluorescent penetrant check on all aluminum parts. After ferrous parts inspection, demagnetize to remove all metal particles.
- (4) Inspect OD and ID surfaces of cylinder assembly, OD surfaces of piston, and ID surfaces of housing for galls, scratches, scoring, corrosion, and worn plating.

NOTE: Aircraft 25-278 and Subsequent and prior Aircraft modified per AAK 79-3; are equipped with a jamnut and lock key type rod end. Aircraft with jamnut type installation should omit steps (5) and (7).

- (5) With rod end threaded into piston, check rod end for looseness. If looseness exists between actuator piston and rod end, remove rod end.
- (6) Inspect threads of piston and rod end. Replace if damaged.
- (7) If threads are not damaged and end play does not exceed 0.004 inch, apply Loctite Retaining Compound No. 680 evenly to threads and install rod end. If looseness exceeds 0.004 inch, replace defective part.
- (8) Check to assure that spring height is at least 2 inches when supporting a 161-pound load.
- (9) Buff out minor scratches in cylinder assembly, plunger, ball housing, piston, and housing. Do not create low spots by excessive buffing or polishing. Use oil stone to remove scratches and polish with crocus cloth. Check tolerances after polishing.
- (10) Replace any parts which are damaged, cracked, excessively worn, scored, corroded, or do not meet inspection requirements or tolerances.

C. Assemble Nose Gear Actuator (See figure 202.)

- (1) Immerse O-rings, backup rings, and felt wiper in MIL-H-5606 hydraulic fluid prior to installation.
- (2) Install D-ring, felt wiper, rod scraper, and retaining ring in housing.
- (3) Place downlock ball housing and spacer shim on piston and slide piston through housing. If a new shim is to be installed, assure that it is same thickness as shim which was removed.

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- (4) With spacer shim and downlock ball housing firmly seated in housing, check to assure that retract and extend port are in same plane and facing same way to within 5°.

NOTE: Step (4) must be performed with center groove in threaded portions of cylinder assembly aligned with housing lockscREW hole.

- (5) Add to or remove laminations on spacer shim or remove spacer shim completely as required to obtain firm seating of spacer shim and downlock ball housing inside of housing.
- (6) Install O-rings and backup rings on piston and plunger.
- (7) Assemble locking balls in downlock ball housing and slide plunger over them to hold their relative positions.
- (8) Place drive spring over cylinder.

WARNING: TAKE CARE TO CONTROL THE SPRING LOAD INSIDE HOUSING WHEN SCREWING CYLINDER ASSEMBLY INTO PLACE.

- (9) Screw cylinder assembly into housing until its inner face firmly clamps downlock ball housing and spacer shim against housing.
- (10) Install lockscREW and safety wire at place provided on housing.
- (11) Install rod end and adjust so that dimension from center of rod end bolt hole to center of cylinder assembly bolt hole is 20.19 inches with cylinder compressed. Secure rod end with attaching parts. (Torque jamnut (if installed) to 700 to 800 inch-pounds.)
- (12) When installing nameplate, steel stamp new part to duplicate data on part being replaced.
- (13) Refinish all externally painted surfaces with one coat of wash primer, one coat of zinc chromate primer, and one coat of aluminized lacquer.
- (14) Lubricate main gear actuator piston rod end felt wiper with MIL-H-5606 hydraulic fluid. Lubrication is accomplished through lubrication hole on rod end of actuator.
- (15) Perform Adjustment/Test of nose gear actuator.

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NOSE GEAR UPLATCH - MAINTENANCE PRACTICES

1. Removal/Installation

- A. Remove Nose Gear Uplatch Actuator (See figure 201.)
 - (1) Place the aircraft on jacks.
 - (2) Depressurize the hydraulic system.
 - (3) Disconnect hydraulic line from actuator. Cap and plug exposed hydraulic fittings.
 - (4) Remove cotter pins, nuts, washers, and bolts securing actuator and remove actuator from aircraft.
- B. Install Nose Gear Uplatch Actuator (See figure 201.)
 - (1) Position actuator and secure with bolts, washers, nuts, and cotter pins.
 - (2) Connect hydraulic hose to actuator.
 - (3) Adjust uplatch mechanism. (Refer to Adjustment/Test.)
 - (4) Remove aircraft from jacks.

2. Adjustment/Test

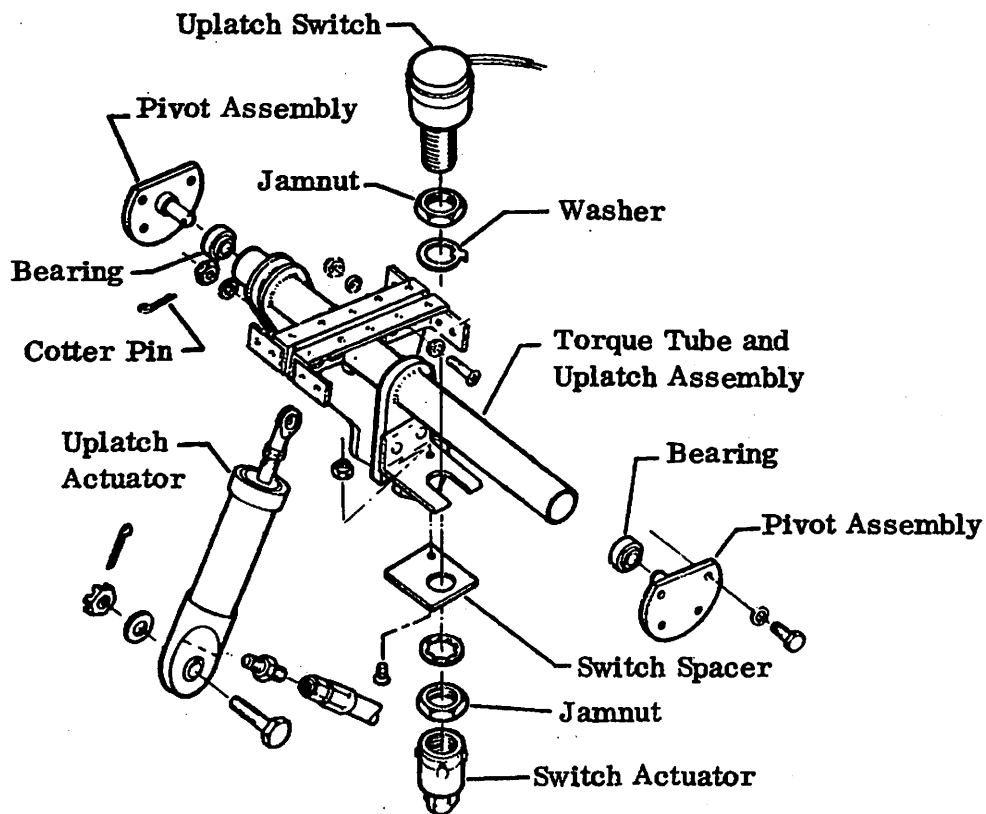
- A. Adjust Uplatch Mechanism (See figure 202.)
 - (1) Place the aircraft on jacks.
 - (2) Disconnect linkage from the nose gear doors.
 - (3) Retract the nose gear.
 - (4) Loosen uplatch roller attach bolt and adjust until uplatch roller is securely engaged in hook with approximately 0.15 inch clearance. Secure uplatch roller bolt with cotter pin.
 - (5) Attach an ohmmeter across uplatch switch leads and adjust inner jam-nut until switch actuates, plus one additional turn.
 - (6) Secure outer jamnut and safety wire per double twist method.
 - (7) Adjust safety cam forward or aft as required to maintain 0.03 inch clearance between cam and uplatch spacer with nose gear retracted.
 - (8) Extend nose gear and attach linkage to doors.
 - (9) Remove aircraft from jacks.

3. Approved Repairs

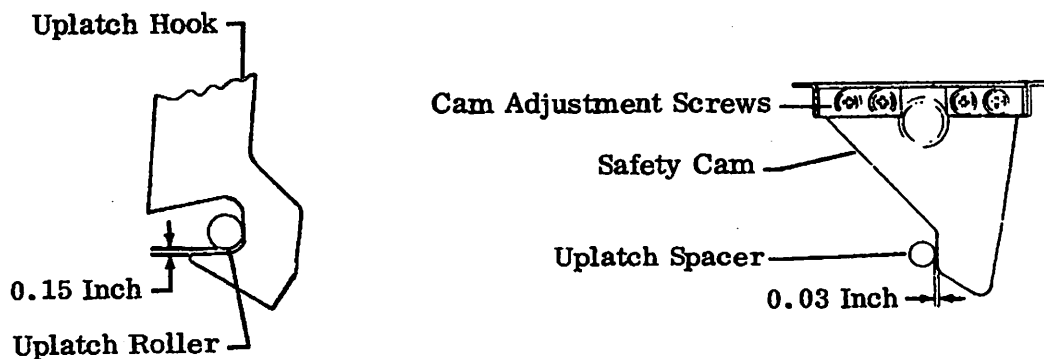
- A. Disassemble Uplatch Actuator (See figure 203.)
 - (1) Cut safety wire and loosen locknut sufficiently to release rod end lock from piston rod. Remove rod end, rod end lock, and locknut from piston rod.

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Nose Gear Uplatch Assembly
Figure 201



Uplatch Mechanism Adjustment
Figure 202

EFFECTIVITY: ALL

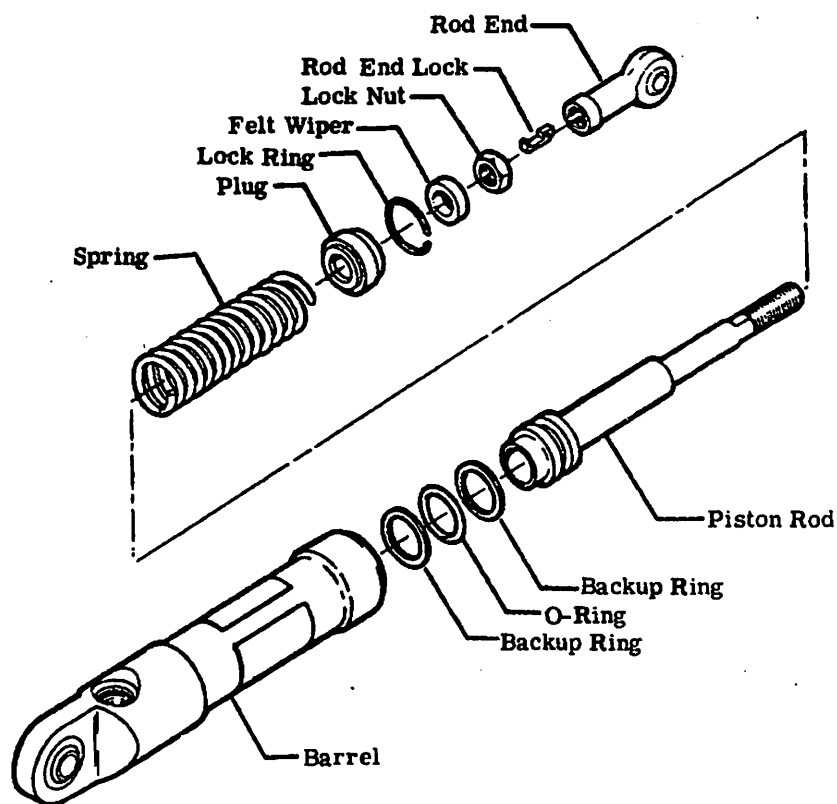
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WARNING: USE EXTREME CARE WHEN REMOVING LOCK RING AND PLUG FROM BARREL. ACTUATOR IS SPRING LOADED TO 40 POUNDS IN RETRACTED POSITION AND COULD SPRING OUT OF CONTROL, CAUSING BODILY INJURY.

- (2) Press plug inboard to allow removal of lock ring. Remove lock ring and slowly release plug.
 - (3) When spring tension is released, remove plug and spring from barrel.
 - (4) Remove piston rod from barrel. Remove O-ring and backup rings from piston rod. Remove felt wiper from plug.
 - (5) Clean all parts with Stoddard solvent or equivalent. Dry parts with clean, dry, compressed air.
 - (6) Inspect all parts for nicks, scratches, or excessive wear. Replace defective parts.
 - (7) Buff out minor scratches using an oil stone and polish with crocus cloth.
- B. Assemble Uplatch Actuator (See figure 203.)
- (1) Use new O-rings, backup rings, and felt wiper at each assembly. Immerse these items in MIL-H-5606 hydraulic fluid prior to installation.
 - (2) Assemble O-ring and backup rings on piston rod and insert felt wiper in plug.
 - (3) Position piston rod and spring in barrel. Install plug on piston rod and push in, compressing spring sufficiently to allow installation of lock ring.
 - (4) Install lock ring and slowly release plug until firmly seated against lock ring.
 - (5) Install rod end lock, locknut, and rod end on piston rod.
- C. Functionally Test Uplatch Actuator
- (1) Connect actuator to a hydraulic source and pressure check at 2250 psi for 3 minutes. Leakage rate should not exceed one drop in the 3-minute period.
 - (2) While actuator is extended, adjust for 9.34 (± 0.015) inch distance between centerlines of rod end and actuator bearing.
 - (3) Remove hydraulic pressure and check retracted length of 8.44 (± 0.045) inches. Safety wire rod end lock in place.

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Uplatch Actuator Assembly
Figure 203

EFFECTIVITY: ALL

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EMERGENCY EXTENSION - DESCRIPTION AND OPERATION

1. **Description**
 - A. The emergency extension system consists of the emergency gear control valve installed on the LH side of the pedestal and the shuttle valves installed at each actuator.
2. **Operation**
 - A. Set the Landing Gear Selector Switch to DN. The emergency gear control valve is manually operated and when the valve handle is down, high-pressure air is applied through the switch actuated valves to the actuators. The main gear inboard doors will open and the main and nose gear will extend to their down and locked positions.
 - B. When the uplatches unlatch, the red gear UNSAFE lights will illuminate; and when each gear is down and locked, that gears' green LOCKED DN light will illuminate. Because air pressure remains applied to the actuators, the uplatch mechanisms will not close and the red UNSAFE lights will remain illuminated.
 - C. Shuttle valves isolate the hydraulic system from the emergency extension system, and reposition from hydraulic flow to emergency air flow when air pressure exceeds hydraulic pressure. Shuttle valves will reposition when hydraulic pressure exceeds air pressure.

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EMERGENCY EXTENSION - MAINTENANCE PRACTICES

1. Adjustment/Test

A. Check Emergency Extension System for Leaks

- (1) Remove emergency air bottle filler cap, loosen valve end nut, and bleed pressure down to approximately 800 psi. This prevents repositioning of main gear actuator, nose gear uplatch actuator, and nose gear actuator shuttle valves, introducing air into hydraulic system, and disturbing O-ring seal when emergency control valves are opened.
- (2) Pressurize hydraulic system to 1500 psi using a ground power cart.
- (3) While system pressure is at 1500 psi, fully depress either copilot's or pilot's brake pedal and set parking brake. This will assure adequate hydraulic pressure is maintained to prevent repositioning of brake system shuttle valves by emergency air.
- (4) Depress emergency gear control handle.

NOTE: When the emergency gear control handle is depressed, system hydraulic pressure will drop momentarily until the power cart returns pressure to 1500 psi. This is due to the door actuator and door uplock actuator shuttle valves repositioning.

- (5) After system has stabilized, monitor emergency air pressure gage (for approximately 3 minutes) for loss of air pressure.
- (6) If loss of air pressure is not evident, bleed air pressure from system by pulling ratchet spring release to allow handle to return to normal.

NOTE: On aircraft equipped with a Kidde emergency gear control valve, the handle will return to the off and latched position when the ratchet spring release is pulled. On aircraft equipped with a Sterer emergency gear control valve the handle will return to the off position when the ratchet spring release is pulled but will not latch. The Sterer valve handle must be manually lifted up into the latched position.

- (7) Service emergency air bottle. (Refer to Chapter 12.)
- (8) Bleed hydraulic system. (Refer to Chapter 29.)

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B. Return Extension System to Normal Operation

- (1) Place aircraft on jacks.
- (2) Bleed air lines by inserting a finger into hole in pedestal just forward of handle and pushing up on ratchet release, allowing handle to return to off position.

NOTE: On aircraft equipped with a Kidde emergency gear control valve, the handle will return to the off-and-latched position when the ratchet spring release is pulled. On aircraft equipped with a Sterer emergency gear control valve, the handle will return to the off position when the ratchet spring release is pulled but will not latch. The Sterer valve handle must be manually lifted up into the latched position.

- (3) Perform necessary maintenance to correct system malfunction.
- (4) With electrical power applied, set gear selector switch to DN position.
- (5) Return system shuttle valves to normal position by pressurizing hydraulic system using a hydraulic power cart.

CAUTION: DUE TO THE POSSIBILITY OF A LARGE VOLUME OF HIGH PRESSURE AIR BEING TRAPPED ON THE HYDRAULIC SIDE OF THE SHUTTLE VALVES, THE HYDRAULIC RESERVOIR COULD BE RUPTURED WHEN THE GEAR IS RETRACTED. THEREFORE, REMOVE THE RESERVOIR FILLER CAP PRIOR TO RETRACTING THE GEAR. THE GEAR SHOULD BE RETRACTED IN A SERIES OF AT LEAST SIX STAGES BY SETTING THE GEAR SELECTOR SWITCH TO UP THEN DN. EACH TIME THE GEAR SELECTOR SWITCH IS SET TO UP, THE GEAR SHOULD BE ALLOWED TO RETRACT FURTHER THAN THE PREVIOUS PARTIAL RETRACTION UNTIL THE GEAR IS COMPLETELY RETRACTED.

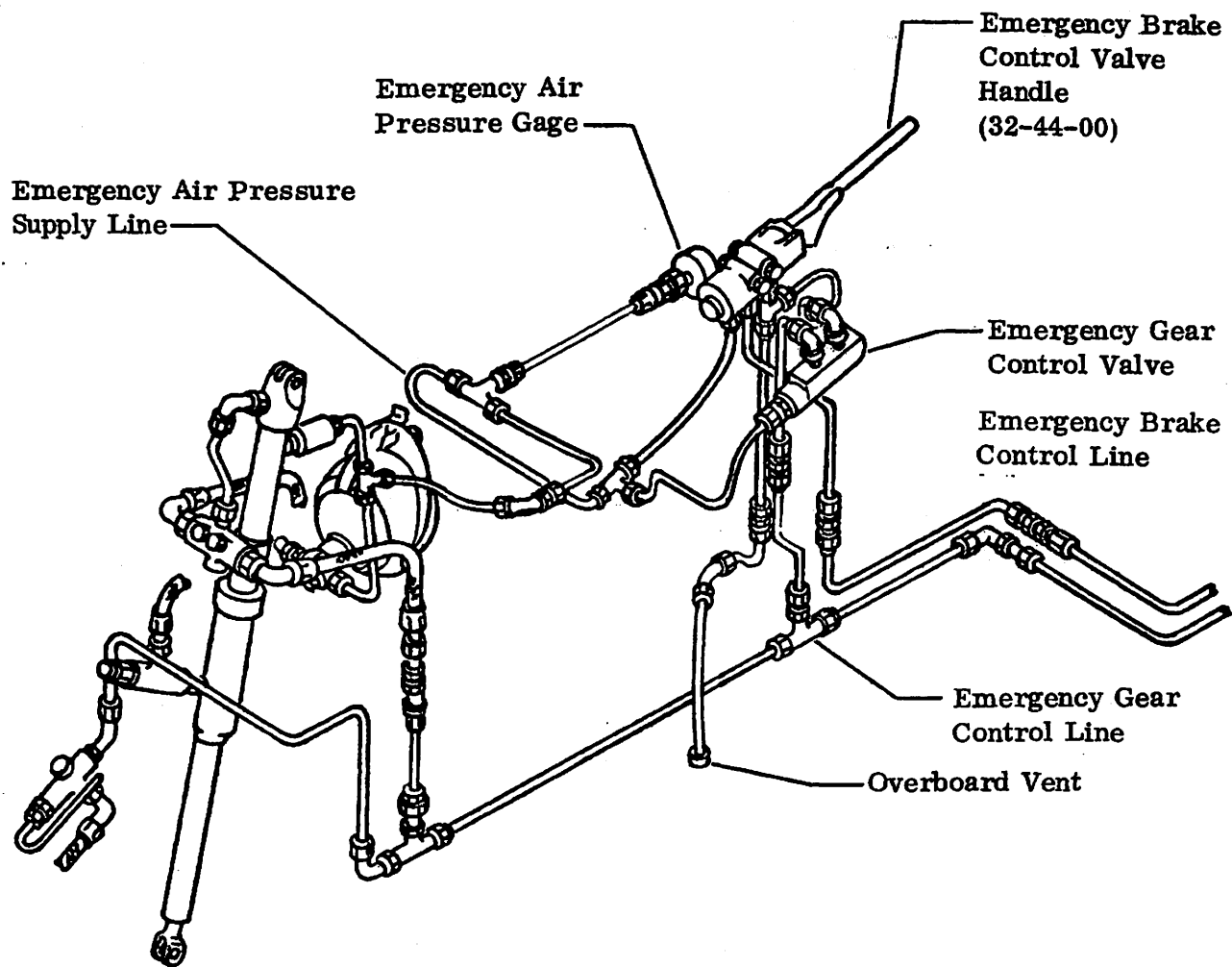
- (6) Lower tailcone access door. Loosen a hydraulic line nut between hydraulic reservoir and vent valve to bleed off reservoir pressure. Retighten nut and remove reservoir filler cap. Assure that reservoir fluid level is above sight glass. Make provisions in tailcone area for possible spurting of fluid from reservoir filler.

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- (7) Retract gear in approximately six stages by setting gear selector switch to UP and then to DN, allowing gear to retract slightly further each time until gear is completely retracted. This will purge trapped air and help prevent fluid from spurting out of reservoir.
- (8) Cycle gear approximately 20 times to assure that all air is expelled from system.
- (9) Service hydraulic reservoir and emergency air bottle. (Refer to Chapter 12.)

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Typical Emergency Extension Components Installation
Figure 201

11-160 W

EFFECTIVITY: ALL

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SHUTTLE VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Shuttle Valve

- (1) Depressurize hydraulic system.
- (2) Disconnect hydraulic and pneumatic lines at shuttle valve. Discard pneumatic B-nut fitting seal.
- (3) Remove attaching parts and shuttle valve from aircraft.

B. Install Shuttle Valve

- (1) Install shuttle valve in its original position with its attaching hardware.
- (2) Connect hydraulic and pneumatic lines to their original fittings. Use a new fitting seal at pneumatic B-nut connection.

2. Adjustment/Test

- A. With outlet port capped and valve connected to a hydraulic test bench, apply a proof pressure of 2250 psi for 3 minutes to each inlet port. There shall be no sign of leakage.

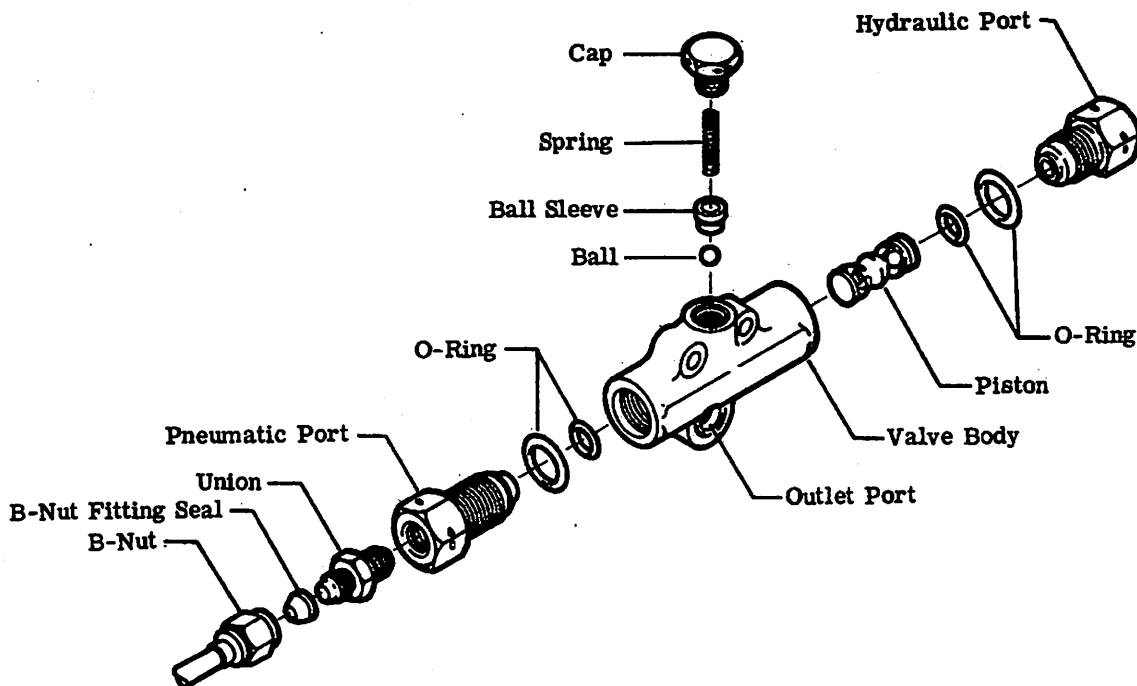
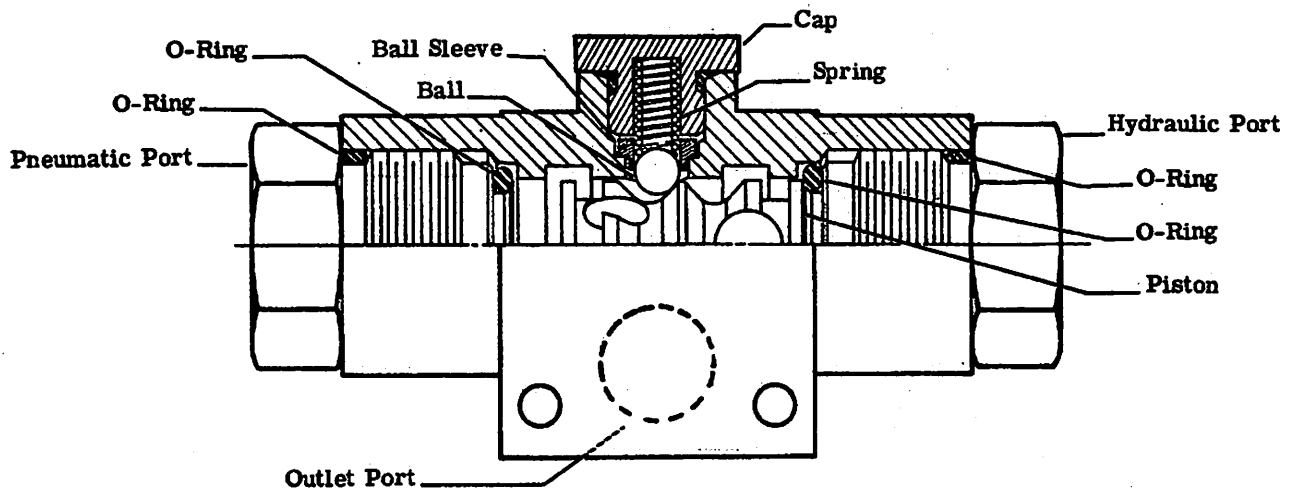
3. Approved Repairs

- A. Due to many configurations of shuttle valves, only general repair instructions can be given.

- (1) Disassemble shuttle valve beginning with cap bolt and/or outlet port. Note sequence of parts removal.
- (2) Discard O-rings, packing, and backup rings (if installed).
- (3) Inspect for scoring, scratches, or corrosion. Buff out minor damage with crocus cloth using caution not to create any low spots.
- (4) Clean thoroughly with Stoddard solvent and blow dry to remove foreign particles.
- (5) Reassemble in reverse order of disassembly using new O-rings, packing, and backup rings (if required). Immerse O-rings in MIL-H-5606 hydraulic fluid prior to assembly.

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Typical Shuttle Valve Assembly
Figure 201

EFFECTIVITY: ALL

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WHEELS AND BRAKES - DESCRIPTION AND OPERATION

1. Description

A. Nose Wheel and Tire

- (1) The nose landing gear wheel consists of two wheel halves. The LH wheel half has a machined groove to accommodate an O-ring seal to retain tire air pressure. The RH wheel half contains an air valve assembly for servicing the tubeless tire. Each wheel half is fitted with a tapered cone bearing. The bearing is rubber sealed to prevent contamination and lubricant loss. (Refer to 32-41-00.)
- (2) The nose wheel is equipped with an 18 x 4.4, 10-ply rated, type VII, tubeless tire. The tire incorporates chines on each side which deflect water and slush away from engine inlets. The use of recapped tires on the nose wheel is not approved.

B. Main Wheel and Tire

- (1) Each main landing gear wheel consists of two wheel halves. The strut side half of the wheel has a machined groove to accommodate an O-ring seal to retain tire pressure. Two fusible plugs release excessive air pressure caused by overheat. The strut side wheel half contains seven slots which are fitted with disc drive keys. The keys drive the rotating brake discs when the wheels are in motion. Each opposite side wheel half contains a valve assembly for servicing tires. Both wheel halves are fitted with tapered cone bearings which are rubber sealed to prevent contamination and lubricant loss. The wheel halves are bolted together to facilitate removal and mounting. (Refer to 32-42-00.)
- (2) In ship sets, each main gear wheel is equipped with either 18 x 5.50 Type VII 10 ply 210 mph rated, or 18 x 5.50 Type VII 10 ply 200 mph rated, or 17.5 x 5.75-8 12 ply 210 mph rated tubeless tires. Use of recapped tires on main gear wheels is not approved.

C. Brake System

- (1) The brake system consists of a primary hydraulic system (refer to 32-43-00), a secondary emergency (air pressure) system (refer to 32-44-00), and an anti-skid system (refer to 32-45-00). The parking brake is a part of the primary hydraulic system (refer to 32-43-00).

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TEMPORARY REVISION NO. 32-8

MANUAL AFFECTED: Gates Learjet Model 25 b/c/d/f Maintenance Manual

FILING INSTRUCTIONS: Insert adjacent to 32-40-00, page 201, dated Apr 30/82.

REASON: Eddy Current inspection of wheel bead seat radius.

INSTRUCTIONS: Add step "C. Eddy Current Inspection of Wheel Bead Seat Radius," to step 1. INSPECTION/CHECK.

1. INSPECTION/CHECK

C. Eddy Current Inspection of Wheel Bead Seat Radius (See figure 201.)

- NOTE:**
- ° Perform Eddy Current Inspection of Wheel Bead Seat Radius in accordance with the current inspection interval specified in Chapter 5 (ref: Temp. Rev. 5-37).
 - ° Procurement of a Hocking Locator UH Eddy Current Crack Detector with pencil probe P/N 29P101 (Mfg. by Hocking Electronics, Temple Hills, Maryland 20748), or equivalent tester/probe is required to perform this inspection. When an alternate type tester/probe is used, refer to the applicable manufacturers instructions for performing Eddy current inspection.
 - ° When using the Hocking Locator tester and pencil probe, use of a Eddy Current Probe Guide P/N 2471017-1, available from Gates Learjet Corp., is recommended to facilitate inspection.
 - ° Wheel halves shall be replaced if any evidence of cracking is noted. Corrosion shall be repaired prior to inspecting. Scratches, nicks, and other types of damage may be repaired, provided damage is within the allowable limits as specified in the appropriate section, this Chapter. Wheel halves shall be reinspected, per the following procedures, after any type repair.
 - ° The following recommended inspection procedure is applicable when using the Hocking Locator UH Eddy Current Crack Detector with pencil probe P/N 29P101, and Eddy Current Probe Guide P/N 2471017-1, only.

- (1) Remove applicable wheel(s) from aircraft and disassemble in accordance with procedures outlined in the appropriate section, this Chapter.

- (2) Clean wheel halves with dry-cleaning solution, Specification P.D. 680 (Stoddard Solvent), or equivalent.

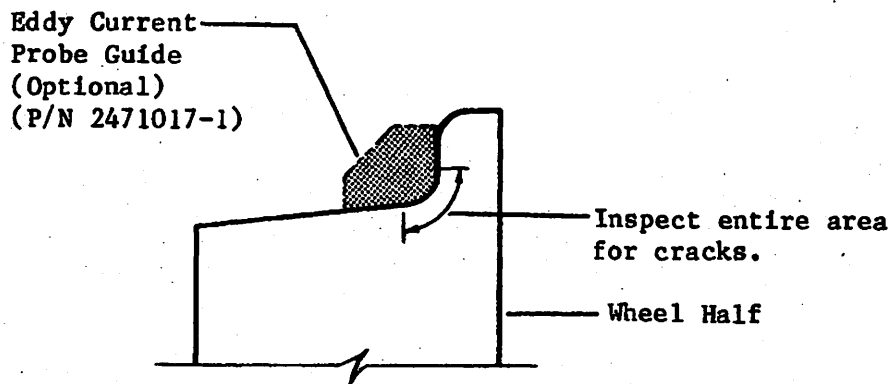
NOTE: ° Paint on bead seat shall be smooth and uniform prior to performing Eddy current inspection. If stripping of paint from wheel halves is required, refer to the appropriate section, this Chapter, for paint removal procedures.

° When inspecting painted wheel half, if an area is suspected of having a crack, strip paint from wheel half and reinspect.

- (3) Utilizing recommended test equipment as identified previously, pre-select Eddy current tester to 500 kHz frequency, and adjust gain to 500.

NOTE: When using equipment other than that recommended, refer to the applicable manufacturers instructions for frequency and sensitivity adjustments.

- (4) Position Eddy current probe guide (P/N 2471017-1) on bead seat radius, and using pencil probe, inspect entire periphery of wheel bead seat. Use each of the four locations on probe guide. Replace wheel half if any evidence of cracking is noted.



**Eddy Current Inspection of Wheel Bead Seat Radius
Figure 201**

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WHEELS AND BRAKES - MAINTENANCE PRACTICES

1. Inspection/Check

A. Inspect Wheel Bolts

NOTE: If wheel bolts are found to be missing, broken, or cracked or if wheels were subjected to rolling with a flat or disintegrated tire, inspection is required.

- (1) Remove from aircraft any wheel which has a missing, broken, or cracked bolt.
- (2) Deflate and disassemble wheel. (Refer to 32-41-00 or 32-42-00.)
- (3) If a single bolt is missing or broken, replace that bolt and both bolts adjoining it.
- (4) If a bolt is cracked, replace cracked bolt only.
- (5) Perform a magnetic particle check of all other bolts.
- (6) If more than one bolt is missing or broken, replace all bolts.

B. Inspect Wheels with Flat or Disintegrated Tires

- (1) If a wheel has rolled with a flat, carefully inspect for deformation in flange areas and tire brinelling and/or chafing damage on flange outside diameter. Replace wheels if damage is in excess of tolerances outlined in applicable Overhaul Manual.
- (2) All wheels where flanges have made direct contact with runway must be removed from service.
- (3) Remove mating wheel from service if distance rolled was sufficient to cause any damage to overloaded tire.
- (4) Inspect mating wheel bolts and nuts for fatigue damage and replace if damaged. In all cases of overloaded rolling, bolts and nuts should be replaced or checked for fatigue damage at each tire change. The over-pressurization relief device, where fitted, or the replaceable diaphragm, as applicable, should be replaced in all cases.

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NOSE WHEEL AND TIRE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Nose Wheel (See figure 201.)

CAUTION: CONE BEARINGS WILL COME LOOSE FROM WHEEL WHEN REMOVING WHEEL FROM AIRCRAFT. DO NOT DROP OR DAMAGE CONE BEARINGS.

- (1) Place aircraft on jacks.
- (2) Remove safety wire, axle nut, and safety lock washer from nose gear axle.
- (3) Remove O-ring and wheel bearing spacer, then slide wheel from axle.

B. Install Nose Wheel (See figure 201.)

- (1) Assemble O-rings and wheel bearing spacer on axle.
- (2) Install inboard cone bearing packed with MIL-G-81322 grease, wheel, and outboard cone bearing packed with MIL-G-81322 grease, on axle.
- (3) Assemble O-ring, wheel bearing spacer, safety lock washer, and axle nut on axle.
- (4) Tighten axle nut to 80 inch-pounds, back off to zero, and retighten to 60 inch-pounds. Check that no wheel side motion exists. Safety wire axle nut and safety lock washer.
- (5) Inflate tire with dry air or nitrogen to pressure specified in Chapter 12. Assure that valve core is not leaking, then install valve cap.
- (6) Check tire pressure after 24 hours; maximum allowable pressure loss is 4%.

2. Approved Repairs

NOTE: Tire tread may be worn down to the cord before replacement. However, a smooth tire will more readily hydroplane than a tire that has tread. The use of recapped tires on the nose gear wheel is not approved.

A. Disassemble Nose Wheel (See figure 202.)

WARNING:

- DO NOT LOOSEN OR REMOVE WHEEL BOLTS UNTIL THE TIRE IS COMPLETELY DEFLATED. FAILURE TO COMPLY CAN CAUSE BODILY INJURY.
- DO NOT ATTEMPT TO REMOVE VALVE CORE UNTIL TIRE HAS BEEN COMPLETELY DEFLATED. THE VALVE CORE WILL BE EJECTED AT HIGH VELOCITY IF UNSCREWED BEFORE AIR PRESSURE HAS BEEN RELEASED.

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- (1) Deflate tire by removing valve cap and installing tire deflator (Schrader P/N 4722 or equivalent). After all air pressure has been released, remove valve core.

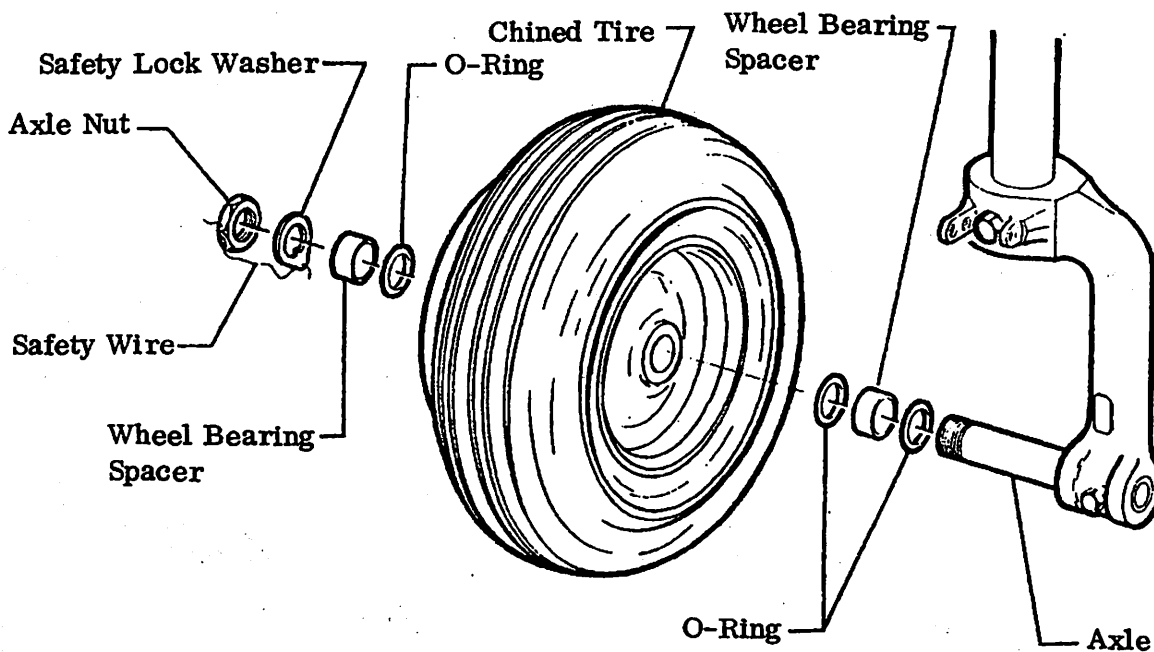
NOTE: Disassemble wheel on a clean, flat surface to preclude damage to wheel halves.

- (2) Break tire beads from both wheel halves. Apply even pressure around entire sidewall as close as possible to tire beads.

CAUTION: ANY FASTENER ACCIDENTALLY TIGHTENED (INSTEAD OF LOOSENED) TO MORE THAN 120% OF RECOMMENDED TORQUE MUST BE DISCARDED.

- (3) Remove self-locking nuts, washers, bolts, and countersunk washers.

NOTE: Do not use power (impact or continuous drive) wrenches for removal of any nuts or bolts made of or attached to any nonferrous material.



Nose Wheel Installation
Figure 201

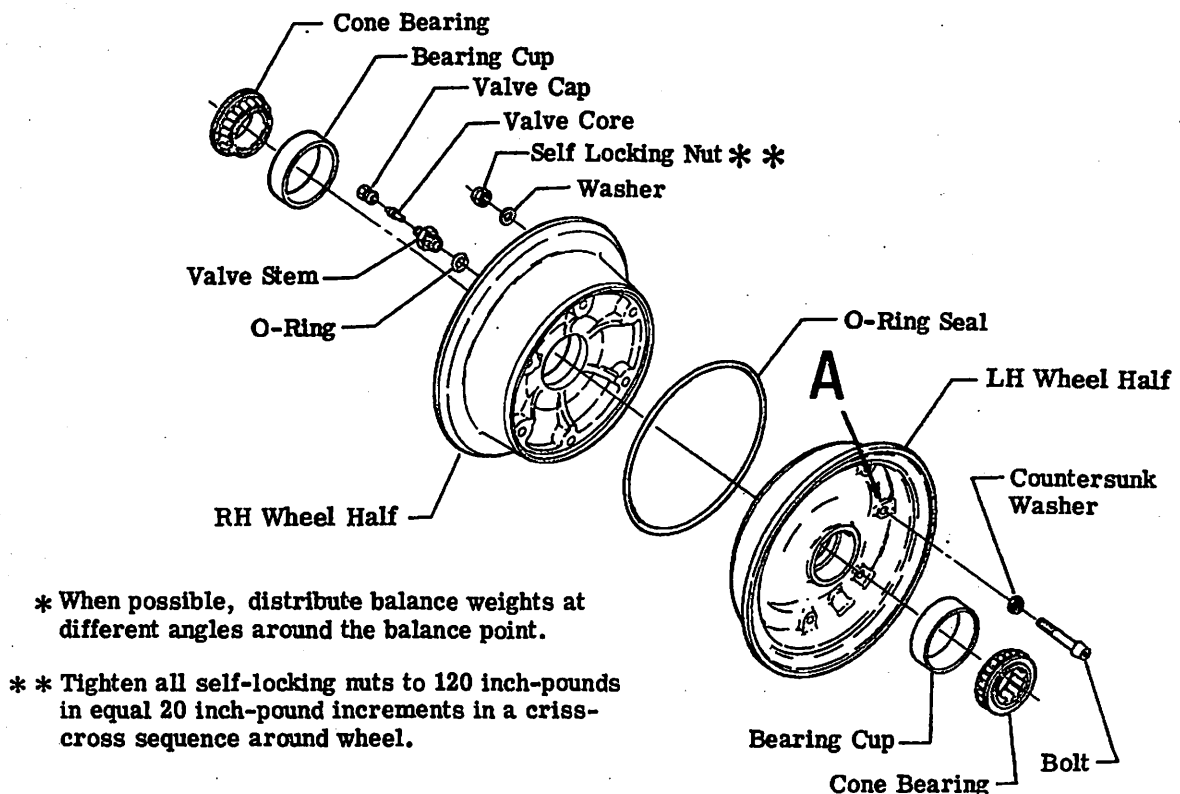
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- (4) Use care when separating wheel halves to prevent damage to wheel O-ring seal. Remove valve stem before heating wheel half for removal of bearing cups.

CAUTION: REMOVE VALVE STEM BEFORE HEATING WHEEL HALF IF REMOVAL OF BEARING CUPS IS REQUIRED.

- (5) Bearing cups are shrunk fit into wheel halves and should not be removed unless replacement is required. To remove, heat wheel half in boiling water or oven to 121°C (250°F) maximum for 30 minutes, then drive cup out with fiber drift pin.

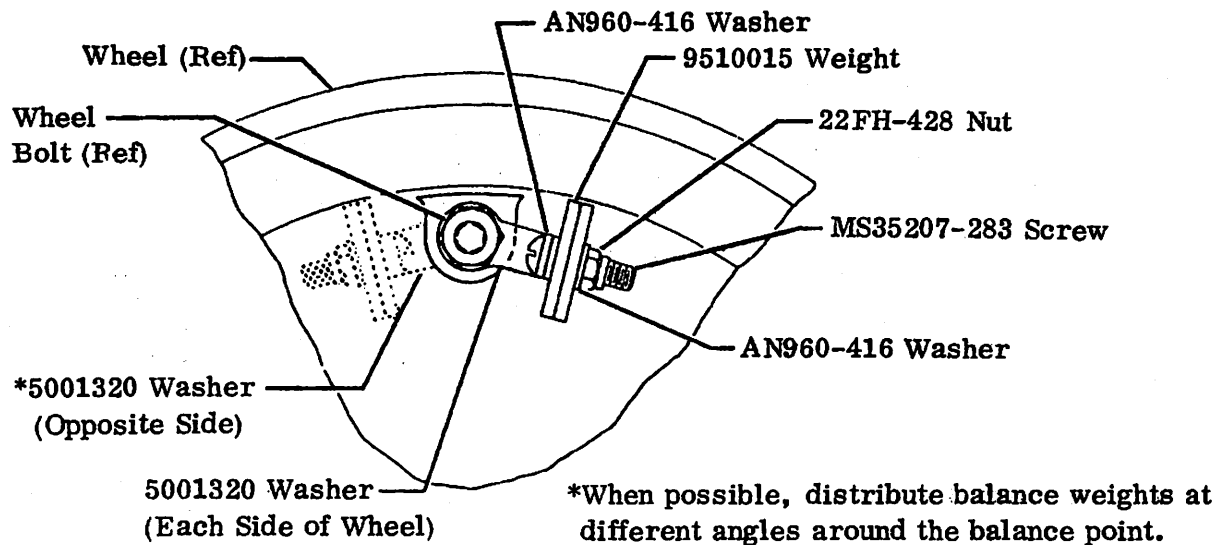


Nose Wheel Assembly
Figure 202 (Sheet 1 of 2)

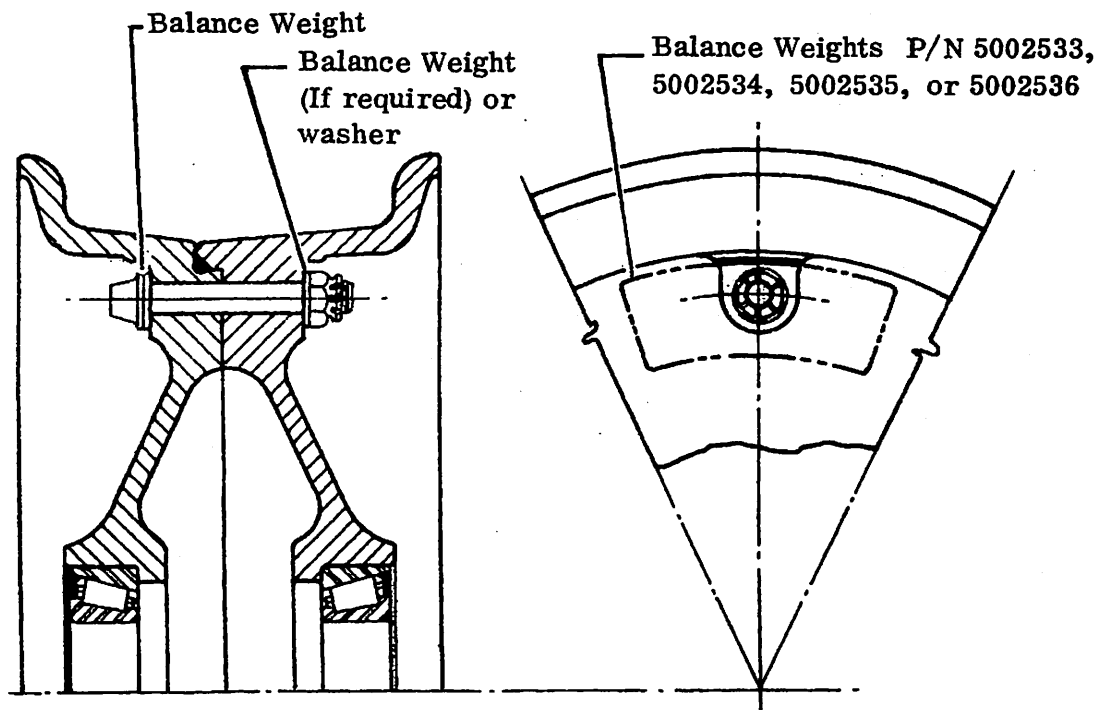
EFFECTIVITY: ALL

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METHOD A BALANCING



METHOD B BALANCING

Detail A

Nose Wheel Assembly
Figure 202 (Sheet 2 of 2)

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- (6) Clean all metal parts with Stoddard solvent and wipe dry with a clean, lint-free cloth.

CAUTION: DURING CLEANING AND PRIOR TO LUBRICATION, PERSONS HANDLING BEARINGS MUST WEAR POLY-ETHELENE (OR EQUIVALENT) GLOVES OR AN ANTI-PERSPIRANT HANDCREAM.

- (7) Clean bearings in a separate, clean container and dry with filtered, compressed air. To prevent bearing damage, do not spin when air drying or allow to soak in cleaning solution.
- (8) If wheel O-ring seal is to be reused, clean with denatured alcohol and dry with a lint-free cloth. Lay seal on a flat surface to prevent stretching while drying.

B. Inspect Nose Wheel Components

- (1) Visually inspect all metal parts for scratches, nicks, cracks, distortion, or other damage. Inspect bead seat radius for cracks or damage using a 10-power magnifying glass. Suspected areas should be thoroughly checked by fluorescent penetrant, dye check, or other suitable means. Replace all damaged parts.
- (2) Small nicks or scratches should be polished out with fine emery paper. Apply chemical film per MIL-C-5541 to bare metal and refinish with one coat of MIL-L-8585 zinc chromate primer and one coat of MIL-L-7178 aluminum lacquer. Refinish wheel register areas with a mist coat of zinc chromate primer only.
- (3) Check threaded parts for stripped, crossed, or damaged threads.
- (4) Inspect wheel O-ring seal for damage, distortion, and wear.
- (5) Check all bolts for deformation and cracks at head junction and first two threads adjacent to shank with a 10-power magnifying glass or by magnetic particle inspection.
- (6) Inspect self-locking nuts. Replace nuts after reusing 10 times or if nuts can be turned past self-locking position by hand.
- (7) Check bearing cups for looseness, rust, pitting, wear, spalling, Brinelling, heat discoloration, and nicks.
- (8) Inspect cone bearing roller surfaces and bearing cage for wear, rust, spalling, scratches, pitting, nicks, dents, distortion, and side and corner wear.
- (9) Replace all rubber parts at each wheel overhaul.

C. Assemble Nose Wheel (See figure 202.)

CAUTION: REMOVE VALVE STEM BEFORE HEATING WHEEL HALF IF INSTALLATION OF BEARING CUPS IS REQUIRED.

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- (1) To install bearing cups, heat wheel halves to 121°C (250°F) maximum for 1/2 hour. Chill bearing cups in dry ice for approximately 1/2 hour.
- (2) Remove wheel half from heat source and immediately install bearing cup by tapping in place with a fiber drift pin. Assure that cup is firmly seated in wheel hub.
- (3) Lubricate valve stem O-ring with a light coat of MIL-G-4343 grease and install on stem.
- (4) Clean valve stem hole in wheel half thoroughly with a clean cloth, install valve stem in wheel half, and torque 100 to 120 inch-pounds.
- (5) Wipe off bead seat area, wheel register, and tire beads with a clean, lint-free cloth dampened with isopropyl alcohol.
- (6) Visually inspect tire for damage with emphasis on bead area.
- (7) Lubricate wheel O-ring seal with a light coat of MIL-G-4343 grease.
- (8) Place machined grooved wheel half on a clean, flat surface and install wheel O-ring seal. Assure that seal is not twisted on installation. Reused seals should be installed as near as possible to their original position.
- (9) Place tire on wheel half so that red balance dot is centered on small weight pad of machined grooved wheel half. Do not disturb wheel O-ring seal.
- (10) Place other wheel half in tire with valve stem at red balance dot and align bolt holes.
- (11) Lubricate bolt and nut threads and bearing surfaces with MIL-T-5544 anti-seize compound.
- (12) Install countersunk washers with countersunk portion of washers facing bolt heads.
- (13) Install bolts and countersunk washers through wheel halves. Attach with washers and self-locking nuts.
- (14) Torque self-locking nuts to 120 inch-pounds in equal 20 inch-pound increments in a crisscross sequence around wheel. Do not use a power wrench.

WARNING: PLACE WHEEL ASSEMBLY IN AN INFLATION CAGE WHILE INFLATING TIRE. PERSONAL INJURY COULD OCCUR SHOULD THE TIRE EXPLODE DUE TO OVER INFLATION OR SHOULD THE WHEEL ASSEMBLY SEPARATE DUE TO IMPROPER ASSEMBLY.

- (15) Install valve core in valve stem and inflate tire sufficiently to seat tire beads.

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- (16) Check that wheel is balanced to 6-8 inch-ounces.
- (17) Pack cone bearings with MIL-G-81322 grease and spread an even coat of grease on cone bearing seals, bearing cups and wheel axle.

NOTE: Persons handling bearings during and after lubrication shall have clean hands coated only with the specified grease or an anti-perspirant hand cream.

D. Nose Wheel Balancing (See figure 202.)

- (1) Place assembled wheel and tire on bubble balancer.
- (2) Wheel and tire assembly must be balanced to 6 to 8 inch-ounces. If not, proceed with step 3.
- (3) If balance does not conform to these limits, assembly may be brought into balance as follows:

NOTE: Two wheel balancing methods are available to correct an out-of-balance condition. Do not attempt to intermix the two methods.

(4) Method A

WARNING: BEFORE ANY ATTEMPT IS MADE TO LOOSEN WHEEL NUTS, VERIFY THAT TIRE HAS BEEN COMPLETELY DEFLATED. IF VALVE IS EQUIPPED WITH A VALVE EXTENSION, REMOVE EXTENSION AND MAKE CERTAIN SECOND VALVE CORE HAS ALSO BEEN REMOVED.

- (a) Deflate tire as indicated in Approved Repairs.

CAUTION:

- REMOVE NUTS AND WASHERS WITH WHEEL ON A CLEAN FLAT SURFACE. TAKE CARE NOT TO NICK, SCRATCH, OR SCUFF PAINT OR PROTECTIVE FINISH OF WHEEL HALVES. DO NOT USE IMPACT OR POWER WRENCHES TO REMOVE WHEEL BOLT NUTS.
- ANY FASTENER ACCIDENTALLY TIGHTENED (INSTEAD OF LOOSENED) TO MORE THAN 120% OF RECOMMENDED TORQUE MUST BE DISCARDED.

- (b) Remove nuts, washers, bolts, and countersunk washers, one position at a time. Do not use a power wrench.
- (c) Replace washers on all wheel bolts with washers (5001320).

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- (d) Lubricate threads of wheel bolts and self-locking nuts and bearing surfaces of nuts, bolt heads, and washers (5001320), with MIL-T-5544 antiseize compound.

CAUTION: APPLYING UNEVEN TORQUE TO NUTS CAUSES BOLT FATIGUE. DO NOT USE IMPACT OR POWER WRENCHES TO TIGHTEN OR TORQUE WHEEL BOLTS AND NUTS.

- (e) Draw nuts up evenly until tight; then tighten nuts in a criss-cross pattern in equal increments of 20 inch-pounds to a final Lubtork of 120 inch-pounds.
- (f) Inflate tire enough to seat the beads, and then to 105 (± 5) psi.
- (g) Place wheel-tire assembly on a static balancer using wheel cones (or an adapter designed specifically for this purpose).
- (h) Determine balance weight point (180° from heavy point) and weight required.
- (i) Secure balance weights (9510015) at points where required. Assure that valve stem extension and/or valve stem cap are installed.

NOTE: Fine balance can be accomplished by rotating balance weight assemblies about wheel bolts.

(5) Method B

- (a) Deflate tire as instructed in Approved Repairs.

CAUTION: HANDLE BEARING CONES WITH EXTREME CARE. MANY AIRCRAFT BEARING FAILURES OCCUR AS A RESULT OF MISHANDLING DURING SERVICING.

- (b) Remove bearing cones.
- (c) Place wheel-tire assembly on a static balancer using wheel cones (or an adapter designed specifically for this purpose).
- (d) Determine balance weight point (180° from heavy point) and weight required.

NOTE:

- When two balance weights are required at any bolt location, place one half total required weight on each side of wheel.
- A maximum of two items per bolt are permitted: either a balance weight and washer or two balance weights.

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WARNING: BEFORE ANY ATTEMPT IS MADE TO LOOSEN WHEEL NUTS, VERIFY THAT TIRE HAS BEEN COMPLETELY DEFLATED. IF VALVE IS EQUIPPED WITH A VALVE EXTENSION, REMOVE EXTENSION AND MAKE CERTAIN SECOND VALVE CORE HAS ALSO BEEN REMOVED.

CAUTION: ANY FASTENER ACCIDENTALLY TIGHTENED (INSTEAD OF LOOSENED) TO MORE THAN 120% OF RECOMMENDED TORQUE MUST BE DISCARDED.

- (e) Loosen all eight nuts; then re-torque all eight nuts to 25 inch-pounds. Do not use a power wrench.
- (f) Remove nut, plain washer, bolt, and countersunk washer, from balance point.
- (g) Replace countersunk washer and/or plain washer on wheel bolt with balance weights where required.
- (h) Lubricate threads of bolt and nut, and bearing surfaces of bolt head, all washers, balance weight or weights, and nut with MIL-T-5544 antiseize compound.

CAUTION: INSTALL BOLT WITH HEAD ON INBOARD SIDE OF WHEEL.

- (i) Install lubricated balance weight on bolt with balance weight countersink against bolt head. Insert bolt through wheel assembly, slip washer over bolt, then secure with self-locking nut.

NOTE: If two balance weights are required at point of off-balance, a second balance weight is to be used in place of washer.

CAUTION: APPLYING UNEVEN TORQUE TO NUTS CAUSES BOLT FATIGUE. DO NOT USE IMPACT OR POWER WRENCHES TO TIGHTEN OR TORQUE WHEEL BOLTS AND NUTS.

- (j) Tighten nuts in a criss-cross pattern in even increments of 40 inch-pounds until final Lubtork value of 120 inch-pounds is obtained.
- (k) Inflate tire enough to seat tire beads and then to 105 (± 5) psi.
- (l) Recheck balance with valve stem extension and/or valve stem cap installed.

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MAIN GEAR WHEEL AND TIRE - MAINTENANCE PRACTICES

1. Removal/Installation

A. Remove Main Gear Wheel

CAUTION: CONE BEARINGS WILL COME LOOSE FROM WHEEL DURING WHEEL REMOVAL. DO NOT DROP OR DAMAGE CONE BEARINGS.

- (1) Place aircraft on jacks. If only a single main gear wheel is to be removed, running opposite tire up on a block (2 x 4) provides adequate clearance for removal.
- (2) Cut safety wire and remove screws from anti-skid generator drive cap.
- (3) Remove anti-skid generator drive cap.
- (4) Remove axle nut screws, axle nut, and key washer.
- (5) Remove wheel from aircraft.

B. Install Main Gear Wheel

- (1) Pack cone bearings with MIL-G-81322 grease and spread an even coat of grease on seals, bearing cups, and wheel axle.
- (2) Install inboard cone bearing on axle, then install wheel and outboard cone bearing.
- (3) Install key washer and axle nut.
- (4) Torque axle nut to 80 inch-pounds, then back off to 0 inch-pounds (without separating parts). Retorque axle nut to 20 inch-pounds. Nut may be advanced a maximum of 60° to obtain a locking position.
- (5) Install axle nut screws into slots and safety wire.
- (6) Install anti-skid drive cap and safety screws as required.
- (7) Inflate with dry air or nitrogen to pressure specified in Chapter 12. Assure that valve core is not leaking, then install valve cap.
- (8) Remove aircraft from jacks, or run tire off block if used.

2. Approved Repairs

A. Disassemble Main Gear Wheel (See figure 201.)

- (1) Remove cone bearings.

WARNING: DO NOT LOOSEN OR REMOVE BOLTS OR NUTS WITHOUT COMPLETELY DEFLATING TIRE. FAILURE TO COMPLY CAN CAUSE BODILY INJURY.

- (2) Remove valve cap, apply a tire deflator (Schrader No. 4277 or equivalent) to deflate tire, then remove valve core.

NOTE: Disassemble wheel on a clean flat surface to preclude damage to wheel halves.

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- (3) Break tire beads away from both wheel flanges by applying even pressure around entire sidewall as close to beads as possible.

CAUTION: ANY NUT ACCIDENTALLY TIGHTENED (INSTEAD OF LOOSENED) TO MORE THAN 120% OF RECOMMENDED TORQUE MUST BE DISCARDED.

- (4) Before loosening self-locking nuts, use a torque wrench to attempt to tighten nuts. Note torque at which nut begins to turn. Discard any nut which turns with less than 60 inch-pounds of torque. Remove self-locking nuts, washers, bolts, and countersunk washers.
- (5) Separate wheel halves, remove tire, and remove grease retainer tube.
- (6) Remove O-ring from register groove of inboard wheel half.

NOTE: Do not reuse old O-ring unless necessary. If old O-ring must be reused, mark position of O-ring in wheel half. Do not damage O-ring. Store O-ring on a clean flat surface; do not hang or stretch it.

- (7) Remove valve stem and O-ring from outboard wheel half.

CAUTION: REMOVE FUSIBLE PLUG BEFORE HEATING WHEEL HALF IF REMOVAL OF BEARING CUPS IS REQUIRED.

- (8) Bearing cups are shrunk fit into the wheel halves and should not be removed unless replacement is required. To remove, heat wheel half in boiling water or oven to 121°C (250°F) maximum for 30 minutes, then drive cup out with fiber drift pin. (Support wheel half at hub, not flange.)
- (9) Clean metal parts with Stoddard solvent and wipe with a clean, lint-free cloth.
- (10) Clean cone bearings in a separate container and dry with low-pressure, filtered air. Do not soak bearings in cleaning solution as seal damage may result. Do not spin bearings when drying with compressed air.
- (11) Place reusable wheel O-ring seal on a flat surface, clean with denatured alcohol, and dry with a lint-free cloth. Do not hang or stretch O-ring.

B. Inspect Main Gear Wheel Components

- (1) Visually inspect metal parts for cracks, distortion, or other damage. Inspect bead seat radius for cracks or damage using a 10-power magnifying glass. Suspected areas should be checked by fluorescent penetrant, dye check, or other suitable method.

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TEMPORARY REVISION NO. 32-9

MANUAL AFFECTED: Gates Learjet Model 25 b/c/d/f Maintenance Manual

FILING INSTRUCTIONS: Insert adjacent to 32-42-00, page 202, dated Apr 30/82.

REASON: Eddy Current inspection of wheel bead seat radius.

INSTRUCTIONS: Change item (1) under B. Inspect Main Gear Wheel Components to read as follows:

B. Inspect Main Gear Wheel Components

- (1) Visually inspect all parts for scratches, nicks, cracks, distortion, corrosion, or other damage. Suspected areas shall be thoroughly checked by fluorescent penetrant, dye check, acetic acid etch, or other suitable means. Perform Eddy current inspection of wheels in bead seat radius. (Refer to 32-40-00, step 1.C. of Inspection/Check.)

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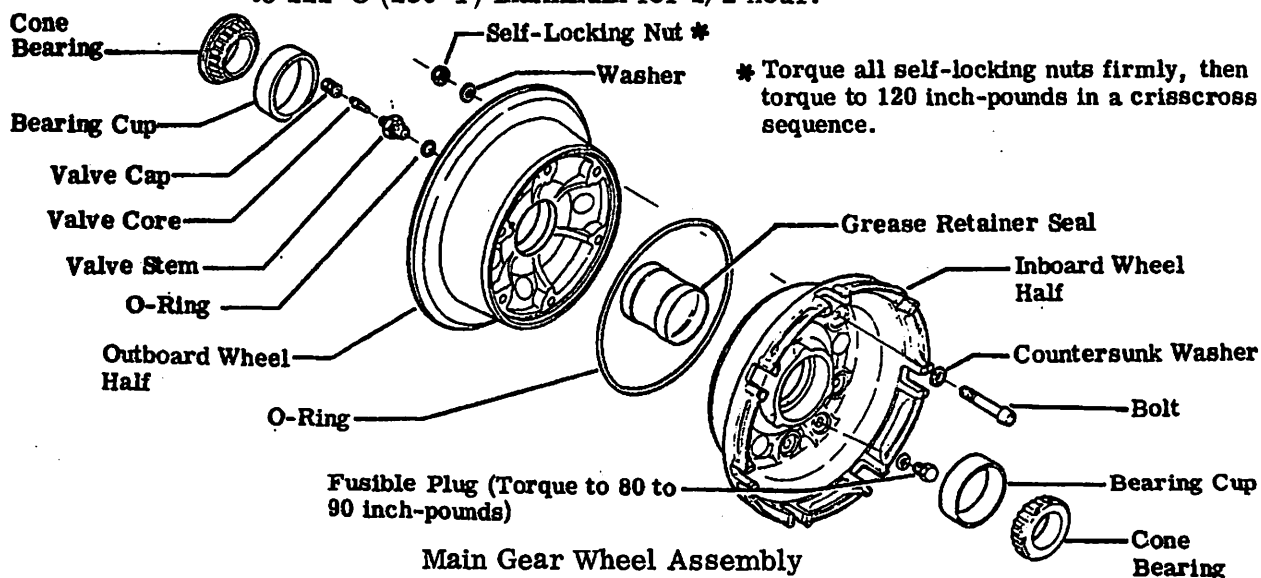
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- (2) Check threaded parts for stripped, crossed, or damaged threads.
- (3) Inspect O-ring seals for damage, distortion, or wear.
- (4) Check bearing cups for looseness, nicks, pitting, and rust.
- (5) Inspect cone bearings for roller wear, cage wear, rubber seal deterioration, nicks, pitting, rust or other damage.
- (6) Check bearing seals for cuts, nicks, or wear.
- (7) Inspect bolts for deformation, cracks at head junction, and first two threads adjacent to shank with a 10-power magnifying glass or by magnetic particle inspection.
- (8) Inspect self-locking nuts. Replace nuts after reusing 10 times or if nuts can be turned past self-locking position by hand.
- (9) Inspect fusible plugs for pitting, rust, and gasket damage.
- (10) Inspect disc keyway liners for wear and tightness. If distance across key slot between liners exceeds 0.690 inch, replace all liners in wheel.
- (11) Replace cracked or distorted castings and parts.
- (12) Polish out small nicks and blemishes with fine sandpaper.
- (13) Apply chemical film per MIL-C-5541 to bare metal areas, allow to dry thoroughly, and refinish with one coat of zinc chromate primer and one coat of aluminum lacquer.
- (14) Refinish bolt bosses and wheel register area (mating surfaces of wheel halves) with a mist coat of zinc chromate primer only.
- (15) Replace all rubber parts at each wheel overhaul.

C. Assemble Main Gear Wheel (See figure 201.)

CAUTION: REMOVE FUSIBLE PLUGS BEFORE HEATING WHEEL HALF IF INSTALLATION OF BEARING CUPS IS REQUIRED.

- (1) To install a bearing cup; heat wheel half in boiling water or an oven to 121°C (250°F) maximum for 1/2 hour.



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- (2) Chill bearing cups in dry ice for approximately 1/2 hour.
- (3) Remove wheel half from heat source and immediately install bearing cup by tapping in place with a fiber drift pin. Assure that cup is firmly seated in wheel hub.
- (4) Lubricate valve stem O-ring with a light coat of MIL-G-81322 grease and place on stem.
- (5) Install valve stem into wheel half and torque to 180 to 200 inch-pounds.
- (6) To install fusible plugs, lubricate O-rings and threads with a light coat of MIL-G-81322 grease.
- (7) Install O-rings on fusible plugs. Then install fusible plugs in wheel half and torque to 80 to 90 inch-pounds.
- (8) Clean wheel register, bead seat area, and tire beads with a clean cloth dampened with isopropyl alcohol.
- (9) Visually check tire and tire beads for damage before installation.

NOTE: Inspect tire tread visually or with a depth gage. A tire should be replaced when the tread has worn to the base of any groove at any location. This can be considered safe as long as the cord below the tread is not exposed. Use of recapped tires on main gear wheels is not approved.

- (10) Lubricate wheel O-ring seal with light coat of MIL-G-81322 grease.

CAUTION: REUSED WHEEL O-RING SEALS SHOULD BE INSTALLED IN THEIR ORIGINAL POSITION. ASSURE THAT SEAL IS NOT TWISTED OR DISTURBED DURING INSTALLATION.

- (11) Place inboard wheel half on a clean flat surface and install wheel O-ring seal.
- (12) Insert grease retainer tube into hub of inboard wheel and seat it against bearing cup.
- (13) Place tire on inboard wheel half with red balance dot equally spaced between two bolt holes.
- (14) Place outboard wheel half in tire with valve stem located at red balance dot and align bolt holes.
- (15) Using MIL-T-5544 antiseize compound (undiluted), lubricate bolt and nut threads, and bearing surfaces of nuts, bolt heads, and washers.
- (16) Install countersunk washers with the countersunk areas facing bolt heads.

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CAUTION: INSTALL BOLT HEADS ON STRUT SIDE. THERE MUST BE A COUNTERSUNK WASHER UNDER EACH BOLT HEAD AND A FLAT WASHER UNDER EACH NUT.

- (17) Install bolts through wheel halves and secure with washers and self-locking nuts.

CAUTION: DO NOT USE AN IMPACT OR POWER WRENCH TO TIGHTEN OR TORQUE WHEEL BOLTS OR NUTS.

- (18) Tighten nuts evenly in a crisscross pattern in equal increments of 40 inch-pounds to final torque of 120 inch-pounds.

WARNING:

- PLACE WHEEL IN AN INFLATION CAGE FOR INITIAL INFLATION BEFORE SEATING BEADS. TIRE AND/OR WHEEL FAILURE MAY OCCUR, CAUSING INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT IF THE TIRE IS OVERINFLATED.
- DO NOT INFLATE TIRE TO FULL OPERATING PRESSURE UNTIL MOUNTED ON AIRCRAFT.

- (19) Install valve core in valve stem and inflate tire sufficiently to seat tire beads.

- (20) Soap check installed parts for leakage. If leakage exists, deflate tire, replace parts, and reservice tire.

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BRAKE SYSTEM - DESCRIPTION AND OPERATION

1. Description

- A. The hydraulic brake system includes four power brake valves located directly below the rudder pedals, six shuttle valves (two under the floorboards and two in each main gear wheel well), two parking brake valves located under the right floorboards between frames 10 and 11, multiple disc brake assemblies, and four hydraulic fuses (one in each brake line).
- B. The hydraulic brake system incorporates an anti-skid braking system which assures the proper amount of brake pressure to each brake. The anti-skid system is explained in 32-45-00.
- C. The parking brake system consists of one parking brake handle (mounted on the LH side of the pedestal) that operates two parking brake valves installed in the main hydraulic brake lines.
- D. Component Description
 - (1) Each multiple disc brake assembly consists of two rotating discs which are driven by radial tangs engaging keyway slots in the wheel. They rotate on either side of a stationary disc keyed to the torque tube. The stationary disc, pressure plate, and back plate provide rubbing surfaces for the rotating discs. The back plate, pressure plate, and torque tube are bolted to the brake housing. The housing contains five brake pistons, four return springs and automatic adjustment units, and an inlet and bleeder port. The piston cavities are interconnected by drilled passages to provide simultaneous piston actuation with equalized pressure on each piston. There is one brake assembly for each main landing gear wheel.
 - (2) Power brake valves are installed forward of each rudder pedal and control hydraulic pressure to the brake assemblies. There are three ports on each valve assembly: system pressure, return, and brake pressure. The valves are manually operated by linkages connected to the rudder pedals.
 - (3) Hydraulic fuses are located in each wheel well adjacent to the anti-skid valves and prevent the loss of system pressure and fluid due to brake line failure. The fuses close when 5.5 (± 1.1) fluid ounces of hydraulic fluid pass through the fuse.
 - (4) The two parking brake valves are interconnected and are manually operated. The parking brake valves are installed in the brake lines and are normally open. When pressure is applied to the brake assemblies and the parking brake valves are actuated, the valves will block off the brake lines, retaining brake pressure on the brake assemblies.

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2. Operation (See figure 1.)

- A. Depressing the brake pedals opens the power brake valves and directs hydraulic fluid from the nose gear down line through the shuttle valves, anti-skid control valves, and brake fuses to the corresponding brakes. An integral braking system is incorporated to prevent the wheels from spinning as the gear is retracted. On retraction, the return landing gear hydraulic fluid passes through a restrictor in the return line to the reservoir. This return line is common to the brakes, and the restrictor creates approximately 100 to 180 psi back pressure in the brake system to snub the wheels during gear retraction. After the gear has retracted, the back pressure returns to normal and the brakes are released. In case of hydraulic system failure, pressure for brake application is provided by the emergency air system. (Refer to 32-44-00.)
- B. The multiple disc brake assembly is actuated by hydraulic pressure. When brakes are applied, pressure enters the housing and acts equally on each piston. Piston movement presses the pressure plate and the rotating and stationary discs against the back plate to produce the braking action. When hydraulic pressure is released, the brake return springs return the pressure plate and piston to their original position. Automatic adjustment of brakes to compensate for brake wear is accomplished as follows. As the brakes wear, the return pins and tubes are pulled through grips by the pressure plate movement. This small movement between pin and grip is equal to the combined wear of all braking surfaces. Thus, when the brake return springs return the pressure plate to the released position, the built-in clearance between the pressure plate and discs remains at a preset constant.
- C. Depressing the brake pedals and pulling the parking brake control handle sets the parking brake. Pulling the parking brake valve control handle closes the parking brake valves and stops the release of hydraulic pressure when the brake pedals are released. With the parking brake handle pulled, the anti-skid system is disconnected to assure that both anti-skid control valves stay closed to preclude pressure bleed-off. Moving the parking brake lever to the off position releases the brake pressure.